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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF THE NETHERLANDS (2002)

THE LAND

Area (1 000 km^2):	
Total	42
Agricultural land	24
Woodland	3

Major cities, 1st January 2002 (thousand inhabitants):	
Amsterdam	736
Rotterdam	599
The Hague	458

THE PEOPLE

Population (thousands)	16 105
Number of inhabitants per km^2 of land	475
Net natural increase (thousands)	57

Employment (thousands, full-time equivalents):	
Total	6 541
Agriculture, fishing and forestry	236
Industry	971
Construction	489
Other activities	4 845

PRODUCTION

Gross domestic product, 2002 (mln Euro)	444 649
GDP per head (EUR)	27 609
Gross fixed investment:	
Per cent of GDP	16
Per head (EUR)	4 529

Origin of GDP (%):	
Agriculture, fishing and forestry	2
Industry	19
Construction	5
Other	74

THE PUBLIC SECTOR

Per cent of GDP:	
Public consumption	20
Current receipts	45
Current disbursements	47

Composition of Parliament (number of seats):	
Christian democratic appeal	44
Labour Party (PvdA)	42
Popular Party for freedom and democracy	28
Socialist Party	9
List Pim Fortuyn	8
Other	19
Total	150

Last general election: January 2003

FOREIGN TRADE

Exports of goods and services (% of GDP)	57
Main exports, 2001 (% of total merchandise exports):	
Food, beverages and tobacco	18
Energy	6
Chemicals	15
Machinery and transport equipment	34

Imports of goods and services (% of GDP)	52
Main imports, 2001 (% of total merchandise imports):	
Food, beverages and tobacco	10
Energy	7
Chemicals	11
Machinery and transport equipment	39

THE CURRENCY

Monetary unit: Euro

Currency units per US$, average of daily figures:	
Year 2002	0.94

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

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The economic situation and policies of the Netherlands were re-iewed by the Committee on 5 April 2004. The draft report was then revised in the light of the discussions and given final approval as the agreed report by the whole Committee on 29 April 2004.

•

The Secretariat's draft report was prepared for the Committee by David Carey, Hubert Strauss, Kristel Buysse, Maria Maher, Douglas Sutherland and Mike Wise under the supervision of Andreas Wörgötter.

•

The previous Survey of the Netherlands was issued in January 2002.

Executive summary

The economic downturn in the Netherlands, like the boom that preceded it, has been more marked than in most other European countries. This is partly attributable to greater exposure to international trade, especially with the US, and to greater sensitivity to international stock price movements. But there are also some structural policies – pension fund regulation, housing and wage setting institutions – that have undermined economic efficiency and contributed to the amplitude of the cycle. Furthermore there has been a marked deterioration in public finances, part of which is structural. Longer term growth trends too are unfavourable. In view of these developments, the main challenges are to reform those policies that have reduced efficiency and increased volatility, put public finances on a sustainable path, and, most importantly, increase trend growth by increasing both employment and productivity.

Reform those policies that have reduced efficiency and increased volatility

Tough zoning restrictions should be eased to make housing supply more price elastic and hence reduce increases and volatility in real house prices. Phasing out high tax subsidies for owner occupiers could also partly reduce volatility and, more importantly, would increase the efficiency of the tax system. The vigorous tightening in pension fund regulation greatly reduces the risk of future solvency crises and the associated sharp increases in pension fund contributions, which depress economic activity and raise the efficiency costs of these contributions. It also should improve pension fund transparency, helping members to plan effectively for retirement. Relaxing strict EPL could speed up labour market adjustment, bringing the economy back to trend more quickly.

Put public finances on a sustainable path

While the economic downturn has contributed to a sharp decline in the general government budget balance in recent years, there has also been a significant structural deterioration. The government's determination to put public finances on a sustainable path by 2007 is welcome as this should minimise the efficiency costs of taxation over time and treat future generations fairly. Careful monitoring will be

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required to ensure that the fiscal targets set in the next coalition agreement, starting in 2007, are compatible with sustainable public finances. In the meantime, the government needs to ensure that its consolidation measures are fully implemented.

Increase labour force participation

The planned reform of the disability scheme is vital for increasing labour force participation. It should significantly reduce flows into the full disability scheme and strengthen work incentives in the partial disability scheme. The partial disability reform should, however, be strengthened to discourage misuse of the scheme to subsidise reduced working time or early retirement. This would entail maintaining incentives for using the remaining earnings capacity and reducing the duration of first-stage (and unemployment) benefits for older workers, respectively. Tax incentives for early retirement schemes should be as fully as possible ended in 2006, as planned. An easing in EPL, as suggested above, could facilitate employment of groups with relatively low participation rates. Barriers to increased hours worked, such as regulations that unnecessarily inflate childcare costs and reduced working time in collective agreements and a high tax burden on labour, should be reduced.

Increase productivity growth

Competition in product markets, which is vital for raising productivity growth, has been strengthened through the Competition, Deregulation and Legislative Quality project (MDW) and the creation of a competition authority (NMa). Competitive pressures should be strengthened by giving the NMa greater investigative and sanctioning powers, as planned, and the power to propose structural remedies in appropriate cases. Unwarranted anti-competitive practices among the liberal professions should also be eliminated and barriers to competition in the network industries should be removed, notably through greater unbundling and privatisation of retail activities. The planned reduction in the administrative burden will also help to strengthen competition, by reducing barriers to business start-ups and the expansion of small businesses, as well as lowering business costs. Tertiary education policies could contribute more to productivity growth by greater reliance on private funding and by making public funding more dependent on performance. Strengthening the corporate governance framework, as planned, will also increase productivity.

Increase cost efficiency in sustainable development

In the area of sustainable development, the government should seek to reduce the costs of climate change policies by equalising marginal abatement costs and should strengthen the use of economic instruments to reduce water pollution and manage natural resources.

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Assessment and recommendations

The main policy challenges are to encourage stronger and more stable growth and to put public finances on a sustainable path

Economic growth has fallen more sharply in the Netherlands since 2001 than in most other European countries. Domestic factors that boosted growth in the late 1990s – wealth effects from housing and stock markets and pension fund developments – have since turned around, accentuating the effects of the international business cycle. A number of institutional arrangements – residential zoning laws and tax subsidies for owner occupiers, plus weaknesses in pension fund regulation – have contributed to these developments as well as undermining economic efficiency. Output has thus fallen from 3 per cent above potential in 2000 to 2½ per cent below potential in 2003. While the severe economic downturn has taken its toll on public finances, there has also been a marked deterioration in the structural budget balance. From a longer term perspective, economic performance has also been disappointing in some respects. Trend growth in GDP per capita has slowed to 1½ per cent, reflecting lower growth in the labour force participation rate and in hourly labour productivity. Productivity growth is low both from a historical perspective and by international comparison. The main policy challenges are thus to: *i*) reform policies that undermine economic efficiency and accentuate cyclical swings in activity; *ii*) put public finances on a sustainable path; *iii*) increase labour force participation; and *iv*) increase productivity growth.

A number of factors that boosted growth in the late 1990s have turned around

As regards the strong cyclical volatility, the relative severity of the economic downturn in the Netherlands is attributable to a number of factors that boosted growth in the late 1990s but have since ceased to do so or indeed have turned negative:

– Wealth effects from increases in real house prices are estimated to have boosted the annual growth rate by

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½ percentage point over 1997-2000, falling to ¼ percentage point over 2001-03. Stock market wealth effects also boosted the growth rate by ½ percentage point over 1997-2000, while the subsequent decline may have reduced it by a similar amount.

– The stock market boom also masked the deteriorating underlying solvency position of pension funds and even encouraged some funds to reduce contribution rates. When stock prices fell, pension funds were obliged to increase contribution rates from below the cost-covering level to above it in order to restore solvency, slowing economic growth in the process.

– The boom in the late 1990s also set in motion forces – notably, a loss of competitiveness – to slow the economy and hence bring it back to trend. Initially, this loss was masked by depreciation of the euro, but once it started appreciating, there was a large loss of competitiveness and deterioration in export performance.

– Movements in the euro exchange rate have had a greater effect on the Netherlands than most other euro area economies owing to its greater openness and exposure to trade with the United States.

– Finally, monetary conditions were too loose in the Netherlands in the late 1990s but now seem to be somewhat too tight.

Economic recovery should build over 2004-05

With some of these factors now starting to unwind, the Dutch economy should benefit from the recovery in the world economy. Exports are expected to pick up in the coming months and gather some momentum as the effects of the euro appreciation gradually fade away and as wage moderation leads to a decline in relative unit labour costs. Better sales prospects, the need for modernising equipment after years of shrinking investment, increasing capacity utilisation and low interest rates will help business investment to grow again beginning in the first half of 2004. Private consumption, however, is the weak spot in the outlook. With pension contribution rates rising, real wages stagnating or even falling and no positive housing wealth effects expected, consumption expenditure growth will only

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resume by the end of 2004 in tandem with improving labour market prospects. GDP growth should rise to slightly above the potential rate (around 2 per cent) and unemployment should stabilise by the beginning of 2005. With output levels remaining far below potential, wage restraint on track and the delayed effects of appreciation of the euro, inflationary pressures will be absent. The main risk to the forecast is of a further appreciation of the euro, which would hamper recovery throughout Europe.

Regulatory and tax reforms in the housing market would increase efficiency and contribute to reducing volatility

The large increases in real house prices in the late 1990s followed by the sharp slowdown continues the pattern of recent decades in which there have been large increases in house prices in relation to disposable incomes and high volatility in real house prices by international comparison, amplifying business cycles:

– The low price elasticity of housing supply has contributed to both large increases and high volatility in house prices. Housing supply has not been very responsive to demand owing to tough zoning rules, which have restricted the supply of residential building sites, to the growing quantity and complexity of building regulations and to the costliness of acquiring private building sites. T*he government should ease these constraints, as it is considering doing*.

– High tax subsidies for owner-occupied housing, especially mortgage financed, and a change in banking policies in the 1990s that increased the amount that could be lent to the increasing number of two-income households, have magnified swings in demand and may have contributed to price volatility. More importantly, the tax subsidies have reduced economic efficiency, notably by raising the excess burden of taxation, and are unlikely still to achieve equity objectives. While the government has taken several steps to limit mortgage interest deductibility when owner occupiers withdraw housing equity, *it should go much further, terminating tax subsidies for additional purchases of owner-occupied housing* (i.e., *grandfathering tax subsidies on existing mortgage loans*). This would also reduce the need for rental subsidies (including through rent regulation) to attenuate the effect of high property prices on rents, reducing the efficiency costs of taxation while still achieving social equity objectives.

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Pension fund regulation is being tightened to reduce the risk of insolvency…

The increase in pension fund contribution rates from below the cost-covering level to above it has increased the "tax-like" element of contributions and hence the economic efficiency cost (*i.e.*, excess burden) of pension arrangements and has accentuated the economic downturn. A new supervisory framework in the context of a new Pension Act that will come into effect in January 2006 will reduce the risk of another large solvency crisis occurring. The Principles of the Financial Supervision of Pension Funds that came into effect in March 2004 (and will be elaborated in the Pension Act) will require adequate buffers and should induce more prudent behaviour. For an average pension fund, the Principles require it to restore capital coverage of its nominal (*i.e.* not indexed) liabilities to 130 per cent within a period of 15 years. This solvency ratio is compatible with the requirement that pension funds should be able to honour their pension liabilities one year ahead with a probability of 97.5 per cent. These Principles should keep the Dutch pension system sustainable.

… and increase transparency

One of the casualties of the solvency crisis has been indexation of accrued entitlements and pensions. Pension fund boards, which are governed by employers and employees, have judged that the contribution rate increases required to maintain full wage indexation would be unduly harmful to labour market performance. Accordingly, they prefer only to commit to nominal pension liabilities, with price or wage indexation usually conditional on fund performance. In addition, they have agreed to a major shift from final salary to average salary schemes in recent years. This has markedly increased trustees' discretion to reduce pension replacement rates if fund returns are poor as not only pensions, but also accrued entitlements may end up being only partially indexed. *Transparency on these issues should be improved* as members need to be able to appreciate their significance and to make supplementary arrangements if necessary. This improvement of transparency should occur now that the Principles have come into effect as they stipulate stringent transparency conditions if a pension fund wants to grant indexed pensions rights while solvency requirements are not fully met. The emphasis in the Principles on transparency regarding

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indexed pension rights will contribute to making Dutch pension arrangements more understandable for pensioners and the general public.

Budget consolidation is moving public finances towards a sustainable path

The second main policy challenge is to put public finances on a sustainable path. The general government budget balance has deteriorated markedly, from a surplus of 1.5 per cent of GDP in 2000 to an estimated deficit of 3.2 per cent of GDP in 2003. Some 2.7 percentage points of this 4.7 percentage point deterioration appears to be cyclical, with the structural balance having declined from broad balance in 2000 to a deficit of 2.2 per cent of GDP in 2003. In order to get public finances back on a sustainable path – defined as one on which the future costs of government programmes can be financed at constant tax rates – the government has announced four large consolidation packages since July 2002 with a cumulative effect of € 20 billion (3.8 per cent of GDP) by 2007, of which about € 15½ billion relate to expenditure reductions. The budget deficit is projected to fall to 0.6 per cent of GDP in 2007. This budget consolidation should also raise medium-term growth prospects as it focuses on expenditure reductions that strengthen work incentives. To guard against the recurrence of any fiscal deterioration in future years, the government has also announced a strengthening in the medium-term budget framework with windfalls being earmarked for debt reduction (unless otherwise decided by the council of ministers in the case of expenditure windfalls) and revenue shortfalls for new debt as long as there is no risk of breaching the limits imposed by the Stability and Growth Pact.

Careful monitoring will be required to safeguard future fiscal sustainability

These measures are projected to raise the underlying EMU balance to 1.1 per cent of GDP in 2007, which is judged by the Netherlands Bureau of Economic Policy Analysis (CPB) to be sufficient to put public finances on a sustainable path. The difference between the actual budget balance and the underlying EMU balance of 1.7 percentage points of GDP is attributable to the future beneficial impact of economic policies on potential growth and the cyclical component of the balance (together amounting to 1.2 per cent of GDP) and to pension fund contributions temporarily

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being above their cost-covering level (0.5 per cent of GDP). There are, however, considerable uncertainties surrounding estimates of the structural balance needed for public finances to be on a sustainable path. On the one hand, a higher surplus could be required as the net fiscal costs of second-pillar pension schemes may have been underestimated by assuming unrealistically high real rates of return. On the other hand, the level of disability beneficiaries in the projections does not take into account the effect of recent reforms, resulting in an overestimate of the balance required for fiscal sustainability. These and other factors will be taken into account when the CPB updates its fiscal sustainability projections in 2006. *Careful monitoring will be required to ensure that the fiscal targets set in the next coalition agreement, starting in* 2007, *are compatible with sustainable public finances. In the meantime, the government needs to ensure that its consolidation measures are fully implemented.*

Steps are being taken to stem the decline in trend growth in GDP per capita

The greatest challenge of all is presented by the need to reverse the growth slowdown by increasing labour force participation and productivity growth. Trend growth in GDP per capita has slowed from around 2 per cent at the beginning of the 1990s to around 1½ per cent in recent years. This decline is attributable equally to lower growth in labour force participation and in labour productivity. The rise in the female participation rate, which entirely accounts for growth in the participation rate, has slowed as the effects of reforms to increase participation wane and as the rates attained reach high levels, making further increases more difficult to achieve. The decline in trend growth in labour productivity per hour (to around 1.2 per cent) reflects a slowdown in trend multifactor productivity (MFP) growth. The authorities are actively pursuing policies to reverse these negative trends. Social transfer programmes and the tax system are being reformed to increase incentives to work. To increase productivity growth, policies are being developed to strengthen entrepreneurship and competition, foster innovation and research, and improve human capital.

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The planned disability pension reform is vital for increasing labour force participation

The largest source of labour that could potentially be activated is disability beneficiaries, who in 2001 numbered slightly less than 1 million (about 9 per cent of the working age population), of whom about 200 000 work part time. Tougher reintegration obligations before applying for disability benefit (DB) and the anticipation of future reforms together with the effects of the current economic downturn contributed to a 25 per cent reduction in inflows into the DB scheme in 2003. The stock of disabled is to be re-tested according to new, stricter criteria in the run-up to the introduction of the new DB scheme, planned for January 2006. In the new scheme, fully disabled persons will only receive a full DB if their disability is assessed to be permanent while it is planned to strengthen work incentives for the partially disabled. In this regard it is important that the requirement to use the residual earnings capacity to the full be implemented, as planned. Strict testing of disability claimants is crucial for the success of the new partial disability scheme (WGA) since there remains a risk of self-selection. *To strengthen the reform of partial* DB, *the government should*: i) *reduce the duration of first stage* (*and of unemployment*) *benefits, which effectively rise with age, for persons in the second half of their career to discourage the continued use of the scheme as a route to early retirement*; *and* ii) *prevent topping up of partial* DB *to ensure that the risk of self-selection is not higher than in the current situation*.

Reintegration pressures are being raised but work incentives need to be strengthened…

Making municipalities fully financially responsible for the payment of welfare benefits from the beginning of 2004 should increase their incentives to favour reintegration over benefit payment. This is because they have to pay back unused money from the reintegration budget they receive but keep unused money from the social welfare benefit budget. *Efficiency and transparency in the reintegration market should be further improved by systematic evaluation of private providers' value added. Moreover, incentives for clients to co-operate actively should be strengthened by giving reintegration firms the possibility of initiating benefit sanction procedures. Poverty traps, which remain substantial despite several recent measures, should be addressed by reducing the level of welfare benefits. The government should also strengthen work incentives by pushing through with its plans to increase the minimum work record required for access to unemployment benefits.*

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… and working time increased

Labour force participation of persons above age 55 is currently below the Lisbon target and well below the level of the best performing countries in the OECD, suggesting that there remains considerable scope for improvement. The termination of second-stage unemployment benefits and the reinstatement of job-search requirements for the older unemployed as of 2004 are welcome steps in the right direction. *The government should also push ahead with its plan to end tax-incentives for early-retirement schemes by* 2006. GDP per capita could also be increased by raising the annual number of hours worked per employee, which is the lowest in the OECD area, partly on account of widespread female part-time employment. *The government should reduce barriers to increased hours worked, such as regulations that unnecessarily inflate childcare costs and reduced working time in collective agreements, while aiming at reducing the tax burden on labour in the medium term*. The government plans to introduce the "life-cycle scheme", a tax-favoured arrangement for employees to save up to 2.1 years of salary to be used at any time in their career, with the details subject to negotiations with the social partners. *The government should make the tax subsidies for employees taking their saved salary just before retirement conditional on continuing to work part-time so as to prevent the scheme from being used as an alternative route to early retirement.*

Easing strict employment protection legislation would facilitate the employment of groups at risk

The measures to increase labour force participation should be complemented by an easing in strict employment protection legislation (EPL), which penalises groups with low participation rates, especially as there are likely to be more job seekers from these groups following the reforms to social transfer schemes. Loosening EPL could also reduce the significant lag in labour-market adjustment to economic conditions. In particular, the government should reduce the cost of dismissals, notably by lowering requirements for third-party authorisation and severance pay.

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Competition law has increased competitive pressures and is set to become more effective with planned reforms

Such activation and employment creating measures need to be complemented by measures to increase productivity. Enhancing competition in product markets is probably the most important means of improving relatively poor productivity growth, especially in non-traded services. The Netherlands has a history of widespread anti-competitive practices but the authorities have been vigorously combating these in recent years. Competition law was strengthened and brought into conformity with EU norms in w1998 and a new competition authority, the NMa, was created. Initially, the NMa focused on a mountain of applications for exemptions from the new *Competition Act*, reflecting the importance of corporatist arrangements in the Netherlands, but it has since been able to turn to driving out anti-competitive practices. It has been particularly active in the construction sector, where a parliamentary inquiry found that there was widespread bid rigging. Following an evaluation of the *Competition Act* and experience in implementing it, *the government rightly plans to give the* NM*a greater investigative and sanctioning powers*, including higher fines for non co-operation. T*he authorities should also give the* NM*a power to impose structural remedies in appropriate cases*, a power that the European Commission will have as from May 2004. The NMa's credibility will be further strengthened when plans to make it formally independent are realised.

Increasing competition in professional services and retail distribution remains a priority

As in many other countries, competition in professional services has long been weak owing to self-regulation by professional bodies and entry barriers that appear to be high. While the authorities increased competition in these sectors over 1994-2002 through the Competition, Deregulation and Legislative Quality project (MDW), they remain in transition. Restrictions on price competition, advertising and permitted business structures are still widespread. T*he government should eliminate unwarranted anti-competitive practices in the light of the re-examination of regulations and laws governing the liberal professions scheduled for this year, as planned*. In retail distribution, planning restrictions in the Netherlands have inhibited the entry and expansion of large-format operators, which has impeded productivity growth in the sector. While Dutch planning and zoning restrictions have been successful in protecting small and specialist retailers located in town

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centres, they also may distort competition and offer considerable incumbency advantages to established retailers, thus creating and maintaining rents. The government is decentralising decision-making power concerning the location of large retail outlets to lower levels of government so that spatial policies can be better customised to local and regional needs and to facilitate dynamism in the retail sector at the local and regional levels. H*owever, the government should keep monitoring this process to ensure that local governments are not unduly responsive to incumbent interests.*

More needs to be done to increase competitive pressures in network industries

Market power by incumbent firms remains a problem in network industries. This is why economic oversight of network industries for the foreseeable future cannot just rely on the application of competition law but requires stronger regulatory oversight promoting competition. Formal independence of the sectoral regulators within the NMa, which will occur when it gains independence, should increase their credibility. In the electricity sector, the major barrier to competition is the absence of ownership unbundling of supply and distribution activities. The government recently decided to unbundle ownership of supply and distribution activities. L*ocal governments should privatise their retail activities after the electricity and gas distribution networks have been split off*. In this transitory period the ownership of the networks will be in public hands, and privatisation of the networks can be considered after network quality has been safeguarded through adequate regulatory measures. T*he authorities should also increase interconnection capacity subject to satisfactory co-ordination with networks in neighbouring countries*, as planned, as the lack of such capacity constrains competition. In the roll-out of broadband internet infrastructure, *the government should decline requests from the incumbent telecoms operator* (KPN) *for financial assistance to take "fibre optics to the home" so as not to turn the market effectively into a monopoly.*

Consumer grievance mechanisms should be improved

The Netherlands is only one of three EU countries without public enforcement of consumer protection rights and protection against unfair commercial practices. S*ectoral* D*isputes* C*ommittees provide an easily accessible dispute resolution mechanism, but should be improved as they do not exist in all sectors and are sometimes slow to settle disputes.* T*he authorities should consider creating small claims tribunals*, which would provide

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consumers with an inexpensive route to settle grievances with suppliers. Before summer, the government aims to set up a consumer protection enforcement agency for collective consumer interests. Also a one-stop-shop will be introduced for individual consumer complaints.

Reducing the administrative burden is a priority

Reducing the administrative burden on firms would also help to strengthen competition by facilitating business start-ups and the growth of small businesses, as the burden falls most heavily on them; it would also increase productivity directly by lowering business costs. The government has committed itself to reducing the current administrative burden by one quarter by the end of 2007. As almost half of the administrative burden is caused by European legislation, the Dutch government will be seeking a reduction of this part. *The government has already launched many initiatives in this area and should push strongly to reach its objective.*

Interaction between basic research and business needs to be improved in the Dutch innovation system

Another plank in the government's strategy to increase productivity growth is to strengthen the national innovation system. The major problems in this system are a relatively low level of private sector R&D and inadequate interactions between science/higher education and industry at a time when such interactions become an even more important driver of knowledge creation, transfer and commercialisation. The government has increased its innovation budget and created the Innovation Council to tackle these problems. The traditional focus on generic R&D support has been widened to incorporate specific subsidies, namely those granted on a competitive basis to joint public-private R&D projects, and *their effectiveness should be closely monitored. Moreover, public support to start-up finance should be more closely targeted to innovative firms to provide starters with an incentive to innovate.*

Reforms in the education system are aimed at increasing the supply of skilled labour

Raising the quality of human capital is also an important means of increasing productivity growth. Vocational education is being strengthened by the move from an attainment-based to a competency-based system of qualifications and the requirement for educational institutions to update their curricula regularly. *The academic output of Dutch higher education could be improved by making funding more dependent on performance criteria. The government should also consider increasing*

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higher education tuition fees and granting student loans with income-contingent repayments to meet demands for increased funding as this is likely to be more efficient and fairer than raising general taxation given that returns on tertiary education are mainly private.

Corporate governance practices need to be improved

Improving corporate governance would also increase productivity by contributing to financial market development, a lower cost of capital for enterprises and stronger company performance. Corporate governance practices have been relatively poor as shareholders have had little power to ensure the accountability of supervisory boards and management has had powerful anti-takeover devices at its disposal. A new code of corporate governance specifies practices that would rectify many of these problems and a new law which has to be approved by the Upper House of Parliament would underpin the code and grant shareholders meaningful rights. T*he government should move quickly to implement the proposed law, which will require listed companies to apply the code or explain their reasons for not doing so, and will also strengthen supervision of financial reporting.*

Climate change policy should be rebalanced to be made more cost effective

Climate change policy has imposed higher energy taxes on small energy consumers, albeit recycling the extra revenues to them, while relying more extensively on voluntary agreements with large companies. T*he authorities should harmonise abatement costs across the economy by changing the regulatory energy tax to an explicit carbon tax for energy users not included in the* EU*'s trading scheme, with the rate set at the level of the expected price of an emission permit.* This would minimise the economic costs of reducing GHG emissions, although increases in revenues from other sources would be required to offset the loss of revenue from carbon taxes. S*upport offered to renewable energy and combined heat and power should not exceed the carbon tax rate.*

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The authorities should strengthen the use of economic instruments to reduce water pollution and manage natural resources

The authorities have been innovative in introducing economic instruments to address water pollution from agriculture. However, *they should seek to reduce the significant costs of implementing the* MINAS *scheme to reduce such pollution*. In the area of natural resource management, protection from saltwater intrusion due to over-exploitation of aquifers is particularly important owing to the low-lying nature of much of the country. Given this threat, *the authorities should ensure that all abstractions of groundwater, including for agriculture, incur the groundwater abstraction charges levied by the central and provincial governments and reflect the full cost of the associated externalities, unless the administrative costs outweigh the benefits*. Management of the Netherlands' substantial hydrocarbon reserves is also an important natural resources issue. In this respect, *the government should evaluate the net present value of different resource management approaches to assess whether or not the "small fields" policy, which favours the production of high-cost reserves from small fields before low-cost reserves from the main Groningen field, should be maintained in the future*.

Summing up

The government is responding to the challenges posed by recent economic volatility and the slowdown in trend growth but needs to do more. To increase economic efficiency and reduce volatility, the government should go further in easing tough zoning restrictions on residential building sites and reducing high tax subsidies for owner-occupiers. Following a structural deterioration in public finances, the government should fully implement the consolidation measures aimed at putting public finances on a sustainable path. Close monitoring will be required to ensure that the targets set in the next coalition agreement, starting in 2007, are compatible with fiscal sustainability. The planned disability benefit reform is vital for increasing labour force participation but should be strengthened to discourage misuse of the scheme to subsidise reduced working time or early retirement. Competition in product markets, which is pivotal to raising productivity growth, should be strengthened by giving the NMa greater powers and eliminating unwarranted anti-competitive practices among the liberal professions, as planned, and by removing barriers to competition in the network industries. Higher education policies could contribute more to productivity

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growth through greater reliance on private funding and by making public funding more dependent on performance. While a wide range of reforms is underway to address the challenges facing the Dutch economy, there is scope to go further in laying a solid foundation for future sustainable prosperity.

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1. Economic developments and policy challenges

Economic growth has been anaemic since the boom of the late 1990s came to an end – growth has gone from well above the euro area average to well below it. The amplitude of the current business cycle is large both by historical and international comparison. A number of factors beyond the control of the Dutch authorities – such as developments in world trade, stock prices and the euro-dollar exchange rate – boosted growth in the late 1990s but have since been a drag on growth. But some structural economic policies – pension fund regulation and housing market policies in particular – have also amplified the business cycle while at the same time undermining economic efficiency. From a longer term perspective, economic growth has also been declining. The trend growth rate in real GDP per capita has fallen to only 1½ per cent, reflecting lower growth both in the participation rate and in labour productivity. These developments are discussed in this chapter together with the policy challenges to improve economic performance that they raise.

The economic downturn and outlook for recovery

Economic growth has slumped, sharply reducing the output gap

Economic growth has fallen sharply in recent years, from an annual average rate of 3.5 per cent over 1994-2000 to only 0.2 per cent over 2000-03 (Figure 1.1). The Dutch economy has gone from being a star performer in the euro area, with average growth exceeding the euro area average by 1.0 percentage point over 1994-2000, to a laggard, with growth trailing the euro area average by 0.8 percentage point over 2001-03. At the beginning of the economic recovery that got underway in the early 1990s, the Dutch economy had less slack than most other European economies (Figure 1.2). It then grew faster in relation to potential, with the result that output exceeded potential by a relatively large margin at the end of the expansion. The output gap at the peak of the business cycle (2000:Q1) was also larger than at the peaks of previous cycles (1980:Q1; 1986:Q1; 1990:Q4). Accordingly, returning to trend entails a deeper slowdown and/or longer period of sub-trend growth in the Netherlands than in past business cycles or most other European countries.

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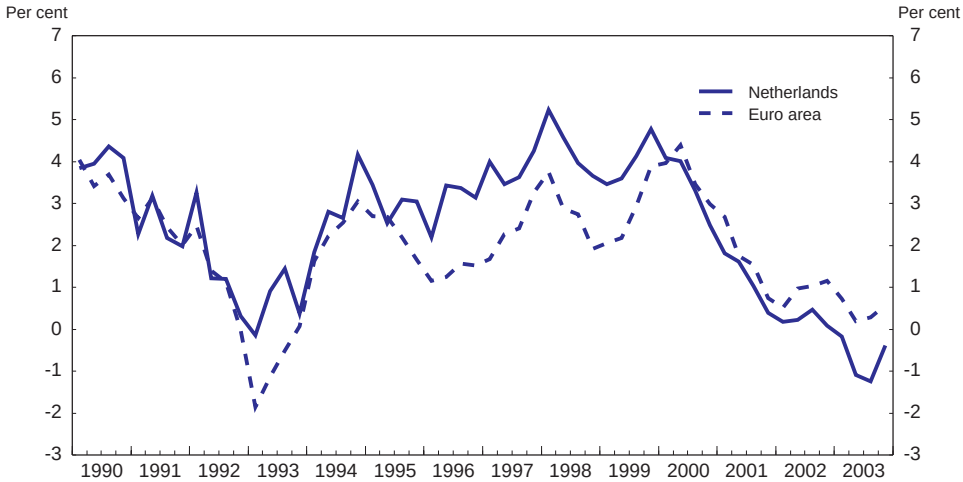
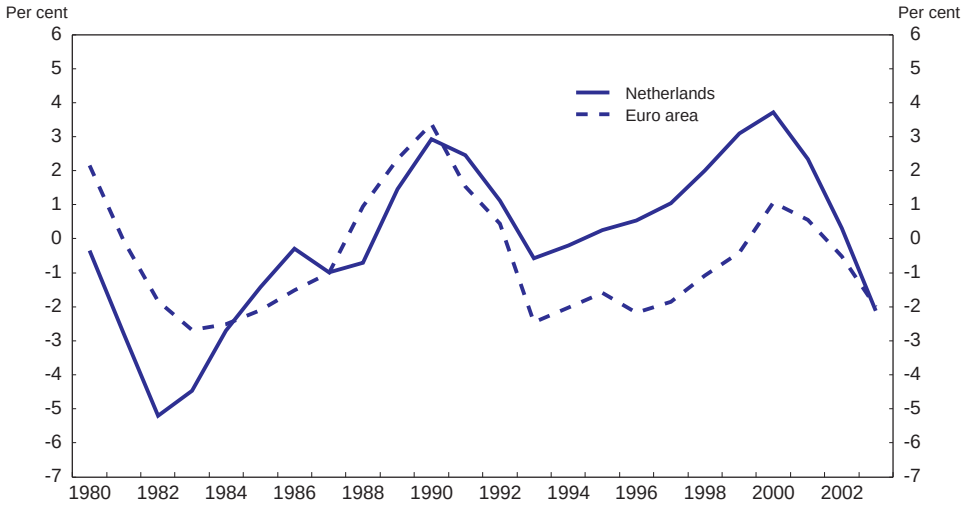
Figure 1.1. **Real GDP growth in the Netherlands and the euro area**[1]



1. Percentage changes from the corresponding quarter of the previous year.
Source: OECD.

Figure 1.2. **Estimated output gap in the Netherlands and the euro area**



Source: OECD.

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A *number of factors that boosted growth in the late* 1990s *have ceased to do so or have turned negative*

Booming house prices boosted growth in the late 1990s but have since stabilised. Real house prices[1] rose at an annual average rate of 10.5 per cent over 1995-2000, by far the highest rate of increase among OECD countries except for Ireland, with increases reaching a peak of 16 per cent in 2000 (Table 1.1). Increases have since slowed sharply, to around the euro area average and well below rates of increase in the United States and the United Kingdom. Wealth effects from the large increases in property prices in the late 1990s boosted private consumption expenditures and are estimated to have increased annual GDP growth by ½ percentage point over 1997-2000. With increases in house prices having fallen sharply since 2001, this contribution to annual GDP growth is estimated to have fallen to around ¼ percentage point. Households withdrew a substantial part (about 40 per cent) of their housing equity increases mostly to finance home improvement and consumption expenditures, contributing to a large increase in debt.[2] Household debt increased from around 109 per cent of disposable income in 1995 to 173 per cent in 2000. It has since only increased marginally (Figure 1.3). This increase and the current level of indebtedness are high by international comparison. Such a high level of indebtedness makes Dutch households relatively vulnerable to adverse shocks. The modest increase in debt since 2000 suggests that Dutch households are now taking a more prudent approach to taking on further debt, weighing on growth in consumption expenditures in the short term.

Wealth effects from stock market movements also boosted growth in consumption expenditures, and hence GDP growth, in the late 1990s but have since been a drag on growth. The Dutch stock market increased at an annual average real rate of 25 per cent over 1995-2000 but has since fallen at around the

Table 1.1. **Real house price increases**
Average annual percentage change

	1995-2000	2001-02
Netherlands	**10.5**	**3.9**
United Kingdom	7.7	10.9
United States	3.0	5.9
OECD[1]	1.9	4.3
Euro area[2]	1.2	4.1

1. Weighted average of OECD countries for which data are available. This excludes Austria, Czech Republic, Greece, Hungary, Iceland, Korea, Luxembourg, Mexico, Poland, Portugal, Slovak Republic and Turkey.
2. Weighted average of euro area countries for which data are available. This excludes Austria, Greece, Luxembourg and Portugal.

Source: OECD.

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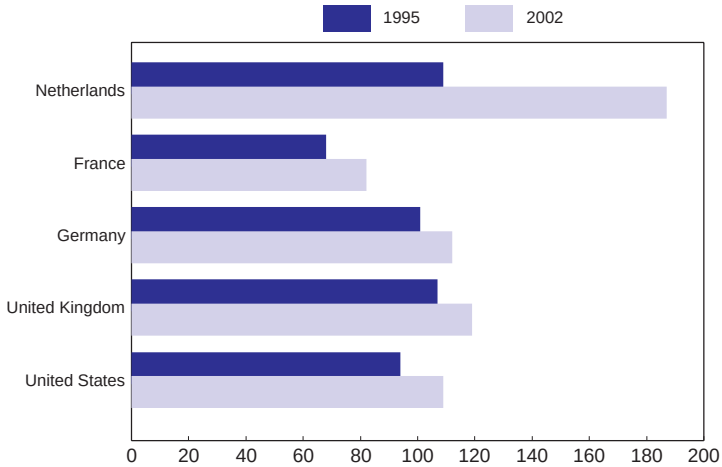
Figure 1.3. **Household debt relative to disposable income**
Per cent



Source: CPB for the Netherlands and OECD (2003) for the other countries.

Table 1.2. **Real stock price increases per year**
Average annual percentage change

	1995-2000	2001-03
Netherlands (AEX)	**24.6**	**–25.4**
Euro area (MSCI ex UK)	8.2	–15.3
United Kingdom (FTSE 100)	11.2	–15.4
United States (S&P 500)	19.1	–13.8

Source: Datastream.

same rate (Table 1.2). This increase is somewhat greater than that in the euro area, as is the decline. Based on simulation results for the effect of a 20 per cent simultaneous change in global real share prices, the rise in share prices in the late 1990s may have boosted the growth rate by up to 0.5 percentage point while the subsequent decline may have reduced it by a similar amount (Table 1.3). Although Dutch firms participate more in the equity of other firms than firms in other European countries, the effect of international stock market movements on the Netherlands appears to be similar to that in other European countries, reflecting the importance of feedbacks through international trade, but less than in the United States.

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Table 1.3. **Simulated effect of a 20 per cent fall in real share prices**[1]

	Real GDP (per cent difference from baseline)	
	Year 1	Year 2
Netherlands	**0.0**	**–0.5**
United States	–1.0	–1.0
Japan	–0.7	–0.8
Germany	–0.4	–0.5
France	–0.3	–0.4
Italy	–0.2	–0.4
United Kingdom	–0.1	–0.7
Canada	–0.5	–0.5
Major 7	–0.7	–0.8
OECD total	–0.6	–0.7

1. Simultaneous fall in real share prices in G7 countries and the Netherlands. Real interest rates, real government expenditures and nominal exchange rates have been kept unchanged to baseline in the simulation. Simulations allow for feedback effects, notably from international trade.

Source: CPB (2003) for the Netherlands; Boone *et al.* (1998) for the other countries.

The boom in international stock markets in the late 1990s masked the deteriorating underlying solvency position of pension funds. Solvency ratios rose to high levels despite contribution rates being below the cost-covering level. The creeping stock market crash since then slashed the pension fund solvency rate to 108 per cent by the end of 2002, far below the level (128 per cent) considered to provide an adequate safety margin against the risk of default on nominal pension liabilities, let alone implicit indexation promises (see Chapter 2). Strong measures have been taken to restore pension fund solvency, entailing widespread suspension of full indexation and increases in contribution rates. Contribution rates have increased from an average of 10.5 per cent of gross wages in 2002 to an expected average of 14.75 per cent over 2004-10. This increase – which was necessary to safeguard the sustainability of the pension system – is estimated to result in a cumulative loss of 1.25 percentage points of GDP by 2007 (Ewijk and van de Ven, 2003 and 2004).

The international stock market boom reduced the cost of equity capital, stimulating business investment. This effect was more important in the Netherlands than in other euro area countries because firms are more reliant on equity financing, as indicated by the relatively high stock market capitalisation in relation to GDP (Table 1.4). This factor, together with trends in capacity utilisation, contributed to higher growth in business investment in the late 1990s than in the euro area and a sharper subsequent contraction (Table 1.5). The turnaround in growth in business investment is closer to what has occurred in the United States and United Kingdom, where firms are also relatively reliant on equity financing, than what occurred in the euro area.

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Table 1.4. **Stock market capitalisation to GDP**
Per cent

	1995	1998	1999	2000	2001
Netherlands	**77.4**	**129.6**	**150.1**	**160.2**	**123.8**
France	31.5	54.5	78.5	99.2	85.7
Germany	21.4	42.6	55.2	64.2	54.3
Italy	17.8	36.3	50.6	61.8	51.0
United Kingdom	115.8	146.0	167.5	170.9	144.1
United States	80.8	135.1	150.3	143.4	122.9

Source: World Bank Financial Structure Database (2004).

Table 1.5. **Growth in the volume of business investment**
Annual average rate, per cent

	Netherlands	Euro area	United Kingdom	United States
1997-99	8.2	6.0	11.0	10.9
2000-03	–2.9	0.7	1.5	–0.4

Source: OECD.

Another factor that stimulated growth in the late 1990s but that has since depressed it is exchange rate developments. The effective exchange rate depreciated by 9 per cent over 1996-2000, considerably more than the euro area average (2 per cent), but has since appreciated by 8 per cent, somewhat more than the euro area average (7 per cent). Effective exchange rate movements in the Netherlands have been larger than in the euro area partly because the Netherlands' is more exposed to trade with the United States and hence to fluctuations in the euro-US dollar exchange rate, which have been relatively large. As a result of this trade exposure, the Netherlands initially received more of a boost from the appreciation of the US dollar in the late 1990s and has since suffered more of a drag from the depreciation of the dollar (Table 1.6).

Growth was also stimulated in the late 1990s by European monetary policy settings that were too loose for the Netherlands. Taylor rule calculations suggest that on average short-term interest rates in the Netherlands needed to be about 3 percentage points higher than they were over this period (Figure 1.4). With inflation and the output gap in the Netherlands having fallen so much in recent years, European short-term interest rates are now too high for the Netherlands. Monetary conditions have tightened by more in the Netherlands than in the euro area since 2001, mainly reflecting the larger fall in inflation in the Netherlands than in the euro area (Figure 1.5).

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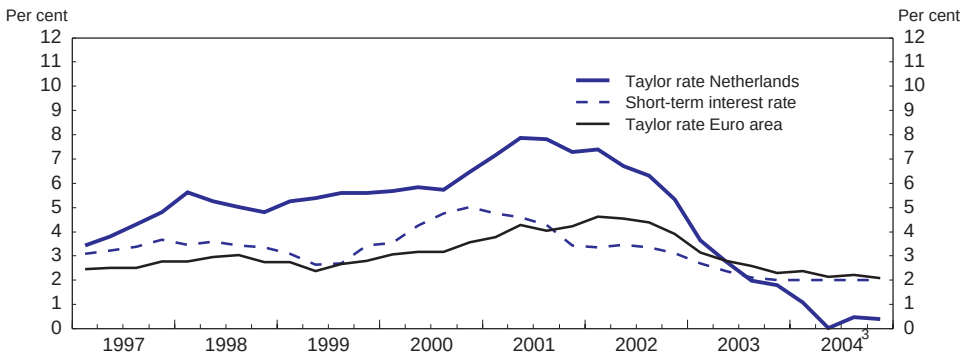
Table 1.6. **Simulation for a 10 per cent US dollar depreciation**

	Deviations from GDP baseline, in per cent				
	Year 1	Year 2	Year 3	Year 4	Year 5
Netherlands[1]	**–0.4**	**–0.4**	**0.0**	**0.1**	**0.2**
Germany	–0.3	–0.4	–0.3	–0.2	–0.1
France	–0.2	–0.4	–0.3	–0.3	–0.2
Italy	–0.2	–0.4	–0.2	–0.2	0.0

1. Simulations for the Netherlands were made on a comparable basis.
Source: CPB (2002, pp. 96-98) for the Netherlands and Dalsgaard *et al.* (2001) for the other countries.

Figure 1.4. **Short-term interest rate[1] and Taylor rates[2] for the Netherlands and euro area**



1. Three month money market rate for the Netherlands.
2. The version of the Taylor rule used here sets the short-term interest rate equal to the sum of the equilibrium real short-term rate (assumed constant at 2 per cent), the core inflation rate (year-on-year, excluding effects of tax changes for the Netherlands), half the output gap and half the deviation of inflation from its target (assumed to be 1.75 per cent).
3. Projections.
Source: OECD.

The long period of recovery and expansion in the 1990s was rich in employment, with total employment growing at an average annual rate of 2.1 per cent over the decade. Initially, demand for labour was met by strong increases in labour supply (brought forward by structural reforms and autonomous trends to higher female participation) as well as by running down unemployment, but as the expansion matured reserves of additional labour shrivelled up. Employment growth actually peaked relative to trend one year before the peak of the economic

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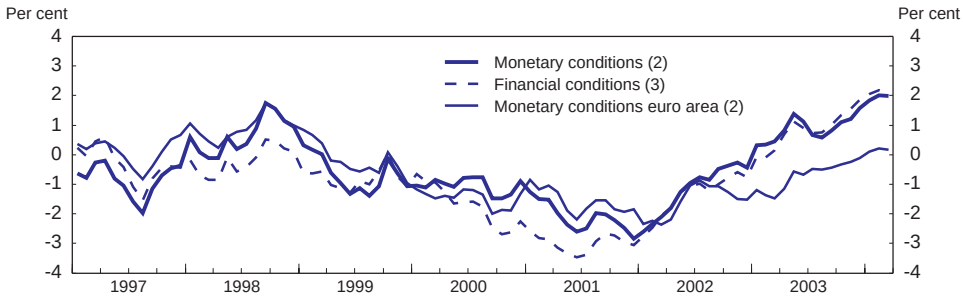
Figure 1.5. **Monetary and financial conditions**[1]



1. Declining (increasing) values mean that conditions are becoming easier (tighter). A value of zero implies that conditions are the same as in the base year (1990).
2. The weights underlying the MCI are 1 for the short-term real interest rate and 0.3 (0.15) for the real effective exchange rate for the Netherlands (the euro area). Thus a decrease in the interest rate by 100 basis points would be as supportive for Dutch monetary conditions as a 3.33 per cent decrease in the real effective exchange rate.
3. The weights underlying the FCI are 0.15 for the short-term real interest rate, 0.62 for the long-term real interest rate and 0.23 for the real effective exchange rate.

Source: OECD.

cycle (2000:Q1), compared with an average two quarters after the peak in previous cycles (DNB, 2002). The unemployment rate fell below the estimated NAIRU of around 5 per cent in 1998 and continued to decline to a cyclical low of 2.0 per cent in 2001 (Figure 1.6).

Wage increases rose markedly once the unemployment rate fell below the structural rate. Together with declining labour productivity growth, the higher wage increases pushed up growth in unit labour costs sharply. Initially, the effect of higher growth in unit labour costs on export competitiveness was more than compensated for by the depreciation of the euro. However, once the euro started appreciating, there was a very large loss of competitiveness and deterioration in export performance (Figure 1.7). High growth in unit labour costs (together with depreciation of the euro) also contributed to an increase in core inflation, which tended to reduce growth in household disposable incomes and hence growth in consumption expenditures, especially once the effects of the 2001 income tax cuts passed. Even so, labour income has fared better than business income, with labour's share in business value added rising since the mid-1990s (Table 1.7). The increase in labour's share since 2000 has been large by international comparison, reflecting labour hoarding, which delayed wage adjustment to economic conditions. The sharp fall in employment growth since mid-2003, together with the wage freeze agreed for 2004-05, should reduce labour's share of business income, laying a more solid foundation for future business investment and employment growth.

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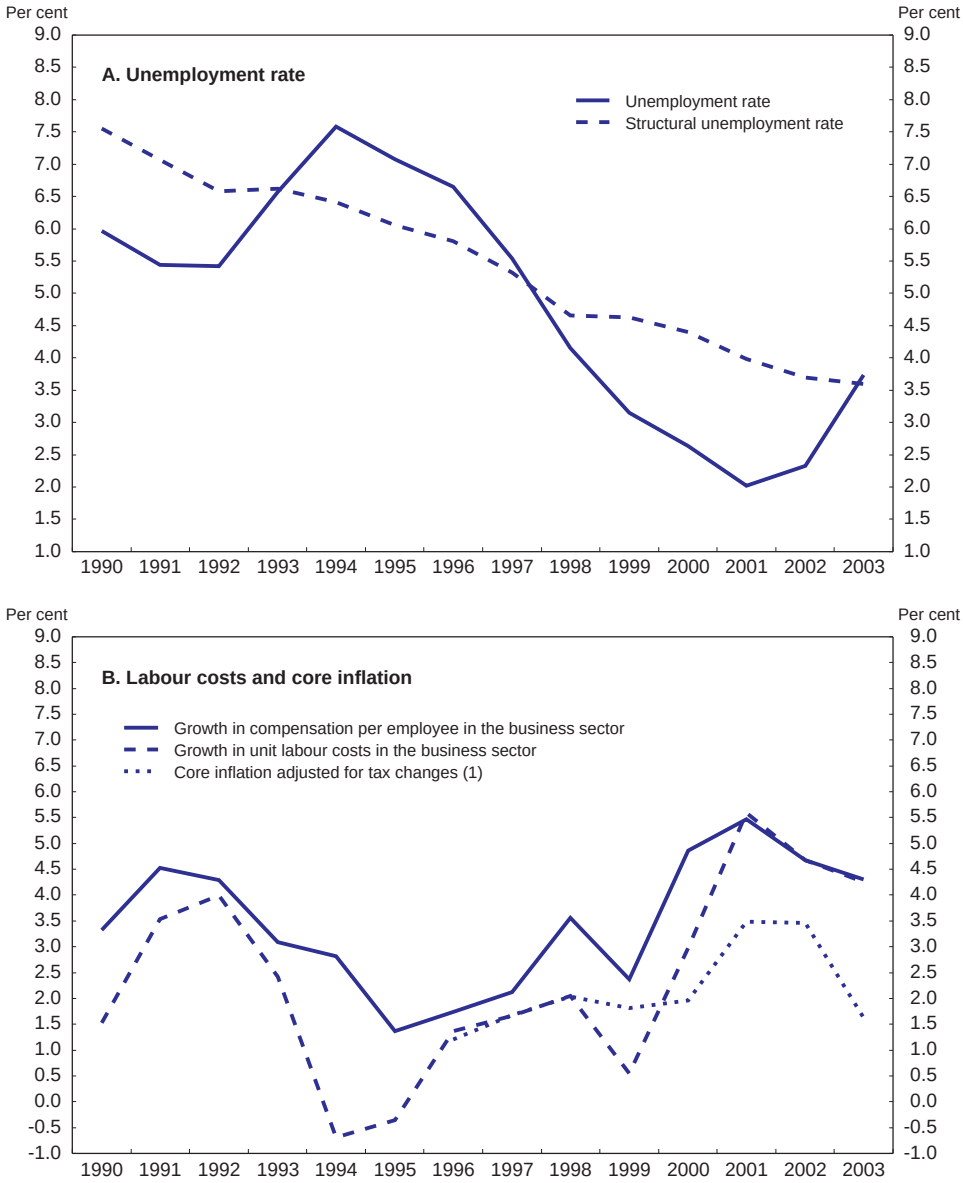
Figure 1.6. **Unemployment, labour costs and core inflation**



1. This series excludes the effects of indirect taxes, government services, energy, fruit and vegetables.
Source: OECD and CPB Netherlands Bureau for Economic Policy Analysis.

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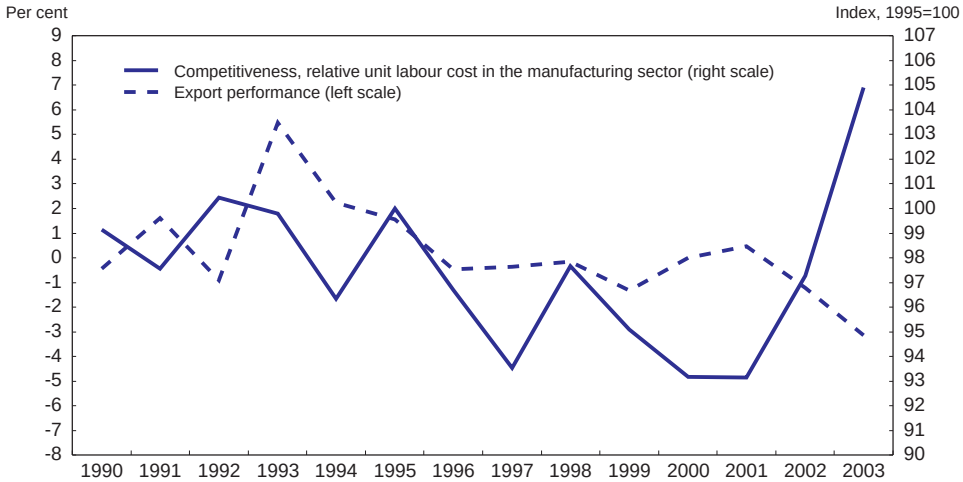
Figure 1.7. **Cost competitiveness and export performance**[1]



1. Export performance is defined as the difference between growth in exports and growth in the relevant export markets.
Source: OECD.

Table 1.7. **Increase in labour's share of business sector value added**
Percentage points

	Netherlands	France	Germany	United Kingdom	United States
1996-2000	0.8	0.3	–0.4	4.0	1.8
2001-03	2.4	1.5	–0.6	–0.5	–2.9

Source: OECD.

Economic recovery should build over 2004-05

With the world economy recovering during 2003, the Dutch economy has finally emerged from recession at the end of the year. After declining in the first half of 2003, GDP stagnated in the third quarter and posted an increase by 1.7 per cent (seasonally adjusted annual rate) in the fourth. While final domestic demand was still lacklustre – as suggested by another decrease in private consumption and a further, albeit slowing, decline in private fixed investment, exports picked up vigorously, although much less strongly than the recovery of world trade given the loss of competitiveness. Stock building also made a considerable contribution. This may point to firming industrial production as suggested by survey data that point to rising order inflows and to capacity utilisation picking up in early 2004

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Table 1.8. **Short-term projections**

	2003	Projections	
		2004	2005
	Percentage change		
Demand and output (volume)			
Private consumption	–1.2	0.4	1.7
Government consumption	2.7	–0.4	0.3
Gross fixed capital formation	–3.2	0.0	2.8
Private sector non-residential	–3.9	–0.0	3.0
Private sector residential	–1.8	2.1	3.9
Public sector	–2.6	–3.0	0.2
Final domestic demand	–0.7	0.1	1.6
Change in stockbuilding[1]	0.2	0.3	0.2
Total domestic demand	–0.4	0.4	1.8
Exports of goods and services	0.1	4.4	6.9
Imports of goods and services	0.6	4.4	6.8
Change in net exports[1]	–0.4	0.3	0.5
GDP at market prices	–0.7	0.9	2.1
Inflation			
GDP deflator	2.8	1.3	0.8
Private consumption deflator	2.0	1.5	1.0
	Per cent		
Others			
General government balance (per cent of GDP)	–3.2	–3.1	–2.9
Unemployment rate[2]	3.5	5.0	5.1
Household savings ratio[3]	11.2	12.5	12.0
Current account balance (per cent of GDP)	1.5	2.6	2.6
Short-term interest rates	2.3	1.7	1.8
Long-term interest rates	4.1	4.0	4.7

1. Contribution to GDP growth.
2. National definition (registered unemployment as percentage of labour force).
3. Including net contributions (actual or imputed) to life insurance and pension schemes.

Source: OECD *Economic Outlook* No. 74.

for the first time in two years. Overall business expectations have steadily improved since mid-2003 and consumers' opinions about future tendency, although still negative, have brightened.

Economic growth is expected to gain momentum during 2004, if only gradually given still low levels of economic sentiment and the tightening of fiscal policy. While past losses in competitiveness continue to weigh on the economy, these were partly offset in 2003 by lower short-term interest rates; monetary and financial conditions have not tightened any more since early 2003 (see Figure 1.5). The upswing in the world economy will help exports to grow for the first time after two years of stagnation (Table 1.8). They will gather further momentum in 2005 as

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the effects of the euro appreciation fade and the effects of wage moderation, expected to last until 2005, contribute to restoring competitiveness; prolonged wage cost moderation, however, would be required to recover the ground lost since 1997. Business investment should turn around by mid-2004, helped by low interest rates, a growing need for replacing obsolete equipment and rising capacity utilisation in export-oriented sectors. Private consumption expenditure will remain weak because of increasing tax and contribution rates, sluggish labour demand and stagnant real wages (albeit supported by the effect on consumer prices of the appreciation of the euro). With aggregate production turning around in the first half of 2004 and output growth matching potential by the end of the year, labour market prospects should gradually improve and unemployment should stabilise. Still low levels of capacity utilisation, declining import prices, moderating wage increases at last and a cyclical pick-up in productivity should underpin falling inflation through 2005. All in all, the recovery is expected to be less strong than after previous recessions as a result of both the domestic factors hampering consumption and the loss of competitiveness.

In view of the imbalances described above, the recovery is surrounded with downside risks. Stretched financial positions make companies and households vulnerable to a stronger-than-expected rise in long-term interest rates. In particular, households, whose leverage is high both by historical and by international standards, would cut back on expenditure should real house prices fall and pension contribution rates post an unexpected sharp increase. In addition, another strong depreciation of the US dollar would prevent the recovery from unfolding, as expected, *via* a pick-up in foreign trade.

Long-term economic performance

Trend growth in GDP *per capita has slowed owing to lower growth in labour inputs and productivity*

Overlaying short-run developments is a slowing in the trend growth rate in GDP per capita from around 2 per cent in the early 1990s to about 1½ per cent a decade later (Figure 1.8). This slowdown reflects lower growth in the labour force participation rate and in hourly labour productivity. The rise in the female participation rate, which entirely accounts for growth in the participation rate, has slowed as the effects of reforms to increase participation wane and as the rates attained reach higher levels, making further increases more difficult to achieve.[3] The decline in growth in trend labour productivity per hour entirely reflects a decline in trend multifactor productivity (MFP) growth (Figure 1.9).[4] The contributions of capital deepening and of improvements in the quality of capital services were similar at the beginning of the current decade to the corresponding contributions a decade earlier. Trend growth in GDP per capita at the beginning of the current decade was around the euro area average but lower than in most of the large

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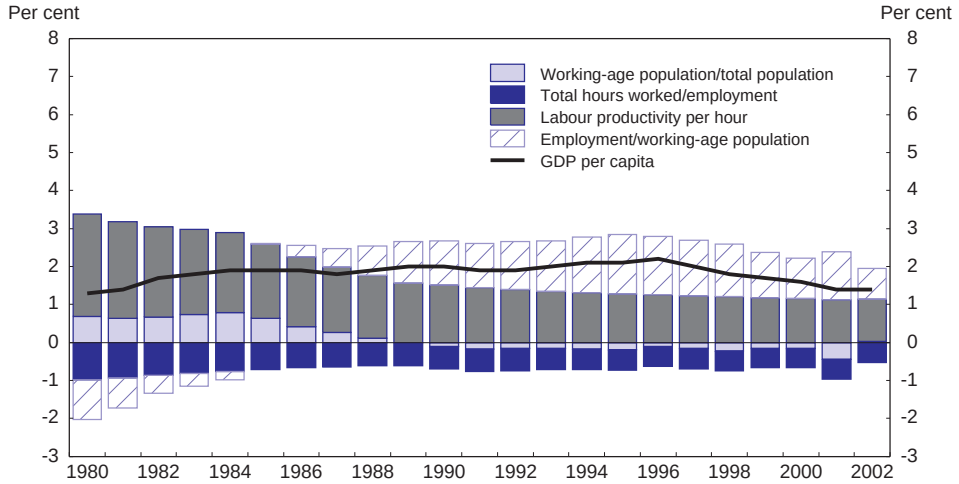
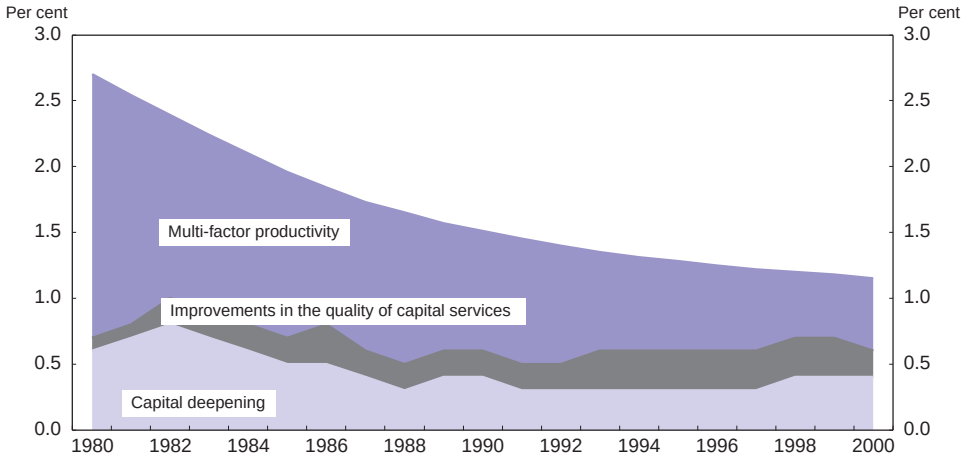
Figure 1.8. **Decomposition of the slowdown in growth in trend GDP per capita**
Trend series, percentage change



Source: OECD.

Figure 1.9. **Contributions to hourly labour productivity growth**[1]
Trend series, 1980-2000



1. Total economy.
Source: OECD.

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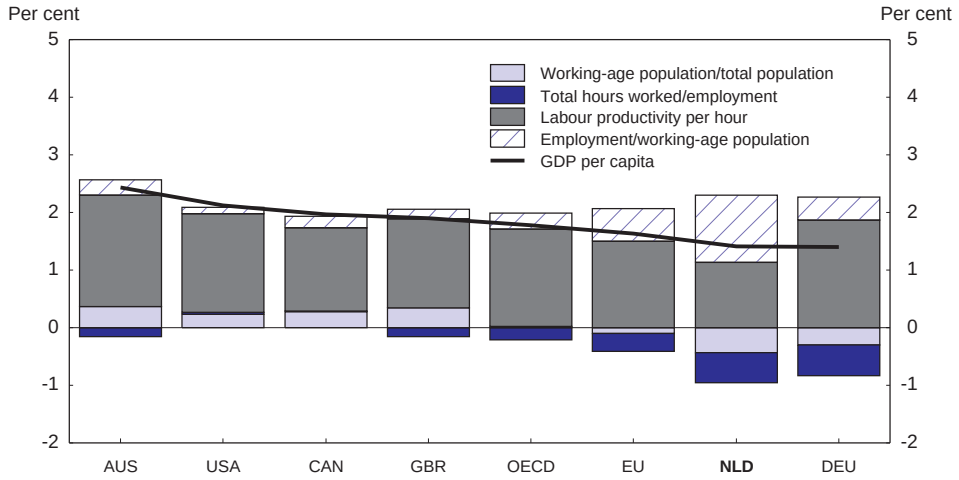
Figure 1.10. **Contributions to growth in trend GDP per capita in selected countries**
Trend series, 2001



Source: OECD.

OECD countries outside the euro area (Figure 1.10). The contribution of labour resource utilisation (trend hours worked divided by population) was high by international comparison, reflecting high growth in the participation rate – this more than compensated for the decline in average hours worked per employee, which was similar to the euro area average but high compared with that in the large economies outside the euro area. Growth in hourly labour productivity (1.2 per cent), on the other hand, was low, reflecting low growth in MFP (0.5 per cent) (Figure 1.11). The contribution of improvements in the quality of capital services to labour productivity growth in the Netherlands (0.2 per cent) was marginally above the euro area average (0.1 per cent) but less than in some countries outside the EU, notably the United States (0.4 per cent) and Australia (0.3 per cent), that have invested more heavily in ICT capital goods. In these two countries, improvements in the quality of capital services contributed 0.2 percentage point more to trend labour productivity growth in recent years than a decade earlier whereas in the Netherlands and most other countries this contribution has not changed.

There remains scope to raise labour inputs

Although the contribution of labour resource utilisation to growth in GDP per capita is already high by international comparison, there remains considerable scope to increase labour inputs. While the employment rate (75 per cent for the population aged 25-64) is comfortably above the EU and OECD averages, it still lags far behind

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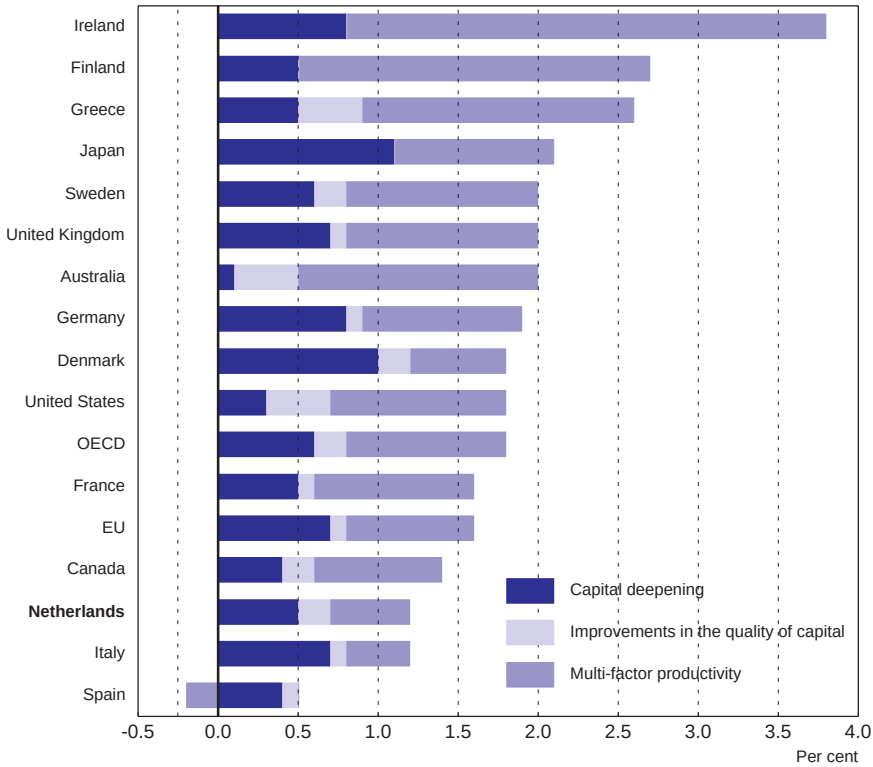
Figure 1.11. **Contributions to growth in trend hourly labour productivity in selected countries**[1]

Trend series, 2000



1. Total economy.
Source: OECD.

the highest rates in the OECD (Figure 1.12). This reflects much lower employment rates for older workers and women than in the best performing OECD countries in these areas. Average annual hours worked per employee in the Netherlands are far short of those in the benchmark OECD countries in this respect – indeed, they are the lowest of any OECD country. Low average working hours in the Netherlands reflect a high share of part-time employment and of reduced working hours for full-time employees.[5] Many households prefer a 1½ income model, which may entail one full-time job and one part-time job or two full-time jobs with reduced working hours, so as to reconcile work and family life better. Employers are normally obliged to accede to employees' requests for reduced working time.

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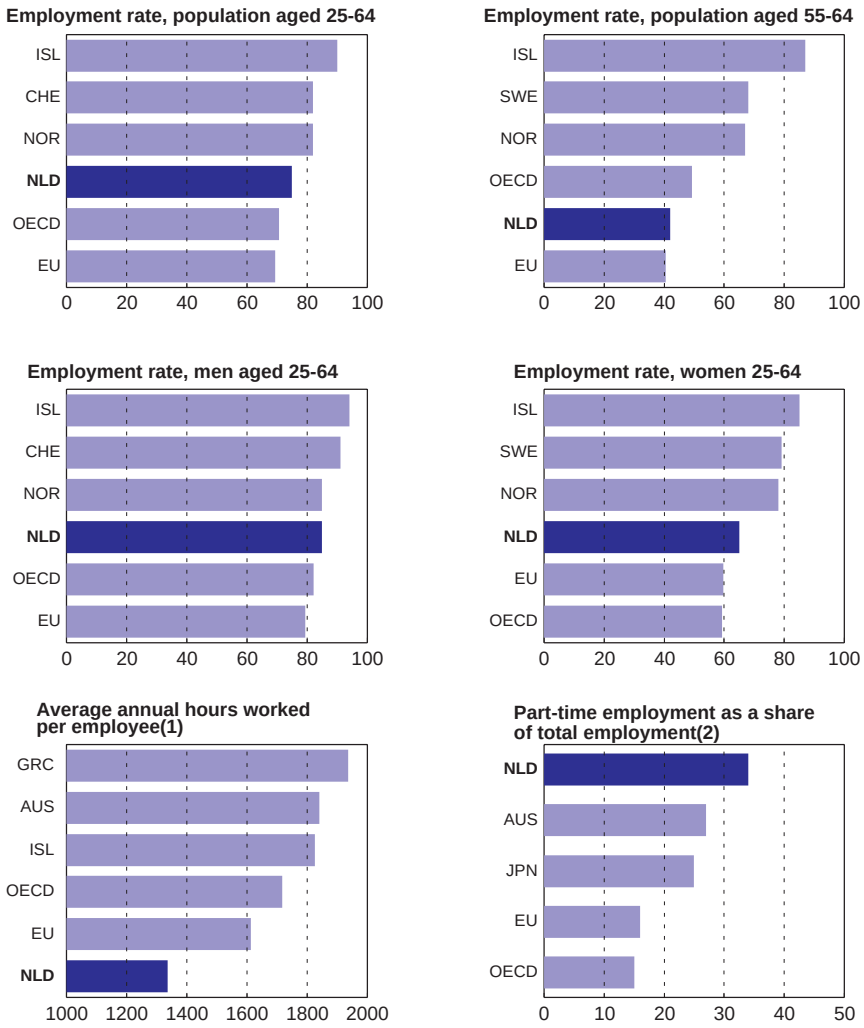
Figure 1.12. **Benchmarking labour inputs**
2002



1. Trend data.
2. This graph does not represent a benchmarking exercise. It shows the three countries with the highest proportion of part-time employment in total employment along with the EU and OECD averages.

Source: OECD.

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Policy challenges

Reforming policies that undermine economic efficiency and have accentuated the amplitude of the business cycle

Regulatory and tax reforms are needed in the housing market to reduce housing costs and the amplitude of swings in the business cycle

Tight zoning regulations restrict the supply of building sites, resulting in inelastic housing supply. This underpins a large trend increase in housing prices relative to disposable income and large fluctuations in prices in response to variations in demand. Tax deductibility of mortgage interest payments reduces the user cost of capital for owner-occupiers, contributing to high prices (but not to the rate of increase in prices if the present value of tax deductions remains unchanged relative to income) and to large fluctuations in prices as households become more or less optimistic about their permanent income prospects. In addition, tax deductibility encourages households to be more highly leveraged than otherwise, amplifying the effect of changes in house prices on net worth. While residential landowners are made better off by large increases in housing prices, this gain is at the expense of other current and future residents who will have to pay higher rentals. Unnecessarily high housing costs reduce purchasing power, discourage labour supply (in the same way as a consumption tax) and lower cost competitiveness. *The policy challenge is to reduce upward pressure and volatility in housing prices by easing zoning regulations and by considering to phase out tax subsidies for owner-occupied housing*. This challenge is discussed in the first part of Chapter 2.

Pension fund regulation needs to be tightened to reduce the risk of insolvency and increase transparency

Pension fund regulation allowed past contribution rates to be below the cost-covering level, leading to the solvency crisis now confronting pension funds, and transparency to be inadequate for members to make optimal life-cycle consumption plans. The measures taken to restore pension fund solvency have increased the "tax-like" element of contributions and hence the economic efficiency cost (excess burden) of pension arrangements and have accentuated the economic downturn. The growing realisation that pensions are likely to be lower than anticipated may also have encouraged households to save more, reinforcing the downturn. *The policy challenge, which the authorities are well on the way to meeting, is to reform regulation to reduce the risk of such a crisis occurring again and to enhance fund transparency*. This challenge is discussed in the second part of Chapter 2.

Putting public finances on a sustainable path

The sharp economic downturn has contributed to a substantial deterioration in the general government budget balance, which has fallen from a surplus of

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1.5 per cent of GDP in 2000 (excluding UMTS proceeds of 0.7 per cent of GDP) to an estimated deficit of 3.2 per cent of GDP in 2003. While much of this deterioration appears to be cyclical, the structural balance has also fallen considerably, from being in balance in 2000 to a deficit of 2.2 per cent of GDP in 2003. Public finances were not on a sustainable path – one that permits the government to meet future obligations, including the budget costs of population ageing, without further increases in taxes or reductions in expenditures. Hence, the government has been obliged to implement an austerity programme while the economy is still weak in order to respect the requirements of the Stability and Growth Pact and to put public finances on a sustainable path. T*he policy challenge is to ensure that public finances are indeed put on a sustainable path by fully implementing the austerity programme and that the fiscal targets set in the next coalition agreement, starting in* 2007, *are compatible with keeping public finances on a sustainable path*. This will contribute to economic efficiency by smoothing taxes over time and hence minimising the efficiency costs (*i.e.*, excess burden) of taxation, which rise more than in proportion to tax rates. It should also enable the authorities to allow automatic stabilisers to operate fully in future downturns. This challenge is discussed in Chapter 3.

Increasing labour force participation

Even though the Netherlands has been doing well in terms of increasing trend employment, employment rates for older workers and women remain well below those in the best performing OECD countries. In addition, average annual hours worked per employee are the lowest in the OECD area. This suggests that there remains considerable scope to increase GDP per capita by raising employment rates for these groups and average annual hours worked per employee. T*he policy challenge is to raise employment rates for older workers and women and to remove barriers to longer average working hours per employee*. These objectives could be met by reforming social transfer programmes and reducing the tax burden so as to strengthen work incentives and by reforming laws, such as employment protection legislation, that weaken labour demand for groups with low participation rates. This challenge is discussed in Chapter 4.

Increasing productivity growth

Hourly labour productivity growth (in trend terms) has fallen to around 1 per cent, which is low by both historical and international standards. This reflects a decline in MFP growth. The Netherlands has not experienced an increase in the contribution to labour productivity growth from improvements in the quality of capital services, in contrast to some other OECD countries that have invested more heavily in ICT capital goods. I*ncreasing productivity growth is the greatest policy challenge facing the* D*utch authorities as this is the most important means of raising living standards in the long term*. Productivity growth can be increased by strengthening

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entrepreneurship and competition, fostering innovation and research and improving human capital. Increasing public sector efficiency and reform of corporate governance arrangements to increase management accountability to shareholders should also raise productivity. Competition policy is discussed in Chapter 5 while the other policies for raising productivity growth are discussed in Chapter 6.

Making better use of economic policy instruments to reduce the cost to the economy of achieving sustainable development objectives

The Netherlands faces major sustainable development challenges in three main areas:

- Greenhouse gas (GHG) emissions – they grew by 4 per cent during the 1990s but the Netherlands is obliged to reduce them by 6 per cent between 1990 and 2008-12 as part of the EU's burden-sharing agreement related to the Kyoto protocol.
- Pollution of water resources – this is also a major challenge owing to the intense pressure from agriculture.
- Sustainable use of natural resources – sustainable management of fresh water resources is important to mitigate the threat of saltwater intrusion, which can irreversibly damage land, owing to the low lying nature of much of the country; and a balanced policy approach to ensure that exploitation of the Netherlands' substantial hydrocarbon reserves is not to the detriment of future generations' welfare.

The major policy challenges are to reduce GHG emissions, improve the quality of water resources and ensure that use of natural resources is sustainable in such a way as to minimise the economic costs or maximise the economic benefits (taking account of the interests of future generations) of achieving these objectives. These challenges, which could be met by making better use of economic policy instruments, are discussed in Chapter 7.

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Notes

1. Real house prices are defined as nominal house prices deflated by the consumption price deflator.
2. 70 per cent of home equity withdrawals since 1998 have been used to finance home improvements and a further 8 per cent financed consumption expenditures (DNB, 2003, p. 44).
3. The female participation rate has increased by 20 percentage points since the mid-1980s to 67 per cent. This increase is attributable to autonomous trends such as an increase in educational attainment as well as measures limiting access to social transfer programmes (in particular disability and welfare) and promoting the transition from such programmes to work. The deregulation and increase in flexible employment contracts also contributed to the increase in the female participation rate.
4. MFP growth is the growth in (total economy) output not accounted for by growth in inputs (labour and capital). Capital inputs are measured by capital services adjusted for quality changes. Labour inputs are measured by hours worked without adjustment for quality changes. Such an adjustment is not made owing to a lack of internationally comparable data. Accordingly, MFP growth includes the effect of changes in labour quality.
5. The example of Australia, which combines the second highest proportion of part-time employment in total employment in the OECD and the second highest average annual hours worked per employee, suggests that a high proportion of part time employment in total employment does not necessarily lead to low average annual hours worked per employee.

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2. Reforms to housing and pension funds to increase economic efficiency and reduce business cycle fluctuations

Housing market polices and pension fund regulation have undermined economic efficiency and amplified the current business cycle. Policies that have the effect of restricting housing supply and high tax subsidies for owner-occupied housing have underpinned large increases in house prices relative to disposable income, reducing living standards, and have contributed to macroeconomic volatility stemming from housing wealth effects. Pension fund regulation allowed pension funds to set past contribution rates too low to cover the costs of the promises being made, leading (together with some other developments, notably the fall in interest rates and the worldwide fall in equity prices) to the current solvency crisis. Contribution rates have now been increased to above the cost-covering level to make up for past shortfalls, increasing the "tax-like" element of contributions and hence their cost to economic efficiency (the excess burden) and accentuating the economic downturn. While the government is taking steps to deal with both sets of problems, it has made much more progress in reforming pension fund regulation than in reforming housing market policies.

Housing policies

Housing price increases have been high owing to weak supply responses

Real house prices in the Netherlands increased at annual average rate of 2.9 per cent over 1971-2002, the fourth highest rate among OECD countries with available data (Table 2.1). Increases in real house prices rose to an annual average rate of 6.7 per cent over 1991-2002, the highest rate in the OECD area. Consequently a large rise in real land prices emerged – real construction costs have risen at a much lower rate. While residential landowners are made better off by such price increases, this gain is at the expense of other current and future residents who have to pay higher rentals.

Normally, high increases in house prices are associated with high income growth. This is because income growth generates more demand for space, increasing

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Table 2.1. **Real increases in house prices, construction costs and household disposable income**[1]

Average annual percentage change

	Real house prices				Ratio of house prices to disposable income				Real construction costs	Real household disposable income	Ratio of house prices to disposable income per capita
	1971-80	1981-90	1991-02	1971-02	1971-80	1981-90	1991-02	1971-02	1971-02		
United Kingdom	3.7	4.7	2.8	3.6	0.9	1.6	0.0	0.8	10.1	2.9	1.0
Spain	2.0[2]	6.3	2.5	3.5[3]	–1.0[2]	3.7	0.2	0.9[3]	10.6	2.8	1.5[3]
Ireland	3.3	–0.6	6.1	3.1	5.4[4]	–2.1	0.7	0.1[5]	10.9	3.0[5]	0.8[5]
Netherlands	**3.7**	**–2.2**	**6.7**	**2.9**	**0.7**	**–4.1**	**4.8**	**0.7**	**5.3**	**2.3**	**1.3**
Australia	1.9	2.3	3.3	2.5	–2.1	–0.3	0.8	–0.4	7.5	3.0	1.0
Belgium	4.1	–1.6	3.9	2.2	0.6	–3.3	2.4	0.0	5.1	2.2	0.2
Canada	3.2	1.8	0.6	1.7	–2.2	–0.7	–1.0	–1.3	5.5	3.1	–0.1
Italy	3.7	2.2	–0.5	1.6		–0.5	–0.8	–0.6[6]	9.8	1.4[6]	–0.5[6]
Norway	0.0	0.9	3.7	1.6	–4.3[7]	–0.8	–0.2	–1.2[8]	5.9	3.0[8]	–0.7[8]
United States	2.2	0.8	1.9	1.6	–1.1	–2.4	–1.1	–1.5	5.3	3.3	–0.5
France	2.6	0.8	0.8	1.3	–0.7	–0.8	–1.2	–0.9	5.8	2.3	–0.4
Denmark	1.7	–1.8	3.4	1.2	0.4	–3.7	2.2	–0.2	6.2	1.5	0.0
Japan	1.8	4.5	–2.3	1.0	–2.5	1.8	–3.2	–1.4	3.4	2.5	–0.8
Finland	–1.8	6.4	–1.7	0.7	–4.8	3.3	–2.8	–1.6	7.4	2.5	–1.2
New Zealand	0.1	–0.7	1.1	0.2		–2.9[9]	–0.9	–1.4[10]	9.1	2.1[10]	–0.4[10]
Sweden	0.6	–0.7	0.4	0.1	–1.2	–1.6	–1.3	–1.4	7.9	1.5	–1.0
Switzerland	0.7	3.5	–3.1	0.1	–0.4	0.9	–4.9	–1.7	1.4	1.9	–1.2
Germany	1.6	–0.7	–0.6	0.0	–1.5	–3.2	–2.5	–2.5	0.8	2.6	–1.6
Average	1.9	1.4	–0.6	1.6	–0.9	–0.8	–0.5	–0.8	6.6	–0.1	2.4

1. All variables are deflated by the private consumption deflator.
2. 1972-80.
3. 1972-2002.
4. 1977-80.
5. 1977-2002.
6. 1980-2002.
7. 1975-80.
8. 1975-2002.
9. 1986-90.
10. 1986-2002.

Source: OECD.

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the price of land for construction. Indeed, the correlation coefficient between increases in real house prices and growth in real household disposable income for the OECD countries included in this analysis is 0.5. But growth in real household disposable income in the Netherlands was actually slightly below average (see Table 2.1). In other words, increases in income are associated with unusually large increases in house prices in the Netherlands. House prices rose at annual average rate relative to household disposable income of 0.7 per cent over 1971-2002, the third highest rate in the OECD area; housing affordability, as indicated by house prices relative to household disposable income per capita, declined by more than in any other country except for Spain. In most countries, house prices actually declined relative to disposable income. Over 1991-2002, the annual average increase in house prices relative to disposable income was 4.8 per cent, by far the greatest increase in the OECD area. The large increase in house prices relative to disposable income in the Netherlands mainly reflects the relatively small effect of housing stock supply in restraining increases in real house prices by international comparison (Box 2.1). The greater easing in access to mortgage finance in the Netherlands than in most other countries, especially in the 1990s, is also likely to have contributed to the relatively large price increase in the Netherlands.[1] Finally part of the explanation of house price increases in the 1990s may be found in a catch-up effect after the strong house price decline in the early 1980s, which initially faded away only slowly.

Tough zoning regulations, especially in rural areas, are a major barrier to increased supply of residential building sites. Municipalities are free to develop their own regulations, but these must comply with spatial policy of central and provincial governments as well as national and EU regulations concerning such things as noise pollution and safety. The government is considering a relaxation of its rules in rural areas with which municipalities must comply when developing zoning regulations and expects to make an announcement in this respect later this year. Cumbersome building regulations and administrative procedures may also restrict housing supply. Building regulations have been growing in quantity and complexity. This has created difficulties for municipalities to deal promptly with building permit applications. There has also been an increased use of the public enquiry procedure, meaning that more time and processing are required to deal with building permit applications. The government is reassessing building regulations ("*herijking* VROM-*regelgeving*") with a view to simplifying them. The government has recently also set up "boosting teams" ("*aanjaagteams*") to solve impediments to housing supply at specific sites or in specific regions.

Volatility in wealth effects from housing has also been high owing to price inelastic housing supply and tax subsidies for owner-occupied housing

When real house prices evolve at a different rate from what households expect, the wealth effects are likely to stimulate or retard growth in consumption

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Box 2.1. Determinants of real house prices

Following Meen (2002), real house prices (G) are assumed to be positively related to real household disposable income (RY) and negatively related to the real stock of dwellings (H) and the real interest rate (RR),[1] it is important to include housing supply in this relationship because failure to do so results in a downward bias in the estimated income elasticity of house prices as income and the stock of dwellings are co-integrated. Estimating this relationship as an error correction model, all explanatory variables have the expected signs and are statistically significant (the housing stock is significant at the 90 per cent confidence level but does not quite reach significance at the 95 per cent level). The long-run elasticity of real house prices with respect to real disposable income (1.94) is lower than those obtained by Meen (*ibid.*) for the United Kingdom (2.51) and the United States (2.71).[2] This, together with lower growth in household disposable income in the Netherlands than in the other two countries (see Table 2.1), suggests that high growth in real house prices is not attributable to strong demand but rather to weak supply responses by international comparison (especially compared with the United States). The long-run elasticity of real house prices with respect to the housing stock (–0.52) is low compared with those for the United Kingdom (–1.91) and especially for the United States (–7.64). This may be because prices are so high in relation to disposable income that demand is price inelastic – Dutch households would find it difficult to economise much more on space in response to an increase in prices whereas American households could easily economise on space in these circumstances. In addition, a much higher proportion of the increase in housing supply in the Netherlands has been for social housing than in the other two countries, which may have a weaker effect on house prices than private housing supply.

Table 2.2. **An error correction model of real house prices**

Constant	–13.549	(–5.4)
In $(G)_{-1}$	–0.434	(–5.4)
In $(G)_{-1}$	0.037	(6.9)
In $(RY)_{-1}$	0.841	(4.1)
In $(H)^{-1}$	–0.226	(–1.9)
RR_{-1}	–0.031	(–5.2)
R^2	0.759	
Standard error of regression	0.045	
DW	2.19	

Note: The dependent variable is Δln G; t-values are given in brackets. The estimation period is 1970-2002 (annual data).
Source: OECD.

1. Nominal variables other than the housing stock were deflated by the private consumption deflator to obtain real variables. The housing stock was deflated by the residential construction price deflator. The housing stock was estimated from investment data using a perpetual inventory model with a 0.85 per cent depreciation rate: this depreciation rate, which comes from ter Rele and van Steen (2001, pp. 57-58), is based on the loss in value due to the ageing of the house, despite carrying out regular maintenance. The interest rate series is for mortgage rates.
2. Meen's estimation periods are 1969(3) to 1996(1) for the United Kingdom and 1981(3) to 1998(2) for the United States.

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Table 2.3. **Volatility in growth in real house prices**[1]
Per cent, 1971-2002

	Change in house price inflation	Difference between actual and OLS[2] trend house price inflation	Difference between actual and HP trend[3] house price inflation
Netherlands	**7.2**	**22.9**	**7.3**
Italy	12.7	15.0	9.2
Spain	9.7	17.9	8.4
Finland	9.4	17.1	9.2
United Kingdom	9.8	14.5	8.9
Denmark	8.7	13.8	7.2
Ireland	5.2	17.3	5.1
Sweden	6.0	14.0	6.3
Norway	6.0	13.7	6.0
Switzerland	5.2	13.7	5.3
New Zealand	6.6	11.3	6.0
Japan	5.9	11.9	5.7
Belgium	4.5	14.1	4.8
Canada	6.9	9.9	5.5
Australia	6.6	8.0	5.1
France	3.7	7.7	3.8
Germany	2.6	5.0	2.3
United States	2.3	4.8	2.3
Average	6.6	12.9	6.0

1. Volatility is measured by the standard deviation. The countries are ranked in descending order of volatility based on a simple average of the standard deviations of the three series.
2. Ordinary least squares.
3. Hodrick-Prescott filter (lambda = 100).

Source: OECD.

expenditures depending upon whether the wealth effects are positive or negative. If it is assumed that households expect real house prices to increase at the same rate as in the previous period, then changes in the real rate of house price inflation will impact on growth in consumption expenditures; this is the formulation in the CPB's consumption equation. Based on this measure, volatility (as measured by the standard deviation in changes in the real rate of house price inflation) in the Netherlands (7.2 per cent) is the sixth highest in the OECD area (Table 2.3). Alternatively, it could be assumed that households anticipate real house price inflation to be equal to the rate of increase in the trend of real house prices. Based on this measure, volatility (22.9 per cent) is the highest in the OECD. Another possibility is that households give more weight to recent observations when forming expectations of trend increases, which could be approximated by increases in a Hodrick-Prescott (HP) filter of real house prices. In this case, volatility is the fifth highest in the OECD. Volatility in the Netherlands is higher than average on all three measures. Indeed, a simple average of the three measures points to volatility being the highest in the OECD.[2]

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Volatile house prices are to be expected when housing supply is price inelastic – variations in demand mainly affect price rather than quantity. But this is aggravated in the Netherlands by high subsidies through the tax system for owner-occupied housing by international comparison (Figure 2.1).[3] These subsidies mainly result from owner occupiers being able to deduct mortgage interest costs from income taxed at the individual's highest marginal rate while paying low taxes (including local property tax) on imputed rental income compared with taxes on other assets; owner occupiers normally declare a large tax loss on their primary residence (ter Rele and van Steen, 2001). In an efficient market, the present value of expected housing subsidies will be capitalised into house prices. If income rises more quickly (slowly) than expected such that households revise up (down) their estimate of permanent income, this additional tax effect on demand will translate into even higher (lower) house prices. In the event that the rise (fall) in permanent income expectations coincides with an increase (decrease) in real interest rates, as might be expected during a cyclical upswing (downturn), this stabilising effect on house prices will also be weakened by the subsidies.

The especially favourable tax treatment of debt-financed owner-occupied housing encourages households to maintain mortgages at high levels, which may also amplify economic fluctuations. It has been estimated (ter Rele and van Steen, 2001) that the subsidy to housing costs for owner-occupiers is only around 10 per cent for equity financing[4] but rises steeply with income for mortgage financing, reaching 30 per cent at a household income of around € 40 000 and 35 per cent at € 80 000. Mortgage financing combined with a capital insurance policy[5] is subsidised even more. If households choose to invest in assets, notably shares, with a different risk profile from that of the mortgage, the increased mortgage financing encouraged by the tax system will raise the risk profile of households' portfolios. This is likely to increase the amplitude of business cycles by amplifying the effects on household wealth of fluctuations in share prices, which tend to be pro-cyclical.

The government has taken several steps in recent years to limit mortgage interest deductibility when owner-occupiers withdraw housing equity: since 1997, they have not been permitted to claim interest deductions on equity withdrawals from their existing residence except for home improvements; since 2001, the period during which mortgage interest payments could be deducted from personal income has been limited to 30 years and, more importantly, a cut in tax rates has effectively lowered tax subsidies for high income earners by about 20 per cent; and, from the beginning of 2004, owner occupiers can only deduct interest on that part of the mortgage that is equal to the home's value minus the equity withdrawal from the former home.

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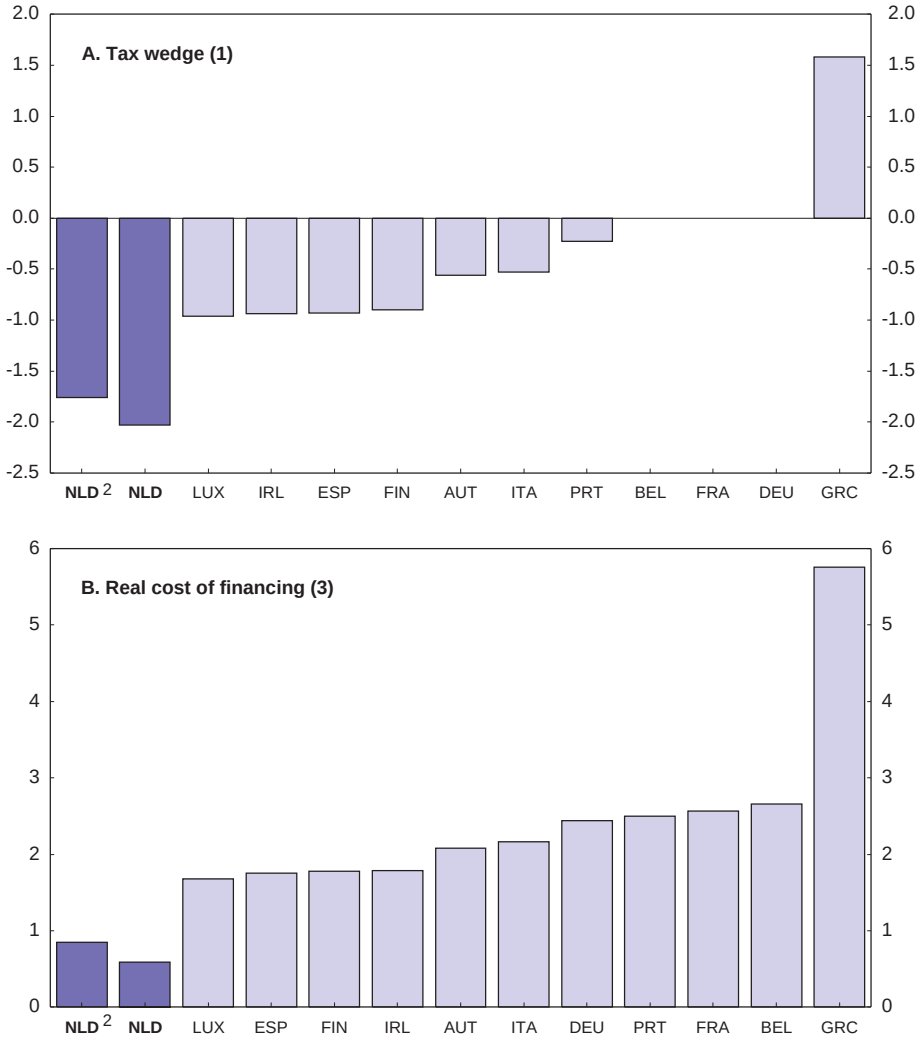
Figure 2.1. **The impact of taxation on housing cost**

Per cent



1. Difference between after-tax and pre-tax interest rate on mortgage loans; 1999 tax rules and interest rates unless otherwise indicated.
2. 2001 tax rules.
3. Real after-tax interest rate on mortgage loans; 1999 tax rules and interest rates unless otherwise indicated. Expected inflation is 2 per cent in all countries.

Source: OECD.

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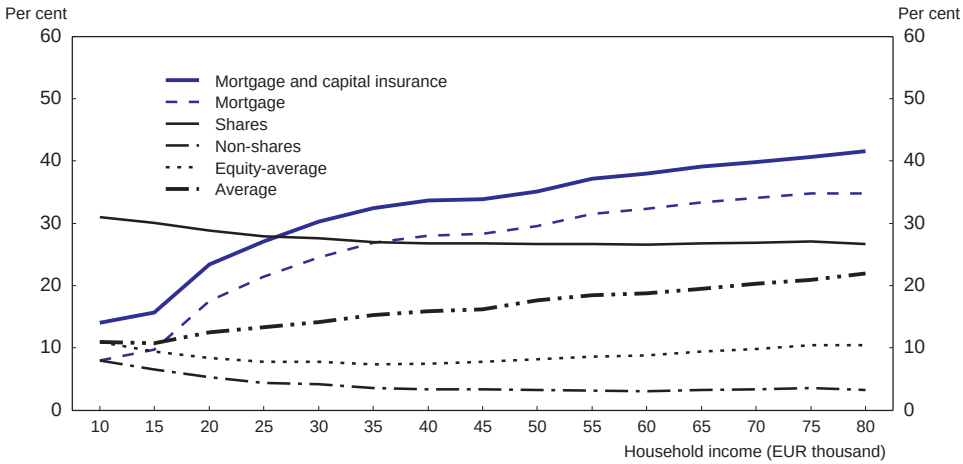
Figure 2.2. **Price subsidy under the various forms of financing**
2001



Source: Ter Rele and van Steen (2001).

Housing subsidies reduce economic efficiency and are not well adapted to achieving their social objectives

Tax subsidies for owner-occupied housing increase demand for housing services but are not very effective in achieving their social objective – increasing real consumption of housing services – owing to the low price elasticity of supply. Rather, such subsidies are capitalised into property values, transferring wealth to home owners from home buyers and tenants insofar as they are not compensated by rental subsidies (see below). High-income homeowners benefit most from these arrangements. Tax subsidies also reduce economic efficiency by narrowing the tax base (Table 2.4), resulting in higher tax rates than otherwise and hence, a higher excess burden, and by increasing financial intermediation activity, drawing resources from activities that would be more profitable in the absence of the subsidies.

High rental subsidies attenuate the effects of high property prices on rents. Part of these subsidies (€ 2 billion) are provided directly by central government in the form of housing allowances (*Huursubsidie*), which are means tested, or through housing subsidies granted to housing corporations (Government subsidy (narrow definition) in Table 2.4, most of which were capitalised in 1995).[6] But most are through lower rents charged by social housing corporations (which is partly possible because they accumulated capital by not passing on to tenants all of the subsidies governments provided them with in the past) and rent regulation.

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Table 2.4. **Housing subsidisation, 2001**
In billions of euros

Owner-occupied sector	
Tax saving from deductibility of mortgage interest payments	7
Tax saving from investing equity in house (the avoided taxation on alternative investments)	6
Tax on imputed rent	–2
Local property tax and transfer tax	–3
Total owner-occupied	8
Rental sector	
Housing allowances	2
Local tax	–1
Government subsidy (narrow definition)	1
Rent reduction by corporations and through regulation	6
Total rental (narrow definition)	2
Total rental (broad definition)	8
Total of both sectors (narrow definition)	10
Total of both sectors (broad definition)	16

Source: Ter Rele and van Steen (2001).

Taken together, high subsidies in the owner-occupied and rental sectors have many drawbacks. First, while the rental subsides attenuate the impact of high property prices on some tenants, they further increase overall taxation and the associated excess burden of taxation. Second, with the exception of housing allowances, the subsidies benefit existing tenants irrespective of their current income – new tenants are usually obliged to pay much higher rentals on the private market. The absence of means testing for these benefits (except at the point of entry into social housing) makes them very inefficient at achieving their social objectives. These arrangements may also reduce mobility for existing tenants, who benefit from subsidies only so long as they retain their current lease.[7] Third, the private rental market is small (Table 2.5). Together with regulations that hold rents for existing tenants below market rates, this means that the supply of private accommodation available for rent must be very small indeed. This would reinforce incentives for households unable to obtain accommodation from a social housing corporation to become owner-occupiers, even though being tenants would make more sense for some households (especially those for whom mobility is relatively important) in a less distorted housing market. The large share of social housing and small share of private rentals, together with rent regulations, also reduce arbitrage between the rental and owner-occupied sectors, accentuating movements in prices for owner-occupied housing. High property transfer fees (6 per cent) also discourage such arbitrage, as well as reducing mobility for owner-occupiers.

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Table 2.5. **Share of forms of tenure at the beginning of the 1990s**

Per cent

	Owner-occupied sector	Private rental sector	Social rental	Other
Netherlands	**46**	**13**	**40**	**1**
United States	67	32	1	–
Great Britain	68	10	22	–
Denmark	56	19	21	5
Germany	40	40	20	–
France	56	21	17	6
Sweden	43	17	22	18

Source: Ter Rele and van Steen (2001).

Assessment

The Netherlands has experienced large increases in real house prices in relation to disposable income in recent decades and high volatility in real house prices. This has amplified business cycles. The low price elasticity of housing supply has contributed to both phenomena. The government should relax the rules in rural areas with which municipalities must comply when developing zoning regulations, as it is considering doing. This, together with the planned simplification of building regulations and the activities of "boosting teams" in solving impediments to housing supply at specific sites or in specific regions should increase the price elasticity of housing supply, attenuating real price increases and economic fluctuations emanating from house price volatility.

High tax subsidies for owner-occupied housing, especially mortgage financed, have reduced economic efficiency, may have accentuated price volatility and are unlikely to have achieved their social objectives. While the government has taken several steps in recent years to limit mortgage interest deductibility when owner-occupiers withdraw housing equity, it should go much further in this direction. Tax subsidies for owner-occupied housing should be phased out. This would entail progressively moving owner-occupied housing (along with insurance policies linked to mortgages) to the third box in the hybrid Dutch personal income tax system, in which capital income is taxed at 30 per cent on an imputed (nominal) return of 4 per cent on net assets.[8] In fairness to existing homeowners, who paid high prices in the expectation of receiving these tax subsidies, the new regime should only apply to incremental purchases of owner-occupied housing (*i.e.*, existing tax subsidies should be grandfathered).[9] This would also limit the decline in real house prices from such a change.

While rental subsidies have attenuated the effect of high property prices for existing tenants to varying degrees, this is not so for new tenants who do not qualify for the housing allowances for low-income households and/or housing from

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a social housing corporation. Indeed, insofar as these subsidies take the form of strict rent regulations for existing tenants, this is at the expense of potential new tenants because it narrows the private rental market, driving up rents. The government should relax rent regulations so that rentals for existing tenants can be adjusted to market rates at regular intervals, such as every three years. This would allow the private rental market to develop from its current atrophied state and would permit social housing corporations to raise rents to market rates for tenants who no longer have low incomes. At the same time, the social housing corporations should be obliged to apply the subsidies that government has provided them with to reducing rents for low income tenants according to a government approved schedule. Such targeting of these subsidies would reduce the amount the government needs to spend on housing allowances, reducing the efficiency costs of taxation. If owner-occupied subsidies though the tax system were to be phased out, rental subsidies could be phased down as well.

Pension funds

The pension fund solvency crisis

Solvency ratios have declined dramatically owing to the stock market crash and falling interest rates

The average pension fund solvency ratio[10] has fallen from 159 per cent at the end of 1999 to 108 per cent at the end of 2002. This is well below the level – around 130 per cent – which the Dutch administration recently considered to be necessary for an average pension fund to reduce to a socially acceptable level the risk of default on pension funds' unconditional liabilities. Few funds already meet this standard (10 per cent of funds representing 10 per cent of assets). A significant minority (45 per cent of funds, including the two largest funds, ABP and PGGM, together representing 40 per cent of assets) had a solvency ratio of less than 105 per cent at the end of 2002. The situation has improved in 2003,[11] but there is still a long way to go.

The sharp deterioration in pension fund solvency since 1999 is mainly related to unfavourable developments in international stock markets, but other factors have also exerted downward pressures on solvency. Given that pension funds are allowed to discount their liabilities at a fixed rate of 4 per cent, high interest rates in the early 1990s provided them with ample reserves to cover implicit or conditional indexation promises. When a secular decline in interest rates set in, pension funds increased the weight of equities in their portfolio in order to compensate for the effect on investment returns without earmarking extra reserves as a buffer against their higher risk profile. This enabled them to maintain their indexation promises without increasing contribution rates to their higher cost-covering level. The high stock market returns in the late 1990s led, in some

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cases, to high expectations about future returns, and some pension fund trustees even granted contribution holidays. In addition, reserves were further depleted by contracting new pension promises. When the stock market crash inflicted heavy capital losses, many pension funds were left with insufficient buffers and contributions that were generally too low even to cover their nominal liabilities. Some pension funds became technically insolvent, although none of them ran into liquidity problems.

Measures to restore pension-fund solvency

The Pensions and Insurance Supervisory Authority has clarified and strengthened its financial supervisory framework

On 30 September 2002 the Pensions and Insurance Supervisory Authority (PVK) sent out a circular letter to all pension funds, clarifying the current financial supervisory framework and outlining the adjustments needed to restore pension fund solvency and to establish more prudent financial management in the short and medium run. This circular letter temporarily provided a basis for the financial supervision of pension funds in 2003, but recently lost its relevance after the Dutch Parliament approved the Principles of the new Supervisory Framework of Pension Funds in March 2004.

Pension funds have cut back on their pension promises while raising contribution rates

In an effort to restore their solvency, pension funds have raised contribution rates sharply, from 10½ per cent of gross wages on average in the private sector in 2002 to an expected 13 per cent in 2003 and 14 per cent in 2004; the corresponding increases in contribution rates in the public sector are from 12 per cent of gross wages in 2002 to expected rates of 15¼ and 19 per cent in 2003 and 2004 respectively (CPB, MEV 2004). These rates now consist of a cost-covering component and a catch-up premium reflecting the need to restore pension fund solvency in line with the requirements spelled out in the PVK's circular letter. The catch-up premium benefits older cohorts at the expense of younger cohorts. Even sharper hikes in contribution rates have been avoided by new measures to slow down the growth of future liabilities, such as the move away from final salary schemes towards average salary schemes, suspension of full indexation of pensions and accrued rights, and more emphasis on the conditional nature of indexation (for example, by tying the level of indexation to the solvency position of the fund). These changes have profound implications for the individual as well as for the economy.

Whereas two thirds of employees participated in final-salary defined benefit schemes before the solvency crisis, with most of the rest in average-salary defined benefit schemes, the switch to average salary schemes since then has been

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such that these proportions have been reversed.[12] This shift provides pension fund trustees with a powerful tool to restore solvency because indexation of accrued entitlements (and pensions) is generally conditional on the fund's financial situation. If accrued entitlements are only partially indexed the resulting pension benefit will be well below 70 per cent of final wages. Individuals in this situation would be entitled to increase tax deductible contributions to third-pillar schemes, putting the government budget under additional pressure and making fiscal sustainability harder to achieve.[13] These changes have been most detrimental to younger workers as their expected contributions exceed the expected benefits in this less generous scheme. The alternative, however, would have been higher contribution rates resulting in an even greater loss of competitiveness and jobs.

Another implication of partial indexation in an average salary scheme is that the average wage to which pension benefits are linked will become a weighted average of all earnings with a higher weight given to recently earned salaries. As a result, the disadvantages associated with a final wage scheme – solidarity benefiting employees with a steep wage profile, strong financial incentives for older workers to prefer early retirement to a gradual exit through a reduction in hours worked, a barrier to labour mobility, and reinforcement of vested interests in seniority-based payment schemes – will be somewhat attenuated but not eliminated.

But the general public still needs to be informed in clear language about these changes

The new pension arrangements have shifted some important risks to participants (and pensioners) because state-contingent indexation[14] of accrued rights and pensions and management decisions by pension fund trustees introduce uncertainty about pension benefit replacement rates. However, the general public does not yet appear to be worried about its future pensions (DNB, 2002).[15] This can partly be explained by a lack of knowledge about pension arrangements, the degree of and rules for indexation, accrued pension rights or the level of future benefits. Pension funds have started to include information about the indexation decision and their financial position in their communication with clients and their professional organisations have also taken initiatives to improve the flow of information. There still appears to be much scope for improvement judging by pension fund surveys of their clients, which suggest that they still have difficulty understanding the information presented to them. In addition, the new Pension Act, expected to take effect in 2006 will supplement these efforts by including clear guidelines on communication and transparency and providing for supervision on compliance with such guidelines. It will be important to monitor whether these initiatives help in gradually improving the public understanding of the Dutch pension system.

It is particularly important to give people changing jobs better access to information at a low cost that allows them to assess the solvency and to compare

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the generosity and indexation ambitions of different pension funds in order to assess the gains or losses in future pension benefits incurred when entering into a new pension arrangement. The Pension and Savings Fund Act (PSW) of 1994 gives employees changing jobs a choice between leaving the accrued pension entitlements with the pension fund of the former employer or transferring them to the fund of the new employer. In addition, it has established the basis and rules for calculating the present value of accrued pension entitlements which can be transferred to the new pension fund with the aim of protecting the value of pension entitlements. However, there is a risk that the number of service years at the new fund that corresponds with the amount transferred is lower than the actual number of years worked, in particular when the new pension fund grants a higher pension entitlement for each year of contribution. Moreover, when pension entitlements are transferred, their future value depends on the level of indexation at the new fund, which needs to be compared with the expected level of indexation at the old fund.[16] Financial losses can also occur when indexation is linked to prices rather than to wages, or when the new fund is less generous in other respects (for example the franchise[17] or pension age). It should be noted however that the current arrangements governing the transfer of pension entitlements has so far not acted as a barrier to labour mobility. The upcoming Pension Act will require employers to clearly communicate all terms of employment, including pension schemes, and to inform prospective employees if their pension fund has substantial solvency problems.

The government has tightened the regulatory framework for the supervision of pension funds

In March 2004, the Dutch parliament approved the Principles on the Regulation of Financial Supervision of Pension Funds. These Principles will be elaborated in the new Pension Act which will enter into effect in 2006. The PVK can take the Principles immediately into account. They reflect a political consensus on the trade-off between the certainty and security of pension provision on the one hand and the economic costs on the other hand. Pension funds investing 50 per cent of their assets in bonds and the other 50 per cent in stocks will have to restore the capital coverage of their pension liabilities to at least 130 per cent within a period of 15 years. Pension funds with a higher (lower) exposure to stock market risks should aim for a correspondingly higher (lower) solvency ratio. Simulations of the solvency test[18] have shown that on average a solvency ratio of 130 per cent is necessary to comply with the requirement that pension funds can still honour their pension liabilities one year later with a probability of 97.5 per cent. In other words, the likelihood that a pension fund will end up with insufficient capital to cover its nominal liabilities in the course of a year is 2.5 per cent, implying that a situation of underfunding occurs once every forty years. The pension liabilities to which the solvency ratio applies include all unconditional

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entitlements (all nominal pension entitlements, all index-linked increases in accrued entitlements and pensions to date and the present value of all unconditional indexation promises). The upcoming Pension Act will include clear criteria for judging when indexation promises are considered to be unconditional.

The new regulatory framework has at least three benefits. First, it ensures some continuity so that at least the accrued nominal pension entitlements can be recovered with a high degree of certainty when a pension fund is closed down, or in the case of bankruptcy of its sponsor.[19] Insurance against such risks is quite important, given that company-based pension funds represent approximately 30 per cent of total assets in occupational pension schemes and that the Dutch government does not guarantee the payment of accumulated pension entitlements if a fund goes bankrupt. Secondly, it is important to maintain full capital coverage of liabilities for most of the time in order to avoid the pension burden being passed on to future generations indefinitely, making the scheme unattractive to them and hence jeopardizing its sustainability. Finally, the stability of the pension funds is enhanced by raising their resistance to negative shocks, thereby reducing high volatility in pension contribution rates.[20] These benefits can only be fully captured in the medium run, after 15 years when pension fund solvency has been restored. In the recovery period, pension funds will remain very vulnerable to negative shocks, in particular to shocks originating in financial markets. On the other hand, shortening the adjustment period would raise the macroeconomic costs of the adjustment and further weaken the Dutch economy.

It has been estimated (CPB, 2004) that both a scheme with unconditional indexation of benefits to prices and a scheme with unconditional indexation of pension benefits to wages[21] would require much higher solvency ratios for full capital coverage than 130 per cent. This suggests that, if pension funds limit their ambition to merely complying with the new regulatory framework by holding their solvency ratio constant at around 130, it will only be possible to partially index future pension benefits and accrued rights in average salary schemes, unless full indexation can be financed in the future on a pay-as-you-go basis by higher contributions or by higher than expected investment returns. For full indexation to resume after 15 years, additional sacrifices in the form of higher contribution rates and/or a prolonged suspension of indexation would be necessary in the short run (CPB, 2004). The same conclusion holds if the government wishes to guarantee nominal pension entitlements with a higher degree of certainty.

The Pensions and Insurance Supervisory Authority is developing a new supervisory framework

The PVK is developing a new financial supervisory framework, which will come into effect in January 2006. This framework will need to be consistent with the new Pensions Act, many details of which are still unknown. Comparison of the

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Principals and the original PVK proposal indicates that the implied solvency requirements of the latter are stricter than those of the former, so there remain many uncertainties. In its current design, the proposed framework essentially consists of three tests:[22] a minimum test, a solvency test and a continuity test (also known as continuity analysis). The minimum test checks whether the present value of the assets exceeds the present value of the contracted liabilities by at least 5 per cent at a given time.[23] The purpose of the solvency test is to verify whether the pension fund can pass the same minimum test one year from now with a high probability (97.5 per cent in the legal framework). Finally, the continuity analysis assesses whether the current financial position and policies (investment policies, risk management strategies, indexation ambitions) are adequate to meet risks in the long run. In the analysis, pension funds will be expected to provide information about their objectives, policies and policy instruments, their projections about the evolution of the solvency ratio as well as the assumptions about the macroeconomic parameters underpinning these projections, and the results of several "stress testing" scenarios. The analysis is an important source of information for the supervisor, as it provides him with valuable insights into the objectives, expectations, threats and policy options available to fund managers. At the same time, it allows pension fund managers to assess whether ambitions and corresponding risks are in a healthy proportion with the existing financial buffers and available instruments, whether reasonable assumptions (*e.g.* in relation to past realizations) are underpinning their expectations, or whether the fund has a financial emergency plan.

Finally, in line with the new International Accounting Standards (IAS), the upcoming financial supervisory framework stipulates that fair rates or market rates as determined by the yield curve for government bonds of high quality be used for calculating the present value of pension liabilities. When valuing liabilities at fair rates, pension funds will need buffers to hedge against the risk of falling interest rates because the resulting increase in the present value of pension liabilities will exceed that in assets insofar as the assets are bonds of shorter maturity than the pension liabilities and/or the assets are stocks. Given that current rates of return on bonds are at a historical low, a further drop would seem rather unlikely. In addition, valuation on a fair rate basis can magnify volatility of the solvency rate.

Assessment

Given the massive shift towards average salary schemes which has occurred in recent years, the possibility to suspend indexation of pensions and accrued rights provides pension funds with a very powerful tool that they are likely to use to respond to financial shocks. This development is desirable from a macroeconomic viewpoint as it reduces the volatility of pension fund contribution

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rates, which operated as a transmission mechanism of financial shocks to the whole economy. On the other hand, along with this move, important risks have been shifted to participants, although the latter are not fully aware of that yet. There is a growing understanding among all parties concerned that communication with participants will need to be more transparent, in particular with respect to indexation. The upcoming Pension Act will include some guidelines on minimum reporting standards, which should make pension arrangements more comprehensible for pensioners and the general public.

Restoring the solvency ratio to a level of on average 130 per cent of unconditional pension liabilities is necessary to create a buffer against financial shocks, but not sufficient to resume full indexation of pensions and accrued rights to wages. The cabinet and the social partners considered the objective of full indexation as too costly for the Dutch economy as it would imply further hikes in contribution rates. Instead, pension funds are given the freedom to choose their own level of ambition with respect to indexation without being confronted with additional solvency requirements, as long as the degree of conditionality of such promises remains within certain limits to be defined in the upcoming Pension Act. An increasing number of pension funds apply a rule linking indexation to the financial position of the fund. Such rules make partial indexation of pensions and accrued rights a likely outcome. As a result, future replacement rates are expected to fall short of 70 per cent of final wages, implying that third pillar (tax-deductible) pension contributions may increase.

The PVK's circular letter in 2002 and its preliminary design of a new financial supervisory framework that will enter into effect in 2006 are both necessary steps in reducing the risk of a solvency crisis, inducing more prudent financial management, monitoring pension fund policies (relative to their objectives) and assessing the weaknesses in the current system. It is felt that these measures together with the recently approved Principles on the Regulation of the Financial Supervision of Pension Funds, which enjoy the support of the social partners, will keep the Dutch pension system sustainable.

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Notes

1. Mortgage lenders in the Netherlands now take both incomes in a household into account when calculating borrowing capacity. The typical loan-to-value ratio in the Netherlands is 90 per cent, the highest in any OECD country for which data are available (OECD, 2004).
2. In the 1990s, volatility was below average on two of the measures and above average on the other. The simple average of the three measures of volatility was only somewhat higher than the average for all the countries considered.
3. Van den Noord (2004) finds a strong correlation between marginal effective tax wedges on owner-occupied housing and the variability of real house prices (gauged by the root mean square of the per cent deviation from trend) across euro area countries. He estimates that the Netherlands has the largest tax breaks and the greatest price variability.
4. The subsidy is higher (around 30 per cent) if the alternative equity investment is shares because of the higher rate of corporate taxation that is due on this form of investment. In the non-share case, the subsidy is small because the alternative investment is taxed only slightly higher.
5. With such a combination, principal repayments are paid into the insurance policy rather than deducted from the outstanding mortgage. This enables the borrower to maximise mortgage interest deductions by not paying off the debt while at the same time accumulating capital in the insurance policy to pay off the debt when the mortgage term expires. Income on insurance investments linked to a mortgage is not taxed.
6. The government transferred € 16.7 billion to social housing corporations in 1995, representing the present value of future subsidies to which they were entitled.
7. Less scrupulous social housing tenants do, however, move but illegally sub-let their flats, thereby continuing to benefit from the difference between their low social housing rental and market rates.
8. Since 2001, income is allocated to three categories in the personal income tax system: *a*) labour income, imputed rental income from the main residence less mortgage interest payments, and social benefit receipts; *b*) income from substantial business interests (meaning a shareholding of more than 5 per cent); and *c*) income from net wealth. Transferring owner-occupied housing to the third box would bring its taxation into line with that of other assets. Imputed rentals would be taxed at 1.2 per cent (30 per cent tax rate on an assumed return of 4 per cent) of the property value net of mortgage debt. For example, the imputed rental taxation on a house worth € 500 000 with a € 300 000 mortgage would be € 2 400 (1.2 per cent of € 200 000).
9. For example, suppose that an owner-occupier moves from a house worth € 500 000 with a € 300 000 mortgage to a house worth € 700 000 with a € 500 000 mortgage. He would continue to include imputed rental income based on the value of the former home

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(worth € 500 000) less interest charges on the old mortgage (€ 300 000) in Box 1 (labour income) of the personal income tax (PIT) system. However, the incremental € 200 000 of housing assets (€ 700 000-€ 500 000) and the associated increase in mortgage debt of the same amount would be included in Box 3 (income from net wealth) of the PIT, which in this case would not generate any extra tax liability as there is no increase in net wealth. Nevertheless, the owner occupier would pay more tax than under the current system because he would no longer be able to deduct interest on the additional € 200 000 mortgage debt from labour income.

10. The solvency ratio is defined as the current value of assets, divided by the present value of accrued future pension liabilities net of investment returns. Under the current accounting standards, which will stay in operation until the end of 2005, nominal liabilities are discounted at a fixed rate of 4 per cent, whereas assets are valued at market rates. In addition, the PVK imposes a ceiling on the rate of return on investments which funds can use, namely a maximal rate of return of 5 per cent on bonds and a maximal average risk premium of 3 per cent.

11. The solvency ratio of ABP and PGGM went up to 109 and 105 respectively at the end of Q4, 2003.

12. Accrued entitlements from the former final salary scheme are generally grandfathered – for employees with such entitlements, the average wage is calculated as the average of the wage earned at the moment of switching to the average-wage scheme and subsequent earnings. In order to make the new scheme more attractive to younger workers, the franchise (see below) is often reduced and the accrual rate increased. Even if accrued pension entitlements were to be fully indexed to wages, which is generally not the case, the resulting pension would normally be lower than would have been the case in the final wage scheme.

13. Fiscal sustainability implies that the future costs of government outlays can be financed at constant tax rates.

14. Several pension funds recently introduced the "staffelsystematiek", which links the level of indexation to the solvency position of the fund.

15. According to the DNB survey, 44 per cent of respondents do not know the type of pension scheme that they are covered by, 61 per cent do not know about their pension rights, and 65 per cent have no idea about the scale of their future benefits.

16. In case no transfer takes place, the accrued rights of employees who no longer contribute (sleepers) are legally required to be indexed at the same rate as pension benefits.

17. Most pension funds aim to pay an old age pension of 70 per cent of final salary (or average salary) after 40 years of contributions. This replacement rate takes into account the first pillar AOW-pension through an adjustment known as the AOW franchise.

18. The PVK has issued a white paper of the solvency test, outlining a standardised method that pension funds can use to calculate the required solvency rates given that unconditional pension liabilities need to be honoured with a 97.5 per cent probability in a year. The underlying idea of the test is that pension funds need to hold buffers against a number of risks, identified in the paper. The capital requirement associated with each risk (denoted by S) depends on the specific characteristics of the pension

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fund and on the parameters given by the regulator, which determine the size of the unfavourable shock. The solvency requirement is calculated as:
√(S interest + S stocks + S real estate)2 + S2 credit + S2 currency + S2 commodities + S2 assets + S2 insurance techniques.

Stress test	97.5 per cent parameters
Interest rates	Depends on maturity
Stocks mature markets	25 per cent decline
Private equity	30 per cent decline
Emerging markets	30 per cent decline
Real estate	15 per cent decline
Credit	40 per cent increase in spread
Currency	20 per cent depreciation with respect to the euro
Commodities	30 per cent decline
Specific investment risks	2x tracking error
Insurance related risks	Fixed at 5 per cent of liabilities

Note that the test assumes a perfect positive correlation between shocks in interest rates, stock prices and real estate prices. The other shocks are assumed to occur independently from each other. The standardised method yields a solvency requirement that corresponds to a solvency ratio of 130 per cent on average.

19. Note that in the Netherlands, bankruptcy of the sponsor does not automatically imply closing of the pension fund as they are separate legal identities.
20. The necessary adjustments in pension contribution rates following a shock in financial markets will become larger over time as the dependency ratio increases. That is because a shock of the same magnitude has to be born by a smaller number of contributors.
21. Real wages are assumed to grow on average by 1.5 per cent annually.
22. The PVK has issued three white papers for discussion and comments. In the papers, each of the tests is described, with lots of possibilities for pension funds or life-insurance companies to fill in some blanks, indicate the best option or give their feedback. A more final version of the financial supervisory framework is expected later this year, and it will be part of the new Pension Act.
23. The 5 per cent buffer is related to general risks.

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3. Fiscal policy

There has been a very substantial deterioration in public finances, from a surplus of 1½ per cent of GDP in 2000 (excl. UMTS) to an expected deficit of almost 3¼ per cent of GDP in 2003. Indeed, it has been necessarily to implement consolidation measures during the economic downturn to avoid the budget deficit from exceeding the 3 per cent limit imposed by the Stability and Growth Pact (SGP). While most of the deterioration in the general government budget balance appears to have been cyclical, the structural budget balance has nevertheless also fallen considerably. The government is actively addressing the need for further consolidation, and measures are aimed at putting public finances on a sustainable path – meaning that it is possible to keep up social security and government expenditure without running into financial problems, and thus having to raise taxes in the future (Ewyk, 2000). The medium-term budgetary framework has also been tightened to reduce the risk of slippage in the structural budget balance in an upturn from recurring. The consolidation measures and the reforms to the medium-term budgetary framework should increase the room for automatic stabilisers to operate fully in future downturns.

Recent developments

The budgetary position has deteriorated markedly, pushing public finances away from a sustainable path

Expenditure as a percentage of GDP declined almost continuously until 2001 (with an interruption in 1995[1]) as expenditure ceilings constrained their growth to be less than that in GDP (Figure 3.1). This declining trend has been reversed since then owing to the exceptionally poor performance of the Dutch economy and high expenditure growth in priority areas. The tax burden declined in the mid-1990s to stabilize at around 41 per cent of GDP until 2000, since when there has been a further decline, mainly attributable to the economic downturn, a tax reform and special circumstances that have reduced tax receipts. Following several years of improvements in the fiscal balance, the general government budget balance fell from a surplus of 1.5 per cent of GDP in 2000 (excluding UMTS proceeds of 0.7 per cent of GDP) to an expected deficit of 3.2 per cent of GDP

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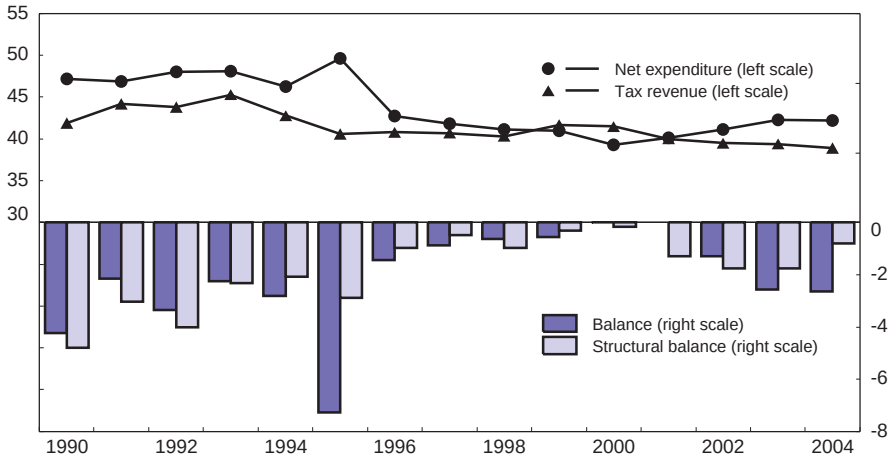
Figure 3.1. **General government finances**
Per cent of GDP



Source: CPB Netherlands Bureau for Economic Policy Analysis, OECD.

in 2003. The deterioration in 2001 and 2002 was mainly due to a deterioration of the structural balance as measured by the Netherlands Bureau of Economic Policy Analysis (CPB),[2] which had fallen considerably from roughly being in balance in 2000 to a deficit of 2.2 per cent of GDP in 2002. The further deterioration of the budget in 2003 was cyclical, with the structural deficit remaining unchanged (CPB 2004) as a result of the implementation of the coalition agreement of Balkenende I (Strategisch Akkoord) which included a fiscal consolidation package and hence prevented a further slippage of the structural balance.

The cyclical downturn was unexpectedly severe, while recent new measures had reduced available budgetary margins

The rules for sound fiscal policy spelled out in the Stability and Growth Pact strongly recommend that countries aim for a structural budget balance close to zero or in surplus so as to allow sufficient room for automatic stabilisers to operate freely over the business cycle without breaching the 3 per cent deficit limit. In the Netherlands, the structural balance had reached a position close to balance in 2000, but with the benefit of hindsight, it appears that fiscal policy at that time resulted in it slipping back to a deficit of 2.2 per cent of GDP in 2002. This evolution which had not been fully anticipated at the time[3] can be attributed

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to a larger than foreseen budgetary claim of the tax reform of 2001, the use of windfall revenues and expenditures, and some non-recurrent one-off factors depressing tax receipts.

Windfalls on the revenue side were used to finance tax cuts in addition to the income tax reform

Revenue windfalls occur when government revenue is higher than in the medium-term budget projections made when each new government is formed and based on cautious macroeconomic projections. Such windfalls were available during the boom of 1999-2000 when economic growth had been much higher than assumed in the medium-term projections. At the time the medium-term budget framework allowed for half of the available revenue windfalls to be used on tax cuts, the other half being allocated to debt reduction. In reality, only 25 per cent of revenue windfalls were used to finance tax cuts, but these cuts came on top of an income tax reform (estimated to have cut taxes by 0.5 per cent of GDP) implemented in 2001 as part of the coalition agreement of 1998. However, revenue windfalls evaporated as economic growth fell below its expected value during the downturn and estimates of potential growth[4] were revised downwards, making the cautious projections at the time of the coalition agreement look somewhat optimistic. In retrospect, it would have been preferable to have allocated all windfalls on the revenue side to debt reduction and none to additional tax cuts.

Windfalls on the expenditure side were used to finance new structural policies

The previous medium-term budgetary framework allowed for windfalls on the expenditure side, which arise when expenditure falls short of that in the medium-term projections, to be used for additional spending. Hence, lower than expected unemployment outlays and interest payments during the boom were compensated for by higher growth in non-cyclical outlays for education, health care, safety and infrastructure while still respecting the ceiling on total expenditure growth. This worsened the structural budget by an estimated 0.5 per cent of GDP between 2000 and 2002. According to the same rules, expenditure overruns in one area have to be offset by lower expenditure elsewhere because ceilings are binding. When unemployment outlays began to rise more rapidly than expected in 2002, consolidation became unavoidable.

One-off factors have reduced tax receipts

The remaining part of the deterioration in the structural balance between 2000 and 2002 – approximately 1 per cent of GDP – can be attributed to one-off factors unrelated to discretionary fiscal policy, which have reduced tax receipts. Pension fund contributions, which are tax deductible, have increased in the context of efforts to restore pension fund solvency, mortgage interest

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deductions have increased following the boom in house prices and cyclically-adjusted corporate tax revenues have been weak, perhaps reflecting the adjustment to more normal levels following the boost in 2000-01 from the exhaustion of loss carry-forwards.

Four consolidation packages have been enacted to put public finances on a sustainable path

In order to put public finances on a sustainable path, the government has enacted four consolidation packages since July 2002 – two of them were part of the coalition agreements of Balkenende I and II in the springs of 2002 and then 2003, respectively, and the other two packages comprised additional measures included in the budgets for 2003 and 2004 – with a cumulative effect of some € 20 billion (3.8 per cent of GDP) by 2007 (Table 3.1). Roughly 80 per cent of the measures involve net expenditure cuts,[5] while the remaining 20 per cent takes the form of tax increases. On this reckoning the reduction in the total tax burden resulting from the 2001 tax reform is only in part offset by the consolidation packages, but if higher contributions for health care and second pillar pensions are also taken into consideration, the 2001 reduction in the tax burden has been more than reversed.

Table 3.1. **Impact of consolidation packages on budget balance**
Per cent of GDP

	2003	2004	2005	2006	2007	Size package in bn euros
EMU-deficit without consolidation	3.0	3.8	3.8	3.0	2.9	
Effect of coalition agreement Balkenende I[1]	–0.7	–0.6	–0.5	–0.5	–0.5	+5
Effect of coalition agreement Balkenende II	0.0	–0.6	–1.3	–1.2	–1.3	+11
Effect of additional measures budget 2004	0.0	–0.3	–0.4	–0.4	–0.5	+4
Actual EMU deficit	2.3	2.3	1.6	0.9	0.6	

1. Inclusive additional measures of budget 2003.
Source: Ministry of Finance, *Miljoenennota* 2004, p. 87.

All packages focus on major expenditure cuts in public and healthcare administration, healthcare coverage, social transfers, and subsidies.

- Wages in the public sector will be frozen during 2004 and 2005, whereas additional savings in public administration costs will arise from efficiency improvements (Box 3.1), staff cuts (a net reduction by some 10 000 is planned, to be achieved mainly through natural attrition) and less money for hiring external professionals. In the healthcare sector, wages will be frozen for two years as well.

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Box 3.1. Reforms to increase the cost effectiveness of public spending

Several steps have been taken to enhance the cost effectiveness of public spending. A systematic evaluation based on cost-benefit analysis of all newly proposed national infrastructure investments was introduced in 2000 and has already resulted in important cost savings and the implementation of better projects. The creation of the Professional Purchasing and Public Procurement Agency (PIA) in 2001 has proven to be effective as well and its lifespan has been extended until 2006.

The government introduced performance-based budgeting[1] (PBB) in 2002. The budget now routinely provides information on the outcome targets for each ministry and an annual evaluation report is published on the third Wednesday in May. The focus on results has also been facilitated by the creation of agencies[2] and semi-independent public entities[3] (ZBOs). Between 60 and 70 per cent of civil servants are currently employed by an agency and this share is expected to increase further to somewhere between 80 and 90 per cent. Parent ministries of the central government enter into contractual arrangements with their agencies and ZBOs specifying the objectives that they are expected to achieve, the corresponding indicators that will be used to assess performance, and the way in which they will be rewarded (sanctioned) for their (lack of) results. These contracts are a substitute for direct input controls, hence the challenge to design contracts in such a way that agencies and ZBOs have an incentive to contribute towards the government's objectives. Like other governments that have sought to move to PBB, the Dutch government has experienced difficulties in implementation.

- First of all, objectives need to be clear and are ideally set by the ministries. However, it has been found that in practice there are often too many objectives with no clear prioritisation and they often take the form of operational output objectives because they are set by civil servants instead of politicians. As a result, objectives reflect the need of the organisations more than the political priorities. On the other hand, the government may not always have an interest in committing itself to clear objectives over a longer period of time because that may reduce the flexibility with which it can respond to unforeseen events. In such cases, *ex post* accountability or benchmarking may be preferable.
- Secondly, it has proved difficult to come up with a limited number of indicators that can adequately measure the efforts put into achieving a goal and/or that are closely correlated with that goal. The development of good performance indicators is a long-term process of learning by doing and learning from other countries. This requires significant investments in the generation and improvement of information. In the case of the recent performance contracts concluded with the police forces, it has been suggested (CPB 2004) that the desired outcomes may not be achieved because the contracts may induce police forces to shift their efforts towards quantifiable tasks that will be rewarded at the expense of more qualitative work that is harder to measure and hence less rewarded. Australia and the United Kingdom have succeeded in improving the performance of their police forces through a system of benchmarking. The Netherlands should consider adopting a benchmarking system as well.

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Box 3.1. Reforms to increase the cost effectiveness of public spending (*cont.*)

– Finally, the enforcement of contracts has been weakened by a tradition in which problems of poor performance are preferably settled through negotiation and consensus. In spite of these difficulties, there is a perception that PBB has enhanced transparency about the government's objectives and about the extent to which these objectives have been realized with a given amount of taxpayer's money. A first major evaluation of PBB is due to take place in 2004.

Human resource management practices in the Netherlands are compatible with a greater focus on results, although it is expensive to make employees or managers redundant. On the one hand, current pay arrangements in the public sector appear to leave some scope for remunerating employees with highly valued skills or outstanding performance. Moreover, the Ministry of Internal Affairs is reviewing the system of pay scales with the aim of raising the weight given to attributes other than seniority in the determination of public sector wages. In contrast, the consolidation packages have imposed restrictions on incidental wage growth. Such controls can probably not be maintained in the medium run in view of the looming skill shortages and the retirement of many civil servants. On the other hand, it is possible to lay off civil servants under certain conditions[4] as part of a major reorganisation, but so far this option has not been used because civil servants are covered by generous unemployment insurance arrangements (relative to private sector employees), and the former employer must pay both the unemployment benefits and the top-ups. Hence termination of the employment contract offers almost no costs savings. In addition, managers of agencies and especially ZBOs usually operate under contracts that are very restrictive with respect to the conditions under which dismissal is possible and that stipulate high termination payments.

1. The approach is known in the Netherlands as "*Van Beleidsbegroting tot Beleidsverantwoording*" or VBTB.
2. Agencies are separate corporate identities under direct ministerial control, subject to supervision and instruction by its core ministry.
3. ZBOs are organisations in charge of the implementation of one or more public policies, for which they receive public funding. They currently operate at arms-length of government, but their statute is under supervision and they may be brought back under closer control of the government.
4. Before laying off a redundant employee, a ministry or agency must first verify if there are vacancies in another ministry or agency that match the profile of this employee. If available, the employee will be given preferential treatment in the hiring decision. If not available, the employee can be made redundant.

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- Important savings are also expected from reductions in the services covered by the basic healthcare package, the introduction of compulsory private insurance for essential curative care services, an increase in contributions to the long-term care scheme (AWBZ) and the introduction of a no-claim deduction. All of these measures are planned to come into effect in 2004 and 2005, prior to the introduction in 2006 of a new health insurance scheme aimed at increasing competition amongst insurers and making them more active purchasers (Box 3.2). The new health insurance scheme is expected to impose a budgetary cost of € 0.6 billion in 2006, falling back to € 0.2 billion in 2007. This negative impact on the budget is due to a package of income support measures, targeted towards lower income groups, which the government will implement in 2006 to compensate for the higher burden on households resulting from a shift towards private financing of health care expenditure. The costs of the compensation packages initially exceed the savings on public health care expenditure, but given that the latter are expected to grow more rapidly, the negative budgetary impact is set to decline in 2007 and will presumably be reversed subsequently.

- As for the social sector, measures will be taken to reinforce the financial incentives for reintegration, which should reduce the number of transfer recipients. In addition, access to disability and unemployment benefits will be tightened (see Chapter 4). Social transfers will be frozen in 2004 and 2005, after which indexation will be linked again to the average of wage increases in the public and market sector.

- Finally, a wide range of subsidies, including rental subsidies, will be cut. These large cuts in what the government considers to be less productive expenditure will to some extent be offset by increases in more productive expenditure, namely the knowledge economy, infrastructure, public order, public health care and the environment, thereby improving the quality of public expenditure.

The tax burden will go up between 2004 and 2007 as a result of tax base broadening measures and some specific tax increases. The former include the gradual phasing out of the deductibility of early retirement contributions, the abolition of tax credits for employee education and the narrowing of mortgage interest deductions, whereas the latter follow from a greater emphasis on environmental taxation. However, lower marginal tax rates on labour income and corporate profits will provide some relief. If considered along with the measures of the 2001 tax reform, there has been a major shift from high marginal tax rates on labour income towards a broader income tax base and higher indirect taxation since 2000, resulting in a composition of tax revenues that is more supportive of economic growth.

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Box 3.2. Healthcare reforms to increase competition amongst insurers and encourage them to be more active healthcare purchasers

The current supply-driven healthcare system does not appear to meet patient demands adequately. This is manifest in limited choice, inadequate coordination and poor matching of supply and demand (qualitative and quantitative) (Ministry of Health, Welfare and Sports, 2003). These problems are increasing in severity as a result of social trends and advances in medicine.

The authorities aim to increase efficiency and responsiveness to patient demands by moving to a managed competition system. Compulsory private health insurance for essential curative care is to replace the existing mixture of social and private insurances in 2006. Insurance companies will not be able to refuse clients or to charge different premiums depending on their risk characteristics and there will be compensation for insurance companies that have higher risk clients. Insurance companies are free to conclude preferred provider arrangements and collective contracts. This system encourages insurance companies to compete for clients notably by being active purchasers of healthcare. A key building block for insurers to become active purchasers is the recent introduction of the Diagnosis Treatment Combinations system (DTC, which is broader than DRG because it includes related specialist outpatient services), which measures the hospital and related services required for a given treatment episode. The new private insurance will be broader than the current health insurance and cover some of the risks presently covered by the public scheme for long-term care (AWBZ).

An important condition for the reforms to meet their objectives is that consumers in fact put pressure on insurers to deliver better value services. This is more likely to occur if consumers have ready access to information on the quality of the services provided by their insurer. But quality is hard to measure, giving insurers an opportunity for advertising-as-persuasion rather than advertising-as-information to reduce their own price elasticity of demand (Appelman, 2002). This suggests that there may be a role for government to play in developing quality indicators while at the same time endeavouring to minimise the administrative burden and risk of tacit collusion that this may entail. Switching costs for consumers should also be as low as possible.

The DTC system may be especially vulnerable to creep (the deliberate and systematic shift in the reported case mix in order to improve reimbursement) as the specialist rather than an independent encoder (under classic DRG systems) assigns the DTCs to patients (Folmer and Mot, 2003). This codification process needs to be monitored. It will be important to find ways of reducing the costs of active purchasing, which may have contributed to high growth in administrative costs in the US healthcare system in recent decades (Woolhandler *et al.*, 2003, p. 773).

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The cumulated effects of the consolidation packages on the actual budget balance are an estimated 0.7 per cent of GDP in 2003, 1.5 per cent of GDP in 2004 and 2.3 per cent of GDP in 2007. As a result, the budget deficit is projected to fall from 2.4 per cent of GDP in 2004 to 0.6 per cent of GDP in 2007.

The above calculations are based on the assumption that the consolidation packages will be fully implemented. However, in the accord with social partners of 14 October 2003, agreement to freeze wages for a period of two years (2004-05)was conditional upon satisfactory progress being made on other sensitive issues, notably early retirement, unemployment and disability. This implies that some planned reforms may be modified, scaled down or dropped altogether. The government expects the direct effect of such adjustments on the budget to be limited, but the positive economic effects of the reforms and hence the expected budgetary improvement (after 2007) would be smaller than initially hoped for. On the other hand, if all reforms pass as planned and succeed in increasing labour participation by more than currently anticipated, then the budget balance will improve by more than expected after 2007.

The experience of recent years demonstrates that a structural budget in balance is not necessarily sufficient to let automatic stabilisers operate freely during a recession and respect the 3 per cent of GDP limit on budget deficits imposed by the SGP. Had there not been any consolidation packages, the general government budget balance would have fallen to an estimated deficit of almost 4 per cent of GDP by 2004 – a 5.5 percentage point of GDP deterioration from 2000. Abstracting from the part of the deterioration of the structural balance that was attributable to the spending of expenditure and revenue windfalls in 2001 and 2002 (roughly 0.8 per cent of GDP) and the tax reform of 2001 (0.5 per cent of GDP), the exceptionally sharp cyclical downturn still reduced the budget balance by approximately 4 per cent of GDP. A target of maintaining a structural surplus around 1 per cent of GDP – as foreseen in the medium-term budgetary outlook – would seem consistent with the objective of assuring full compliance with the SGP without the need for fiscal consolidation during future downswings and is in line with the CPB's estimate of what is required for long-term sustainability.

And the medium-term budgetary framework has been tightened

In order to reduce pro-cyclical fluctuations in the structural balance, the government has revised the medium-term budget framework to prevent spending of cyclical windfalls on the expenditure side and structural tax cuts financed by windfalls on the revenue side. This will ensure that windfalls are automatically earmarked for debt reduction. However, the government has not provided an exhaustive list of which expenditure, and in particular which part of unemployment benefits and interest payments, should be classified as cyclical, nor has it given any clear criteria to determine under which conditions such windfalls can be spent.

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Many elements of the previous framework have been maintained: multi-annual budget plans are based on cautious economic projections, a strict separation between expenditure and revenues applies, expenditure is sub-divided into three categories (central government, social security and health care), and debt reduction is aimed for.

On the revenue side, automatic stabilizers are allowed to operate freely, so windfalls and setbacks are reflected fully in the budget balance. However, these rules apply only if the actual budget deficit does not exceed 2½ per cent of GDP and if the structural deficit (according to the EU definition) does not exceed ½ per cent of GDP (or is decreasing by at least half a percentage point per year). If these conditions are not satisfied, then the government will implement additional measures.

On the expenditure side, growth ceilings are determined in real terms[6] for each year and for each of the three sectors as well as for total expenditure. Setbacks in one sector should in principle be compensated within the same sector; in exceptional cases other solutions should be found. Hence expenditure growth caps in one sub-sector can occasionally be breached, but the ceiling on total expenditure needs to be respected at any time. Expenditure windfalls cannot be used to finance new measures; this would require a decision by the ministerial council.

Non-tax receipts (such as royalties on natural gas extractions) are treated as a sub-division on the expenditure side but a special regime is applied. Non-tax windfalls can only be used to compensate for setbacks in another area of non-tax receipts.

Medium term prospects

The government considers that the consolidation packages are sufficient to put public finances on a sustainable path…

The government's objective to reduce the budget deficit to 0.6 per cent of GDP by 2007 is estimated to be sufficient to put public finances on a sustainable path. According to the CPB method, the actual deficit of 0.6 per cent of GDP corresponds with a structural surplus of ½ per cent of GDP since actual output is projected to still be below potential in 2007. The difference between the actual structural balance in 2007 and that consistent with sustainable public finances, estimated to be a surplus of approximately 1 per cent of GDP (CPB, 2003d), can be accounted for by the fact that (tax-deductible) pension fund contributions are temporarily above their cost-covering levels to rebuild pension fund solvency. The structural balance is expected to improve by another ½ per cent of GDP to 1.1 per cent of GDP when pension fund contributions return to their cost-covering level after 2007. Achieving the structural budget surplus consistent with the sustainability of public finances means that on unchanged policies public debt would fall substantially (from 53 per cent of GDP in 2003 to 15 per cent in 2025), reducing interest payments by enough to pay for population-ageing related budget costs.

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… but this may be more demanding than envisaged

It is worth noting that calculations on sustainability are always subject to great uncertainty and very sensitive to the assumptions made about the evolution of the exogenous parameters. For example, given current knowledge, the assumed real rates of return on pension fund assets beyond 2007 – 4 per cent on bonds and 8½ per cent on equities – used in the CPB's fiscal sustainability calculations of 2000, might be too high. They are certainly higher than implied by current prices in financial markets, which appear to be consistent with real rates of return on bonds and equities of 2 per cent and 4-5 per cent, respectively. Lower rates of return would have a negative effect on the budget as (tax-deductible) contributions to pension schemes would need to be higher and/or (taxable) pension benefits would be lower. If instead these lower real rates of return were used and, following the CPB, it is again assumed that pension funds hold 60 per cent of their assets in bonds and the remainder in equities, the structural surplus in 2007 consistent with sustainable public finances would turn out to be much higher than 1 per cent.[7] In addition, the spontaneous improvement of the budget balance, occurring when pension fund contribution rates fall back to their cost-covering level, would be smaller. As current real rates of return are low by historic standards, there is a reasonable chance that they will increase somewhat. However, it should be borne in mind that the benefits to pension funds of such a development would be partially offset by the capital losses they would incur on the re-rating of their current asset holdings.

Another downward risk is related to the possibility of increased (tax-deductible) contributions to third pillar schemes by persons in schemes (notably average salary with partial indexation of accrued entitlements – see Chapter 2) where the maximum permissible pension for which tax deductions may be made (70 per cent of final salary) is unlikely to be reached.[8] More rapid declines in mortality rates for older persons (aged 65 and over) than currently anticipated would also have severe adverse consequences for public finances both directly through outlays for first pillar pensions and healthcare and indirectly through the need to raise pension-fund contributions and/or reduce second pillar pensions. These costs – and costs occurring as a result of the other risks mentioned above – could be attenuated by increasing the official retirement age and the number of years of contributions required to receive a full occupational pension in proportion to the increase in life expectancy. On the other hand, no account has been taken of the impact on the structural balance consistent with fiscal sustainability of the increases in participation rates after 2007 which are likely to result from recent implemented and planned structural reforms. For example, the recent reform of the disability scheme will contribute towards the sustainability of public finances by strengthening the work incentives for the partially disabled (resulting in higher income tax receipts) and by reducing the number of new entrants into the scheme (resulting in lower outlays). In general, higher labour participation will lower the structural surplus consistent with fiscal sustainability.

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Given that the best estimates of the exogenous parameters underlying the sustainability calculations are likely to vary over time, the CPB will review its calculations every few years. The next comprehensive update of the study on fiscal sustainability in the light of ageing is planned for 2006 and will be published prior to the next elections in 2007. As mentioned earlier, there is a risk that the outcome of this update will require the next government to implement additional consolidation measures in order to ensure that public finances are back on a sustainable path. However, the current government has based its fiscal policy on the most recent calculations, which suggest that public finances will be back on a sustainable path by 2007. In order to maintain a stable fiscal policy framework, the government does not plan to respond to in-between changes in the values of one or more of the exogenous parameters affecting the sustainability calculations.

Assessment

Slippage in the structural budget balance in 2001-02 has aggravated the current fiscal problems. In order to prevent this from recurring, the government has strengthened the budgetary framework, earmarking cyclical windfalls primarily for debt reduction (unless otherwise decided by the council of ministers in the case of expenditure windfalls) and financing setbacks by new debt as long as there is no risk of breaching the terms of the SGP. Although these measures are most welcome, they should be accompanied by a target to maintain a positive structural balance, as evidence has shown that a mere structural balance at the peak of a business cycle is not enough to guarantee the full operation of automatic stabilisers during a recession.

The four budget consolidation packages enacted since 2002 are aimed at putting public finances on a sustainable path by 2007 and should be implemented completely. The government's commitment to this objective is very positive, given that this strategy smoothes taxes, and hence minimises the excess burden of taxation, and provides for an equitable burden sharing between current and future cohorts of the costs of population ageing. Recent efforts to consolidate the budget have been very significant. Nevertheless, additional efforts may be needed in the future if the packages cannot be completely implemented this time or if a higher structural surplus in 2007 than 1 per cent of GDP is required for the sustainability of public finances. The government has based its fiscal policy on the most recent estimates made by the CPB of the structural balance needed in 2007 for public finances to be back on a sustainable path. There are, however, considerable uncertainties surrounding these estimates which are related to the evolution of key parameters such as the rate of return on pension fund investments, the mortality rate, and the labour force participation rate. Hence, careful monitoring and regular updates of the sustainability calculations are necessary to ensure that fiscal targets set by future governments are compatible with the goal of sustainable public finances.

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Notes

1. The high level of expenditure as a percentage of GDP in 1995 can be attributed to a large, one-off transfer from the government to the housing corporations, amounting to 17.5 billion euro or about 5 per cent of GDP. The transfer reflects the price paid by the government to get out of its obligation to continue to subsidise housing corporations in the future.
2. The estimates by the CPB are used here in order to be consistent with the forecasts for the medium term (2004 to 2007) which the OECD does not produce but which are needed to assess the sustainability of public finances. It is important to note that the CPB method assumes that budget balances respond to changes in economic activity with a lag of three quarters of a year.
3. Recent estimates (MEV 2004) have resulted in a downward revision of the structural deficit.
4. In the CPB method potential output is determined by structural productivity growth and potential labour supply; the latter depends on the structural participation rate and the value of the NAIRU, which is in turn determined by a bargaining model. In such a model, an increase in the tax burden on labour due to higher pension fund contribution rates feeds into a higher value for the NAIRU and hence a lower estimate for potential growth. Moreover, both structural productivity growth and the structural participation rate has been revised downwards since the beginning of 2001, although the latter was partly due to the change in the period considered (2003-06 versus 2004-07).
5. This is the estimate of the government which defines net expenditure cuts as the difference between total expenditure cuts and expenditure increases due to new measures. The CPB books € 3½ billion of the expenditure cuts as tax increases instead. The disagreement follows from a different interpretation of the budgetary impacts resulting from a shift towards more private financing of health care expenditure in 2006. The government books the amounts associated with cuts in healthcare coverage and increases in co-payments on the expenditure side, whereas the CPB books them as higher tax burdens on households.
6. The multi-annual budgetary framework determines the real growth ceilings for the years the government is expected to be in office. These ceilings are converted into nominal growth ceilings in the annual budget, using the deflator of domestic demand.
7. The required structural surplus for fiscal sustainability rises by 0.9 percentage point of GDP for every percentage point reduction in the assumed real rate return on pension fund assets (van Ewijk, 2000). The assumed 60-40 split between bond and equity holdings corresponds to pension funds' current asset allocation.
8. The group to whom this applies and that actually shifts contributions to third pillar schemes is expected to be small, so that this downside risk is considered moderate.

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4. Increasing labour force participation

The activation of potential labour inputs represents an important element of the policy mix to limit the fall in potential growth associated with the demographic ageing process. Women will continue to make the largest contributions to growth in labour inputs over the next ten years, albeit smaller ones than between 1985 and 2000. While this trend is to a large extent autonomous, there is scope for increasing employment in the second largest group of the inactive population, persons with a disability benefit. The government has embarked on a comprehensive reform of the disability system, complemented with associated measures to strengthen work incentives for unemployed persons, which together promise encouraging results even in the short-to-medium term. Disability benefits account for the inactivity of almost 20 per cent of persons aged 55-64, the tax-favoured early retirement schemes (VUT and pre-pension), long-lasting unemployment benefits and social assistance together making up for another 20 per cent, suggesting that there is significant scope to increase activity in this age group by discouraging use of these routes to early retirement (Figure 4.1). For the labour market as a whole, ensuring that the recent steep increases in the unemployment rate do not become structural and that low-skilled persons are not trapped on social benefits represents a challenge for the government. Finally, female labour force participation is relatively low in terms of hours worked, reflecting a strong part-time preference. However, the latter is also shaped by economic factors such as the tax system and the availability of childcare and other household services for working families.

A comprehensive reform of the sickness and disability schemes would have significant employment effects

Three decades of huge inflows and recent signs of improvement

There are almost one million persons receiving a disability benefit (DB) (9 per cent of the working age population), making this group the largest potential pool of inactive labour to be activated through social policy reform (Figure 4.2). The law on work disability (*Wet arbeidsongeschiktheidsregelingen*, WAO)[1] grants a disability benefit (DB) in case of a partial or full loss in earnings capacity

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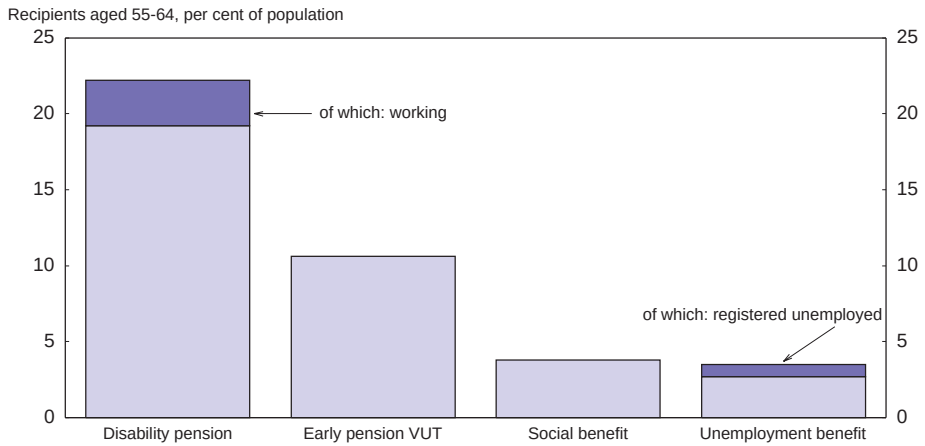
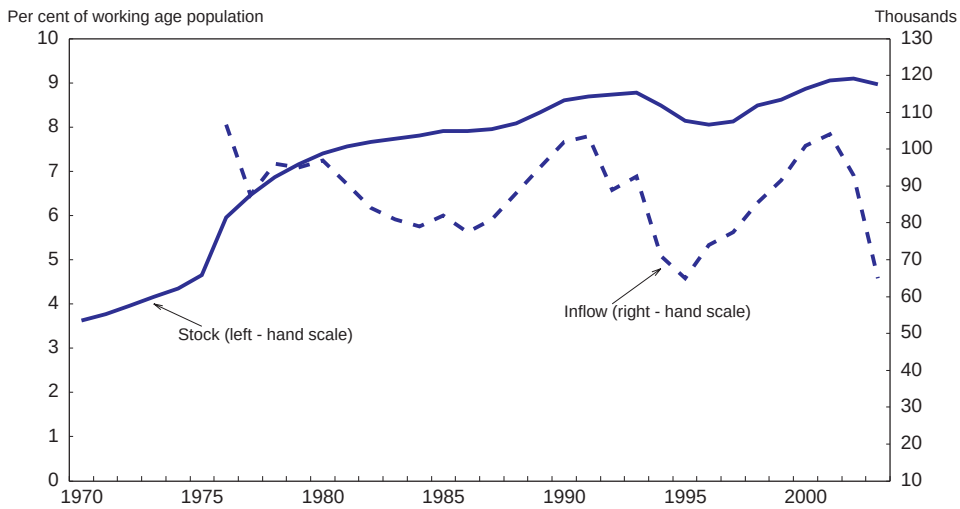
Figure 4.1. **Contributions of benefits and public pensions to inactivity at age 55-64**



Source: CBS (2004).

Figure 4.2. **Disability numbers from an historical perspective**



Source: CBS (2004) and UWV(2004b).

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regardless of the cause of this loss. The system has been reformed several times but without achieving a sustainable reduction in the number of beneficiaries. More recently, however, clear signs of improvement became visible. In 2003 the total number of beneficiaries fell for the first time in seven years. Inflows came down by one-quarter, reflecting a decline in sickness benefit (SB) take-up in 2002 that continued in 2003. Part of the decline is due to institutional improvements. First, a law on improved gate-keeping came into force on 1 April 2002, introducing stricter mutual obligations on reintegration efforts for employees on long-term sickness benefit (SB) and their employers.[2] Second, the five public DB agencies were merged into a single one, UWV (*Uitvoeringsinstituut Werknemersverzekeringen*), on 1 January 2002, reducing the influence of sector interests on the DB authority.[3] Third, the effect of experience-rating in WAO premiums at firm level (Pemba), introduced in 1998, grew stronger as premiums became more and more differentiated for medium-size and large firms.[4] However, the other part of the decline is likely to be cyclical. A cooling labour market usually leads to lower sickness numbers as employees fear losing their jobs. Moreover, inflow numbers also depend on societal debates on the disability problem (UWV, 2004b).

Policy challenges for a growing generation of partially disabled

The number of persons receiving a partial disability benefit (DB) took off during the nineties.[5] Whereas the first wave of these typically withdrew from the labour market, more recently a new group of working recipients of partial DB has emerged and increasingly shaped the policy debate. In 1999 already 50 per cent of partial DB recipients were working (Thio and De Jong, 2002, p. 12), increasingly so with the same employer. The importance of partial DB has kept growing over recent years, especially among women and among persons in the second half of their working lives (Figure 4.3).[6] As a consequence, providing the sick and their employers with strong reintegration incentives and an independent, high-quality disability assessment are obvious challenges for disability reform. However, as a precondition for successful reintegration, the benefit system should also become less attractive. According to a recent OECD study on disability policies in 20 countries (OECD, 2003a), the Netherlands is one of the most generous countries in terms of coverage and benefit levels, the two most powerful predictors for the incidence of DB across countries (Figure 4.4). Therefore, restricting the uptake of full DB to really severe cases, strict testing and barring access to partial DB for the mildest cases are additional core challenges.

Current comprehensive reform proposals and the position of the social partners

To lock in the current broad consensus for tackling the disability problem the government has worked out and partly implemented a comprehensive strategy to meet the challenges outlined above, building on the Donner II

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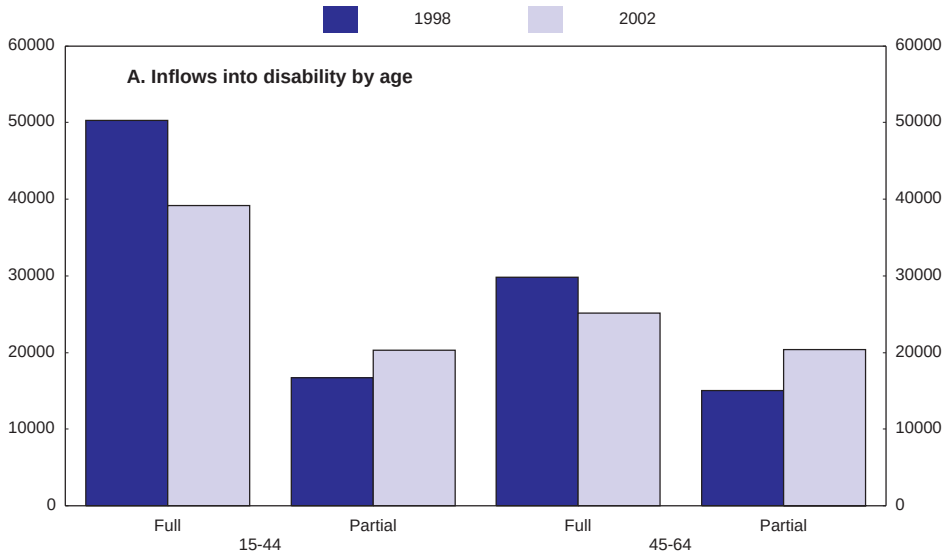
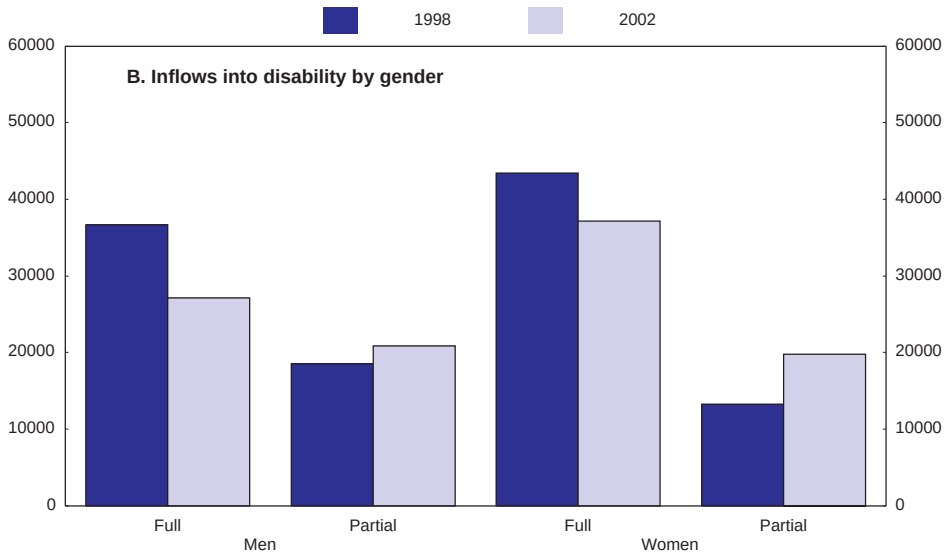
Figure 4.3. **Partial disability has kept growing**





Source: CBS(2004) and UWV (2004b).

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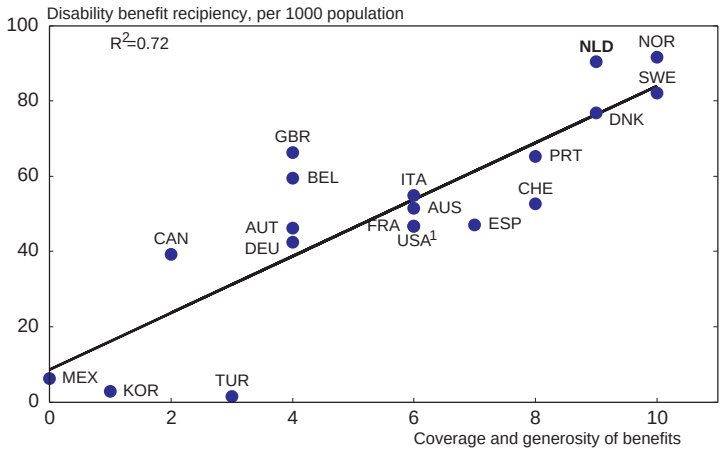
Figure 4.4. **Coverage and generosity determine benefit recipiency levels**
Benefit coverage and benefit generosity 2000 *vs.* benefit recipiency outcome 1999



1. The scatter plot is the same value for the USA and FRA.
Source: OECD (2003f), p. 135.

proposals (*Adviescommissie Arbeidsongeschiktheid*, 2001). Apart from reducing access generosity, the objective is to mobilise labour supply among beneficiaries. In a first step this implies making the group of full DB recipients as small as possible. In a second step the resulting growth in partial DB recipients is to be channelled towards part-time work through financial incentives. The key elements of the reform to be introduced in 2006 are (MinSZW, 2003a, b, and 2004):

- stricter medical definition of full disability and restriction to permanent cases; and
- raising the minimum loss in earning capacity to qualify for partial DB from 15 to 35 per cent and making the level of the partial DB dependent on the use of the residual earnings capacity.

The duration of the sickness benefit (SB) period was extended from one to two years on 1 January 2004, implying that there will be very little inflow into the WAO in the year 2005.[7] Moreover, the criteria for assessing still executable jobs will be made stricter. The authorities think it is necessary to reach a lasting consensus with the social partners on the reform in order to avoid the embedded labour participation incentives being distorted by voluntary benefit top-up agreements as in the past.[8] In the tripartite Autumn Accord 2003, the social partners committed themselves not to top up the second year of SB and to work towards

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bringing down work disability inflows to below 25 000, the government's objective for 2006. If these conditions are met, the DB replacement rate will be increased from 70 to 75 per cent and the Pemba fine abolished. However, important issues have not yet been agreed. They are likely to be negotiated jointly and the results presented in a possible tripartite Spring Accord 2004. In March, the government finalised its reform plans, taking into account the advice of the Social and Economic Council (SER, 2004). Among other changes from earlier plans, this led to putting the introduction of a separate scheme for work injuries and occupational diseases on hold. It has also been decided to reassess the stock of existing DB recipients according to the new stricter rules[9] but to leave the DB calculation and payment rules unchanged for them.

Access conditions to a full disability benefit are being tightened

Only fully and permanently disabled persons will have access to the new WAO. The definition of full disability continues to be an 80 per cent loss in earnings capacity. The remaining earnings capacity is based on the three best paid functions the applicant is still able to perform. A function will already be considered as being executable if only three jobs are available in that function (compared with seven jobs today).[10] This implies that for many persons more better-paid functions can be taken into account so that the remaining earnings capacity is assessed to be higher, which reduces their measured degree of disability.[11] Furthermore, the assessment of whether or not a given job is feasible is becoming stricter.[12] This tightening of the all-occupation assessment mainly affects low-wage earners, the largest fraction of the WAO. The UWV, which will use the low-inflow year 2005 to reassess those beneficiaries most likely to be reclassified and found able to work, estimates at 50 per cent the share of today's full DB recipients that would be rejected under the new WAO (UWV, 2004b). Closing WAO access to temporary cases is also expected to reduce the inflow significantly given the current probability of 46 per cent of being reclassified partly disabled or found recovered within the first five years on full DB. The government plans to establish a list of average recovery durations for all the sicknesses assessed by gatekeepers (MinSZW, 2004).[13] Persons with sicknesses lasting less than five years on average will not normally be allowed to enter the new WAO. With these decisions the government maintained its definition of "permanence" against the more lenient proposal by the SER of letting fully disabled persons with a "low chance of recovery in the long run" flow into the new WAO. On the other hand, the government made a major step towards the SER reform proposals by giving up the much stricter work capacity criterion[14] in the definition of full disability. While according to UWV simulations both aspects of the SER proposal taken together would have cast doubts on bringing the number of new WAOers below 25 000 per year in 2006 (Table 4.1), maintaining the government's definition of permanence makes it likely that this goal will be achieved.

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Table 4.1. **Projected inflows into full disability under the SER proposal**[1]
Thousands of persons

	Low scenario[2]	High scenario[2]
Baseline	66	80
Second SB year and demography[3]	–1	–2
Persons 15-35 per cent income loss	–13	–16
Persons with >35 per cent income loss	–15	–17
Fully disabled current definition	37	45
SER proposal		
Stricter all-occupation assessment[4]	–0.3	–0.3
Fully disabled (permanent and temporary)	37	44
Share of persons judged permanently disabled[5]	0.65	0.65
Fully and permanently disabled persons[6]	24.2	28.7
2010	26.4	31.6
2020	29.5	35.6
No reform		
Fully and partially disabled in the WAO	65	78
2010	71	85
2020	81	97

1. Figures all refer to 2006 unless otherwise stated.
2. The low scenario takes the WAO inflows of 2003 as the basis, the high scenario averages inflows of 2002 and 2003.
3. The second year of SB is assumed to lower inflows by 15 per cent as many persons recover fully or partly during their first year in WAO. Demography almost offsets this effect as cohorts with a higher inflow probability grow stronger.
4. Effect of expecting a part-time worker to do more hours than in the past if he is able to do so in a new function. The planned relaxation of the definition of an "executable function" (three available jobs instead of seven), which would bring inflows further down, is not taken into account because the SER had not proposed it.
5. Under the government's stricter definition of permanence the probability of being accepted in the new WAO is assessed to be 60 instead of 65 per cent.
6. Assuming an inflow probability of 60 per cent reflecting the government's stricter definition of permanence would bring the number of fully and permanently disabled persons to 22.3 thousand persons in the low scenario and 26.5 thousand persons in the high scenario.

Source: UWV (2004b) and OECD.

A *new scheme for the partially disabled, with stronger work incentives*

A new Return-to-work scheme (WGA, *Werkhervatting Gedeeltelijk Arbeidsgeschikten*) is to be introduced for persons not found fully and permanently disabled who are no longer able to earn more than 65 per cent of the peer income of persons with comparable levels of education and work experience. Beneficiaries will receive a wage subsidy if they work and a benefit otherwise (Table 4.2). While the former covers 70 per cent of the difference between the old wage (up to a ceiling) and the new wage, the latter amounts to 70 per cent of the old wage during a first stage (duration of which is age-dependent, as for unemployment benefit) and to 70 per cent of the minimum wage times the loss in earnings capacity thereafter ("second stage"). This implies a 70 per cent withdrawal rate at the first stage but a strong work incentive at the second stage.[15] For incomes between the

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Table 4.2. **Partial disability benefit entitlements under the planned WGA**[1, 2]

	First Stage[3]	Second Stage
"Working"[4]	0.7*(final wage[5]-new wage)	0.7*(final wage[5]-new wage)
"Not Working"[4]	0.7*final wage[5]	0.7*minimum wage*degree of disability

1. All wage concepts are on a daily basis.
2. At any stage, the beneficiary is entitled to a separate, partner-income tested supplement topping the household's income up to the social minimum.
3. The duration of the first stage (0 to 5 years) depends on the beneficiary's age and employment history.
4. At the second stage, beneficiaries are only classified as working if they use their residual earnings capacity to the full.
5. Medium wage from at least 30 jobs representing the three best paid functions the beneficiary could have got before becoming disabled (OECD, 2002f, p. 172). The upper limit is € 37 000, around 2.4 times the minimum wage.

Source: Min SZW (2003d).

minimum wage and the threshold for the benefit calculation, the wage subsidy at the second stage tends to be the more generous than today's DB (excluding private top ups) the younger the beneficiary, the higher the pre-disability wage and the more severe the reduction in earnings capacity.[16]

Employers have to cover the risk of partial disability under the WGA. They can choose between private insurance companies and the public insurer (UWV) or may even carry the risk themselves. In the public system premiums will be differentiated at the firm level to avoid that employers with a low-risk profile systematically opt out of the public system, making it expensive for the remaining firms.[17] The government will investigate whether there is a need for mitigating differences in premiums during private insurers' necessary capital build-up. Whether a claimant is covered by private or public insurance, benefit conditions are fixed and the UWV will remain in charge of assessing the degree of disability in any case. These decisions represent the outcome of an intensive debate on whether or not to fully privatise the insurance of the first five years of partial disability, as initially proposed by the government.[18]

Plans to introduce a new scheme for work injuries and occupational diseases have been put on hold

Following the advice of the SER, the government has put on hold plans to introduce a new scheme for work injuries and occupational diseases, the EGB (*Extra Garantieregeling Beroepsrisico's*). By creating the EGB the authorities wanted to give up the Dutch specificity of a single scheme for disability regardless of its causes. Another reason was that making the WGA benefit conditional on employment behaviour is difficult to reconcile with international agreements (ILO Treaty 121) in case of work injury. In most OECD countries the risk of work injury and the general disability risk are separated because they justify a different funding structure, whereby costs for work injuries are to be born by employers (OECD, 2003a, p. 168). As the aspect of compensation for damage adds to the

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income protection function, capacity reductions from work injuries are usually compensated already at a low minimum level of disability (Annex 4.A1). Moreover, the government is concerned by the growing tendency of employees engaging in often long and costly court battles against their employers in case of injuries or diseases, which would become worse with tighter access to DB.[19] The SER's counter-proposal consists of embedding the injury scheme as a system of specific top-ups to the new WAO and WGA while having the same basic benefit structure for occupational and other disability risks (new WAO DB for the fully and permanently disabled, and a minimum benefit or wage subsidy for the partly and fully but temporarily disabled).

Assessment

The government should implement the stricter criteria planned for full and permanent disability to ensure that the intended fall in inflows, which is plagued by huge uncertainties, is achieved. Medical assessment of "permanence" in disability cases is also subject to uncertainty, making it desirable to maintain the possibility of some periodic reassessment of beneficiaries. To reduce inclusion errors from assessing DB applicants as permanently disabled prematurely, a list of diseases considered as temporary and their average recovery periods should be established and enforced throughout the country, as proposed by the government. The planned extension of the all-occupation assessment represents a welcome adjustment to modern employment biographies that will be ever less confined to narrow functions and time slots.

A narrow and well-kept gate to the new WAO is also crucial for the success of the new partial disability scheme (WGA). The proposed benefit structure would provide for a stronger mobilisation of remaining work capacity among partial DB recipients. However, while work incentives in the WGA benefit are very positive in the second stage, they are weakened by a high benefit withdrawal rate in the first stage. Duration of the first stage should be shortened for persons in the second half of their carrier to prevent the WGA from becoming an alternative route to early retirement.[20] Moreover, since the scheme is still rather generous for those who work enough hours, there remains the risk of self-selection as recently diagnosed by De Jong and Thio (2002). To ensure that the replacement rate does not become even higher for many working partially disabled, the government needs to prevent the currently widespread benefit top-up payments by social partners from being applied to the WGA wage subsidy. The labour market effects of the partial disability reform will set in gradually because the new system only applies to those becoming partially disabled after 2005.

The quality, strictness and impartiality of the disability assessment by the UWV is more than ever a challenge to counter the risk of a growing, albeit working, population of partial DB recipients. The UWV should further develop its

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quantitative toolkit for assessing its contribution to gate-keeping, identifying high-risk groups. This would allow for benchmarking UWV regions against each other, with benefits and sanctions attached to the outcome. While remaining under the authority of the MinSZW, the UWV should publish the results of its benchmarking exercises as this would lend it credibility as a responsible gatekeeper and make discretionary policy intervention less likely in the future.

Setting up a separate work injury scheme (EGB) would have several advantages. First, it would end the Dutch tradition of neglecting disability causes that has contributed to the dramatically high stock of DB recipients. Second, a separate system with differentiated premiums would promise more effective accident prevention. The government should continue to weigh these advantages against the risk of a poorly designed EGB: that of creating an attractive benefit alternative for persons refused a full DB or not willing to fully use their remaining capacity under the WGA. To avoid this risk, any separate scheme should be limited to work injuries and the internationally agreed list of occupational diseases, and the assessment should be carried out by the UWV rather than a new body.

Overall, if the reform is implemented along the lines proposed by the government, there is a realistic chance of mobilising a significant amount of additional labour thanks to more partial DB recipients returning to work. It is not certain, however, that the sum of inflows into the new WAO and the WGA will be significantly lower than past WAO inflows. Keeping the system financially viable may require shortening the first stage of the WGA, which would also help to keep partial DB recipients closer to the labour market. Strict testing of DB claimants is also crucial for the success of the WGA. It would limit the risk of self-selection, as would ruling out top-ups to the WGA wage subsidy.

Fostering employment of benefit recipients, older workers and women

Incentives for a swifter return to work would help to limit structural unemployment…

The duration of unemployment benefits (UB) for persons in the second half of their career is long by international standards (Figure 4.4) and benefit access is relatively easy (Annex 4.A2). Combining the UB with unemployment follow-up benefit[21] has provided workers aged over 57½ with a route to early retirement. To end this situation, the government has re-instated job-search requirements for the older unemployed on 1 January 2004 and has abolished the follow-up benefit (unemployment assistance) for new inflows into the UB scheme. The government wants to combine these measures with an exemption from the basic contribution to the DB scheme for firms keeping older workers. Even after these reforms the Netherlands would remain one of the most generous OECD countries in terms of UB duration for workers aged over 45, reducing the incentive for investing in employability and the effectiveness of reintegration services. The

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government should therefore go further by reducing the maximum benefit duration and making it depend more on the actual work history than on age. This is all the more pressing because tighter access to disability and early retirement schemes may increase the demand for UB. Due to the close link to the first stage of the partial DB, a shorter UB duration would also allow for a faster transition to the second stage of partial DB with its embedded work incentives. As far as younger workers are concerned, the government intends to abolish the special benefit for persons with short work records and deny access to UB for persons having worked less than 39 out of the past 52 weeks. The possible drawback of these measures, an increase in young social benefit recipients, is small given the success of the vocational training system in preparing young persons for the job market (Chapter 6) and the low incidence of youth unemployment. Therefore, the benefits of stronger incentives for successful training and uninterrupted work records should be largely dominant.[22]

... as would more efficient placement services...

Private placement services for persons in unemployment or on social assistance have become increasingly important over the past years, making the Netherlands one of the pioneers in the field, together with Australia (see Annex 4.A2). The average placement probability has improved markedly since the extension of private placement services but improvements still need to be made. First, market transparency needs to be enhanced in order to enable the UWV and municipalities to make better purchasing decisions. Second, the uniform contract duration of one year may be too short for difficult client groups given their risk of re-entry into unemployment and the dependence of placement chances on services delivered in the past (Grubb, 2003, p. 368). Finally, the placement success also depends on client compliance, which would be enhanced by giving providers the possibility of initiating benefit sanctions. The authorities should consider placing the payment of unemployment benefits under the responsibility of reintegration firms, boosting their incentives to provide effective reintegration.[23] The provider's benefit needs could then become a parameter in the bidding process.

... and smaller poverty traps

High benefit replacement rates and the tax wedge represent barriers to labour market participation of low-wage earners. The government has taken various measures to reduce poverty traps in 2004. The general labour income tax credit, introduced in 2001, and the special working parents' tax credit will be increased by almost € 200 and € 81, respectively. Furthermore, local taxes will be lowered, including the abolition of the local housing tax. In addition, the authorities will abolish special municipal income support schemes.[24] Nevertheless,

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benefit withdrawal rates remain high at the low end of the wage distribution, raising the question of whether the budgetary means for tax credits should not be targeted to the lower end of the income distribution and then be phased out over the middle range of incomes. The authorities are opposed to this, pointing to negative macroeconomic effects that originate in the relatively flat income distribution. Given the high number of persons earning up to 1½ times the minimum wage (the ceiling for most family and housing support schemes), the smaller unemployment trap would be more than offset by a larger poverty trap as persons in low-paid work would be discouraged from increasing hours worked or upgrading their skills. Therefore the authorities should strengthen general work incentives and gradually bring down the tax burden as fiscal room for manoeuvre becomes available. There is also scope for making social benefits less generous, especially for households without children. This would prevent persons from getting locked in to social benefits after the tightening of access to unemployment benefits.

Closing the tax-subsidised route to early retirement

Early retirement pensions, disability benefits and long-lasting unemployment benefits together provide a powerful explanation for the decline in participation rates of workers between their early and their late fifties and the sharp fall once they are over 60. In the Netherlands, the drop in participation from the second oldest to the oldest cohort was more than 60 percentage points in 1999, in line with what could be expected given the strength of the work disincentive stemming from the VUT, the most widespread tax-favoured pay-as-you-go early retirement scheme (Figure 4.5). The government wanted to withdraw tax facilities from all early-retirement schemes introduced in the future and maintain them for existing schemes on the condition that they are transformed into actuarially neutral pre-pension schemes. The project was put on hold in order to reach wage moderation in the autumn Social Accord but also because the government wanted to include both PAYG and funded early retirement schemes in the arrangement and also to close loopholes in the third pillar (abolition of the "bridging annuity"). Present government plans are to end tax favoured early retirement schemes by 2006, but the exact conditions, including those concerning the creation of a "lifecycle" scheme and any transitional arrangements for existing schemes will be decided shortly in the context of a possible tripartite Spring Accord. The government should push ahead with these plans so as to reap the benefits of higher employment among older workers and react in a timely way to the increasing cost of ageing. While the social partners could still offset these measures by increasing contributions to second pillar schemes with a standard retirement age of 65 but the possibility to retire earlier, the incentive to do so would be reduced because the pension would then be actuarially adjusted. There is, however, a risk that the "lifecycle" scheme ("*levensloop regeling*") becomes an alternative route to early retirement. According to the government proposal, it would allow employees to

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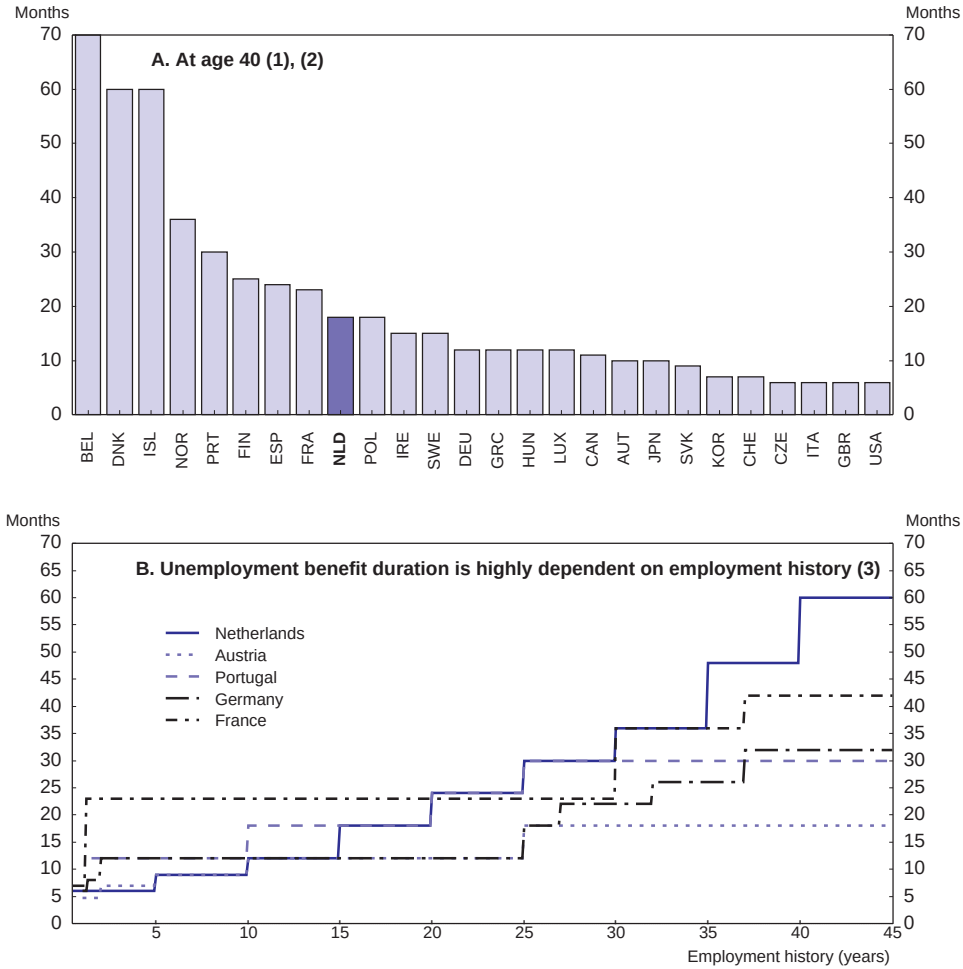
Figure 4.5. **Unemployment benefit duration in selected OECD countries**



1. Work record of 18 years, single worker.
2. Unlimited duration for Belgium. Duration independent of age or work record in Iceland, Norway, Ireland, Canada, Korea, Czech Republic, United Kingdom and United States. Duration independent of age or work record but longer duration for older workers in Luxembourg, Switzerland, Italy and Denmark.
3. Not accounting for initial access conditions and reduced follow-up benefits (unemployment assistance). As in many countries the unemployment benefit duration is also based on age, the figure assumes labour market entry at age 20 and uninterrupted work or insurance record.

Source: OECD Benefits and Wages 2002; European Commission (2003); and Unédic (2004) for France.

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put aside up to 1½ monthly salaries each year tax free for use later in life, up to a maximum of 2.1 years. The government should make the tax subsidies conditional on continuing to work part-time for persons taking their saved salary just before retirement. Finally, the government should also consider indexing the legal retirement age to life expectancy, which might grow faster than assumed in the past.

Less restrictive EPL *would favour the demand for underrepresented groups*

The ambitious objective of increasing employment among the elderly, partly disabled, unemployed and recipients of welfare benefits requires that barriers to labour market entry of these under-represented groups be lowered. Strict employment protection legislation (EPL) is such a barrier because it creates huge insider-outsider problems (Annex 4.A3). Another negative effect of strict EPL is that it prevents timely adjustment of employment and wages to economic conditions. The features that stand out in the Dutch dismissal system are the high influence of third parties when employees are made redundant *via* the Labour Office and high severance pay otherwise (see Annex 4.A3). The government is preparing a bill to limit the maximum severance pay to one year's salary. Given that the age-dependent structure of UB duration already compensates for the higher depreciation risk of job-specific human capital, this measure would be welcome to reduce high labour adjustment costs for companies. The government should also consider reducing the influence of third parties for the CWI channel in the dismissal process because it is likely that these third parties on balance put a higher weight on protecting jobs of existing employees (insiders) than on increasing the employment chances of groups with low employment rates (*e.g.* ethnic minorities, persons with disabilities) by allowing for a higher job turnover. Therefore it is unlikely that the CWI dismissal channel leads to correct internalisation of the negative externality of a dismissal.[25] Allowing for differentiation in unemployment insurance contributions according to the number of persons dismissed *via* the CWI channel would be a more efficient way of accounting for the negative external effects of a dismissal decision.

Facilitating higher labour force participation of women

The employment rate of women is well above the OECD and the EU average but relatively low in terms of hours worked and the share of part-time work among women in employment is the highest in the OECD. While this choice may be partly shaped by genuine cultural differences, economic incentives pertaining to the tax system and the availability and cost of childcare and other household services also play a role (Annex 4.A4). Some progress has been achieved in making the decision of couples on how to allocate their labour force more neutral since the switch to individual taxation and the general tax credit on labour income in 2001. To further stimulate female labour force participation, the

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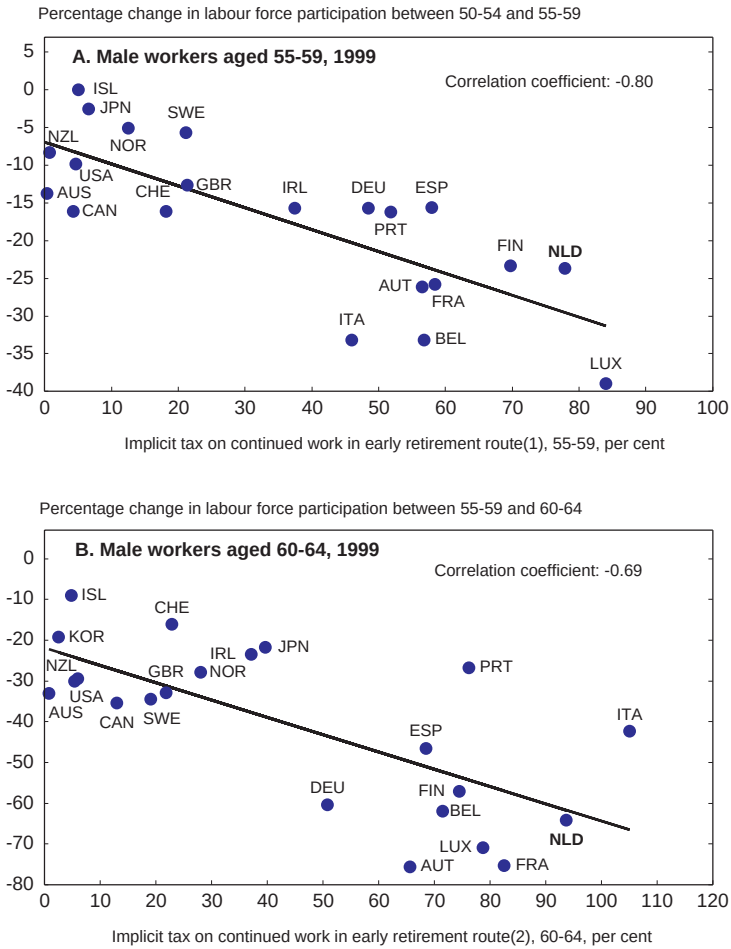
Figure 4.6. **Disincentives from routes to early retirement and the participation decision**



1. The early retirement route is modelled as the unemployment benefits/assistance pathway into retirement with the exception of Ireland, where the modelling refers to the pre-retirement allowance, and Luxembourg, where disability benefits were considered given their widespread incidence among pensioners. In those countries where it was considered that no early retirement scheme could be widely used to withdraw from the labour market before the minimum pensionable age (Australia, Canada, Iceland, Italy, Japan, Korea, New Zealand, Norway, Sweden, Switzerland, United States) the retirement scheme considered in the chart is simply the "regular" old-age pension system.
2. Implicit tax rate computed as in Panel A except for the Netherlands where the VUT scheme is used.

Source: Duval (2004).

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"working parents' tax credit" has been lifted and will be highest for the lowest-earning partner from 2004 onwards. Nevertheless, full-time participation is particularly low among women with low and medium attainment levels, reflecting, *inter alia*, the unemployment traps discussed above.[26] These traps are also the most likely reason behind the lack of low-skilled household services.[27] The number of childcare facilities used to be one of the lowest in the European Union (OECD, 2001, p. 144) but has doubled from 1998 to 2003. As prices also rose sharply,[28] waiting times have disappeared in most areas. The government is preparing a bill that regulates the financing and guarantees the quality of childcare. One of the goals of the bill is to improve the functioning of the market for childcare by:

- providing working parents with an income-related subsidy bound to pay for formal childcare and family-home care from private suppliers replacing the patchwork of existing transfers to parents and municipal subsidies to suppliers; and
- formulating quality regulations at the national level, replacing the often tight and diverse regulations at lower levels of government.

The bill should be implemented because it would lower barriers to market entry (stimulating supply of childcare) and make the transfer dependent on actually working (stimulating demand).[29] In designing the national regulations the government should take care not to make quality standards overly strict but should rather rely on market forces to lead to a differentiated supply. This would allow families with more modest incomes to pay for childcare, and hence participate in the labour market.

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Notes

1. There are three different disability schemes in place in the Netherlands. The WAO covers disability of employees and the unemployed, WAjong is the scheme for young persons, and WAZ covers disability of the self-employed. The latter is being abolished by 1 July 2004.
2. Insufficient reintegration efforts, as shown by a reintegration report, lead to sanctions or refusal of the claim, prolonging the SB period at the employer's expense. In turn, employers may dismiss non-compliant employees.
3. UWV reports to the Minister of Social Affairs (MinSZW) and agrees with it on performance targets. UWV also conducts pilot projects to improve its functioning, *e.g.* with respect to the comparability of individual assessment results. A new selection protocol for reassessments was implemented (UWV, 2003, p. 9) and may have contributed to the steep rise in the share of DB outflows in 2003 (UWV, 2004a). UWV is also evaluating stricter assessment of young women who are four times as likely to obtain a DB than the average, often on grounds of psychological problems.
4. The "Pemba fine" was already replaced with a uniform contribution rate (2.4 per cent of the wage bill) for firms with less than 25 employees in 2003.
5. The first generation of WAOers was granted full DB even if they had lost only part of their earnings capacity.
6. The partial DB was initially granted to help impaired persons to shift to part-time employment rather than dropping out of the labour market. However, there may also be an opposite labour-destruction effect if the nature and degree of disability are not exactly observable, leading to self-selection. According to recent empirical evidence both effects coexist in the WAO. The positive relationship between the probability of staying employed and the probability of obtaining a partial DB shows causality in both directions (Thio and de Jong, 2002, p. 65). Specifically, those resuming work immediately after their year of sickness have a much higher chance of getting a partial DB than persons with a weak labour market position, all other conditions being equal.
7. The inflow in 2005 will not be zero because in a limited number of "evident cases" (persons with a full disability having reached a stable condition without a chance of recovering), the entry to the WAO will be possible before the end of the SB period.
8. Current examples of collective agreements reducing work incentives in many sectors are the top-up of SB to 100 per cent of the previous wage in the first year and the top-ups of first stage DB to 85 per cent and of the second-stage DB to the first-stage level.
9. An exemption will apply to persons already exempted from periodical medical re-evaluation and to persons older than 54 years on 1 July 2004.

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10. The official computer system used in the assessment of available jobs does not contain all jobs existing in the Netherlands but only a representative sample.
11. For a number of persons this may imply that they will be assessed as being only partially disabled, whereas they were found to be fully disabled in the old system.
12. A previous part-time worker is in principle eligible to do jobs with more hours if he is able to execute the function. It will not be possible to reject a function because the usual hours of work are outside the slot worked in the past. Non-possession of easy-to-learn computer or language skills is no reason not to qualify for a function.
13. The lists are indicative, *i.e.* insurance doctors may let "temporarily" sick persons access the new WAO nevertheless if convincing proofs for the need to deviate from the standard can be delivered.
14. According to the government's initially planned work capacity criterion, "A fully and permanently disabled person is someone who as a direct and medically objectively observable consequence of sickness is permanently unable to earn income from work" (MinSZW 2003c, p. 5). This definition came very close to that recommended by the D*onner* II Commission (A*dviescommissie* A*rbeidsongeschiktheid* 2001, p. 5) and leant against the notion of a loss in earnings capacity that is currently in force and is defended by the social partners (SER, 2002 and 2004).
15. The subsidy rate of 70 per cent of the wage loss helps to avoid unemployment traps for more severely disabled low-wage earners. A person previously earning the minimum wage and who has lost 70 per cent of her earnings capacity can achieve an income equal to 79 per cent of the minimum wage by working compared with 49 per cent when not working (topped up to 70 per cent if the household has no other income).
16. For example, a 40-year old person with a pre-disability income of twice the minimum wage who is 50 per cent disabled and fully uses her remaining earnings capacity of 50 per cent will receive a wage subsidy that is one-third higher than today's partial DB, allowing for an increase in combined gross wage and benefit income of more than 10 per cent.
17. Experience-rating is a challenge for temporary work agencies that face an increased risk of sickness and disability of their members due to the concentration of low-skilled, foreign and older workers as well as free-rider problems as to prevention measures at the actual workplace. The government has decided to limit total SB outlays but has shrugged off the sector's demands, supported by the SER (SER, 2004), to limit the SB-period for temporary workers to one year and to exempt the temporary work sector from experience rating in the WAO.
18. A competitive private market could foster effective reintegration because of insurers' incentives to avoid inflow into WGA (Kist and Keuzenkamp, 2003). The advantages of a seamless reintegration trajectory covering the SB and DB periods, faster learning and dissemination of best practices and private insurers' greater ability to find incentive mechanisms that reduce deadweight losses were also mentioned. There were, however, questions about insurers' incentives to reintegrate more severe cases of partial disability. But the government's main concern was that the private system would not be cheaper than the public system, even in the long run the (MinSZW, 2004).
19. Provisions in Part VI of the European Social Security Code declaring a private health insurance system (planned for 2005, see Chapter 3) incompatible with patient payment for rehabilitation after work accidents are a third reason for introducing the EGB.
20. In addition, the same job search criteria should be enforced as for unemployment benefit recipients.

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21. This benefit is minimum-wage related and is neither income-means nor capital-means tested. An unemployed person was entitled to this benefit for two years after the salary-related UB has expired or until retirement if they were aged 57½ years or more when they became unemployed.
22. The new criteria would still take due account of the more scattered work biographies of persons on temporary work contracts. The latter have been made more attractive with the 1999 Act on flexibility and security that allows for their renewal during up to three years.
23. Integrating the three functions of labour market policy (placement and counselling; benefit payment; labour market programme management) is a long-standing OECD recommendation (OECD, 1994).
24. A new agency will coordinate all income-dependable social arrangements with a view on lowering the administrative costs of social assistance.
25. The externality, mainly consisting of not compensating the dismissed worker for the opportunity cost of building up job-specific human capital during a long tenure, should not be over-stressed. The social partners have a high degree of co-operation in determining training needs of the employed workforce, and the participation in continuing training is high by international standards.
26. When looked at by level of education, it turns out that the Dutch incidence of full-time work of prime-age women is only 32 per cent for low attainment levels, rising to 43 per cent for medium- and 57 per cent for high attainment levels (Burniaux *et al.* 2003, Table A2.8).
27. There appears to be substantial demand for low-skilled household services according to recent evaluation of a subsidy that brings the hourly wage for house-cleaning personnel in line with that prevailing in the informal sector.
28. Demand for care personnel rose most strongly at a time of tight labour markets.
29. While unconditional child benefits have a negative effect on female employment in general, a bound childcare subsidy affects full-time employment positively (Jaumotte, 2003).

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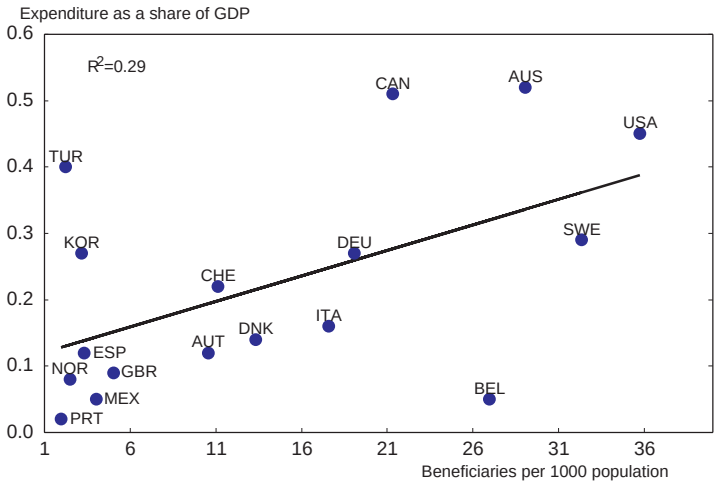
Annex 4.A1

Work injury schemes in OECD countries

In March 2004 the Dutch government decided to put on hold earlier plans for the introduction of a separate insurance scheme for work injury (WI) and occupational disease. Arrangements in other countries could serve as a guide for the design of such a system in the Netherlands in the event that the government eventually decides to implement one. Most OECD countries have a separate WI scheme but display huge differences as to the number of benefit recipients and the total cost of the scheme (Figure 4.A1.1), reflecting variation in coverage, access conditions and generosity levels. Among the 20 countries covered in the OECD study on disability and reintegration (OECD, 2003a), coverage by the compulsory scheme is complete for employees everywhere except in Canada and Italy, but some countries also include certain groups of self-employed (France, Germany) or even the entire

Figure 4.A1.1. **Stock of beneficiaries and cost of work injury schemes in OECD countries**

1999



Source: OECD database on programmes for disabled persons.

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labour force (Austria, Sweden, Turkey). In a few countries important groups outside the labour force are also covered such as students in Germany and social benefit recipients in France. Differences in financing medical rehabilitation measures and the first weeks of sickness leave after an accident help explain why total costs of schemes vary even when the number of beneficiaries is comparable. While varying across countries, access to a WI benefit is generally granted at a lower level of impairment than that to DB (Table 4.A1.1).

Another varying aspect (not shown) is the possibility and occurrence of re-assessments of the health status. The law provides for adjustment of WI benefits to account for changes in the health status in many countries within the framework of temporary benefits, but in some of them changes are limited to cases in advantage of the beneficiary (*e.g.* Portugal in the first ten years, Belgium after three years). Enforcement varies across countries and some (like Norway) usually renounce to re-testing altogether. To limit the danger of "locking" the victim into a given degree of capacity reduction too early, the benefit is often made permanent only after an observation period of varying length (*e.g.* one year in Spain, up to two in Austria, three years in Belgium) or may be defined as temporary from the outset (possible *e.g.* in Germany and Sweden). For other aspects differences are small or inexistent. In most countries WI insurance premiums are paid by employers, access to benefits does not require fault on the employer's side, and there is no minimum contribution period. With a few exceptions there is no waiting time before receiving the first benefit, unlike in most DB schemes. Receiving a WI benefit is generally compatible with rehabilitation. In some countries, the two go together as rehabilitation is a requirement for benefit receipt (or *vice versa*).

While finding legal definitions for work accidents and injuries resulting from them is not too difficult, occupational diseases are much more controversial and a potential source of benefit inclusion or exclusion errors. Many countries avoid inflow numbers getting out of control by sticking to a positive list of well-defined occupational diseases, *e.g.* the List of Occupational Diseases Recommendation (ILO, 2002). It provides a breakdown into diseases caused by agents (chemical, physical, biological), diseases by target organ system (respiratory, skin, musculo-skeletal), occupational cancer etc. Most countries with separate WI insurance follow this recommendation, sometimes specifying (as Germany does) that to be eligible workers must be exposed to the causes of an occupational disease to a significantly greater extent than the average population. This latter requirement and that of observable causality normally rule out giving consideration to psychosomatic, psychological and motivational troubles and to widespread problems the insured may be able to avoid by a reasonable amount of own prevention (*e.g.* back pains, slipped disc etc. for office staff).*

* Only in a few countries is the set of diseases covered the same in DI and WI (France, Poland, Spain, Turkey), yet with a much lower capacity loss threshold for WI benefits than for DB in France.

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1]

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
AUS[5]	(2) All cases where employer's "fault" can be established	Medical-vocational (level of impairment using guidelines from the American medical association) Usually 10% (5) Same definition; no contributory DB scheme (5)	Work impairment tables GP assesses disability status, insurer determines level of impairment	BTF ≥ 0.95w (1st half-year) 0.65 w ≤BDF ≤0.9 w BDP = d*BDF Lump sum payment for certain injuries under some circumstances	Work income: no (benefit withdrawal rate 100%) Same for means-tested DB	Compulsory employer premiums 0.25% to more than 15% of payroll (average 1.85-3.45% of payroll), risk- and firm-experience rating Public and private providers
AUT	(4) All work-related disabilities (accident, injury, disease)	Medical (reduction in work capacity) 20% (students 50%), more than 3 months (4) Same definition, 50% (3)	Doctors of the accident insurance authority	BF = 0.8 w if d = 100% BF = 0.67 w otherwise 20% top-up per dependent child BP = d*BF	Work income: yes DB: yes	Uniform employer premiums 1.4% of payroll Public insurance bodies (self-governing)
BEL	(2) All work-related disabilities (accident, injury, disease)	Medical-vocational (permanent earnings capacity loss) No explicit minimum rate (5) Loss in earnings capacity for usual occupation 66.6% (2)	Victim's socio-professional status taken into account Doctors from insurance administration, but agreement from GP needed	BTF = 0.9 w (first 3 years) BDF = w BTP = w – new wage BDP = d*w	Work income: yes OAP: no	General employer contributions (0.3% of payroll) Risk-dependent WI premium, plus uniform 1.10% for OD Community insurance funds, private insurers; control by public interest organisations

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1] *(cont.)*

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
CHE	(2) Occupational accidents and diseases Non-occupational accidents (employees ≥ 8 hours)	Medical-vocational *De jure* 1% *De facto* 10% (5) Same definition 40% (4)	Main insurer assesses d in collaboration with DB and WI authorities that assess d separately, taking each other's opinion into account	BDF = BTF = 0.8 w BDP = BTP = 0.8*d*w BT granted until final entitlement decision is made and rehabilitation completed	Work income: yes, combined income up to w (otherwise reassessment of d) WIB complementary to DB or OAP up to 0.9 w	Mandatory WI scheme: Employer premiums, 0.04-17.2% of payroll (on risk) Semi-public agency (55% of employees); 40 private firms
DEU	(4) Work and commuting accidents and injuries, OD (ILO list); costs of treatment and prevention	Medical-vocational Permanent WIB based on loss in earnings capacity, 20% (5) Medical-occupational (work capacity lost), 25% (5)	Medical-vocational experts of benefit administration	BT = 0.8 w (after SB, up to 1½ years if no hope for reintegration); transitory benefit during rehabilitation BDF = 0.67 w BDP = d*BDF	Work income: yes DB: no (priority to WIB, topped up if DB higher)	Employer contributions Rated by sector and firm- risk (average 1.3% of payroll) Semi-public branch organisations; State for public sector employees and non-employed
DNK	(2) All work accidents and injuries, and occupational diseases	Medical (in stating injury) 5%, permanent (5) Vocational (stating WIB) 15% loss in earnings capacity (5) Medical (workers over 50: also social reasons), 50% (3)	Injury: disadvantages in everyday life Earnings loss: age, occupational prospects, education and rehabilitation possibilities GP (assessment); board of industrial injuries (control)	Permanent injury compensation up to $62 500 for d = 100% (lump sum) BF = 0.7 w	Injury compensation and work income: yes WIB and work income: up to residual capacity Contributory OD benefit: yes Public DB: no	Employer premiums Accidents and temporary injuries: private insurers OD and permanent WI: Labour-Market OD Fund (independent, self-governing)

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1] *(cont.)*

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
ESP	(2-3) No difference with disability programme, except for disease origin	Medical-professional 33% capacity loss in usual occupation (4) Same definition 33% (4)	Partial disability: capacity loss in usual occupation "Total/absolute" disability (F/A): unable to carry out usual obligation/ any work Doctors from social security or mutual accident companies	Occupational sickness: BTF = 0.75 w (up to 1 year) BDA = 0.85 w BDF = 0.47 w (as for DB) +50% by employer if health or safety regulations breached BDP as a lump sum	Work income: yes with social security authorisation (possible review of d and WIB) DB: no (one and the same benefit)	Employer premiums 0.81% to 16.2% of payroll (average 1.98%) Public insurance; private companies may pay out WIB, do rehabilitation, prevention
FRA	(3) Injuries from work and commuting accidents	Medical-vocational 1%, permanent (5) Loss in earnings capacity 66.6% (2)	Nature of the disability, victim's physical and mental state, work capacity lost GP assesses medical status; insurance doctors determine level of incapacity	BT = 0.8 w BD = [d + ½ (0.5 – d)]*w 40% top-up for d = 100% requiring assistance Lump sum amount for disability of less than 10%	Work income: yes	Employer social security contributions Uniform component (commuting, general expenses), risk-rating by sector (establishment if ≥ 200 employees) Generally private insurers

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1] (*cont.*)

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
GBR	(2) Disablement from work accident or one of over 70 listed industrial diseases Employer liability compensation scheme (civil damages, liability and negligence established)	Medical 14% (normal) 20% (deafness) 1% (respiratory diseases) (5) Medical (personal capability relevant for work) (1)	WI resulting in loss of physical or mental faculty compared with that of a healthy person (schedule of degrees of disablement) Separate injuries may be aggregated Assessment by WIB agency	BF = 1.6*DB, £109/week BP ≈ d*BF + 20-120% for long-term care Liability insurance: huge variation in compensation (on average similar to WIB)	Work income: yes DB or other not means-tested benefits: yes Compensation under liability insurance: yes	State programme funded by general taxation Liability compensation: employer's own responsibility
ITA	(1) Work-related injuries leading to >3 days of absence from work	Medical 11% (5) Same definition 66.6% (2)	Legally trained doctors from the national work-injury insurance authority assess disability	BT = 0.6 w for 87 days, 0.75 w thereafter BDF = w, if 0.8 < d < 1 BDP = d*w	Income from work: yes DB: no	Employer premiums 0.5%-16% of payroll (average 3%), risk-dependent Public insurance monopoly
NOR	(3) Mandatory liability insurance outside the public system compensates economic loss caused by occupational disability	Medical-vocational Economic loss: 30% of earnings capacity Non-economic loss: medical degree of injury 15% (5) Medical, 50% But loss in earnings capacity determines DB level (3)	National insurance administration decides on benefit claims (may delegate to regional and local offices) Liability insurance: assessment by private insurers (often relying on national insurance administration)	Non-economic loss: max. ¾ of $3 750 ("basic amount") BDF = 0.65 w for APW; strong redistribution (as for DB) Liability insurance tops up sum of disability and non-economic-loss benefits to w	Work income and BDF: no Work income and compensation for non-economic loss: yes Disability and injury schemes integrated (*de facto*, not many persons on WIB roles)	General social security contributions (uniform rates) Voluntary for self-employed: 0.4% of income Liability insurance: risk-rated employer premiums Public monopoly for national scheme; private insurers for liability insurance

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1] *(cont.)*

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
PRT	(3) Mandatory liability insurance for WI and accidents (OD part of social security)	Medical "More than 0%" (5) Medical-vocational 66.6% loss in earnings capacity (2)	Clinical services of insurers company assess d, legislated National Table for Incapacity Labour Court analyses (corrects) proposal Medical Commission (representing victim, insurer, court) settles disagreements	Total incapacity: For work: BD = 0.8 w (top-up for dependence) For usual occupation: 0.5 w ≤BD ≤0.7 w If fault on employer: BD = w BTF = 0.7 w BDP = 0.7*d*w	Work income and total incapacity for any work: no DB: no (priority to injury benefit, topped up by difference if DB > WIB)	Liability: Employer must purchase liability insurance OD: Uniform contribution (0.5% of payroll) Premiums risk-dependent, on average 3% of payroll Private providers of liability insurance in private sector
SWE	(4) Injury as result of an accident or other injurious influence at work that leads to a reduction in work capacity	Medical-economic (reduction in earnings capacity) 6.7% (5) Medical Daily hours person can work 25% (5)	National social insurance office decides based on own and GP's assessments	SB = w until d said permanent BDF = w up to about $33 000/year BDP = BTP = d*w ⅔ of WI victims are on DB (redistributive, 0.7 w for APW)	Work income: no DB: no WIB provided as supplement to salary if WIB > DB	Uniform employer premiums 1.38% of payroll Largely public insurance, but private insurers covering related damages, *e.g.* pain, suffering and disfigurement
TUR	(3) Work accidents and disabilities	Medical (lost work capacity) 10% (5) Same definition 66.6% (2)	Total disability if d ≥ 60% Social security and health facilities assess d; supervision by Ministry of Labour	BT = 0.5 w (if hospitalised), 0.67 w otherwise BDF = 0.7 w BDP = 0.7*d*w	Work income: yes DB or OAP: yes, (100% of the higher, 50% of the lower)	Same scheme for WI and DB Employer premiums 1.5% to 7% of payroll (average 2.5%), risk-dependent

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Table 4.A1.1. **Work injury programmes in selected OECD countries, 1999**[1] *(cont.)*

Country	Coverage: Persons (in brackets)[2] Risks	Definition of incapacity Minimum level required for benefits and generosity index (in brackets) Comparison with disability program: Definition, minimum level (its generosity)[3]	Assessment structure: Determinants of disability level Assignment of responsibility	Benefit levels[4]	Combination with work income or other benefits	Funding and execution
USA[5]	(2) All work place injuries, related medical costs and rehabilitation Protects against civil litigation	Medical-economic (ability to engage in sustained remunerative employment) No minimum 1 week's work lost (5) Medical-economic (unable to earn more than $740) (1)	Medical decisions: GP; impairment guidelines (American Medical Association) to state d Permanent WIB decision by Industrial Commission after re-evaluation	BF ≈ 0.67 w BP = d*BF, often as lump-sum settlements	Work income and WIB: no Work income and lump-sum settlement: yes DB: yes	Employer premiums, risk- and experience-rated except for small businesses; self-insurance for large employers Mainly private companies; State schemes to cover poor risks (monopoly in 6 states)

1. Acronyms: DB: disability benefit; OD: occupational disease; WI(B): work injury (benefit); SB: sickness benefit; APW: average production worker; GP: general practitioner; LF: labour force; OAP: old-age pension; w: monthly basis for the calculation of the benefit level or beneficiary's monthly earnings before the injury (often similar or same amount); d: degree of disability as defined by WI scheme.
2. Numbers in brackets mean: (5) total population; (4) Entire LF plus specific groups (*e.g.* students); (3) Entire LF or all employees and voluntary self-insurance for self-employed; (2) all employees; (1) specific groups of employees.
3. Following OECD (2003a), p.186, the generosity index takes a value of 5 if the minimum degree of disability (d) to qualify for benefits is less than 25%, 4 for d from 25 to 40%, 3 for d from 41 to 55%, 2 for d from 56 to 70%, 1 for 71 to 85% and 0 from 86 to 100%.
4. B denotes "benefit", D permanent, T temporary, F full, and P partial. Lump-sum payments are only mentioned if beneficiary cannot choose regular benefits. In many countries a choice between lump-sum and periodic payments exists for cases of less severe losses in work incapacity.
5. WI insurance organised at regional level, prevailing tendency reported.

Source: OECD database on programmes for disabled persons.

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Annex 4.A2

The market for placement services

In the current system an unemployed person first has to contact the public employment office (Centres for Work and Income, CWI) that provides nation-wide information on vacancies and classifies jobseekers into target groups according to their distance to the job market. Laid-off workers eligible for unemployment benefits (UB) are referred to the UWV and those eligible for social assistance to municipalities. The benefit authority (UWV) makes competitive tenders covering one year for lots typically of 100 persons sorted per target group, sector and region. Bidders specify the cost per client and a "guaranteed" placement rate. The remuneration of the winner consists of a commencement payment and a bonus upon placement that together cover the specified cost if the guaranteed placement rate is achieved. In 2002 there were 41 providers of reintegration services, with an increasing market share for newcomers (Lundsgaard 2002, p. 110).

While provisions for income policies have become the sole responsibility of the central government, local governments have got more autonomy in finding the best way of reintegrating social benefit recipients. Municipalities took over full financial responsibility of their social benefit budget on 1 January 2004 as a result of the Work and Social Assistance Act. The law also stipulates that they have to tender out at least 70 per cent of the separate activation budget received from central government to private reintegration service providers and must pay back unspent money from this budget. As a result, municipalities are encouraged to reintegrate rather than confining themselves to paying social benefits. While they remain free as to how they spend the remaining 30 per cent of the activation budget, *e.g.* whether to use subsidised municipal jobs or small wage subsidies to regular employment, the central government wants to reduce the number of jobs on the secondary labour market (2¼ per cent of all jobs) as beneficiaries' chances of re-entering the regular job market were found to be small. During 2003 a temporary subsidy of € 17 000 per worker was granted for turning subsidised jobs into regular employment.

As the success of placement services is highly dependent on clients' initial conditions, more needs to be known about the net value added by providers. Specifically, better information on *ex ante* matching probabilities and evaluation of placement results would limit problems of creaming and under-investment into difficult clients. Creaming on the one hand and under-investment on the other occur in the absence of an optimal fee structure with a negative fixed fee (accounting for placements without input) and a high placement fee (reflecting the social value of resumed work, Grubb 2003).[1] Given the remuneration structure based on average client cost, these problems are currently accentuated by heterogeneity within contract populations (de Koning 2003, p. 19). Resolving the issue puts high demands both on profiling techniques and empirical performance evaluation. UWV is currently developing a public performance benchmark based on company and client characteristics that will become operational in 2005. The resulting information, especially on disadvantaged

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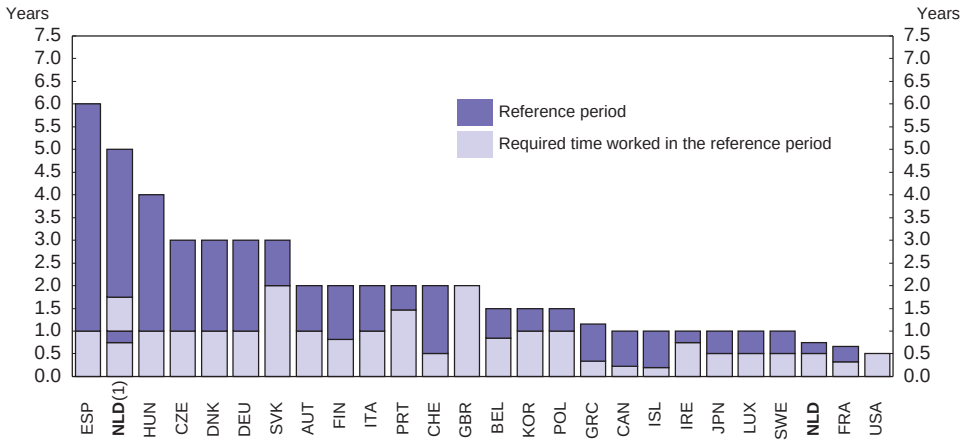
Figure 4.A2.1. **Work record required for access to unemployment benefits**



1. Current reform proposal entails having worked 39 out of the past 52 weeks and 52 days in each of four of the past five years.

Source: OECD, Benefits and Wages 2002.

unemployed, is particularly relevant for municipalities that need to enhance their purchasing decisions.[2] Further improvement in market transparency would be achieved by introducing a revocable "quality card" awarded to firms that achieve their placement promise in more than 80 per cent of contracts, as suggested by the Council for Work and Income.

The success of a market-driven reintegration system crucially depends on clients' incentives to comply and thus on the level of unemployment benefits (UB) and social benefits. As to the former, access to a benefit requires having worked during 26 out of the past 39 weeks and 52 days in each of four out of the past five years as well as not being responsible for one's unemployment. Except for these conditions, duration is determined by age. The benefit replacement rate is 70 per cent up to an income ceiling. A special short-term benefit (six months) exists for persons with an insufficient work history (*i.e.* not having worked in four out of the past five years) who receive 70 per cent of the minimum wage. This is also the amount of unemployment assistance that lasts for two years after the end of the UB period. Implementing the plans for tighter access to UB would make the Netherlands become one of the stricter countries in terms of the proportion of time worked in the immediate past (Figure 4.A2.1). Obligations for benefit recipients include availability for a job, sufficient search and applications, and accepting suitable work they can commute to.[3] Unemployed civil servants enjoy more generous benefit conditions, with replacement rates between 80 and 100 per cent in the first year, falling only gradually to 70 per cent, and much higher duration.[4] The latter does not lead to a major loss in labour inputs under current public management settings because termination of employment contracts is rare for civil servants (see Box 3.1).

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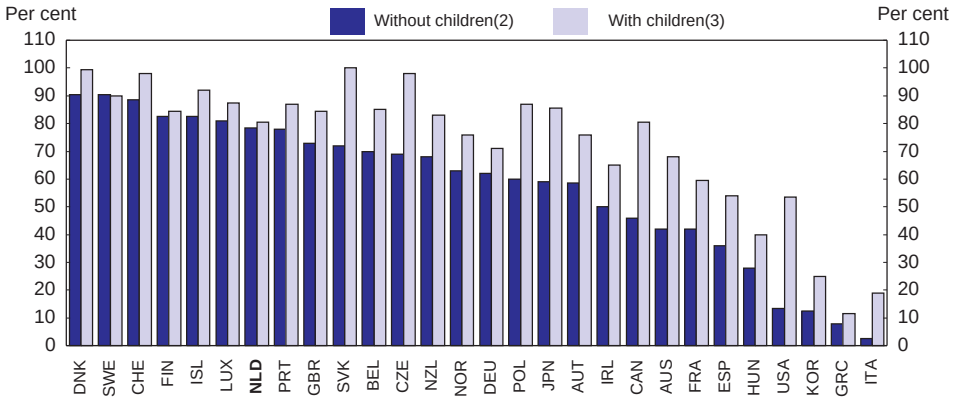
Figure 4.A2.2. **Net replacement rates for long-term benefit recipients**[1]



1. As compared to work income at 66.7 per cent of the APW level after tax and including family and housing benefits.
2. Average of two household types (single; married couple).
3. Average of two household types (couple; lone parent).

Source: OECD Benefits and Wages (2002), p. 36.

In encouraging labour market participation of low-wage earners, the government has shifted its policy focus from demand to supply side measures. The reduction in employers' contributions on wages up to 115 per cent of the minimum wage (SPAK) was suspended from 2003 to 2007 as it apparently led to high deadweight losses and did not appear to reduce the number of vacancies for low-skilled workers despite a high number of social benefit recipients. The authorities concluded that the problem was a lack of labour supply. As pointed out in the last Survey, unemployment benefit replacement rates and the tax wedge are rather high by international comparison (OECD 2002, pp. 78-79). Moreover, net replacement rates on long-term benefits are high for low-wage earners, as in a number of countries (Figure 4.A2.2). Special municipal benefits and rent subsidies push up the replacement rate further, resulting in a loss in disposable income for a household with children taking up employment at the minimum wage (Table 4.A2.1).

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Table 4.A2.1. **Evolution of the poverty trap**

	Level 2003	Level 2004
Unemployment trap[1]		
2 adults, 1 income, with child(ren)	–3½	–1¾
1 adult	–½	3½
1 adult, with child(ren)	–4½	–6¼
"Re-entry trap"[1, 2]		
Partner at minimum wage	15	18½
Marginal burden		
2 adults, 1 income, with child(ren)		
Minimumplus	91¾	93¼
Modal income	81¼	81
2*modal income	54¾	55
Double income		
Minimumplus + ½*minimumplus	25¾	24¼
Modal + ½*modal	34	32¾
2*modal + modal	48	48

1. Percentage change in disposable income after entering employment at minimum wage, from receiving minimum level benefits. Figures include effects of municipal benefits and regulations, and rental subsidies.
2. Percentage increase in household income of a family with one adult earning the minimum wage, and the non-earning partner taking up employment at an income level of half the minimum wage.

Source: Min SZW.

Notes

1. The optimal fee structure has the disadvantage of high financial risks, excluding small providers from the market. As an alternative to negative fixed fees, UWV grants separate funding of vocational training needs for persons with a high distance from the labour market (input subsidy).
2. "Each municipality has its own way of dealing with [the procurement of reintegration services] and is reinventing the wheel". Also, "few municipalities have a follow-up system on the basis of which individual clients can be monitored" (De Koning 2003, pp. 19-20).
3. Suitability criteria become stricter as duration increases. Refusing a suitable job offer leads to a cut in the benefit corresponding to the number of hours specified in the offer.
4. Benefit duration for civil servants depends on seniority and on a multiplicative combination of age and seniority. A civil servant who loses his job at age 46 with 21 years of service gets UB until he is 65.

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Annex 4.A3

The Dutch dismissal system and effects of EPL on (un)employment

There are two channels for terminating employment contracts in the Netherlands – *via* the Labour Office (CWI) or third party intermediation from a local court. If termination occurs *via* the Labour Office (CWI), it has to fulfil a number of criteria (last-in first-out principle, non-discriminatory composition of the list of persons, etc.) and must be based on sufficiently important reasons (such as no work whatsoever available in the firm and incapability unless caused by illness) to obtain the necessary consent from dismissal advisory boards, consisting of employers' organisations and unions. Moreover, the MinSZW has issued rules to protect specific categories of employees from dismissal, *e.g.* older workers. Termination *via* third-party intermediation from local court is faster, and limits the number of parties involved in the conflict. However, this procedure is also more expensive, as the judge's severance pay formula, by which most cases are settled, provides for one month of salary per year of service. Judges deviate from the formula according to the circumstances of the case.[1] Although the allegation of unfair dismissal has, in principle, to be proved by the dismissed employee, in practice courts shift it to the employer who has to prove that the dismissal was fair. The court channel of contract termination has grown more important over the 1990s and has stabilised at about 50 per cent of all cases since 1998. In 1999, legislation covering termination of employment contracts on economic grounds was somewhat simplified (Act on flexibility and security). Now a shortened procedure (1 to 2 weeks) applies and the time spent on initiating a termination is withheld from the notice period. On the other hand, more and more often severance pay is also granted under the CWI procedure, suggesting that there has not been a significant reduction in overall EPL strictness.[2] The influence of third parties, which is the highest in the OECD, and the amounts of severance pay in the court channel make the largest contributions to overall EPL strictness in the Netherlands.

Strict EPL and special protection for certain groups create huge insider-outsider problems because firms prefer not to hire persons whose capability of fulfilling the requirements of the job is more uncertain. As a consequence, unemployment is concentrated with specific vulnerable groups (*e.g.* ethnic minorities, long-term unemployed, persons with disabilities), a result well documented in the empirical literature (see OECD 1999 for an overview).[3] Furthermore, in combination with the long duration of UB, high severance pay provides dismissed workers with little incentives for a quick return to the labour market. Reducing the strictness of EPL on permanent work contracts is particularly pressing in the current context of overhauling benefit systems (disability, unemployment of inexperienced younger and older workers), which will lead to a need for integrating an increased number of outsiders. The generally negative effect of strict EPL on average employment (Nickell, 1997)[4] increases the risk that a larger part of these groups will voluntarily withdraw from the labour market. This makes adjustment of EPL for regular contracts particularly desirable in the Dutch context of a generous UB, a high tax wedge (tending to increase the unemployment rate, Elmeskov

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et al., 1998 and CPB, 2003) and low regulation of part-time contracts (discouraging voluntary quits or leading to them not being re-filled, Boeri, 1999). While the effect of EPL on the overall long-term unemployment rate is less clear-cut in the empirical literature,[5] strict EPL reduces the speed of labour input and wage adjustment to economic conditions (Jackman *et al.*, 1996), putting higher strains on firm survival during recessions, which may accentuate the size of business cycle fluctuations. Labour market flexibility will become increasingly important as faster technological developments and globalisation are leading to more frequent sectoral shifts in the economy. The move to new technology is not always feasible with in-house training of the existing workforce alone but will increasingly require re-modelling the skill-composition on production lines in order to reap the benefits of new technologies on productivity.[6]

Notes

1. For instance, an employee with a long tenure and little or no continuing training during his career will generally receive a higher severance pay to account for specificity of human capital that may weaken his labour market position.
2. The OECD indicator of EPL strictness were published in 1999 (OECD, 1999). They are being updated and will be presented in the OECD E*mployment* O*utlook* 2004.
3. Strict EPL reduces inflow to and outflow from unemployment while increasing job-to-job mobility, a clear disadvantage for persons on unemployment (Bertola and Rogerson, 1997). Chances for outsiders are further reduced by the fact that with strict EPL quits occur less frequently and are more often not re-filled (Boeri, 1999). As a consequence, the incidence of long-term unemployment increases (Nickell and Layard, 1998). Strict EPL also partly explains the success of temporary work contracts but tends to increase the proportion of part-time and temporary work that is involuntary (Grubb and Wells, 1993).
4. Also see Scarpetta (1996) who finds that stricter EPL implies higher non-employment rates.
5. Elmeskov *et al.* (1999) and Lazear (1990) find a positive effect of EPL strictness on the unemployment rate, whereas in Nickell and Layard (1998) the EPL variable has no significant effect and the effect on the unemployment rate is found to be ambiguous in Blanchard and Portugal (1998).
6. The OECD Growth Study has found that the overall impact of EPL on MFP is negative (OECD, 2003b).

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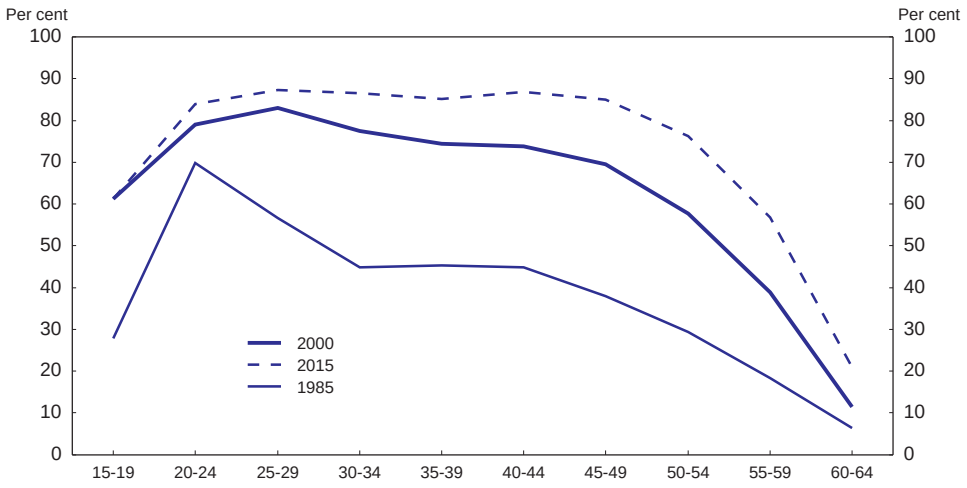
Annex 4.A4

Trends in female labour force participation

The one-and-a-half earner model (often with the woman working part-time) has become the most popular form of family labour supply. The share of women working part-time because they did not want a full-time job is the highest in the European Union (European Labour Force Survey, 2001). The economic literature distinguishes between demand and supply models to explain the success of (female) part-time work (Jaumotte, 2003). Demand models centre on firms covering peak-time needs, those preferring part-timers because they are cheaper and firms retaining productive female workers facing family responsibilities. Among the factors on the demand side, firms' acceptance of part-timers to fill full-time vacancies was certainly of particular importance during the tight labour market of the 1990s. On the supply side, conditions for working part-time are as attractive as for full-time work. The median hourly wage is virtually equal to that of full-timers, part-timers have the same social security coverage, enjoy the same EPL rules and have been assimilated to full-timers by a

Figure 4.A4.1. **Female labour force participation will grow at a slower pace**

Past and future participation profiles of female cohorts



Source: Burniaux *et al.* (2003).

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Table 4.A4.1. **Predicted labour force participation of women aged 25-54 and share of full time, 2025**

In per cent

	Participation rate	Share of full time
Australia	78.4	62.9
Austria	75.6	73.3
Belgium	83.2	65.3
Canada	85.7	78.6
Czech Republic	81.5	96.0
Denmark	85.0	83.7
Finland	87.7	90.7
France	79.0	76.8
Germany	80.7	64.8
Greece	77.4	90.8
Hungary	70.7	96.0
Iceland	95.9	71.0
Ireland	80.9	67.9
Italy	64.5	75.9
Japan	68.3	62.4
Korea	59.6	91.3
Luxembourg	79.1	70.5
Mexico	58.4	74.2
Netherlands	**82.2**	**44.1**
New Zealand	74.5	67.6
Norway	88.6	71.4
Poland	77.8	84.9
Portugal	79.4	88.9
Slovak Republic	84.8	97.3
Spain	73.2	84.7
Sweden	80.4	82.1
Switzerland	87.0	53.6
Turkey	15.5	81.6
United Kingdom	79.4	61.4
United States	78.2	87.5
OECD average (unweighted)	76.4	76.6

Source: Baseline scenario from Burniaux *et al.* (2003).

legal ban on discrimination since 1996. Switching from full-time to part-time is relatively easy and happens often, as employers' possibilities to restrict such a move are limited.

According to recent OECD analysis based on demographic trends and the extrapolation of generational changes in female employment behaviour, women will continue to make significant contributions to the growth in labour inputs over the next ten years, albeit smaller ones than between 1985 and 2000 (Figure 4.A4.1). Looking ahead to 2025, a simulation of female labour force participation rates holding the institutional environment in each country fixed reveals that the Netherlands could become a victim of its huge part-time success. While overall participation would indeed remain higher than the OECD average, pure full-time participation would fall way behind the average, implying lower growth in total labour inputs (Table 4.A2.1).

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5. Product market competition and economic performance

Overview

There is a well-identified empirical connection between the intensity of competition in product markets and better productivity performance (OECD, 2002a). Indeed, over the past decade the Netherlands has undertaken a number of structural reforms in product markets, primarily aimed at increasing its productivity growth. These reforms, even those implemented only recently, have been assessed as resulting in a more competitive environment for businesses, stimulating economic efficiency and technological progress.[1] The service sector, where competitive pressures are weaker, has experienced relatively poor labour productivity growth over the 1990s. Planning restrictions are inhibiting competition in the retail sector and there is considerable scope to eliminate practices that restrict competition in professional services, even though both are relatively liberalised in the Netherlands. Reforms in some network industries (*e.g.* electricity, gas and telecoms) are also recent and market power on the part of incumbent firms remains a concern. Reforms that would increase competitive pressures in these sectors include enhancing the powers of regulators in network industries and eliminating barriers to entry. In general, further service sector reforms should be matched with effective enforcement of the new competition law.

This chapter assesses the role product market competition, and those policies that impact upon competition, may have played in the performance of the Dutch economy over the past decade and what further measures might contribute to enhancing growth. While recognising that considerable progress has been made, the analysis primarily seeks to identify areas where policies continue to impair performance. The main links between stronger competition and macroeconomic performance are reviewed in the first section of this chapter, while the second section lays out the competition legislation framework and introduction of the Competition Act in 1998. In the third section, competitive conditions, regulation and recent reforms are analysed for a wide range of non-manufacturing sectors. This includes service industries that are competitive, such as retail distribution and professional services, and network industries containing non-competitive segments, such as telecommunications, electricity and gas. A concluding section draws on the analysis to provide a set of policy recommendations.

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Product market competition and economic performance

Over the 1990s the Netherlands had relatively good economic performance at the aggregate level. The levels of GDP per capita and GDP per hour worked in the Netherlands are above average, with the latter being very high and exceeding that of the United States. Average GDP growth between 1990 and 2001 was also above both the OECD and EU averages. This strong aggregate growth performance can be explained primarily by the growth in employment, which was well above that of other countries (Table 5.1). However, the Netherlands position is slowly being eroded due to low productivity growth rates. The poor productivity growth performance can in part be explained by the labour market developments over the decade as less productive workers were employed.

While productivity growth in the Dutch manufacturing sector has been comparable to that of other OECD countries, growth rates in services were lower than in most other OECD countries (Figure 5.1). Only France, Spain and Japan had lower productivity growth in services during the 1990s. A sectoral breakdown shows in almost all service sectors the Netherlands had relatively weaker productivity growth as compared with other OECD countries (Table 5.1). Performance was particularly poor in construction, where the Netherlands had negative productivity growth, and in wholesale and retail trade, where productivity growth was one of the lowest in the OECD. Productivity growth was also relatively weak in transport, storage and communication and in electricity, gas and water supply.

Indicators of the intensity of product market competition

Although it is difficult to classify markets according to the strength of market forces, the degree of product market competition may be gauged from jointly considering a number of imperfect proxy measures. The measures of product market competition presented below primarily look at the manufacturing sector but some non-manufacturing sectors are also considered. Manufacturing industries are grouped into four categories. A distinction is made between low R&D and high R&D industries; and between fragmented industries, which are those industries that are less concentrated and characterised by a large number of firms, and segmented industries, which are more concentrated industries characterised by a smaller number of relatively large firms.

Overly stringent product market regulations can have an impact on the strength of competition in domestic markets either by exerting direct control on economic activities or by maintaining high barriers to trade, foreign direct investment and entry into domestic markets. Various barriers to entrepreneurial activity (*e.g.* restrictions on market access or administrative burdens and red-tape on firms) can also inhibit competition and discourage entry by both domestic and foreign firms. In international comparison the Netherlands' institutional and regulatory arrangements in general appear to be favourable to competition.

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Table 5.1. **Output, employment and productivity**

1990 to 2001

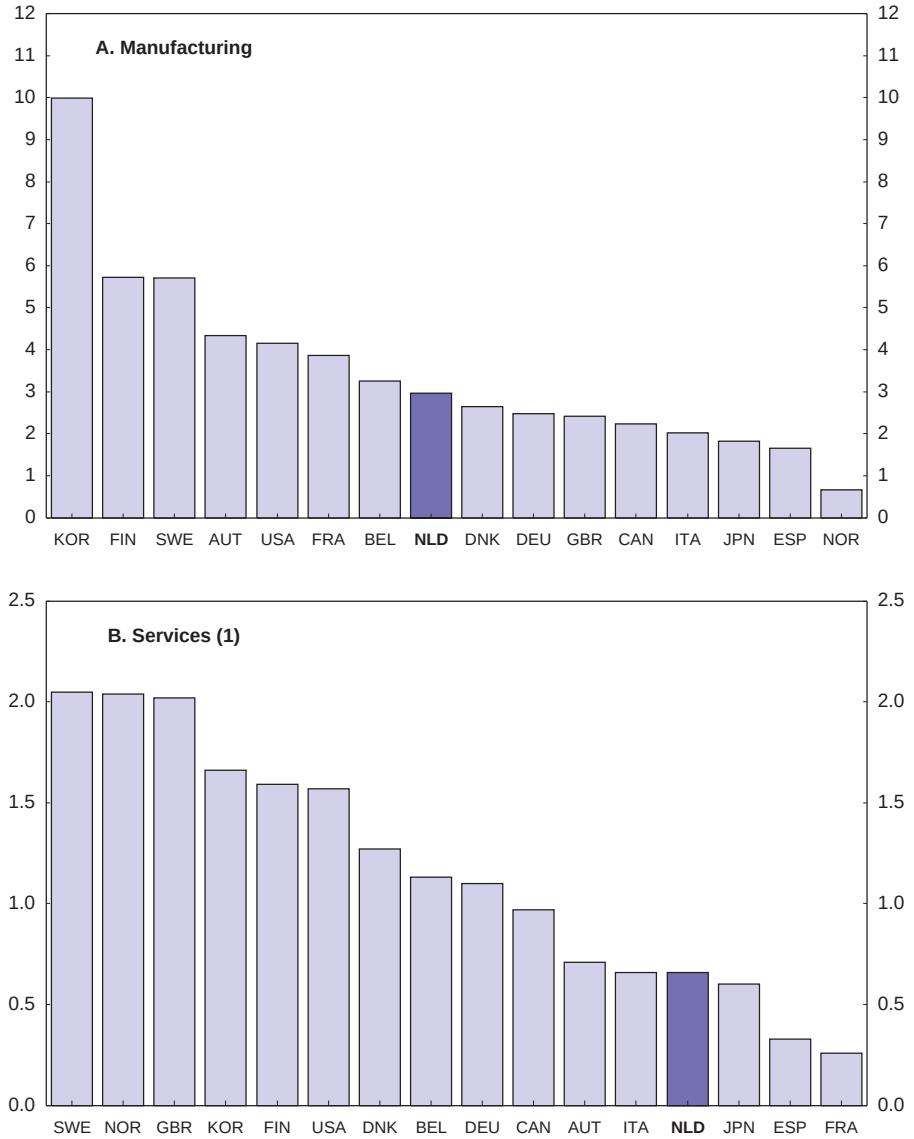
	Netherlands	Australia	Belgium	Denmark	France	Germany[1]	United Kingdom	United States	OECD	European Union
Average GDP growth	2.9	3.3	2.1	2.2	2.0	1.5	2.3	2.9	2.6	2.2
of which:										
Labour productivity per employee	0.7	1.9	1.6	1.9	1.1	1.7	1.8	1.6	1.8	1.6
Employment	2.1	1.4	0.6	0.3	0.8	–0.2	0.4	1.3	0.8	0.6
Labour productivity growth per employee in selected industries										
Total manufacturing	2.5	2.4	2.7	2.6	3.3	2.4	2.6	3.3	n.a.	n.a.
Electricity, gas and water supply	3.3	5.9	4.6	2.7	3.0	5.4	10.1	1.2	n.a.	n.a.
Construction	–0.8	0.6	0.3	–0.8	0.1	0.0	2.5	0.0	n.a.	n.a.
Wholesale and retail trade; restaurants and hotels	0.5	1.9	1.2	2.1	0.8	–0.5	1.7	3.3	n.a.	n.a.
Transport and storage and communication	2.9	4.4	1.2	3.9	2.6	7.6	3.9	2.7	n.a.	n.a.
Memorandum items:										
GDP per capita[2]	83.1	75.5	78.5	83.1	75.0	74.9	74.9	100.0	71.6	72.6
GDP per hour worked[2]	114.6	78.7	113.9	95.2	113.2	98.8	84.7	100.0	83.3	95.3

1. Average growth 1992-2001.
2. 2001 levels, PPP based 2001, USA = 100.

Source: OECD.

Figure 5.1. **Labour productivity growth per employee in manufacturing and services**

Average annual, per cent, 1990-2000



1. The services sector covers ISIC classes 50-99.

Source: OECD STAN database.

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Focusing on regulations that restrict competition and market mechanisms (*e.g.* economic and administrative regulations and barriers to trade and FDI), OECD indicators of regulation suggest that in 1998 the economy-wide regulatory stance of the Netherlands was comparatively friendly to competition (Figure 5.2A).[2] Despite this relatively favourable economy-wide regulatory stance, the Netherlands had an average regulatory stance in important service sectors (*i.e.* the utilities and transport sectors) (Figure 5.2B).

In general, mark-ups, a frequently used gauge of market power, and thus competitive pressures, appear to be relatively low in the Netherlands. In fragmented manufacturing sectors,[3] estimated mark-ups are below average indicating that competitive pressures in these sectors seem to be quite intense (Figure 5.3). Somewhat higher than average mark-ups in segmented manufacturing industries[4] could indicate that there are problems with competitive pressure in segmented industries. The strength of competitive pressures depends to a large extent on how exposed industries are to international competition. Import penetration rates indicate that Dutch firms face relatively stronger competitive pressure from foreign firms than their counterparts in other OECD countries (Table 5.2). Only Belgium has a higher import penetration rate in total manufacturing. A sectoral breakdown shows that competitive pressures are strong in all industries, including the segmented sectors, and reflect the fact that the Netherlands is an extremely open economy.

Barriers to entry through restrictions on foreign direct investment may also act as an impediment to competition. The Netherlands, along with the United Kingdom and Ireland, has some of the lowest restrictions to FDI amongst OECD countries (Figure 5.4). Partly reflecting the low restrictions, inflows of FDI into the Netherlands are comparatively high, with only Belgium-Luxembourg, Ireland and Sweden having higher inflows (Figure 5.5). The Netherlands also has the second highest outflows of FDI amongst OECD countries. The relatively high mark-ups in segmented sectors in conjunction with the high import penetration rates and FDI inflows, signifying strong competitive pressures from foreign firms, would seem to suggest that Dutch firms in these sectors may be more efficient than their competitors in world markets.[5]

In contrast to manufacturing, there are numerous restrictions to entry and competition in some of the non-traded sectors in the Netherlands. During the latter part of the 1990s, the Netherlands implemented a number of important reforms in product markets aimed at increasing competition (these reforms are discussed in the sections below). Coinciding with these reforms, productivity growth over the 1996 to 2001 period has improved in a number of non-manufacturing sectors (*e.g.* electricity and gas, construction, wholesale and retail trade and transport and communications) (Figure 5.6). However, while there was an improvement in productivity growth in services in the second half of the 1990s,

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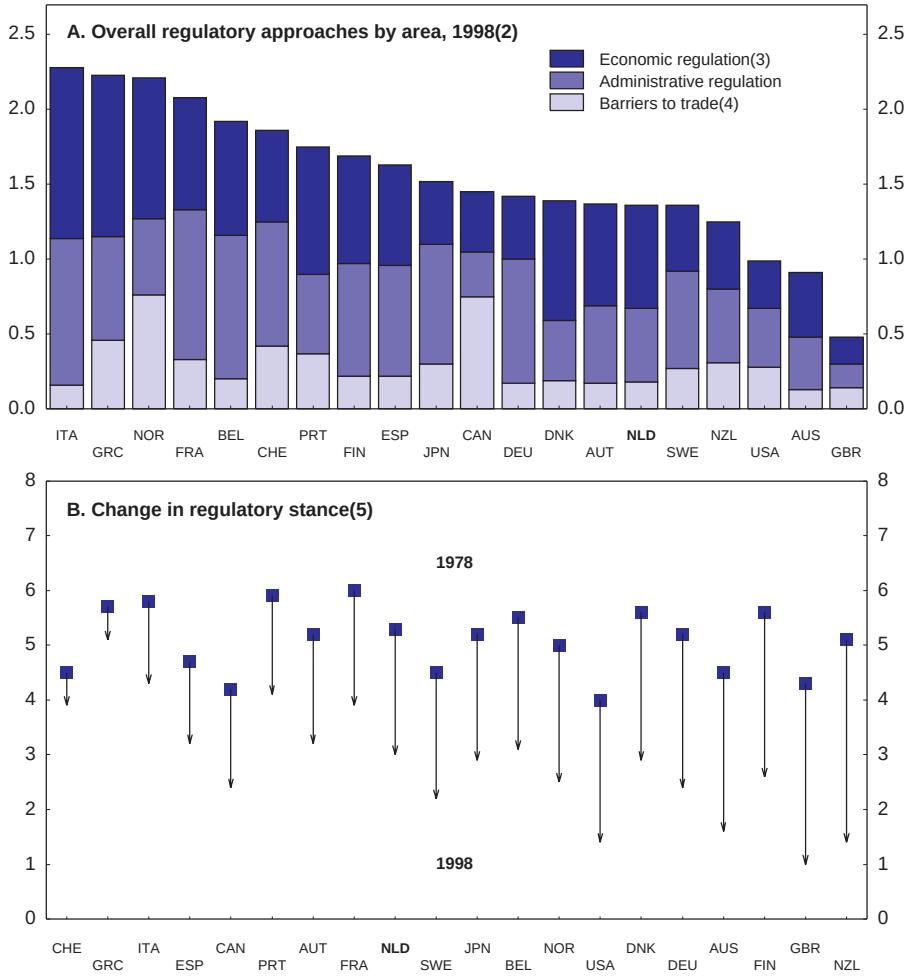
Figure 5.2. **Indicators of product market regulation**[1]



1. The regulatory stance is measured by a synthetic indicator ranging between 0 (least restrictive) and 6 (most restrictive).
2. Indicator of economy-wide product market regulations.
3. Includes barriers to competition and state control.
4. Includes trade and FDI restrictions.
5. Reports changes in the regulatory stance in seven non-manufacturing industries (gas, electricity, post, telecommunications, passenger air transport, railways and road freight) between 1978 and 1998.

Source: Nicoletti *et al.* (2001); Nicoletti and Scarpetta (2003).

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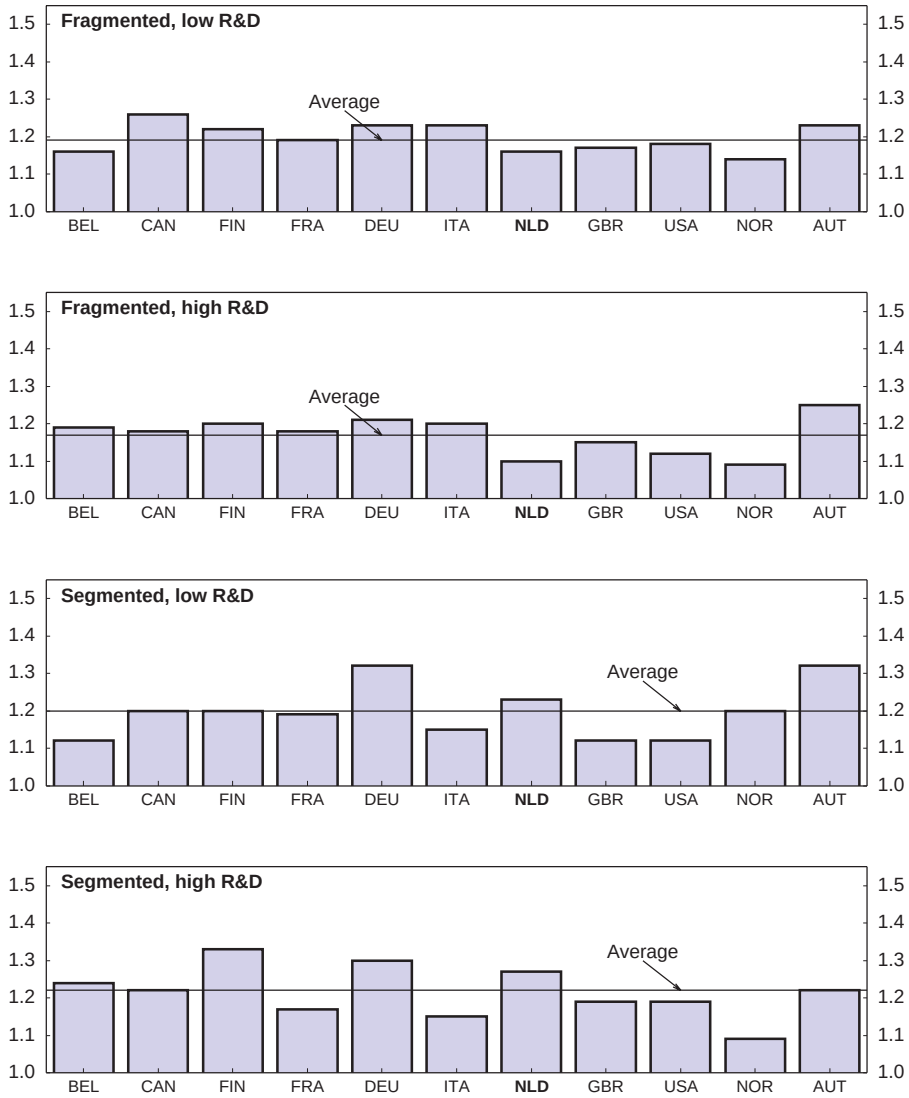
Figure 5.3. **Average mark-ups in manufacturing by market structure**
1981 to latest available year[1, 2]



1. For the Netherlands data is from 1987 to 2002.
2. The average mark-up is an unweighted average of the available mark-ups. ISIC, Rev. 3 classification.

Source: OECD STAN database.

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Table 5.2. **Import penetration by manufacturing industry Latest available year**

	Netherlands	Canada	France	Germany	Italy	United Kingdom	United States	Austria	Belgium	Denmark	Sweden	Norway
	2001	2000	2002	2001	2001	2000	2001	2001	2002	2002	1999	2002
Total manufacturing	80.2	52.6	36.5	40.8	30.5	44.8	23.1	64.8	103.9	68.2	44.3	48.2
Fragmented, low R&D												
Textiles	92.8	62.4	52.3	89.6	22.2	54.0	27.2	81.6	125.4	126.3	91.3	74.9
Wearing apparel	153.6	38.8	57.0	78.7	22.5	65.2	53.3	96.7	119.9	159.7	112.2	90.3
Leather products and footwear	163.5	77.9	86.7	89.0	44.0	86.2	79.5	115.6	n.a.	126.7	99.7	95.5
Wood products	51.8	15.6	22.6	19.4	16.5	32.0	12.7	24.3	68.6	50.9	13.1	25.9
Paper and pulp	73.2	n.a.	38.3	43.4	26.3	35.2	10.4	45.0	87.3	64.3	24.8	51.7
Printing and publishing	12.0	n.a.	8.8	7.2	7.1	8.0	2.5	25.9	22.3	14.5	9.2	13.5
Non-metallic products	29.1	36.8	19.1	19.6	8.5	17.9	13.5	26.4	41.4	31.6	31.6	25.1
Fabricated metal products	27.8	32.8	14.0	15.3	7.4	16.6	8.9	39.3	42.1	31.8	20.9	37.9
Fragmented, high R&D												
Medical precision and optical instruments	69.8	n.a.	46.4	64.8	60.8	63.5	22.7	96.1	169.2	107.8	47.8	57.6
Machinery and equipment	162.5[2]	84.1[1]	54.3[4]	36.6[4]	37.6[4]	57.0	26.4	77.2	152.1	66.6	52.4	70.4
Furniture manufacturing	n.a.	56.8[1]	41.8[4]	44.0[4]	20.2[4]	38.8	38.5	54.4	n.a.	51.2	n.a.	51.9
Segmented, low R&D												
Refined petroleum, coke	56.6	10.8	18.0	27.3	16.4	21.4	17.9	36.5	44.4[3]	35.2	44.7	n.a.
Basic metals	101.3	44.6	45.6	45.2	44.4	46.9	22.0	57.1	86.3[3]	81.8	48.8	65.7[3]
Shipbuilding and repairs	n.a.	58.6	27.6	49.8	33.8	21.0	6.8	n.a.	29.2[3]	47.6	13.9	42.1[3]
Rubber and plastic[1]	75.6	42.6	30.5	29.1	22.3	26.6	12.1	65.9	102.7[3]	56.4	52.8	61.9[3]
Food, beverages and tobacco[1]	37.1	17.5	18.4	19.5	19.7	20.5	6.3	27.1	50.3[3]	40.0	21.9	14.3[3]
Segmented, high R&D												
Chemicals	72.5[3]	57.3	51.1	57.7	48.6	53.3	20.4	91.0	129.1	83.4	69.6	n.a.
Pharmaceuticals	78.7[3]	52.5	44.7	83.6	49.1	63.1	18.7	108.8	144.8	67.7	51.7	64.3[3]
Office and computing machines	n.a.	108.3	99.3	108.6	93.0	105.7	68.2	123.2	47.4	149.2	102.7	121.4[3]
Electrical machinery	262.5[3]	82.2	46.2	32.4	27.0	53.6	46.6	91.7	97.3	61.0	90.2	64.0[3]
Radio, TV and communication equipment	121.6[3]	74.3	73.6[3]	107.0	60.5	90.8	41.8	78.6	109.9	132.0	50.3	72.5[3]
Motor vehicles	106.7[3]	76.4	36.6	35.0	61.8	59.0	36.5	117.9	149.2	107.8	43.0	95.0[3]

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Table 5.2. **Import penetration by manufacturing industry Latest available year** (*cont.*)

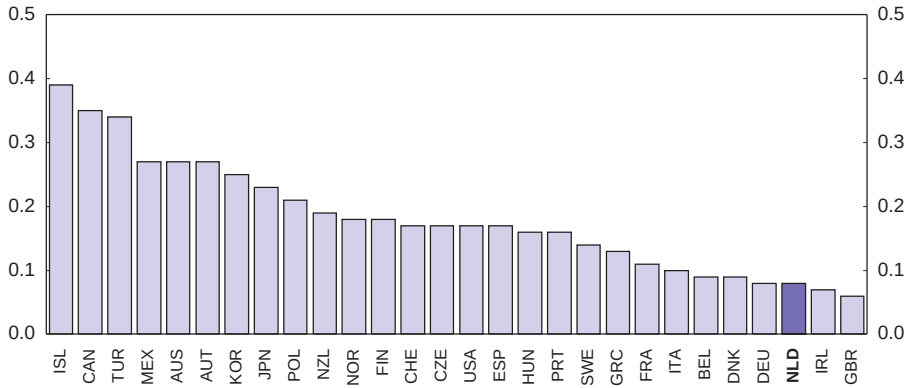
	Netherlands	Canada	France	Germany	Italy	United Kingdom	United States	Austria	Belgium	Denmark	Sweden	Norway
	2001	2000	2002	2001	2001	2000	2001	2001	2002	2002	1999	2002
Aircraft	n.a.	82.8	47.0	155.7	74.2	87.4	30.2	n.a.	77.5[3]	. .	83.2	108.4[3]
Railroad equipment	n.a.	37.6	41.7	42.2	41.2	43.6	20.3	55.0	94.2[3]	133.5	37.7	85.4[3]
Other transport equipment	68.2[3]	67.5	44.0	108.1	55.1	72.2	25.4	145.4	79.8[3]	83.9	57.7	54.5[3]

Note: Imports as a percentage of domestic demand (estimated as production minus exports plus imports). Values greater than 100 can occur when exports exceeded production because of the inclusion of re-exports, *i.e.* products that are imported and then re-exported without any further transformation.

1. Mix of fragmented and segmented sectors.
2. 1999.
3. 2000.
4. 2002.

Source: OECD STAN Database.

Figure 5.4. **FDI restrictions in OECD countries**[1]
2000



1. Includes limits on foreign ownership, restrictions on foreign personnel and operations freedom, and screening requirements. The indicator ranges from 0 (least restrictive) to 1 (most restrictive).

Source: OECD.

Dutch labour productivity in these sectors still grew slower than productivity in most other OECD countries. Productivity growth in the Dutch business services sector has also been poor, contributing to the weak performance of aggregate labour productivity growth (Kox, 2002).

The poor productivity performance of the Dutch service sector is partly explained by the insufficient use of ICT and (until recently) disappointing little investment in ICT. However, ICT-using services have contributed very little to aggregate productivity growth in the Netherlands (Figure 5.7). Wolfl (2003), in a recent study measuring and comparing productivity growth performance in service sectors, finds that poor productivity performance originates precisely in those service industries where reforms or competition are thought to have been weaker. This suggests that there is considerable scope for the Netherlands to improving productivity growth by improving competition in product markets, particularly in service industries where restraints to competition are comparatively high.

Potential macroeconomic effects from regulatory reform

The macroeconomic benefits in the Netherlands of regulatory reforms that would increase product market competition are significant. The propagation and channels through with regulatory reform affects the economy depend on a number

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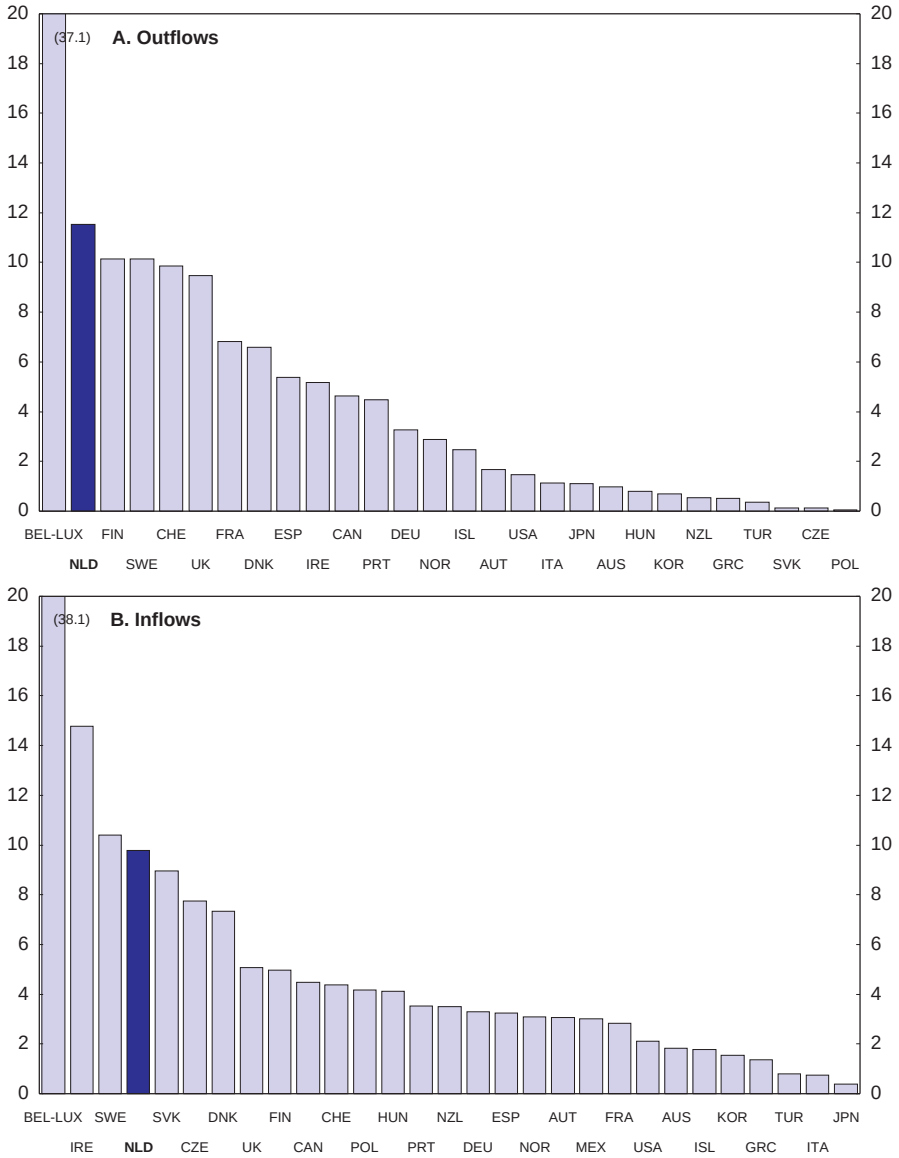
Figure 5.5. **Foreign direct investment outflows and inflows**
Per cent of GDP, average 1997-2001



Source: OECD, International Direct Investment Statistics.

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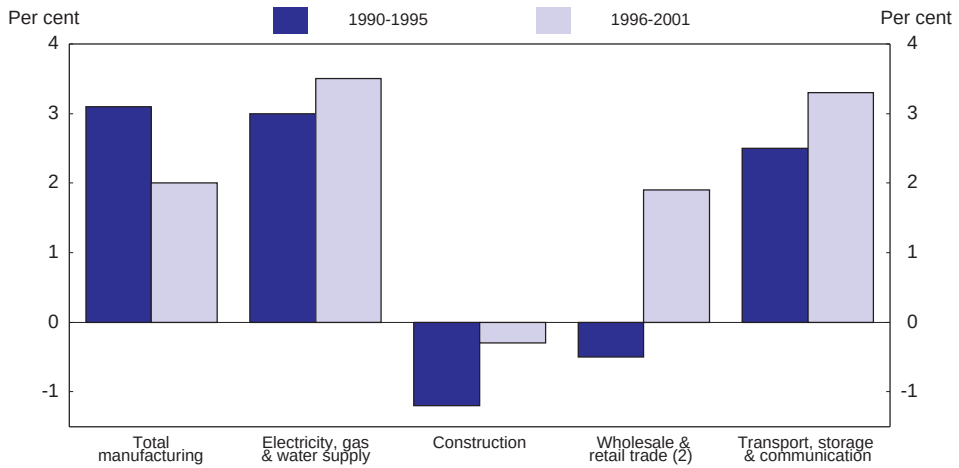
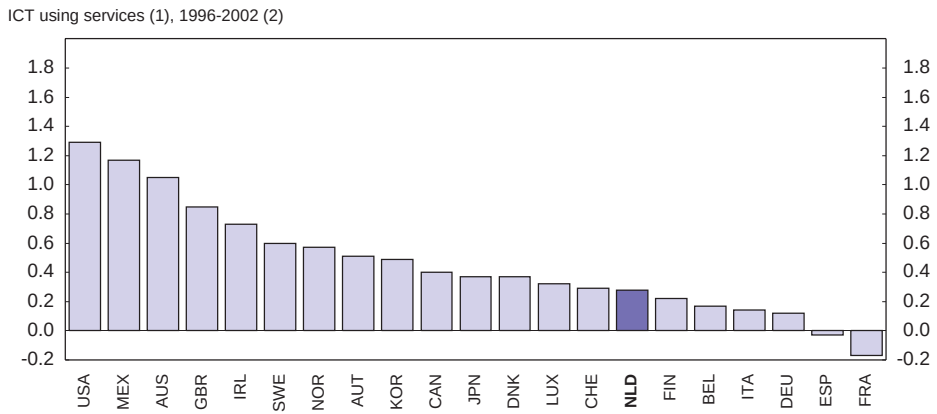
Figure 5.6. **Labour productivity growth in selected industries**[1]



1. Average annual growth in labour productivity per worker.
2. Hotels and restaurants are included in wholesale and retail trade.

Source: OECD STAN database.

Figure 5.7. **ICT and productivity growth**



1. Contribution to aggregate labour productivity growth.
2. 1996-2000 for Ireland, Norway and Switzerland. 1996-2001 for France, Germany, Japan, Mexico, Netherlands, United Kingdom and the United States.

Source: OECD.

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Box 5.1. Economy-wide effects of regulatory reforms

Regulatory reforms that increase product market competition within a sector improve that sector's economic performance through a number of channels: these static gains are further enhanced by dynamic effects.

- Sectoral reforms change relative prices, improving overall resource allocation and consumer welfare.
- Reforms that increase competition reduce price-cost margins thus lowering price and expanding output in the sectors concerned. This in turn, may diminish the scope for rent-sharing, thereby putting downward pressure on wages in those industries.
- Reforms force firms to reduce slack in the use of input factors (*i.e.* reducing X-inefficiency), enhancing labour and/or capital productivity.
- In addition to these static effects, a more competitive environment stimulates efforts to innovate and adopt new technologies, which raises productivity growth.

Quantifying the possible magnitude of regulatory reform on sectoral performance is bound to be subject to considerable uncertainty, which is only multiplied in the assessment of economy-wide effects. For example, reduced rent sharing (stemming from lower mark-ups) might have favourable spill-over effects on wage formation more generally. Furthermore, propagation of sectoral effects into the wider economy also depends on the labour market. The initial effects of a sectoral reform may be a reduction in employment in the sector concerned, which has to be employed elsewhere in the economy – highlighting the importance of a flexible labour market in maximizing the economy-wide effects of reforms.

of factors and assessing the impact of regulatory reform is a complex task (Box 5.1). Nevertheless, an attempt is made to quantify the potential effects of further reforms. Following the approach taken by Nicoletti *et al.* (2001) and Nicoletti and Scarpetta (2003), synthetic indicators of regulatory stance are included in regressions of aggregate performance variables. This method is appealing because it does not require any assumptions about the character of reforms or *ad hoc* assumptions regarding the impact of reforms on price-cost margins and productivity. Assuming the Netherlands were to align its economy-wide regulation to that of the least restrictive EU country,[6] it is estimated that the long-run employment rate would increase by 1.5 percentage points and that over ten years multi-factor productivity (MFP) growth would be 0.11 per cent higher per year (Table 5.3). Aligning state ownership and industry specific regulations to that of the best performing EU country would further increase annual MFP growth by 0.74 and 0.3 per cent respectively. This implies an increase in annual MFP growth of over

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Table 5.3. **Potential effects of further regulatory reforms in the Netherlands**[1]

	Long-run employment rate (percentage point increase in level)	Multifactor productivity growth over 10 years (per cent increase in annual rate)
Effect of easing economy-wide regulation	1.47	0.11
Effect of easing industry-specific regulation	–	0.3
Effect of reducing state ownership	–	0.74

1. Alignment of regulation on least restrictive EU country in 1998. Effects estimated from the results of panel regressions relating the employment rate and multifactor productivity to regulation and other variables.
Source: Nicoletti *et al.* (2001) and Nicoletti and Scarpetta (2003).

1 per cent, equivalent to an 11.5 percentage point increase in the level of MFP after 10 years. While the magnitude of such gains must necessarily be rather uncertain, there is clearly significant potential for improving performance, though it is also evident that comprehensive reforms in both product and labour market would be required to achieve such results.

Competition legislation and enforcement

The Netherlands long had a reputation as a "cartel paradise". Industry co-operation was encouraged within a corporatist structure and the law to control and prevent restraints on competition was tolerant and ineffective. It thus marked a major change in policy direction when a completely new law was enacted in 1998, and a new enforcement authority, the NMa, was set up. Competition law in the Netherlands now follows the EU approach of prohibiting restrictive agreements (subject to criteria for exemption) and abuse of dominance. Changing the law did not by itself change market behaviour, of course, and habits of non-competitive accommodation persist.

The NMa made it a priority at first to complete the process of transition from the previous regime, which meant deciding over 1 000 applications for exemption from the new prohibitions. In this process, the NMa tended to rely on formal classifications, perhaps more than on careful market analysis, in order to finish the task quickly. Nevertheless, because its attention was concentrated on deciding these applications, as late as 2001 the NMa still only had six investigations of potential violations of the law. In 2000, widespread price fixing was uncovered in the construction industry (Box 5.2) and a Parliamentary inquiry criticised the NMa for its inaction. The NMa responded to Parliament's demands with a special cartel task force, whose first major case results were announced at the end of 2003.[7]

The NMa's experience in these cartel investigations, with tactics of concealment or destruction of evidence, revealed weaknesses that need to be addressed in its powers to get information and compel compliance. Legislation is

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Box 5.2. Cartels in construction

Collusion in the construction industry shows how a tradition of co-operation can restrict competition in non-traded goods and services, even in a small open economy. Transport is a high proportion of the cost for standardised, low-tech, and perishable products such as asphalt and concrete, so markets are necessarily local, limiting the number of suppliers who need to collude. The industry is vertically integrated and independents can be punished by cutting off their supply. Competition is mostly in terms of price – indeed, government procurement rules may even require that price be the only significant competitive consideration.

In the Netherlands, rules against bargaining over bids simplified collusion greatly, and public procurement officials appeared to tolerate and support the industry's arrangements and helped to discourage entry from outside, even from elsewhere in the Netherlands. Co-operation actively excluded competition from abroad, sometimes through reciprocal market division and threats of boycott. Firms would sometimes team up, not because they were too small to handle a project alone, but in order to clear their cartel pooling accounts with each other. The industry's historic habits of co-operation persisted despite the new competition law, and even after the NMa denied applications for exemption in some cases and after it publicly launched its anti-cartel enforcement program aimed at the construction industry. Although the overall economic impact of this collusion has not been determined, it was estimated that construction cartel bids were increasing prices by about 8 to 9 per cent.

Source: Tweede Kamer der Staten-Generaal (2002).

in progress to increase the NMa's investigation powers (and adapt them to the new EU enforcement system), two of which are expected to become effective in 2004.[8] Other legislation in process also needs attention. The NMa is an agency of the Ministry of Economic Affairs, and the Minister has the power to issue instructions. No such instruction has ever been given, but nonetheless the NMa should be made formally independent to increase its credibility. While the government approved legislation to change the NMa to an autonomous administrative organisation ("ZBO") in 2000, it is still awaiting final action in Parliament.[9]

Sanctions available against violations of the law appear generally adequate, with some exceptions. The NMa still lacks the power to go after members of an association for violations by the association; however, the new EU regulation includes a pass-through rule for associations, and the Netherlands will pick this up. The NMa does not yet hold individual executives accountable. Substantial fines are now being assessed against firms both for illegal restrictive agreements (such as the mobile phone cartel resulting in fines totalling

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€ 88 million and veterinary products with fines of € 10.5 million) and for abuse of dominance such as denial of access to the electric power grid (a case that resulted in a fine of € 6 million). The NMa's leniency program has finally produced some publicly-announced results in several of the latest construction cartel cases.[10] The NMa has the power to limit monopolists' exploitative prices, which it has done in cases involving cable TV and airport charges. But it does not have the power, in infringement cases, to order structural changes such as divestiture to separate competitive operations from monopoly functions.

Scope and objectives of competition law in the Netherlands

In interpreting the law and setting priorities, the NMa takes a "consumer welfare" approach. It considers the likely impact on the economy of the conduct being addressed, and of correcting it, and the importance to consumers, as well as the "seriousness" of the violation. The principal targets, other than energy and construction, have been in services such as health care, financial services, professions and tendering. The emphasis on consumer effects and benefits discomfits some small business interests, who would prefer that enforcement focus on the structure of the economy and the relative positions of larger and smaller market participants. Claims about the alleged power of large buyers are often complaints about commercial disadvantage rather than anti-competitive effects, and the Dutch government is sceptical that these issues require new legislation. Nonetheless, in recognition of this concern, the NMa in 2004 plans to look into charges that buyer power distorts competition.

Despite its "consumer welfare" conception of competition law and enforcement, the NMa does not have any direct responsibilities concerning consumer protection and marketing practices such as misrepresentation. Institutions for consumer protection in the Netherlands are comparatively informal, and may be comparatively ineffective as a result. The present system presumes that consumers and businesses will work out disputes about contracts and quality directly with the aid of Disputes Committees, which are organised by industry. But some sectors do not have dispute resolution committees, and some are slow. In telecoms, the Disputes Committee takes more than six months to decide complaints. The government is examining the possibilities to improve individual dispute resolution and before summer 2004 will present new plans for consumer protection. These plans envisage setting up a "collective interest" consumer protection public enforcement agency, which will focus on unfair commercial practices, both with regard to cross-border and national inquiries. Also a one-stop shop will be introduced for individual consumer complaints.

The competition law applies in regulated sectors. The Netherlands has experimented with an institutional model for co-ordinating general competition principles and sector regulation, by establishing sectoral regulators as parts of the

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general competition authority. This began with electric power, where the sector regulator, DTe, was created formally as a chamber within NMa and a transport chamber also within NMa is now functional.[11] A separate sectoral health care regulator is now being set up and it is planned to transform this regulator into a "healthcare chamber" within NMa in 2008. Plans to transform the telecoms regulator, OPTA, into a similar chamber have been put on hold for the time being.[12] The model of sectoral-regulator-within-competition-authority approach avoids problems associated with regulatory capture and ensures that competition principles are applied consistently across sectors; however, competition law does not appear to be well suited to deal with some of the particular competition problems that arise in network industries (see below).

Regulatory policies

During the 1990s the Dutch government has brought in a number of structural reforms aimed at increasing competition in product markets and moving away from the Dutch "corporatist" model. These reforms include the introduction of the competition law discussed above and regulations for public tendering. But reforms in many important areas have only recently been implemented. In 1994, the government initiated the Competition, Deregulation and Legislative Quality (MDW) programme, focusing on competition, deregulation and the "quality" of laws in a number of sectors such as retail (*e.g.* shop opening hours), taxis and accountants. Notary services were dealt with in a separate but similar project. More recently, the government has launched an interdepartmental commission for the structure and regulation of markets. All competition related issues will be addressed by this Commission, which should limit the effectiveness of lobbying the department responsible for legislation or regulation of a particular sector, and possibly make more transparent the tradeoffs between competition and other objectives.

Administrative burdens on firms and red tape, which can discourage entry by both domestic and foreign firms, have also been eased.[13] Since these reforms, there has been an increase in start-ups and no signs of an increase in bankruptcy rates.[14] However, in spite of recent reforms the administrative burden on Dutch start-ups is comparatively high, both in terms of cost and time (Figure 5.8). Only Belgian and Italian firms faced longer waiting times and the mandatory costs involved to set-up a Dutch company are also relatively high and considerably greater than that of benchmark countries like the United Kingdom. Recognising that administrative burdens in the Netherlands are comparatively high, the government has recently committed itself to substantially reducing this burden (Box 5.3).

The remainder of this section examines developments and outstanding problems in retail distribution, professional services, telecommunications, electricity and gas. Zoning and planning restrictions are most likely responsible for the

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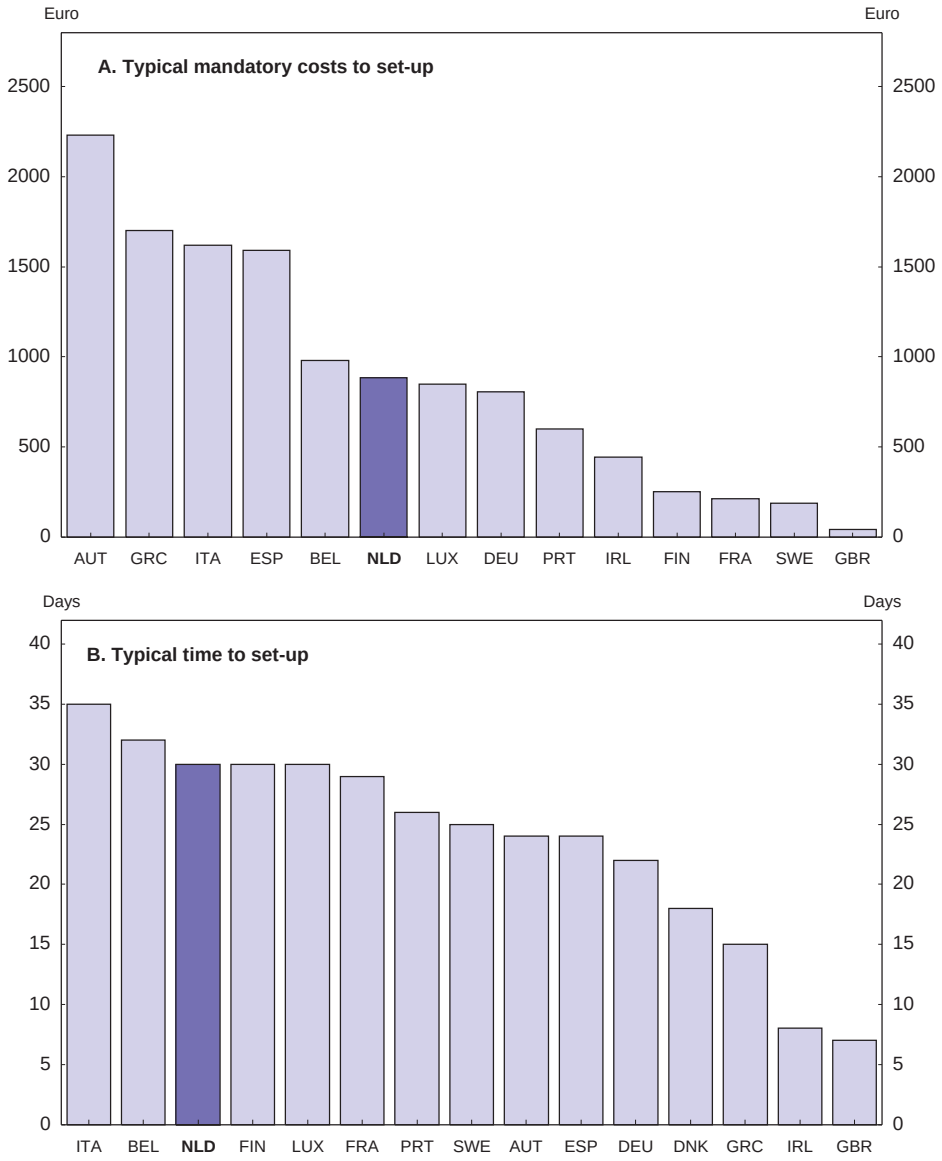
Figure 5.8. **Administrative burden on business start-ups**[1]
2002



1. Typical time and costs involved in setting up a private limited company.
Source: EC (2002).

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Box 5.3. Reducing administrative burdens on firms

The government has firmly committed itself to reducing the administrative burden on firms by 25 per cent of the current level between 2004 and 2007. The current burden has been estimated at € 12.46 billion in 2002 (EIM 2003), with small firms being disproportionately affected. However, a recent stock-taking exercise carried out by each central government ministry indicated that the total burden is as high as € 17 billion, with € 8 billion caused by European legislation and € 9 billion by Dutch legislation. This difference between the two estimates can be explained in part by adjustments in the definition and by economic growth, entailing larger obligations. Nonetheless, the government intends to stick to its objective. As almost half of the administrative burden is caused by European legislation, the Dutch government will be seeking a reduction of this half to contribute to reaching its 25 per cent target. While this will be a challenge, many initiatives have been planned to achieve this objective. These initiatives will be coordinated by the Ministry of Finance in cooperation with the Ministry of Economic Affairs and include:

- Each ministry has been asked to draw up an inventory outlining all opportunities to reduce the administrative burden. In addition, ceilings on administrative costs caused by each ministry will be imposed and gradually tightened, so that additional burdens resulting from new regulations will need to be compensated by reductions elsewhere within the same ministry.
- The individual efforts by the ministries will be complemented by a cluster approach, emphasizing coordination between ministries. The idea is that there is ample scope for reducing administrative burdens by streamlining administrative forms and procedures, minimizing wasteful duplications of efforts to meet information requirements, bundling of different licence applications, and eliminating conflicting regulations. With respect to the latter, the government had designed a temporary special website where firms could post their complaints about conflicting regulations and suggest solutions. At the end of 2003, over 800 conflicts had been mentioned.
- In order to prevent new regulation from unnecessarily raising the administrative burden, the Advisory Board on the Screening of Administrative Costs (ACTAL) was created in 1999 to assess all proposed government legislation and regulation for its impact on administrative costs and, where possible, to propose less burdensome alternatives.
- ICT is expected to make a major contribution to reducing administrative costs through several applications. First, the government is currently working on the introduction of a national one-stop-shop for businesses, a single point of entry on the internet where businesses can access information, forms and services provided by various public agencies. Second, the government is building a transaction gateway to facilitate the transmission of information between the government and businesses, with the idea that large savings on data collection and transmission can be generated by greater sharing of data within and between government agencies. Third, an electronic register for basic business identity information of all companies and organisations in the Netherlands is under construction.

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Box 5.3. **Reducing administrative burdens on firms** (*cont.*)

– Finally, the Ministry of Economic Affairs has proposed a minimal (one-to-one) implementation of EU legislation when transposed into national law, in order to implement European directives in a timely manner. As to decreasing the administrative burden this has the positive side effect of implementing EU legislation without unnecessary national procedures. In addition, the Netherlands intends to make the reduction of the administrative burden caused by European legislation one of its priorities during the upcoming EU presidency later this year.

continued gap in performance in retail distribution relative to other OECD countries. In professional services, numerous restrictions to competition remain, mostly due to the anti-competitive effects of regulations by professional bodies. There is vigorous competition in the telecommunications sector between telecom operators and cable companies, nevertheless, market power on the part of the incumbent is still a concern and competition has yet to take hold in the energy sector. Local government ownership of distribution networks may be a barrier to entry and further structural unbundling would also be warranted. While recent reforms in network sectors provide a formal framework of competition, they do not always implement the structural changes and regulatory provisions that are needed for actual competitive pressures to arise.

Retail distribution

The distribution sector, which includes the wholesale and retail sectors, is often subject to a host of regulatory restrictions in OECD countries. Boylaud and Nicoletti (2001) constructed an indicator of the restrictiveness of regulation in retail distribution in 1998. Their indicator suggests that the Netherlands had one of the least restrictive regulatory environments in 1998 (Figure 5.9). Contributing to the relatively favourable regulatory environment was the liberalisation of shop opening hours in 1996 which allowed stores to open between 6 am and 10 pm on weekdays and Saturdays. While shops are generally closed on Sundays, municipalities can allow shops to open on one Sunday a month.[15] An evaluation of the Shop Hours Act in 1998 found that extended opening hours were positively valued by the majority of consumers, due to the increased flexibility.[16] In addition, employment in the retail sector increased by 7 000 jobs in 1997 and there were no major bankruptcy developments among small shops.

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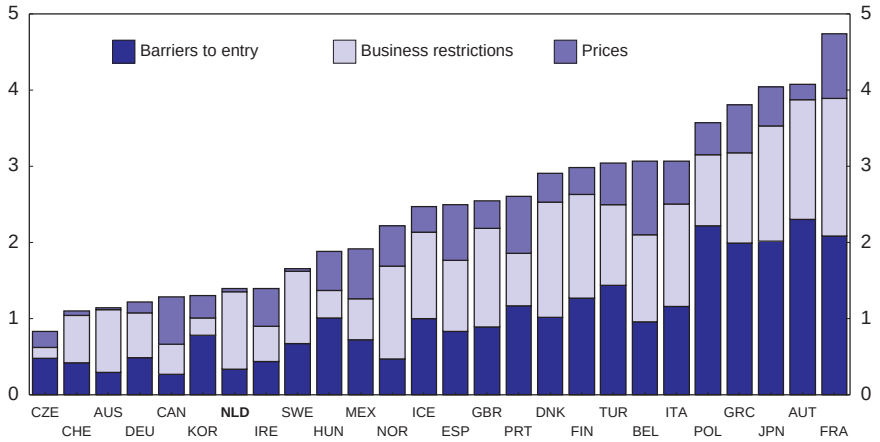
Figure 5.9. **Summary indicators of regulation in retail distribution**
1998[1]



1. The scale of indicators is 0-6 from least to most restrictive.
Source: Boylaud and Nicoletti (2001).

In spite of reforms undertaken during the 1990s, labour productivity, both growth rates and levels, have been particularly poor in the distribution sector in the Netherlands, although they did pick up in the latter part of the decade. Labour productivity growth rates in this sector were one of the lowest in the OECD (Table 5.1). One of the factors contributing to the poor labour productivity growth performance in the wholesale and retail trade sector in the Netherlands is that the use of ICT has been less than in other countries (OECD, 2002b). The sector in the Netherlands is characterised by below average outlet density with an average number of employees per enterprise that is above the EU average (Table 5.4). It might be expected that the low outlet density and higher than average number of employees per outlet is conducive to high productivity levels due to economies of scale. However, productivity levels or value added per person employed is 20 per cent lower than the EU average, with only Portugal and Spain having lower productivity.[17] A possible explanation for this is that compared with other countries, the Netherlands still has a fairly high proportion of retailing by smaller stores with a lower level of productivity.

The retail distribution sector benefits from economies of scale and scope and in many countries, including the Netherlands, this is manifest in the ongoing process of structural change involving larger retail outlets, consolidation into retail

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Table 5.4. **Key structural features of the retail distribution sector**
2000

	Outlet density[1]	Employees per enterprise	Value added per employed person[2]	Value added per unit of labour costs[2]
Austria	43	7.7	108	98
Belgium	80	3.5	109	95
Denmark	47	8.1	103	99
Finland	46	5.0	132	110
France	64	4.2	133	104
Germany	35	9.0	113	116
Ireland	36	9.3	95	
Italy	130	2.2	81	72
Netherlands	**54**	**8.5**	**80**	**117**
Portugal	150	2.5	44	81
Spain	133	2.8	73	97
Sweden	65	4.3	130	88
United Kingdom	36	14.2	99	123
European Union	71	6.3	100	100
Norway	68	6.0	112	98
Switzerland	56	6.8	201	

1. Number of enterprises per 10 000 inhabitants.
2. EU = 100.
Source: Eurostat, New Cronos.

chains and greater concentration and vertical integration. However, concentration in the food retail distribution sector in the Netherlands is still around the EU average (Figure 5.10A).[18] Concerns have also been expressed about the buying power of retailers and its effects on economic welfare (Dobson *et al.*, 2001).[19] In many countries, including the Netherlands, buyer groups are prominent and the concentration ratios, while reflecting consolidation in the retail markets, do not give a full picture of the concentration facing suppliers in retail procurement markets. When such buyer groups are taken into account, concentration in the Netherlands is considerable higher, over 70 per cent, with only Denmark and Sweden having higher five-firm concentration rates that are adjusted for buyer groups (Figure 5.10B).

There are concerns that concentration, consolidation and buyer power may lead to a lessening of competition (Dobson *et al.*, 2003). However, the link between concentration and competitive pressure is complex in retail distribution, and the NMa has appeared to be unwilling to discourage consolidation activity in retailing, recognising the possible efficiency benefits.[20] The scope for anti-competitive behaviour is also limited by the threat of entry and by increasingly mobile consumers. Estimated mark-ups in wholesale and retail distribution in the Netherlands are slightly below the average for the OECD countries for which data are available (Figure 5.11).

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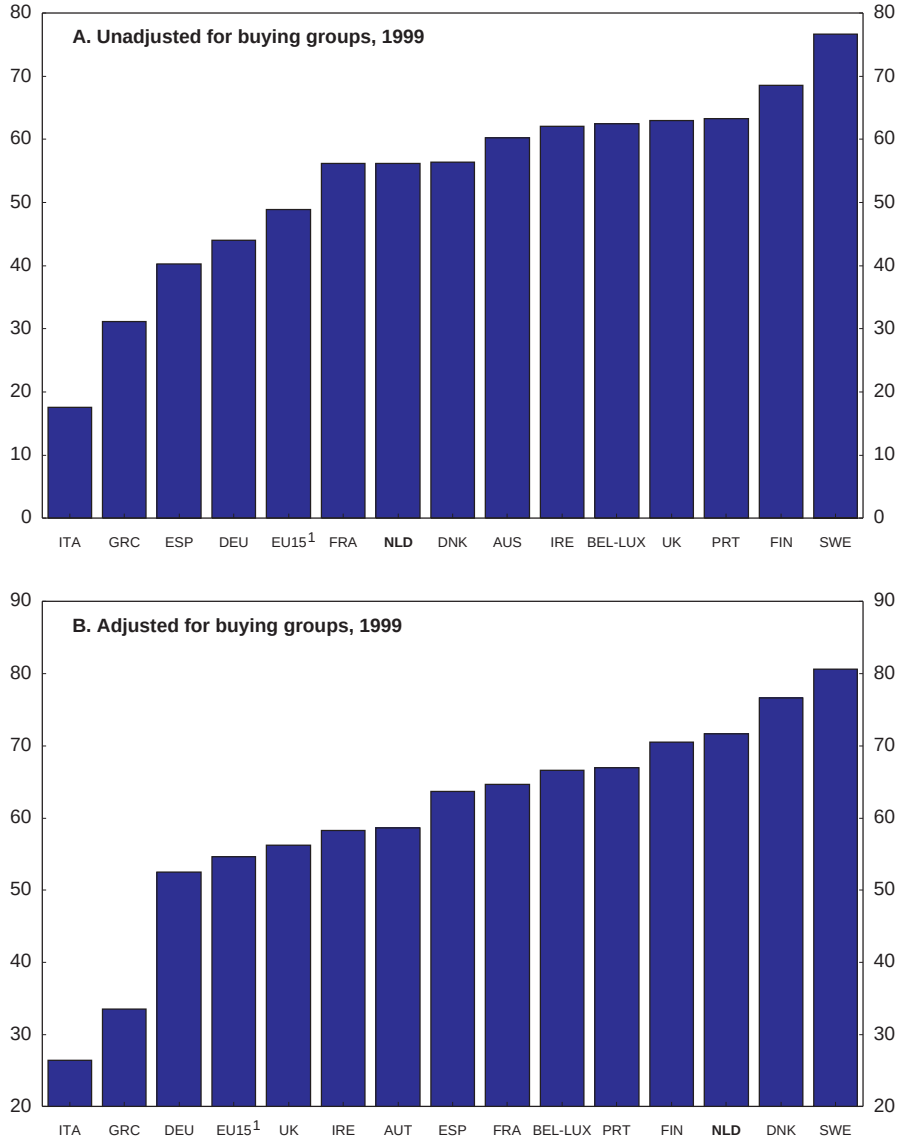
Figure 5.10. **Five-firm market concentration in food retailing in EU countries**



1. Weighted average.
Source: Estimates based on data from Corporate Intelligence on Retailing's European Retail Handbook. Reported in Dobson *et al.* (2001).

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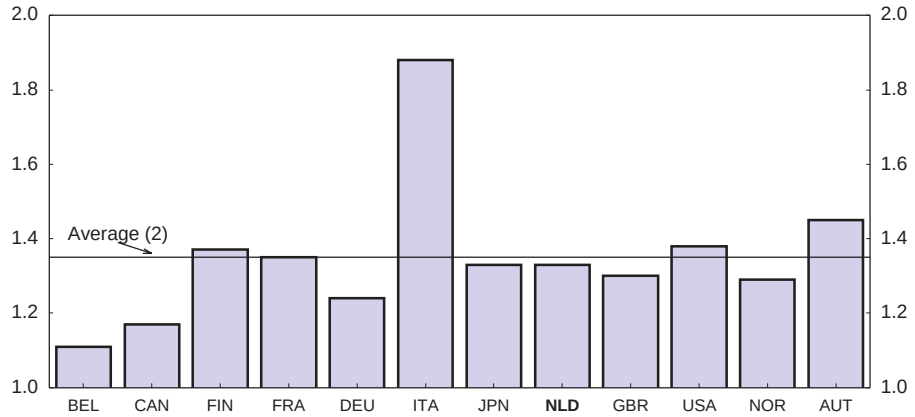
Figure 5.11. **Mark-ups in wholesale and retail distribution in selected OECD countries**

1981 to latest available year[1]



1. For the Netherlands data are from 1987 to 2002.
2. The average mark-up is an unweighted average of the available mark-ups. ISIC, Rev. 3 classification.

Source: OECD STAN database.

Larger retail outlets, providing one-stop shop services, offer greater convenience and lower prices arising from improved efficiency and resulting cost savings that are passed on to consumers. It is often claimed, however, that there is a cost for consumers as large stores will lead to a decline in traditional retailing offering specialisation or location convenience. Evidence seems to suggest that these fears are exaggerated and that small shops can survive because consumers are willing to pay a premium for their services (Dobson and Waterson, 1999). Nevertheless, the retail sector in many countries, including the Netherlands, is characterised by planning and zoning restrictions that limit market entry, often with the objective of protecting shopping in town centres. The Dutch central government has detailed rules for the establishment of large shopping centres or mega stores on the edge of towns through the PDV/GDV policy (*i.e.* policy on peripheral and large-scale retail outlets). The establishment of retail outlets on city fringes, or of very large shopping centres, is closely curtailed by these restrictions. In the mid-1990s, growing demand for "superstores" led to the opening of large stores but only at locations with good connections to public transport and in the "urban junctions" (MDW, 2001). Such planning restrictions, however, also serve to distort competition. They offer considerable incumbency advantages to established retailers by restricting new entry, thus creating or maintaining rents. The

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planning restrictions also have adverse effects by restricting efficient operation, thus limiting innovation and competition in retail format strategies to the detriment of productivity improvements and cost savings that could then be passed on to consumers.

The PDV/GDV policies were recently examined in the context of the MDW project. As a result, decision making power concerning the location of large-scale stores will be decentralised to lower tiers of government. It is thought that local authorities can provide more customised services and are more responsive to the needs of their communities. In future, central government will only lay down the outlines of policy, and municipal and provincial authorities will determine the details themselves. However, the devolution of regulatory powers to regional levels may be problematic for competition in the sector. Experience from other countries (*e.g.* Italy and the United Kingdom) suggests that entry by large formats may become more difficult as local authorities may be more inclined to protect town centres and thus less likely to grant planning permission. However, Dutch authorities perceive this risk to be small as lower levels of government are more susceptible to local and regional consumer pressure. Competition between local and regional governments to offer an attractive and dynamic shopping climate is another counteractive force, and planning procedures offer sufficient checks and balances. However, the government should keep monitoring this process to ensure that local governments are not unduly responsive to incumbent interests.

Professional services

Professional services are usually subject to pervasive regulation, including not only exclusive exercise of certain functions and entry and access requirements, but sometimes even recommended or fixed prices and restrictions on advertising and business structure or residency requirements. Intervention is often explained by the need to correct market failures, which are mostly due to information and transactions costs. Such regulation can be in the interests of both consumers and the profession if it improves service quality and prevents market failure.[21] There is little empirical evidence to suggest that the pervasive set of restrictions found in professional services improves consumer welfare (Nguyen-Hong, 2000; OFT, 2001; Paterson *et al.*, 2003). In practice these restrictions have been correlated with higher prices and less innovation, without improving quality.[22] These results support the view that restrictive regulatory frameworks and self-regulation by professional bodies, rather than supporting the needs and interests of consumers, are often used by the professions to obtain economic rents.

Labour productivity growth in the Dutch business services sector, which includes professional services, has been relatively weak in the 1990s (Kox, 2002; van der Wiel, 2001).[23] Over the past decade, through the MDW programme, a number of professional services (for example notaries, pharmacists and real estate

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agents) were examined and, as a consequence, a number of reforms were introduced. Recommended prices were removed for lawyers in the late 1990s and the low level of regulation of legal services in the Netherlands has resulted in a sector that is characterised by a low number of firms and relatively high employment and turnover per firm, indicative of a higher degree of concentration in the market. This process however has not been associated with high market power (Paterson *et al.*, 2003). The abolition of entry restrictions for real estate agents has led to an increase in new entrants, lower prices for real estate transactions and more flexible provision of services. In notary services, entry barriers have been relaxed, there are no longer a fixed number of establishments, and tariffs are now completely free.

A recent study for the European Commission examined the differences in a host of regulations governing a range of professional services (Paterson *et al.*, 2003). In comparison with other EU countries, the Netherlands is assessed as having a low or moderate degree of regulation in the accountancy, legal, architectural, engineering and pharmaceutical professions. The only other countries with such a profile are Ireland and the United Kingdom (Table 5.5).[24] In spite of such a low regulatory index, a moderate level of restrictions still exists in the accountancy, legal and pharmaceutical sectors.[25] The Netherlands maintains a number of restrictive rules in these professions and barriers to entry also arise from mandatory membership in professional orders (Table 5.6). In addition, other professional services not covered by the Paterson *et al.* study are still subject to pervasive

Table 5.5. **Regulation indices in professional services**[1]

	Accountants	Legal	Architects	Engineers	Pharmacists
Austria	6.2	7.3	5.1	5.0	7.3
Belgium	6.3	4.6	3.9	1.2	5.4
Denmark	2.8	3.0	0.0	0.0	5.9
Finland	3.5	0.3	1.4	1.3	7.0
France	5.8	6.6	3.1	0.0	7.3
Germany	6.1	6.5	4.5	7.4	5.7
Greece	5.1	9.5	n.a.	n.a.	8.9
Ireland	3.0	4.5	0.0	0.0	2.7
Italy	5.1	6.4	6.2	6.4	8.4
Luxembourg	5.0	6.6	5.3	5.3	7.9
Netherlands	**4.5**	**3.9**	**0.0**	**1.5**	**3.0**
Portugal	n.a.	5.7	2.8	n.a.	8.0
Spain	3.4	6.5	4.0	3.2	7.5
Sweden	3.3	2.4	0.0	0.0	12.0
United Kingdom	3.0	4.0	0.0	0.0	4.1

1. The higher the degree of regulation (intensity), the higher the respective figure (within a range from 0 to 12). All the regulation indices with a value of 5 or higher are shown in dark grey boxes, indices between 2.5 and 4.9 are in light grey boxes, and, those below 2.5 have a white background.

Source: Paterson *et al.* (2003).

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Table 5.6. **Regulation of entry and conduct of professional services in the Netherlands**

	Self regulation or governmental regulation	Educational and training requirements	Restrictions on fee setting	Restrictions on advertisement	Compulsory membership	Residency requirements
Accountants	Both	Yes	No	No	Yes	No
Lawyers	Both	Yes	No	Yes	Yes	No
Notaries	Both	Yes	No	No	Yes	Yes
Architects	Both	Yes	Yes	Yes	No	No
Financial professionals	Both	Yes	No	Yes	No	No
Real estate agents	Self-regulation	No	No	No	No	No
Process servers[1]	Both	Yes	Yes	No	No	Yes

1. Gerechstdeurwaarders.
Source: Ministry of Economic Affairs.

regulation, including the exclusive exercise of certain functions, entry and access requirements, and restrictions on prices, advertising and permitted business structures.[26] Considerable scope still exists therefore to ease restrictions on price competition, advertising and on permitted business structures where professional rules and government regulation prevent multi-disciplinary practices.[27]

Competition in the professional services sector in the Netherlands remains rather weak due to self-regulation by professional bodies. Self-regulation by professional bodies or associations, with all the problems of regulatory capture, is still the norm and raises concerns regarding independence and the effectiveness of such regulatory bodies.[28] While business practices in this sector have been "cosy", this is gradually changing. The exemption from the Competition Act that these rules previously received has now lapsed. At the moment these markets are still in transition and in light of the competition concerns the NMa has defined this as a priority area for 2004.[29] The NMa will examine whether or not professional regulations (*beroepsverordeningen*) are a restraint to competition, and will also examine government regulations and legislations in this context.[30] It is intended that an inventory will be made of the problems in professional services with the aim of "re-regulating" the regulations of professional bodies and the government.

Network industries

Network sectors in the Netherlands (*i.e.* electricity, gas, water, transport and communications) account for 8.6 per cent of value-added and 6.2 per cent of employment, and for a large share of intermediate inputs. Performance in these sectors is therefore important and can impact overall economic performance. There is now a solid body of cross-country evidence that liberalisation policies in

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network industries have led to higher productivity, better quality and, often, lower prices.[31] There is a lack of empirical evidence as regards the impact of reforms in the Netherlands as most of the product market reforms in telecommunications and the energy sector are fairly recent. In any event, capturing these benefits is not straightforward and close attention needs to be paid to the design of reforms (Gonenc *et al.*, 2001). While embodying the sectoral regulators with the NMa avoids regulatory capture of sector-specific regulators and allows competition concepts to be applied consistently across sectors, competition law tools are often not well suited to deal with some of the competition problems in these sectors. Stronger structural remedies (*e.g.* divestitures) may be required to effectively promote competition (Newbery, 2002a). However, the NMa, and regulators, currently do not have the power to implement structural remedies on companies that have abused, or may abuse, their dominant positions. This erodes the ability of sector-specific regulators to promote competition.

Government policy in these sectors may often conflict with the aim of promoting competition. Privatisation plays an important role in eliminating possible conflicts of interest between regulators and the firms they are regulating. The distribution networks in electricity and gas should also be privatised since local government ownership can act as a barrier to entry and makes the regulators task more difficult. The Dutch government also retains golden shares, which it should relinquish, in the incumbent telecoms and postal operators. Although the government has minority stakes in these companies, the golden shares grant the government special rights *e.g.* a veto over the companies' decisions.[32] Golden shares, by hindering the market for corporate control, strongly reduce the positive effects of privatisation and are a strong disincentive to investment. For these reasons, the European Commission is suing the Dutch government over their golden shares. The role of the regulators also needs to be strengthened, perhaps by granting regulators stronger structural remedy powers.[33] Strong and independent regulation of network industries does not imply that sectoral regulators are "making policy" but it does ensure that the promotion of competition is a primary objective.

Post and telecommunications

In the telecommunications sector, an independent Post and Telecom Authority, OPTA, is charged with promoting competition and the interests of consumers, and with regulating the telecommunications and postal sectors. OPTA's responsibilities include dispute settlement, approving interconnection and retail tariffs, and supervising the postal concession. Its powers are limited however since it cannot take action on its own initiative and has to wait for an official complaint from market participants. OPTA should, like many other independent regulatory bodies, have the power to initiate actions on its own. OPTA's responsibilities overlap with those of the Competition Authority. This situation has not

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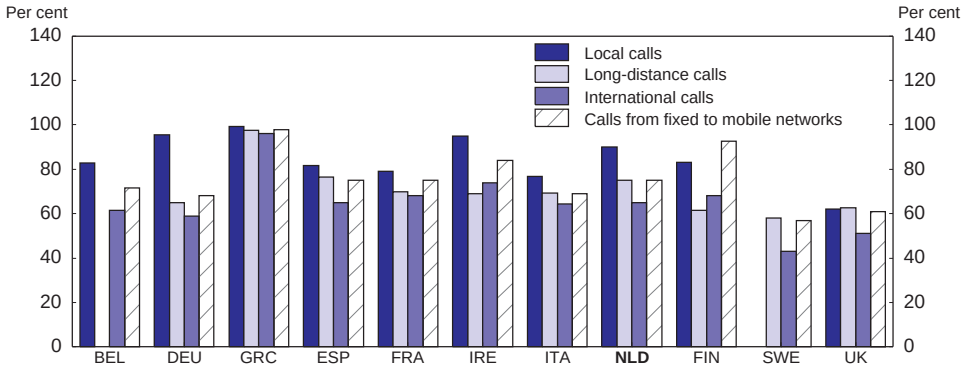
resulted in major frictions and the Telecommunication Act stipulates that OPTA must consult with the Competition Authority when it wants to use the general competition law.[34] Plans for integrating OPTA within the NMa are on hold for the moment, with no plans for proceeding with the reorganisation.

The regulatory regime in the Netherlands has been characterised by structural measures that aim at improving competitive conditions, many of which were implemented in advance of EU legislation. These include carrier pre-selection, number portability for fixed networks and which has recently become available for mobile customers, and local loop unbundling (LLU). In line with EU Directives, telecoms operators are no longer required to obtain a license to operate and only have to register. While the Netherlands regulatory regime in telecommunications is pro-competitive, the incumbent telecoms operator, KPN, nevertheless maintains a dominant position in all fixed voice telecommunications markets (Figure 5.12). Competition is beginning to take off and new entrants in the Netherlands had the second highest share of access lines in the EU, after the United Kingdom (OECD, 2003). OPTA has taken significant action to ensure competitive access to KPN's local loop – LLU prices are regulated and subject to cost orientation and monthly tariffs and connection charges are well below the EU average (Table 5.7).

Figure 5.12. **Estimates of incumbent operators' market share**
Retail revenue, December 2002[1]



1. In Belgium local-calls does not exist as a separate category from long-distance calls. The figures for the Netherlands refer to March 2002, before the introduction of CPS for local calls on 1 August 2002. The figure for Finland local calls is the combined market share of Sonera, Elisa, and Finnet. Finland's figure for long-distance and international market include Sonera only and not Finnet.

Source: EC (2003a).

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Table 5.7. **Prices for unbundled local loop**
€, 2003

	Full unbundling		Shared access	
	Monthly rental	Connection	Monthly rental	Connection
Belgium	11.9	54.9	2.3	54.9
Denmark	8.3	44.8	4.1	104.4
Germany	11.8	56.6	4.8	74.9
Greece	10.6	36.1	5.3	47.0
Spain	12.3	20.0	3.5	27.0
France	10.5	78.7	2.9	78.7
Ireland	16.8	121.5	9.0	123.4
Italy	8.3	32.0	2.8	44.5
Luxembourg	15.8	185.6	7.5	196.2
Netherlands	**9.9**	**33.9**	**2.3**	**44.1**
Austria	10.9	54.5	5.5	109.0
Portugal	12.0	84.1	3.0	88.2
Finland	14.1	218.0	6.7	105.0
Sweden	11.4	167.6	5.4	119.7
United Kingdom	14.8	128.3	6.4	170.5
EU average	11.5	68.2	4.2	80.6

Source: EC (2003a).

The cost of internet access for consumers in the Netherlands was comparatively high in 2002 when dial-up was still the principal medium of internet access (Figure 5.13). Since then vigorous competition has developed between cable companies and telecoms operators in the provision of internet access (flat fee) and this has led to the third highest take-up of broadband in the EU, behind Denmark and Belgium. Dutch households also have the highest internet use in the EU (MEZ, 2002). Heavy users of internet access have switched in large numbers to DSL providers where good DLS-offers are available at € 25 per month. Dial-up users *i.e.* generally the light users, will normally have monthly expenses below this amount. The high internet use and broadband penetration are in large part due to the measures for local loop unbundling that have been put in place and access fees that have been lowered, resulting in a large number of companies using unbundled access for providing broadband internet access. The monopoly power on the part of broadband internet service providers that was identified in the last OECD *Economic Survey of the Netherlands* has thus declined significantly in the past year. KPN, however, is currently lobbying the government for financial assistance in the roll-out of broadband infrastructure – taking fibre optics "to the home". While the government does not intend to offer financial support, government control of the incumbent along with a substantial shareholding (see below) may act as a barrier to entry and discourage other providers from investing in fibre optics "to the home". This

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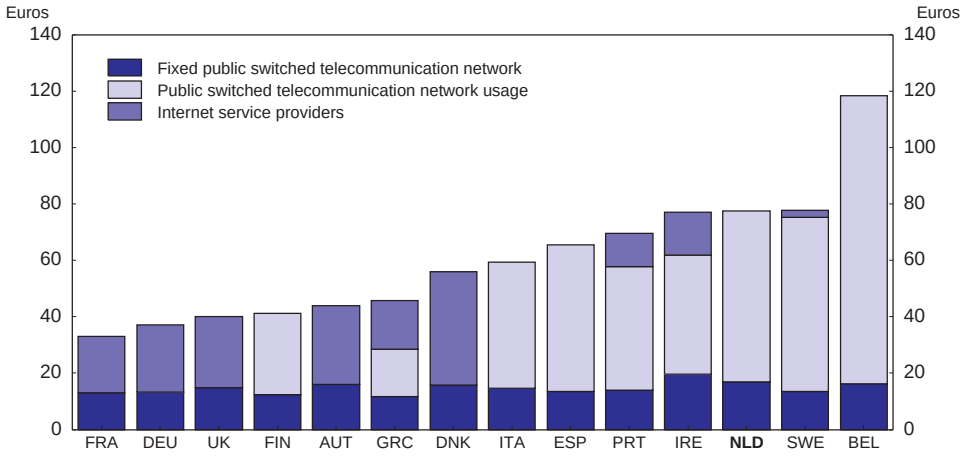
Figure 5.13. **Cost of internet access**[1]
September 2002



1. For 40 hours at day-time discounted Public switched telecommunication network rates, including VAT.
Source: OECD, *Communications Outlook* 2003.

pressure should be resisted and the government should let the market determine the outcome – leaving private parties the responsibility for investing in broadband infrastructure and making the associated technological choices.

Increased competition, in combination with pro-competitive regulatory measures, has resulted in important benefits for consumers and significant price reductions (MEZ, 2003b). By August 2002, residential and business telephone charges were amongst the lowest in the EU, although mobile charges remained on the high side (Figure 5.14). Mobile termination rates are not published and are determined by commercial negotiation. While there is lots of competition in the mobile market, there are problems (as in other countries) with termination charges, since mobile operators have a monopoly on calls terminated on their networks. In response to these problems, at the end of 2002, OPTA published guidelines as to the maximum reasonable charges mobile network operators would be allowed to set and obliged all mobile operators to lower their terminating tariffs to cost-oriented levels. When these decisions were annulled in court, the NMa took up the case but abandoned their investigation after the mobile companies lowered their termination charges in December 2003. Within two years these levels will decrease to half the original amount and will be in accordance with the European average.

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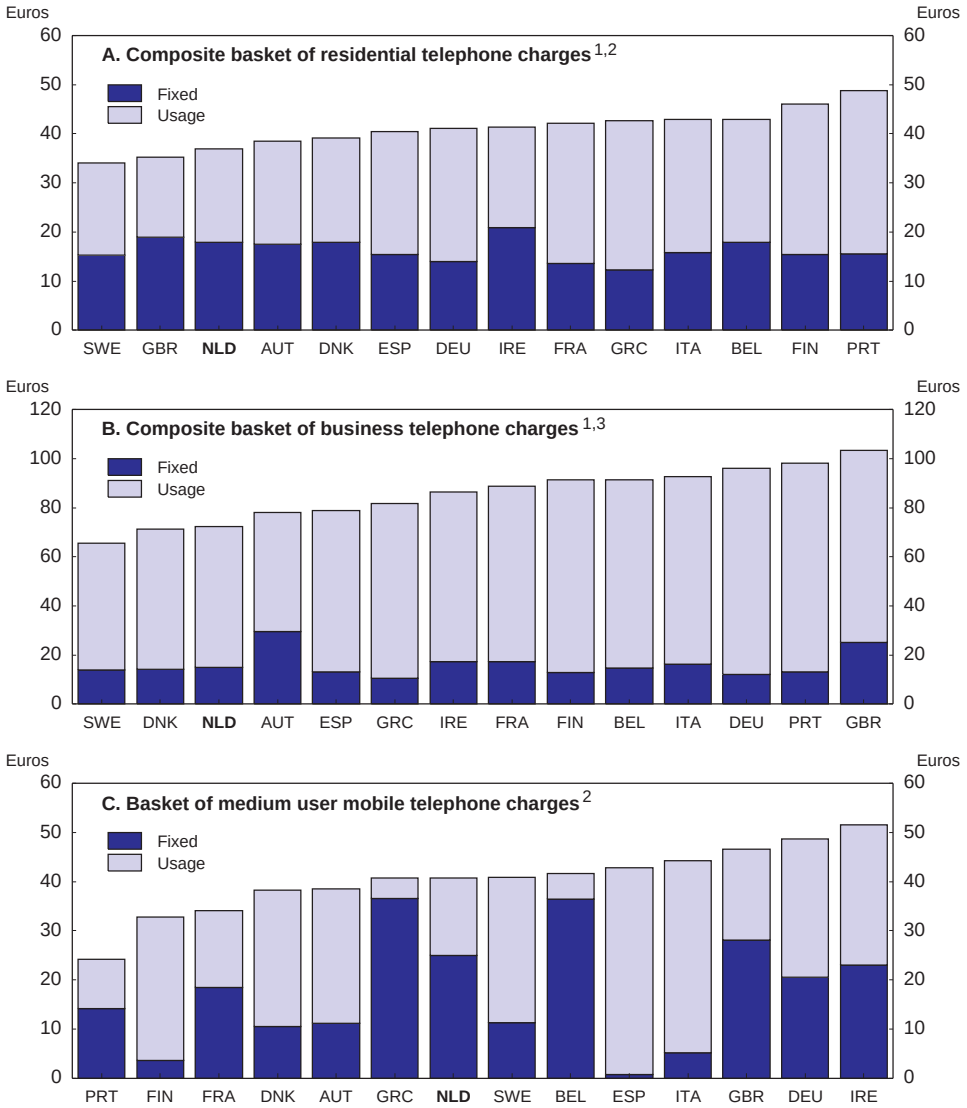
Figure 5.14. **Average monthly telephone charges**
August 2002



1. Composite basket includes international calls and calls to mobile networks.
2. Including tax.
3. Excluding VAT.

Source: OECD, *Communications Outlook* 2003.

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There is a debate about whether to completely liberalise postal services. While the EU Postal Directive aims for 2009 to open up European postal markets, the Netherlands intends to liberalise in 2007. The business segment of this market is competitive, but there are outstanding concerns regarding rural areas where postal services are provided in conjunction with financial services. OPTA has advocated abolishing the postal company's (TPG) monopoly arguing that there are no prevailing economic or social reasons for maintaining the current postal monopoly (OPTA, 2001). The Ministry however has delayed further liberalisation until Germany and the UK do so in 2007. The position of the Ministry is that this will ensure a level playing field in (at least parts of) the postal market. Otherwise, foreign companies would be able to enter the Dutch market whilst the opposite would not be possible. More importantly, the government still owns 34.8 per cent of TPG and the EC has just sued the Netherlands over the control it maintains in TPG and KPN through its golden shares. The government should eliminate these golden shares, and sell its remaining holding in these companies.

Energy sector

Reform of the electricity and gas sectors was launched with the Electricity Act 1998 and the Gas Act 2000. The responsibility for implementing these Acts has been assigned to the Office of Energy Regulation (DTe). Unlike most EU countries, the competition authority deals with dispute settlement in the sector (Table 5.8). Although DTe is organisationally subordinate to the director general of the NMa, it

Table 5.8. **Competencies and resources of energy sector regulators**

	Network access conditions	Dispute settlement	Staff number	Annual budget 2002 (€ m)
Austria	R(elec)/R (gas)	R/R	45	9
Belgium	R/R	R/R	68	15
Denmark	R/R	R/R	30	3
Finland	R/R	R/R	15	1
France	R/R	R/n.a.	80	9
Germany	N/N	C/C	n.a.	n.a.
Greece	M/n.a.	R/n.a.	43	4
Ireland	R/R	R/R	31	6
Italy	R/R	R/R	86	18
Luxembourg	M and R	R/R	2	n.a.
Netherlands	**R/H**	**C/C**	**55**	**6**
Portugal	R/n.a.	R/n.a.	52	7
Spain	M/M	R/R	153	19
Sweden	R/R	R/R	33	3
United Kingdom	R/R	R/R	330	58

Notes: R = regulator responsible, M = ministry responsible, C = competition authority, N = not regulated, H = hybrid, n.a. = no regulator.
Source: EC (2003b).

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acts independently and has its own enumerated powers which were considerably extended in 2001. DTe's regulatory powers include, among other things, the issuing of licences for the supply of electricity and gas to captive consumers; setting service quality standards; setting tariffs and conditions for network access; and determining supply tariffs for captive consumers.

In order to facilitate the establishment of competition in these sectors, unbundling is crucial since vertically-integrated incumbents can impede the functioning of the market through cross-subsidisation and discrimination in network access (EC, 2003). Insufficient unbundling may form a barrier to competition and numerous studies argue that legal and management unbundling is not enough and that further separation is warranted.[35] While initially the Electricity Act did not require full structural separation of generation and transmission, the transmission system has now been vertically separated from both upstream and downstream activities. The owner and system operator, TenneT, is fully state owned and offers regulated access to the transmission network.[36] Regional distribution networks are legally and organisationally unbundled from the supply business of distributors (Table 5.9).[37]

Cross country comparisons of three-firm concentration ratios (CR_3) show that the Dutch electricity generation market is less concentrated than that of other European countries, with the three largest generation companies having a market share of 59 per cent in 2000 (Table 5.9).[38] Only the United Kingdom, Austria and Finland have a lower concentration ratio. In the downstream market, data for 2000 suggests that there are 33 licensed electricity retail companies in the Netherlands, half of which have no ownership ties with the distribution companies and half of which are vertically integrated with distribution (EC, 2003). In addition, the regionally-based vertically-integrated distribution and retail companies are traditionally owned by local councils and provincial governments. The top three suppliers had a market share of 48 per cent in 2000. There is also some vertical integration between the upstream and downstream markets and two of the largest upstream generation firms are also the country's largest retailers and own distribution networks.

In gas, there is management unbundling of the transmission network and the distribution system is legally unbundled from the supply business (Table 5.10). The state owns 50 per cent of the gas transmission network, and Shell and ExxonMobil each own 25 per cent. Negotiations for reform are under way between Shell, ExxonMobil, EBN (the state-owned energy company) and the government. Strict segregation of trading and transport activities is necessary in connection with further liberalisation of the gas market. The Dutch downstream gas company Gasunie has been unbundled and Gasunie was reorganised into a transport and a trading arm on 1 January 2002. Since this date, Gasunie Trade and Supply is involved solely in the supply of gas and the network operation is

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Table 5.9. **Electricity market indicators and implementation of the EU Directive**

2002[1]

	Declared market opening (per cent)	Unbundling		Large users switching suppliers[3] (per cent)	Concentration (CR3)		Concentration HHI[5]	Potential competition from imports[6]	Network access charge[7]	
		Transmission[2]	Distribution[2]		Generation	Retail sales			Medium voltage	Low voltage
Austria	100	L	A	20-30	45	67	2 028	21	20	65
Belgium	52	L	L	2-5	96	53	6 118	25	15	n.a.
Denmark	100	L	L	50[4]	78	38	4 018	39	15	25
Finland	100	O	M	n.a.	45	33	2 472	22	15	35
France	34	M	A	10-20	92	90	9 606	12	15	50
Germany	100	L	A	20-30	64	50	1 756	11	25	55
Greece	34	L	A	0	97	100	10 000	12	15	n.a.
Ireland	56	L	M	10-20	97	90	9 418	7	10	40
Italy	70	L	L	>50	69	72	5 560	14	10	n.a.
Luxembourg	57	M	A	10-20	n.a.	100	8 158	100	20	n.a.
Netherlands	**63**	**O**	**L**	**20-30**	**59**	**48**	**1 814**	**19**	**10**	**35**
Portugal	45	L	A	5-10	82	99	4 008	30	15	n.a.
Spain	100	O	L	10-20	83	94	2 466	4	15	45
Sweden	100	O	L	n.a.	90	47	2 538	29	10	40
United Kingdom	100	O	L	>50	36	42	1 044	3	n.a.	40

1. The information to 2002, except for concentration measures, 2000.
2. Unbundling concerning operators. A = Accounting, L = Legal, M = Management and O = Ownership.
3. 1998-2001.
4. 2001 only.
5. HHI in electricity generation. A market is generally considered competitive with a HHI < 1000; moderately concentrated with 1 000 < HHI < 1 800; and highly concentrated with a HHI > 1 800.
6. Per cent of domestic generating capacity.
7. Estimated average charge, €/MWh.

Source: EC (2003b), AEEG (2002).

Table 5.10. **Gas market indicators and implementation of the EU Directive**

2002[1]

	Declared market opening (per cent)	Unbundling		Large users switching suppliers (per cent)	Concentration % of available		Gas release programme	Network access charges large users (€/MWh)[4]
		Transmission[2]	Distribution[2]		Gas (CR1)	HHI[3]		
Austria	100	L	L	< 2	80	7 598	No	n.a.
Belgium	59	L	L	n.a.	n.a.	10 000	No	1.0-2.0
Denmark	35	L	L	2-5	90	2 841	No	2.5
France	20	A	A	20-30	90	5 932	No	2.0-5.0
Germany	100	A	A	< 2	54	2 405	Planned	2.0-7.5
Ireland	82	M	M	20-30	n.a.	5 883	No	1.5-2.5
Italy	100	L	L	10-20	75	4 916	Yes	2.0-4.0
Luxembourg	72	A	A	5-10	100	10 000	No	1.0-1.0
Netherlands	**60**	**M**	**L**	**30-50**	**80**	**2 634**	**No**	**0.5-1.0**
Spain	100	O	L	20-30	57	9 761	Yes	2.0-2.5
Sweden	47	A	A	< 2	100	10 000	No	3.5
United Kingdom	100	O	O	>50	50	894	Yes	1.5-3.0

1. The information to 2002, except for the concentration measures, 2000.
2. Unbundling concerning operators. A = Accounting, L = Legal, M = Management and O = Ownership.
3. A market is generally considered competitive with a HHI < 1 000; moderately concentrated with 1 000 < HHI < 1 800; and highly concentrated with a HHI > 1 800.
4. Estimated range, rounded to the nearest € 0.5/MWh.

Source: EC (2003b), AEEG (2002).

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conducted by Gasunie Transport Services. The incumbent's supply and trading activities however are not legally separated from its transport and network activities providing scope for discrimination against new entrants or competitors. With the new Gas Act, the incumbent's supply and trading activities will be legally separated from transport system operation activities. In 2000 and 2001, the DTe issued terms and tariffs for access to gas networks and storage and is striving to improve the cost-reflectiveness of these tariffs and conditions. However, an ongoing concern is that the incumbent supplier benefits from its dominant position and access to critical reserves and facilities.

Electricity prices, with the exception of large industrial users, are above the EU average whereas gas prices are below the EU average (Table 5.11). Reforms are still too recent and incomplete to have had a noticeable impact on prices, but there seems to be considerable scope for prices to fall, particularly in the electricity sector. The electricity market was regulated with regard to price formation until 2000 by an agreement (the "Protocol") between the four major generators and the distribution companies, stipulating from 1997 till the end of 2000 mandatory sales of electricity at fixed prices (CPB, 2003). As of October 2002, the Dutch retail electricity and gas markets were opened, respectively, to 63 per cent and 60 per cent of consumers. Household electricity and gas markets have not been opened up to competition and plans to fully liberalise the markets at the end of 2003 have been delayed.[39] A proposal has recently been sent to Parliament to open up the markets to small users as of the 1st of July 2004. Household retail tariffs are currently regulated but this will cease to be the case once the market is opened to competition. Where competition has been introduced this has led to a comparatively high proportion of large users switching suppliers, indicating that there are benefits to be captured from competition. Already 20 to 30 per cent of large electricity users and 30 to 50 per cent of large gas users have switched suppliers (Tables 5.9 and 5.10).

Effective competition in the Dutch electricity markets still has to be established. Congestion on interconnection capacity at the borders is a problem and relaxing these constraints (*e.g.* through investment in cross-border transmission capacity by the government owned network operator in co-ordination with neighbouring networks) would increase competition from foreign suppliers. Although an electricity exchange market, the Amsterdam Power Exchange (APX), was established in 1999, it has had a slow start, so its potential positive effects have not been felt yet.[40] There are also concerns regarding the scope for price manipulation and declining liquidity on the APX. A report by the DTe's Market Surveillance Committee found that the APX spot market lacks liquidity *i.e.* the number of buyers and sellers is still low, allowing trading parties to influence prices *via* their transactions in the exchange (Newbery *et al.*, 2003). The development of spot prices on the APX shows high volatility, with the highest monthly price in January 2000. However since then there has been a tendency towards

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Table 5.11. **Electricity and gas retail prices**

Second half of 2002, before taxes

	Electricity (euros/MWh)			Gas (euros/GJ)		
	Households	Industry		Households	Industry	
		Large industrial users[1]	Small commercial enterprises[2]		Large industrial users[3]	Small commercial enterprises[4]
Austria	93.1	59.8[5]	96.5	11.6	4.8	8.1
Belgium	111.1	58.1	129.9	13.8	4.3	7.4
Denmark	84.4	43.3	66.6	8.1	4.3	8.1
Finland	70.2	36.5	56.6	n.a.	4.4	n.a.
France	92.3	48.7	86.1	13.5	3.7	7.5
Germany	124.9	51.7	128.6	13.6	5.1	7.5
Greece	58.0	50.0	87.0	n.a.	n.a.	n.a.
Ireland	88.3	64.8	127.4	14.4	4.6	7.1
Italy	141.8	73.5	100.6	10.6	4.6	9.3
Luxembourg	115.1	38.3	121.7	11.6	5.4	6.5
Netherlands	**98.2**	**48.5**[6]	**106.4**[7]	**9.9**	**2.8**[8]	**6.1**[7]
Portugal	122.3	55.6	99.9	15.0	4.4	9.6
Spain	85.9	46.6	98.6	12.8	4.0	7.8
Sweden	68.8	25.8	35.7	11.6	3.5	7.1
United Kingdom	97.4	47.0	86.1	9.6	4.1	6.0
European Union[9]	96.8	49.9	95.2	12.0	4.3	7.5

1. Eurostat category Ig (annual consumption: 24 000 MWh).
2. Eurostat category Ib (annual consumption: 50 MWh).
3. Eurostat category I4-1 (annual consumption: 418 600 GJ).
4. Eurostat category I1 (annual consumption: 418.6 GJ).
5. 1999h1.
6. 1999h2.
7. 2001h2.
8. 2000h1.
9. Unweighted average.

Source: Eurostat.

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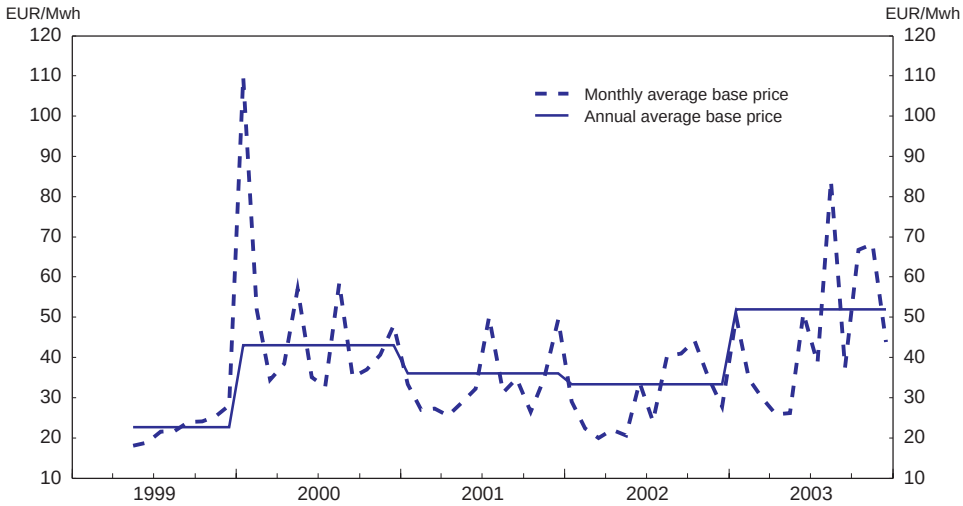
Figure 5.15. **Spot market prices on the Amsterdam Power Exchange**



Source: Amsterdam Power Exchange.

lower monthly spot prices as well as lower annual averages (Figure 5.15).[41] There was some increase in the latter part of 2003 but there is no indication that this was due to market manipulation, and instead reflects a reduction in excess capacity and should provide a signal for new investment.

The level of market power among existing generating companies in the Dutch market, combined with a lack of liquidity in wholesale markets, can impede new entry. Further vertical integration and consolidation in the Dutch electricity market may lead to further declines in market liquidity, resulting in increasing barriers to entry. The report by Newbery *et al.* (2003) to the Market Surveillance Committee recommended that such changes be taken into account by the competition authorities when defining markets and making decisions about mergers and regulation. The structure of the electricity sector plays a crucial role in the development of the wholesale electricity exchange market – a vertically integrated industry has less need for a wholesale market than one where generation and supply are entirely separated. The benefits of introducing competition in the retail market will not be fully realized in this case.[42]

Local government ownership of electricity and gas distribution network companies can also inhibit the development of competition, particularly if the distribution companies also engage in retail activities. While many local governments want to privatise, the Minister of Economic Affairs is responsible for privatisation policy.

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The plans of the former government to privatise the networks have been postponed and are currently under review.[43] The government's position is that privatisation is intermingled with the decision making process regarding the networks and that competition and security of supply considerations are both important. The failure in the California electricity market is often used as an argument against distribution network privatisation and raises the fear that private companies are not inclined to invest in network capacity (Speck and Mulder, 2003). However, the failures in the California market were in large part due to regulatory failure (Box 5.4) and not related to privatisation; and security of supply considerations can be addressed with proper regulation of private distribution companies.[44] The Energy Council, one of the advisors to the government, recently advised against privatisation of the electricity and gas distribution networks, and against the separation of the distribution networks from the other activities of the companies that own the networks.[45] While the advice of the Energy Council may be in the interest of energy companies, it is not in the long term interest of consumers or of promoting competition in the sector. Ownership unbundling of distribution networks from retail activities and the privatisation of the retail businesses should be undertaken to remove barriers to entry and encourage competition. The government has recently announced plans to separate ownership of the distribution networks from retail activities which may then be privatised. Privatisation of the networks can be considered after network quality has been safeguarded through adequate regulatory measures.

Summary and recommendations

In general, competitive pressures appear to be relatively strong in the Netherlands. Sectors that are exposed to international competition (*i.e.* the traded goods sector) are doing well but productivity growth has been low in sectors protected or sheltered from competition (*e.g.* non-manufacturing sectors). In fragmented manufacturing industries, mark-ups are lower than average suggesting that firms have little market power in these sectors. Above average mark-ups in segmented manufacturing sectors may be a cause for concern. However, when combined with the relative openness of the Dutch economy, with few restrictions to trade and FDI, the higher than average mark-ups in these sectors could be interpreted as suggesting that firms are more efficient than their counterparts in other OECD countries. Product market competition in important service sectors has increased in the past decade thanks to privatisation, market opening, and deregulation followed by appropriate re-regulation and institution building. However, this process is still ongoing and despite liberalisation, important restrictions to product market competition still exist in some areas.

The new competition law marked an important shift away from the Dutch "corporatist" model under which industry co-operation amongst firms was encouraged. However, co-operation is so embedded in the culture that it makes

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Box 5.4. Regulatory failure: an example from the California electricity market

California restructured its electricity market in 1998, allowing all consumers to choose their supplier. Electricity generation was deregulated with transmission and distribution remaining regulated functions. In order to encourage competition in generation, the vertically integrated utilities were encouraged to sell their generation capacity, selling about half of their original generating capacity. The utilities were also required to sell their generated electricity to the state's wholesale power exchange, and to purchase electricity at spot market prices. Operational control over the transmission system was turned over to an independent system operator to ensure that utilities did not favour their own generation facilities over competing generators in providing transmission access.

While there are a host of problems that contributed to the California electricity crisis,the regulatory framework put in place at the time of restructuring contributed significantly to the crisis (Borenstein, 2002). The restructuring plan, while deregulating wholesale rates, froze retail rates for a transition period of four years or until utilities recovered their stranded costs.Furthermore, utilities were forced to buy nearly all their power through the spot market and were initially not allowed to enter into long-term contracts with generators. Long-term contracts, however, are a standard feature of electricity markets, allowing buyers to hedge against price increases and sellers to hedge against price decreases. Under these circumstances, incentives for investment in generating capacity were greatly diminished, resulting in a lack of adequate generating capacity, and few energy retail providers found it profitable to enter the market (OECD, 2001).

The combination of frozen retail rates, the absence of long-term contracts and a lack of generating capacity spelled disaster for the utilities in the face of rapidly rising wholesale prices in the summer of 2000. There was a clear need to let prices increase in order to reduce energy demand and encourage new entry and investment. Instead of allowing prices to adjust to let markets clear, regulators at the time chose to introduce rationing instead, which acted to exacerbate the problems. The presence of market power can also aggravate these problems, and in this case sellers were able to exercise significant market power (Joskow and Kahn, 2001).

California's Market Surveillance Committee has since recommended immediate imposition of real-time pricing for all large industrial and commercial consumers and for residential customers as soon as possible. The restriction on long-term arrangements with generators was also abolished in December 2000. The introduction of real-time pricing, long-term contracts, and plans to reduce transmission bottlenecks have provided incentives for the introduction of new capacity. The introduction of market mechanisms, and regulatory oversight to curb market power on the part of generators, should produce an electricity market that operates in a smoother and more cost-effective manner.

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enforcement of the Competition Act difficult. Reducing administrative burdens and red-tape, a priority of the government, should lift major impediments to firm entry, stimulating innovative activity and growth. In spite of regulatory reforms that have led to comparatively liberal retail distribution and professional services sectors, barriers to entry remain important impediments to competition. Further reforms are particularly warranted in these sectors given their poor performance compared with other OECD countries. While incumbents retain dominant positions in network industries, and further restructuring is warranted, competition is slowly emerging as regulatory reforms have opened up these sectors to competition. The liberalisation of telecommunications has been a success and the market is very dynamic with strong competitive pressures from cable operators, although effective competition has yet to emerge in certain segments of the market (*e.g.* mobile communications). Implementation of many reforms in the energy sectors is hampered by local government ownership. Ownership separation of network and supply activities along with privatisation of retail should be undertaken to reduce barriers to entry. Privatisation of the networks can be considered after competition and network quality have been safeguarded through proper regulatory measures. Furthermore, the government should relinquish its golden shares in the telecoms and postal sectors. Box 5.5 provides a summary of recommendations that follows from the findings presented in this chapter.

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Box 5.5. Recommendations for increasing product market competition

Competition legislation and enforcement

The previous law to control and prevent restraints on competition was tolerant and ineffective, and industry co-operation was encouraged within a corporatist structure. The introduction of a completely new competition law in 1998 and a new enforcement authority, the NMa, signalled a major change in policy direction. In order to underpin this change in direction further reforms would be welcome. In particular:

- The NMa is currently an agency of the Ministry of Economic Affairs. While the government approved legislation to change the NMa to an autonomous administrative organisation (ZBO) in 2000, it is still awaiting final action in Parliament. The NMa should be made formally independent with no further delay.
- Aspects of the sanctions system should be strengthened. The NMa lacks the power to go after members of an association for violations by the association. Where restraints are imposed *via* an association, the NMa should have the power to assess fines on their members themselves. Providing for criminal penalties and individual sanctions would improve enforcement in areas such as professional services.
- The NMa should be given the power to implement structural remedies on companies that have abused, or may abuse, their dominant positions.

Regulatory policies

Competitive industries

Regulatory reforms in distribution and professional services are particularly warranted given the poor performance of these sectors. While there are no general exemptions to the Competition Act, other legislation or regulation often creates barrier to competition. In particular:

- Further deregulation of retail distribution is required to ensure effective competition. While regulatory reforms have led to the liberalisation of shop opening hours, planning restrictions, which are being devolved to lower levels of government, are important barriers to entry and impede productivity improvements. Local governments should examine the appropriateness of planning restrictions and their impact on competition.
- Regulatory reforms in professional services need to go further if competition objectives are to be attained. The role of professional associations should be reduced and limited to evaluation of professionals' capacities.
- The determination by law of recommended price schedules and unjustified restrictions to advertising should be abolished. Legislation should also permit a wide range of organisational solutions and ease the rules on the incompatibility between professions.

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Box 5.5. Recommendations for increasing product market competition (*cont.*)

Network industries

While great strides have been made in liberalising network industries, reforms in the telecommunications, electricity and gas sectors are very recent and important obstacles to competition remain. Priorities in this area include:

- Continue with privatisation by selling the remaining government holdings in telecommunications and post and eliminate golden shares. Privatisation of distribution networks in electricity and gas by local governments could be considered after having separated the networks form commercial activities and having safeguarded competition and quality of the networks through adequate regulatory measures.
- In telecommunications, the government should resist pressure to intervene in the roll-out of broadband infrastructure – taking fibre optics "to the home" – and should let the market determine the outcome.
- In the energy sector, restructuring of dominant firms and a reduction in their market shares is desirable from a competition point of view. More competition can be achieved through vertical separation as this ensures non-discriminatory access to essential facilities and the current use of legal or operational separation is insufficient. Ownership separation of distribution and supply activities is recommended.
- Strengthen the enforcement powers of the sectoral regulators to enable them to more effectively address the problems associated with market power on the part of dominant firms.

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Notes

1. Berenschot (2002).
2. A limitation for current cross-country comparisons is that the data in Figure 2 refer to 1998. The Secretariat is now in the process of updating these indicators.
3. Fragmented industries are characterised by small firms and low entry barriers associated with low sunk costs; as market size grows, so does the number of firms.
4. Segmented market structures are characterised by large firms and significant entry barriers associated with high sunk costs; as market size grows, the number of firms tends to remain unchanged.
5. Segmented sectors in the Netherlands (*e.g.* chemicals, petroleum, iron and steel, electronics) are sectors in which Dutch companies are often large multinational enterprises and compete internationally.
6. The simulations take 1998 as the base year, and estimate the impact on employment and multi-factor productivity if the Netherlands were to align their regulatory stance to that of the least restrictive EU country in 1998.
7. Fines were announced in the first big cases against 22 companies for price fixing, market division, and bid rigging in large infrastructure projects, road maintenance, and other areas, totalling € 100 479 900.
8. For example, NMa would be able to enter private homes to obtain evidence (executives were found to have deliberately kept documents in their houses, knowing that NMa did not have power to enter). Further, substantial fines could be imposed on enterprises and on individual executives for not complying with NMa investigations. The power to seal an office continuously, not just during non-business hours, and to temporarily take documents away if necessary to make copies is expected to become effective in 2005.
9. The legislation would also change the structure of NMa, replacing the single Director General with a 3-person Board of Directors. That would provide some assurance against arbitrariness that may be considered prudent for a more formally independent body.
10. To emphasise certainty and create a clear advantage to being the first one to come forward, the first informant is assured of immunity, if it is not the leader of the cartel and the NMa has not yet started an investigation. If the NMa has already started investigating, this first informant is still assured at least a 50 per cent reduction in fine. But for later informants or for the leader, lenient treatment is not guaranteed, and if granted it could amount to no more than a 50 per cent reduction.
11. Its principal concern so far has been keeping public operators to the terms of their franchises and preventing unfair competition with private providers. Piloting will be part of the office's responsibilities as of January 2005.

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12. DTe was already connected to the Ministry of Economic Affairs, but telecom was not.
13. Prior to 2001 a business license was obligatory under the Establishment Law of 1996 for every activity except the "free occupations" (lawyers, accountants, architects and advisors). For various activities in the retail trade and services, prospective entrepreneurs also had to meet the standards defined in the General Entrepreneurs Skills qualifications. A course of study lasting 6 months to a year was often required to obtain the relevant diploma (AOV). After recent reforms that lowered the obstacles to setting up a business, many activities described as "basic businesses" (mainly in retail trade, catering and craft) need simply to register and no longer require an AOV.
14. In 2000, there were 65 000 start-ups as compared to 55 000 in 1999 and the rate of bankruptcy remained unchanged at 2 per cent. There is no evidence, therefore, that the reduction in qualifications has resulted in inefficient entry (EC, 2002a).
15. In practice, shops in the centre of large cities such as Amsterdam, Rotterdam and The Hague are open every Sunday.
16. Research showed that 63 per cent of adult consumers visit shops in the evening hours and that sales increased by 4.9 per cent due to evening hours and 2.2 per cent due to opening on Sundays.
17. However, value added per unit of labour costs is one of the highest in the EU, possibly reflecting comparatively low unit labour costs arising from a more flexible labour market.
18. In food retailing, the combined market share of the five largest retailers is 56 per cent, slightly higher than the EU average of around 50 per cent. However, unlike most other EU countries, the market structure of the sector in the Netherlands is dominated by a single firm, Ahold, with a market share of around 30 per cent. Sweden is the only other EU country to have a market structure characterised by a dominant firm. The market structure of most other countries can be characterised as duopoly, oligopoly or unconcentrated (Dobson *et al.*, 2003).
19. Apart from the ability to extract discounts from suppliers, buyer power may manifest itself in the contractual obligations (vertical restraints) which retailers may be able to place on suppliers. Examples include: listing charges (where buyers require payment of a fee before goods are purchased from the listed suppliers); slotting allowances (where fees are charged for store shelf-space allocation); unjustified high contribution to retailer promotional expenses; and insistence on exclusive supply. See Competition Commission (2000) for a detailed list of examples.
20. On the one hand, greater concentration in the retail market may benefit consumers through lower retail prices arising from increased buying power on the part of retailers. On the other hand, if superior trading terms by leading retailers reinforces competitive advantages over smaller rivals, further consolidation might lead to market power in the retail market.
21. However, restrictions on competitive practices such as price competition and advertising or nationality requirements do not explicitly address the issue of quality and can have a negative impact on competition. For example, recommended prices may facilitate the co-ordination of prices amongst service providers and can mislead consumer about reasonable price levels.
22. OFT (2001) provides an overall review of the empirical evidence. Nguyen-Hong (2000) examined the effects of regulations on price-cost margins in engineering services and found that regulations led to an increase in prices on the order of 10 to 15 per cent in countries with the most restrictive practices. And Paterson *et al.* (2003) found a negative correlation between productivity and the degree of regulation, and no evidence that less

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restrictive regulation led to a lower quality of services. The most dynamic professions, in terms of growth and market consolidation, were found in countries where professions were less regulated. The authors also found that countries with a high degree of regulation tend to have relatively higher turnover from fees, indicating higher mark-ups.

23. This may be due in large part to the poor use of ICT in Dutch business services relative to better performing countries.
24. The index for the degree of regulation includes both entry and conduct regulations that may be determined by government or by professional bodies. Entry regulations include qualification requirements, membership in a professional body, and rules on reserved areas of practice. Conduct regulations include regulations on prices or fees, advertising, location and diversification restrictions and restriction on forms of business practice.
25. The architectural profession in the Netherlands has restrictions on fee setting and advertising that were not picked up in the Paterson *et al.* study and therefore underestimates the level of restrictions found in this profession.
26. Some of the most glaring examples include a national monopoly of interpreters and translators, which is financed by the government and with fixed rates for services, and harbour officials, which belong to a single partnership which has a national concession until 2015.
27. The Dutch government recognises that although the Netherlands scores comparatively well in terms of the indicators constructed, there still exist a number of restraints to competition and problems for consumers (MEZ, 2003a).
28. The Dutch professional services sector is characterised by active trade associations and very strong ties with political bodies, in which the interests of incumbents are often represented in national legislation.
29. NMa press release, 23 January 2004.
30. Since the CIF decision of the European Court of Justice, Competition Authorities are also able to examine government regulations or legislation that permits conduct in violation of Article 81of the EC Treaty.
31. See OECD (2001) *Economic Studies: Special Issue on Regulatory Reform*, No. 32 which thoroughly reviews the literature and adds more evidence on the relationship between regulation and performance in these sectors. The OECD Reviews of Regulatory Reform also constitute a rich source of information on the effects of industry-specific reforms on performance.
32. The Netherlands now holds minority stakes of 19.4 per cent in KPN and 34.8 per cent in TPG.
33. There is a strong need in these sectors for sector-specific regulation and remedies with oversight by an independent sectoral regulator. The NMa can apply structural remedies only in merger cases.
34. The Competition Authority is not subject to exceptions so that it has jurisdiction in the telecommunications sector in addition to OPTA. The NMa and OPTA have concluded a protocol governing the interpretation and implementation of overlapping responsibilities.
35. For example, see Newbery (2002a; 2002b) and Brunekereeft (2002).
36. Prior to liberalisation, the electricity sector in the Netherlands was vertically integrated and the four largest generation firms owned the transmission network. With the liberalisation of the energy market, the Parliament demanded that the national high-voltage

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grid (TenneT) be brought under government control. Negotiations since 1998 between the Ministry of Economic Affairs and the electricity producers resulted in the so-called OEPS Act of 21 December 2000. Among other things, the act stipulated the dissolution of SEP, the former co-operation organisation of Dutch electricity producers; and set out rules for the assignment of rights and obligations after the termination of SEP and compensation of related costs. The act also obliged the State to buy TenneT from SEP. After protracted negotiations, the government bought TenneT at the end of November 2001.

37. Oversight of the distribution networks is undertaken by independent regional distribution grid operators.
38. However, following a recent takeover, there are now five major generation companies (as opposed to six) with a corresponding increase in CR_3 to 67 per cent.
39. The market for green electricity has already been fully liberalised.
40. APX is a full subsidiary of TenneT and was formed to enable day-ahead spot trading to take place.
41. The reduction in average price has been accompanied by a steady increase in the amount traded over the APX, especially since the "Protocol" ended at the end of 2000, corresponding to around 15 per cent of net Dutch electricity consumption in 2002 (Speck and Mulder, 2003).
42. Vertical integration between generation and supply makes entry into downstream markets difficult for new entrants who don't own generation plants. The Swedish electricity producer, Vattenfall, recently declared that it would exit the Dutch electricity market and that it would not renew existing contracts. The firm argued that there are too few suppliers in the market (Vattenfall does not own any electricity generation plants) and that wholesale prices were too volatile, thus making operations too risky. A proposed law implementing the new EU Electricity Directive will create possibilities to oblige generators to offer part of their electricity over the spot market if liquidity is deemed to be inadequate. This should make entry into downstream market for entrants without generation facilities easier.
43. The policy rules on privatisation and the concept legislation, which were introduced in January 2001 and May 2002, have been withdrawn. Privatisation has also been made dependent on the full liberalisation of gas and electricity, which is now due to take place on 1st July 2004. Because of the recent decision to postpone the full liberalisation by six months, the prohibition to privatise is also extended by six months until January 2005.
44. A regulator could require private companies to comply with capacity increases in order to ensure security of supply. For example, capacity expansions deemed necessary by the regulator could be made part of the license requirement.
45. The position of the Energy Council is that this would only result in further fragmentation and weakens the position of the Dutch energy sector (Energieraad, 2003).

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6. Policies to enhance innovation and productivity growth

In view of labour inputs growing more slowly owing to demographic ageing, fostering productivity growth, the slowness of which has been a major concern of the government, is a key challenge for the Dutch economy. As described in Chapter 1, raising the volume and effectiveness of innovative activities and improving the quality of human capital under growing public financing constraints are very important means to achieve this goal. Improving corporate governance, which is a major reform project for the government, would also help to increase productivity by ensuring that managers get better incentives to reach the optimal combination of factor inputs and make responsible investment decisions motivated by medium-to-long term considerations of company growth.

Fostering innovation, research and entrepreneurship

There are indications that the Dutch Innovation system is losing momentum. Besides low business sector R&D spending and few knowledge workers, this is mainly due to lacking interaction between science/higher education and industry and low innovative activity by SMEs. As a consequence the government is giving high priority to science and technology policy. Three of the key challenges that have been identified are (OECD, 2004a):

- increasing the incentives and improving the institutional frameworks for co-operation between public and private actors of innovation;
- streamlining and improving the public support schemes for innovation and a complex innovation governance system; and
- improving the attractiveness of the Netherlands as a top location for re-searchers as well as R&D and other innovative activities.

Business sector R&D *spending is low*

Total R&D expenditure at 1.9 per cent of GDP in 2001 is below the OECD average and slightly below the EU average (see Chapter 5). While the level of public R&D expenditure, representing roughly half the total R&D expenditure, compares well

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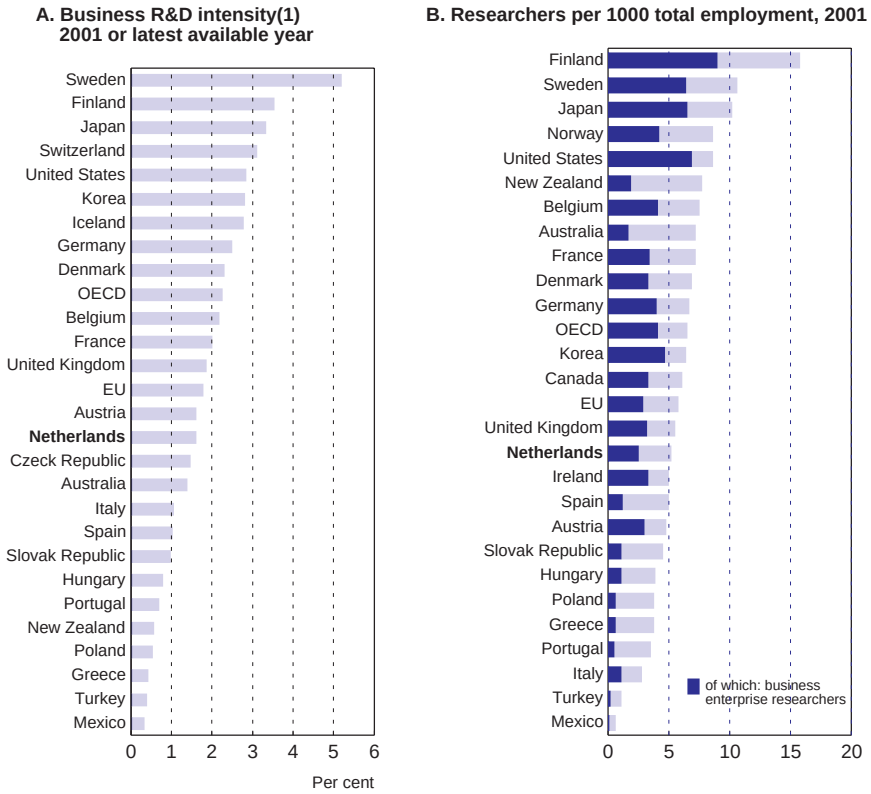
Figure 6.1. **Business R&D is low, reflecting a low number of researchers**



1. Business enterprise sector R&D expenditure as a percentage of value added in industry.
Source: OECD *STI Scoreboard*, 2003.

with that in other countries, business sector R&D, albeit on the rise during the 1990s, lags further behind the OECD and EU averages (Figure 6.1). The biggest seven companies (the Dutch-owned multinationals) account for about 60 per cent of all company R&D (Cornet and Rensman, 2001), implying a strong innovation position in the electronics, chemicals, pharmaceuticals and food sectors. While concentration of expenditure on some big players is also observed in other countries, the rising share of companies with no R&D activity at all is a concern for the authorities.[1] The share of R&D carried out by SMEs is low by international standards in spite of relatively strong tax incentives, mainly the WBSO, a tax allowance for R&D wage costs totalling 0.15 per cent of GDP in 2001 (Figure 6.2). This may explain why, despite the relatively low number of R&D workers (see Figure 6.1), no general upward pressure on wages for

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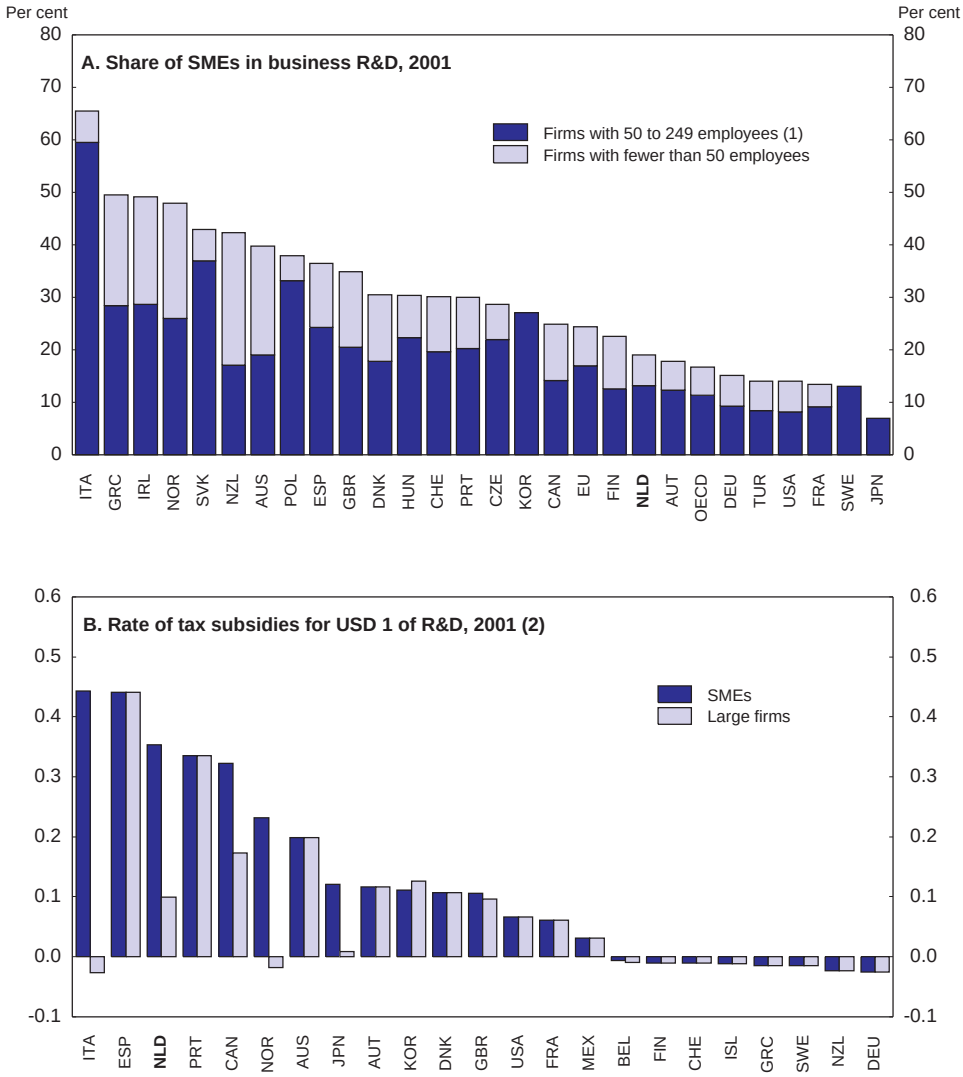
Figure 6.2. **Low R&D investment in SMEs in spite of large tax incentives**



1. For the Netherlands and Norway, 50 to 199 employees instead of 50 to 249. For New Zealand, 50 to 99 employees instead of 50 to 249 employees. For Japan and Korea, fewer, than 299 employees.
2. Tax subsidies are calculated as 1 minus the B index. For example, in the Netherlands, 1 unit of R&D expenditure by large firms results in 0.35 unit of tax relief.

Source: OECD *STI Scoreboard*, 2003.

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R&D worker was observed. During the 1990s most Dutch firms appear to have mainly focussed on cost competitiveness. The sharp rise in labour costs at the end of the economic boom has demonstrated the limits of this approach and has led the government to emphasize the need for competing on product cycle leadership as a more sustainable motor for economic growth in the future.

The WBSO has, however, contributed to an increase in the share of R&D carried out by SMEs from 16.4 per cent in 1994, the year of its introduction, to 22.6 per cent in 2001, when SMEs used about two thirds of the WBSO budget of € 337 million (0.08 per cent of GDP). To increase private R&D further, the government has decided to step up the tax allowance by € 100 million as of 2007. Econometric analysis has shown a positive effect of the WBSO on the share of private R&D, especially among SMEs: for each € 1 of wage tax deduction, somewhat more than € 1 is spent on extra R&D. This effect is in line with the international literature on the subject.

Strengthening the interfaces in the national innovation system

In contrast to low innovative activity in the business sector, the quality of Dutch university research, when measured by international citation indices and the degree of international co-operation in producing scientific papers, is excellent. The authorities have concluded that one major weakness in the national innovation system is a lack of technology transfer from universities, which are very good at creating new knowledge but poor at disseminating it, to SMEs, which lack the willingness or ability to absorb it. The Innovation Council, a newly created commission chaired by the Prime Minister and comprising the major stakeholders in innovation policy, government, academia and industry, is analysing ways of strengthening the links between production and commercial use of knowledge while maintaining the strong research position of universities. To stimulate the transfer of knowledge the government has started a programme on knowledge circulation between higher professional education and companies, especially SMEs.

The nexus between science and industry

Several instruments to promote science-industry relationships have been in operation since 1996 (OECD, 2004a). First, co-financing was introduced in large public research institutes (the Netherlands Institute of Applied Science (TNO) and the five Large Technological Institutes). Second, four Leading Technological Institutes (LTIs) were created. These are an institutional set-up for close public-private co-operation on fundamental research in applied sciences (polymers, food, telematics, and metals). The projects are jointly defined and co-funded by the Dutch government and Dutch business and industry for a period of several years. According to a recent interim assessment, the LTIs are becoming leading institutes in their specific areas of research, which have a strong counterpart in the country's manufacturing base, so public subsidisation was recently prolonged for several

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years. In addition, there have been two instruments set up by the Ministry of Economic Affairs since the 1980s: Innovation-Oriented Research Programmes, which promote technical-scientific research and its application by business and stimulate companies and research institutes to develop joint knowledge investment plans; and the Open Technology Program of the research foundation STW, which stimulates high-quality university research projects with high user involvement and good prospects for utilisation and research yield.

Public-private partnerships received € 222 million in 2003, equivalent to 6.3 per cent of the total Science and Technology (S&T) budget. They have recently seen their funding increased sharply with the government reserving about € 800 million for the period 2003-10, underlining the willingness to move from subsidies for individual companies to stimulating technological partnerships. Sixty-seven research projects with a total investment volume of € 3.6 billion have been submitted to compete for the money that will be allocated according to their quality.[2] The bulk of this sum will be allocated to universities, taking the form of competitive grants for research projects in co-operation with the private sector. The applicability of research for innovation will become one of the criteria in performance-based funding of universities.

A major concern is the low extent of patenting and licensing by universities despite the pro-active attitude and awareness campaigns by the National Bureau of Intellectual Property. The government considers that organisational, jurisdictional and financial problems currently prevent the research community from using patents more effectively. While maintaining the principle that patents belong to universities and research institutes (not to individual researchers), it wants to increase the number of spin-offs by allowing universities to sell patent rights to businesses or license them out and develop specific institutions furthering technology transfer to the private sector, *e.g.* a national network of patent experts who would advise research institutions on the optimal use of their inventions. The number of spin-offs is reportedly higher at universities that have an active policy of using their own innovations and valorising commercial activity, for instance by renting out laboratory facilities at attractive rates over weekends. Another major concern is the need to make the public knowledge infrastructure more dynamic (OECD, 2004a). The government plans to make research grants to the thirteen public universities more dependent on performance than under current arrangements, which entail a fixed allocation between them. In addition, a government commission is investigating the value added of the public institutes for applied research.

The government promotes high-tech start-ups through the *TechnoPartner* action programme, which aims at improving the start-up climate by:

- a seed facility to support the bottom end of the Dutch venture capital market, thereby helping high-tech start-ups to satisfy their capital re-quirements at an early stage;

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- the Knowledge Exploitation Subsidy Agreement (SKE) with as its objective the quicker utilisation of scientific knowledge by high-tech start-ups both inside and outside knowledge institutes [universities, HBO (higher vocational education) institutes, and publicly financed research institutes; and
- the *TechnoPartner* platform offering information and expertise and creating and updating an inventory of the obstacles faced by high-tech start-ups.

These activities are complemented by policies that increase student awareness of entrepreneurship such as business plan contests, matchmaking events between universities and business, and the Dreamstart agency advising potential starters (Waasdorp, 2002, pp. 34-35).

Overall, the government has traditionally pursued a generic rather than specific approach to innovation policy by putting a stronger financial accent on tax subsidies for researchers and providing moderate financial incentives to innovative entrepreneurship. Weight is given to reforming the institutional arrangements in favour of more innovation. The four LTIs, which represent one of the purest forms of public-private partnerships, have been an important exception from this general tendency. The OECD peer review of the LTI programme concluded that it is a proven good practice in mobilising public and private research towards common objectives of high importance for the economy and society (OECD, 2004a). The four LTIs perform well, are based on a sound rationale and are implemented efficiently, namely through competitive selection and a combination of high industry involvement with the pursuit of public interest (*e.g.* public access to research results). Nevertheless, the OECD review also found that the national innovation system would benefit from making existing LTIs less dependent on subsidies as they mature and re-allocating the funds to creating new LTIs in a new round of competitive selection and to improving incentives for LTIs to strengthen their linkages with government-sponsored international co-operative ventures.

Stepping up the budget for competitively awarded public-private research co-operation more strongly than the tax subsidy for researchers will add more specific subsidies to the policy spectrum. While picking winning projects is a delicate issue and puts high demand on evaluation techniques and independence of the commissions in charge, there may be considerable benefits from enhanced knowledge transfer and from increasing mobility of researchers between the public and the private sectors, *e.g.* graduates partly doing their PhD research in private firms and stays of public researchers in companies. The government intends to develop a policy enhancing the mobility of scientists between the public and private sectors, as announced in its White paper on the shortage of scientists and engineers (MinEZ, 2004). Although there are no formal restrictions to private-public staff mobility, existing disincentives for longer-lasting

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stays or mixed careers (*e.g.* loss of civil-servant status, different pension and benefit rules) should be analysed and reduced. Given the simultaneous increases in the WBSO and subsidies to specific research projects, the government should closely monitor the comparative performance of generic versus specific subsidies to private R&D.

International integration of the knowledge infrastructure

Dutch multinationals have moderately increased their domestic R&D expenditure over the 1990s while strongly increasing R&D spending abroad, reflecting a growing internationalisation trend in R&D.[3] Attraction of foreign R&D on Dutch territory is more gradual. As a result, already one-third of patents with Dutch authorship are held on inventions made abroad, whereas only one-fifth of all inventions made in the Netherlands are owned by foreigners (Figure 6.3). The authorities should favour the location of both national and foreign R&D capacity by fostering conditions conducive to a large and highly qualified R&D workforce, up-to-date knowledge of existing researchers and the possibilities for new firms to locate in areas where positive spillovers from and to existing facilities can be exploited.

To reap the benefits of knowledge acquired abroad and enhance its absorption by Dutch research institutions and companies, the government has provided the new policy instrument for R&D co-operation with an additional subsidy for projects involving partners in different countries. Therefore, it is important that the government meets its objective of reducing barriers to immigration of knowledge workers by simplifying administrative procedures (one-stop shop, one procedure and one single document for knowledge workers), as announced in the "Deltaplan Bèta-Techniek" (MinEZ, 2004). Incentives for excellent foreign graduates in science and engineering to extend their stay in Dutch universities to a PhD thesis and beyond would further increase the attractiveness of the Netherlands as a location for R&D thereby triggering additional inflows of R&D-related inward FDI. Such a policy would be greatly facilitated by improving the funding base for universities (see below).

Strengthening entrepreneurship

The rate of firm creation has more than doubled over the period 1987-99 (Prince, 2002, p. 11) and the birth rate (net of mergers, splits or other ownership changes) is in line with that in most other EU countries (Brandt, 2003). However, firm growth in the first years (*e.g.* as measured by employment gains) is very modest. While a dynamic process of entry and exit is important to introduce new technologies and business practices, firm growth is crucial in raising overall productivity. Increasing both birth rates and the proportion of fast growing businesses is challenging and may entail a trade-off (Storey, 2002). The government should address barriers to firm growth that mainly arise from restrictive product and

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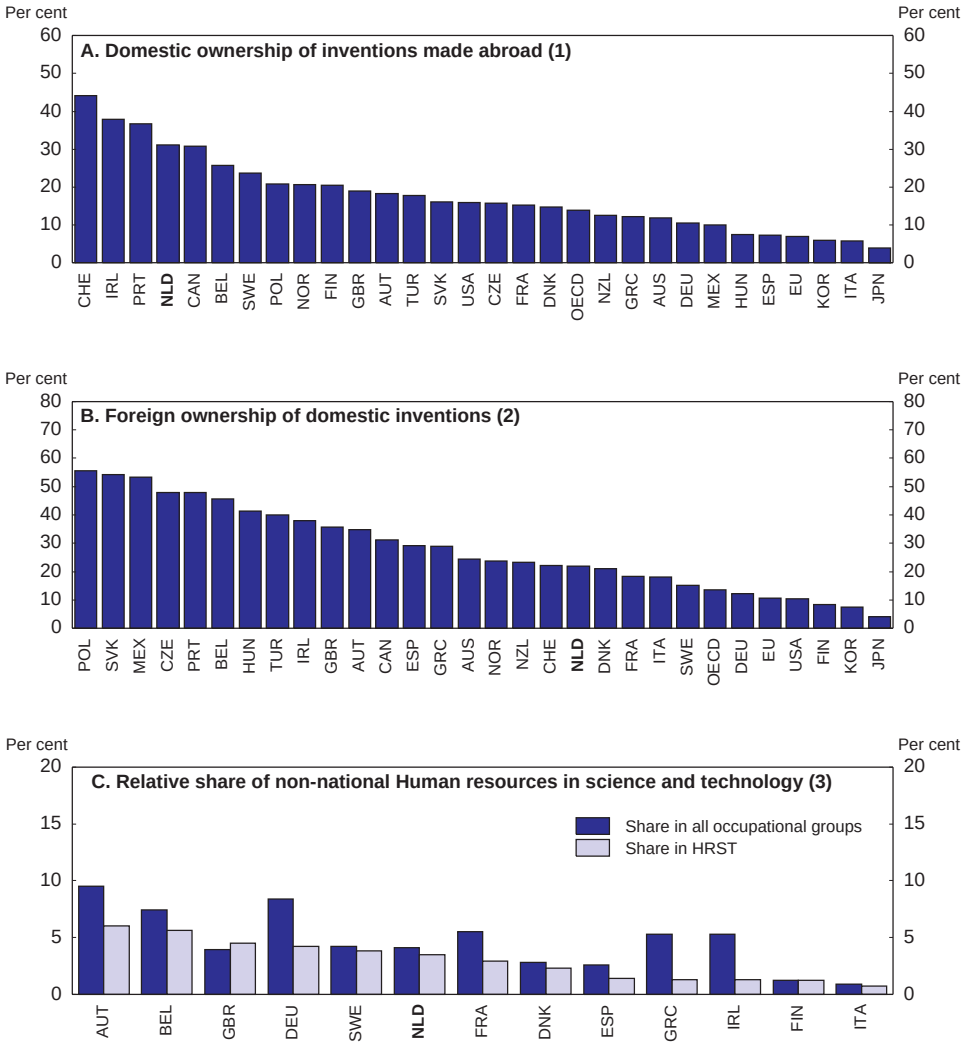
Figure 6.3. **International aspects of innovative activity**



1. Share of patent applications to the EPO invented abroad in total patents owned by country residents. Priority years 1997-99.
2. Share of patent applications to the European Patent Office (EPO) owned by foreign residents in total patents invented domestically. Priority years 1997-99.
3. Employment in the European Union, 2002.

Source: OECD *STI Scoreboard*, 2003.

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labour market regulations. It should also reduce the cost of administrative regulations and foster the intensity of competition in product markets, namely by reducing self-regulation mechanisms at sector level that may act as a barrier to new entrants (see Chapter 5). Barriers to exit also tend to hamper firms from growing rapidly, namely tight EPL and bankruptcy regulations (OECD 2003, *Sources of Economic Growth*). While the latter has been addressed in 2001 with the reform on the bankruptcy law favouring amicable settlements, relaxing labour laws and creditors' position towards firms in financial difficulties, and attenuating the stigma of failure (Waasdorp, 2002), the relatively tight EPL should be relaxed. Over and above the positive labour reallocation effects discussed in Chapter 4, this would provide young firms with an incentive to grow by lowering the potential cost of shrinking at a later stage.

Barriers to firm growth are particularly pervasive in business services that have become increasingly important as a source of intermediary inputs for all market industries. While business service providers make contributions to productivity growth by fostering ICT use and helping management in client firms to re-organise businesses, the productivity performance of the business services sector itself has been weak. Many firms fail to reach the minimum efficient scale or to grow to a size where investment in own R&D and adoption of production technologies created elsewhere would pay off (Kox, 2003).[4] The market for business services is characterised by information asymmetries and client-specificity, resulting in imperfect competition. Due to lacking comparability between business practices a niche player supplying superior services has difficulties in taking over markets from competitors as long as he has not built up a reputation for his concept. The need for own reputation represents a natural barrier to firm growth that adds to the institutional barriers mentioned above. In addition to lowering the latter, the government should consider ways to overcome market imperfections. Signalling devices such as certified best business practices or short-term patents could be one such possibility. In the United States where the business services sector achieved an impressive turnaround in productivity during the late nineties, the Patent Office has granted a fast increasing number of patents on business methods and software in each year since 1998 (OECD, 2003a). While the causal link to productivity growth from these developments is still debated, they should be closely followed. The government should also continue to support initiatives at EU level aiming at completing a single market for services in the European Union and implementing the associated directives with a stronger view to minimising the deviation from common business practices in most partner countries rather than that from past national practice.

Among start-up enterprises, those that undertake to launch new products and processes are exposed to higher risk and therefore often fail to come into existence due to borrowing constraints. There is a general credit guarantee system granting credit guarantees to all SMEs with satisfactory future prospects (up to the

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difference between the collateral asked for by the bank and the company's own available security). Although some positive discrimination in favour of start-ups and innovative firms has been introduced in 2002, granting unconditional guarantees lowers the incentives to become more profitable and may act as a disincentive to innovate. Given that there is no evidence for the often claimed lack of loan capital to SMEs (Wagenvoort, 2003) and that a clear economic case for capital subsidisation can only be made if there are positive external effects such as innovation spillovers, state aid for SMEs should be targeted to young innovative SMEs companies, specifically those financed by venture capital markets, which would leave the selection problem to markets rather than bureaucracies. Furthermore, state aid such as credit guarantees or loan subsidies should be designed in a way that distorts as little as possible the incentives for entrepreneurs to deliver a maximum of own effort. Since even well-designed and targeted subsidies – rather than only solving the problem of borrowing constraints – lead to entrepreneurs and venture capitalists taking more risk than otherwise (Gebhardt and Schmidt, 2002, pp. 251-52), the benefits of more innovations (from higher consumer surplus and an improvement in locational competitiveness) have to be carefully assessed against the costs of higher tax expenditure.

Improving the quality of human capital and the appropriateness of vocational training

Besides product market competition and a high rate of product, process and organisational innovation, providing for a high skill level of the workforce is another key pillar in raising labour productivity. It would also help the Netherlands to remain attractive as an R&D location by increasing the absorptive capacity for research results obtained elsewhere. As far as the non-tertiary workforce is concerned, a good command of a number of basic skills will be increasingly important for life-long employability in a context of more frequently changing workplaces. However, in the short run a good match between training and job realities would increase productivity in the early years of careers, implying that the system of vocational training has to offer an appropriate mix between basic knowledge and specific skills.

Furthering the quality of tertiary education

Dutch universities rank high in terms of the quality of both teaching and research output. However, when adjusting for the quantity and costs of inputs, their performance is average according to recent research (Pomp *et al.*, 2003). The government has started shifting the policy focus towards greater demand-orientation. Ongoing reform efforts to increase university performance and channel students towards programmes enhancing their human capital in efficient ways include increasing flexibility, *e.g.* by recognising partial diplomas or competencies

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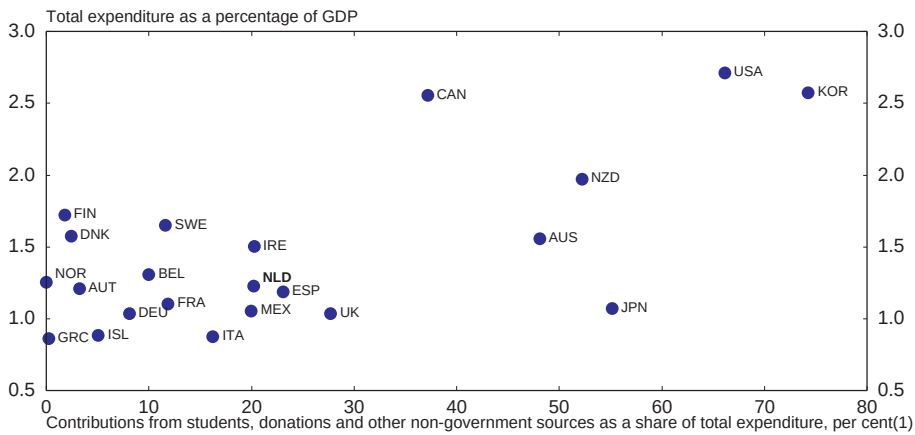
obtained outside school/university and by the introduction of a bachelor-master structure in June 2002. Other measures aim at enhancing transparency, for example the introduction of an accreditation system. The results of programme accreditation processes and assessments by review committees are published. Finally, the government wants to increase international co-operation in defining common standards for accreditation and reduce operational regulations surrounding programme supply.

While maintaining the principle of mostly public financing for higher education, the government also wants to lighten regulation on tuition fees to allow for moderately more freedom in pricing. Increasing funding for universities will be needed in the future to enable them to cope with a growing share of graduates in society and with the growing internationalisation of the scientific community. Universities will need to attract top-level national and international staff to remain competitive in the fields of research and higher education. Given the growing constraints on public finances, tertiary education will increasingly have to raise funds from private sources. At the moment the Netherlands takes an intermediate position both in terms of total spending on tertiary education and the share of non-public sources (Figure 6.4). The total level of spending on tertiary education tends to be higher in countries that have substantial contributions from students, donations and other non-government sources. Those countries that have been able

Figure 6.4. **Expenditure on tertiary education institutions**
2000



1. Net of tuition fees paid by government.
Source: OECD, Education Database and Statement of Financial Performance for tertiary education institutions in New Zealand.

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to channel more than 2 per cent of GDP into tertiary education – the United States, Korea, Canada and New Zealand – all raise a substantial share of funding from these alternative sources (Box 6.1). At about one-fifth, this share is higher in the Netherlands than in a number of OECD countries, but is well below the share in the countries with high total spending on tertiary education. When taking living costs into account, which are partly subsidised by grants, the share of government in financing the total cost of tertiary education comes close to 90 per cent (CPB 2003, p. 146).[5] As in most other European countries, this subsidy level is beyond what may be warranted for economic reasons such as positive externalities, borrowing constraints for students and their risk-aversion against accumulating debt.

An efficient way of fostering competition between universities and making the system more demand-driven would consist of allowing universities to set the tuition fees paid by students relatively freely and get financial autonomy over the funds raised, thereby reducing the share of direct subsidies in their budget. The remaining public subsidy should depend on performance criteria more strongly than they do today, for instance by increasing the share of competitive grants, as the government already plans (OECD, 2004a). Students would have a stronger incentive to vote with their feet, picking the universities that best fit their education needs and provide the best value for money. In addition, universities should be allowed to charge fees for tertiary education with a more consumptive character (evening and holiday classes for persons outside initial education). More competition would put pressure on universities to optimise curricula while higher fees would stimulate students' willingness to demand higher value for money.[6] Competition would also be fostered if new universities could qualify for public funding under the same conditions as existing ones.

Such a system would, however, require removing liquidity constraints from students with a lower-income background in order to preserve general accessibility of the tertiary education system. It would also require mitigating the risk of debt default for individuals investing in their own education. As there is generally no private market for student loans, the government acts as a lender. In view of the empirical findings that private returns to tertiary education are high and that the difference between social and private returns is much smaller in tertiary than at earlier stages of education, extending repayable student loans would be an adequate way of increasing funding both on efficiency and on equity grounds.[7] Making repayment contingent on graduate incomes would ensure that students do not have to worry about the risk of credit default due to circumstances beyond their control. The Australian experience shows that broad access to universities can be granted without expensive public support (Canton and Venniker, 2001).

While lifetime income of graduates is generally found high enough to sustain a student loan system – with some variation depending on sex as well as the type of tertiary education and the subject chosen – the repayment capacity may be

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Box 6.1. University fees and student loans in selected OECD countries[1]

Contributions from students and other non-government sources play an important role in financing tertiary education in the United States, Korea, Canada and New Zealand, the countries where total expenditure on tertiary education is the highest in the OECD (see Figure 6.4). In Australia total tertiary education expenditure is also high and private sources cover about half of it. Whereas most of the OECD countries charging tuition fees allow for some variation across subjects and across universities, the Netherlands and the United Kingdom have so far applied a common level of fees. However, the UK government has made proposals, backed by the OECD and agreed, in principle, by a vote in Parliament in January 2004, that would allow universities to charge up to £3 000 per annum per student, thereby increasing total tertiary education funding by up to 0.15 per cent of GDP. In countries where the level of tuition fees varies across subjects, it generally reflects differences in programme costs and/or in graduate income perspectives from these programmes.[2] In the United States fees vary substantially across universities. In private universities fees can reach that several times the national average (US$ 6 000).

Many OECD countries operate some form of public student loans to finance tuition fees and/or living costs. However, for only a few countries, including Australia, Iceland, New Zealand, Norway and Sweden, do student loans amount to 0.2 per cent of GDP or more, with New Zealand having the highest level of student loans equivalent to ½ per cent of GDP. Student loans are in some cases tied to the payment of tuition fees, as with the Higher Education Contribution System in Australia or the proposed Graduate Contribution Scheme in the United Kingdom. Alternatively, as in the Netherlands, New Zealand and the United States, the same conditions apply whether a student borrows to finance tuition fees or living costs. Countries with no or limited use of loans sometimes make significant use of public grants, in some cases targeted at students from low-income families. Also, loans can be supplemented by public grants as in the United Kingdom, where the introduction of the Graduate Contribution Scheme will be paralleled by a new grant to students from low-income families supplementing the existing student loan for living costs, which is available at a maximum £ 3 905 per year to students from low-income families studying away from home. Sometimes the dividing line between loans and grants can be blurred, *e.g.* in the Netherlands and in Luxembourg where money initially given as loans is converted to a grant provided students meet minimum academic criteria.

In Australia, New Zealand and in the United Kingdom the speed of loan repayments depends on graduate income. This is a way of reducing the risk borne by individuals investing in their education. A more bureaucratic alternative is hardship procedures that exist in the Netherlands and in most other countries and may to some extent replicate income contingency. In Sweden, repayments on loans to cover living costs are initially independent of income but graduates are entitled to have repayments reduced to 5 per cent of their income without, however, prolonging the repayment period. The income threshold at which repayment begins for the UK maintenance loans is currently £10 000, which at roughly half the gross earnings of the average production worker, is close to the thresholds applied in Australia and in New Zealand. As part of the proposed changes the threshold will rise to £15 000 to equal that of the planned Graduate Contribution Scheme.

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Box 6.1. University fees and student loans in selected OECD countries[1] (*cont.*)

Some student loan schemes imply a significant subsidy from low interest rates. In New Zealand loans are interest free while the person is a student and 7 per cent thereafter. At current interest rates, this implies a subsidy of roughly one third of the value of lending. Australia and the United Kingdom apply a zero real interest rate throughout the loan period, implying a subsidy rate of about 50 per cent. However, broad interest rate subsidies are an inefficient instrument of granting broad access to university as they are also paid to those who do not need them. Applying an approximate market rate as in the Netherlands (2 percentage points above long-term government bonds) or a slightly subsidised rate as in the United States and Sweden would free resources for more targeted pro-access policies.

Equity of access is a central concern in education funding policies. Most studies find little or no adverse equity effects from student or graduate contributions to the cost of tertiary education in countries that have combined an increase in tuition fees with an increase in student loan facilities. In New Zealand, for example, participation in tertiary education has almost doubled since the introduction of student loans in 1992, and the share of young people entering tertiary education is today the highest in the OECD. The share of students from Maori and Pacific ethnic groups has even increased from 9 to 24 per cent, mainly as a consequence of making loans available to students of any approved tertiary institution and subsequent market entry of small, innovative education providers. On the other hand, countries trying to widen access by introducing or increasing grants to cover student living costs have had only limited success. Australia abolished tuition fees and introduced grants in the mid-1970s; re-introduced tuition fees in 1989 based on income-contingent loans and increased them significantly in 1997. The abolition of tuition fees did not lead to a significant increase in the share of lower socio-economic status students. Nor did their reintroduction of fees change the socio-economic mix of students. Studies from the UK Universities and Colleges Admissions Service indicate that the replacement of grants with loans and the introduction of tuition fees in 1998 did not affect the social-class mix of entrants to university adversely, and the proportion of ethnic minority entrants and women has grown since then.

1. The cross-country comparisons in this box draw on Annex 4.A1 of OECD (2004b) and on Lundsgaard and Jurner (2004).
2. There is, however, some debate on whether to subsidise those courses judged as important for promoting growth. For example, the New Zealand government is considering re-orienting public funding in order to increase the number of students in engineering, mathematics and computer-science.

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limited by a flat wage distribution and progressive income taxation. Nevertheless, simulation results in Jacobs (2002) show that an income-contingent student loan system with a significant amount of risk sharing is feasible for the Netherlands and would meet the target of removing obstacles for university access more efficiently than current arrangements (Box 6.2). If the government saving from lower social transfers to students are used to reduce labour-distorting taxes, the potential negative effect of higher private contributions on labour supply by the higher educated could be eliminated (CPB, 2003, p. 148). Recently, an advisory commission on the future of student grants appointed by the State Secretary for Education, Culture and Science gave a favourable assessment of an income-contingent loan system in principle but pointed to considerable practical difficulties. In view of the growing importance of high-quality tertiary education institutions, the government should nevertheless consider introducing such a student loan system in order to increase total spending on education and reduce its reliance on public funding.

The functioning of vocational training has improved

The finding of generally high reading and mathematic literacy levels for the Netherlands (OECD 2001, PISA) put the workforce in a good position to meet the challenge of faster technological and organisational change. However, a good match between the contents of vocational training and actual job requirements is also needed in order to raise overall labour productivity. Over the past years the authorities have addressed this challenge by overhauling the vocational education system with a view to making it also more demand-driven. The qualification structure for a total of 900 professional activities is jointly defined by nineteen sector-specific "knowledge centres", which are composed of companies and specialised educational institutions. These centres have set up detailed descriptions of qualification profiles, based on competencies (*i.e.* knowledge, skills and behaviour) rather than knowledge alone. Competencies acquired outside initial vocational education will be recognised more easily than under the old attainment-based systems, thus increasing inter-job mobility within the broad domain chosen. Educational institutions have to update their curricula on a regular basis to make the qualifications they award respond to these market and experts-driven profiles. The qualifications are not intended to act as an objective entry barrier into jobs as most low-to medium-skilled jobs continue to be accessible without the formal qualification. These reforms are a welcome step as they are likely to preserve the good track record of the Netherlands in terms of high employment and low unemployment rates among young people.

However, several problems persist in primary and secondary (vocational) education. First, despite the economic downturn there continues to be a large shortage of teachers, mainly for demographic reasons. Initial steps to counter the problem included specific pay-rises for those groups of teachers for which the

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Box 6.2. Effects of a social student loan system for the Netherlands

The CPB has calculated the effects of introducing various loan systems on graduate repayment rates and public expenditure (Jacobs, 2002, Jacobs and Canton, 2003, and CPB, 2003). An individualised loan system (no insurance or solidarity elements) with a subsidy rate of only 25 per cent for the tertiary education system as a whole (instead of 88 per cent today) would lead to huge differences in repayment rates, ranging from 1.4 per cent of income for a man graduating in business/economics at university (WO) and getting in the highest income quintile to almost 15 per cent for a woman with a diploma from applied university (HBO) in the lowest quintile.

To transform the system into a social loan system, some insurance against credit default is introduced. It is assumed that every student participates in the scheme, which eliminates the problem of adverse selection. The default risks may be born by the participants of the system (risk-pooling) or paid from general taxation (risk-shifting). In the case of risk-pooling, the uniform repayment rate depends negatively on the extent of insurance (*i.e.* the risk premium on the interest rate). Lower risk premiums mean that high-income graduates would pay off their debt relatively faster and leave the system earlier, resulting in a higher average repayment rate for the remaining debtors. For example, with a subsidy rate(s) of 50 (25) per cent the repayment rate would increase from 3.2 (4.8) per cent of income to 4.9 (7.5) per cent if the risk premium is lowered from 4 to 1 per cent. The most extreme case of redistribution within the risk-pooling arrangement is the graduate tax, a uniform percentage of income paid to the scheme until the age of 65, which brings the average repayment rate down to 2.9 (4.4) per cent. The government's gross saving, which is independent of the risk premium, would amount to 0.4 (0.7) per cent of GDP for a subsidy rate of 50 (25) per cent. Net savings would be lower (0.36 and 0.54 per cent of GDP, respectively) because graduates' marginal tax rates including credit repayment would be higher than today, leading to lower labour supply and an associated loss in income tax revenues. This negative effect can be eliminated by using the gross saving to reduce taxes that distort labour supply.

In the case of risk-shifting the saving for the government becomes dependent on the amount of the relative share of the risk borne by participants. A higher risk premium for graduates means that less solidarity has to come from general taxation, implying higher government saving. In case of a 4 per cent repayment rate and a subsidy rate of 50 per cent, the gross saving is 0.5 per cent of GDP for a risk premium of 4 per cent and 0.4 per cent of GDP for a risk premium of 2 per cent. Reducing government subsidies to 25 per cent raises the saving only marginally (to 0.6 and 0.5 per cent of GDP, respectively) because with higher indebtedness among graduates the credit default rate also increases such that public debt remission payments partly substitute for formal subsidies to tertiary education institutions. There is a stronger moral hazard element in a system with risk-shifting than with risk-pooling, calling for differentiation of repayment rates according to the duration of education and/or lending restrictions to students who exceed the normal duration of their education.

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shortages are largest. Second, the percentage of dropouts (early school-leavers) is relatively high in the Netherlands (above the EU and OECD averages and far above the US level). The policy for strengthening vocational education is geared towards reducing the number of dropouts and having more pre-vocational secondary education (VMBO) certificate holders transfer to upper secondary vocational education (MBO) and MBO certificate holders to higher professional education (HBO). Third, schools often complain about the plethora of laws, regulations and administrative obligations, including regulations set by the Ministry of Education, other departments and the local government. This administrative burden not only consumes considerable time and energy, drawing capacity away from teaching, but also tends to hamper creative solutions for existing problems. The government has planned a major campaign to significantly reduce the administrative burden on schools.

Corporate governance

Empirical studies (*e.g.* OECD 2002) have found that financial market development, and in particular stock market capitalization, stimulates economic growth. Moreover, a number of empirical studies indicate that aspects of corporate governance, especially shareholder rights, lead to deeper capital markets. At the company level there is also a strong relation between corporate governance and performance (for the Netherlands, De Jong *et al.* 2000) and the cost of capital for enterprises. However, recent developments have highlighted the weaknesses of the Dutch system of corporate governance, leading to excessive remuneration packages and termination payments for executives at the expense of shareholders value, excessive risk taking, a loss of trust in the quality of financial statements and reluctance to invest in corporate stocks or bonds. All of this is particularly damaging as stock market capitalization rates in the Netherlands are high. With agreement on a new code for corporate governance (Tabaksblatt code) and new laws still under consideration which cover governance and financial reporting, the government hopes to align management interests more closely with those of shareholders and, as a result, to restore investors' confidence.

Corporate governance practices in the Netherlands were well below international standards

Shareholders have traditionally been very weak in the Dutch corporate system as a result of legal requirements and the (excessive) use of takeover defences. As a result, measures of the quality of corporate governance have been below those prevalent in a number of other countries (Deminor, 2004). Firms with more than 100 employees and a turnover greater than € 12.5[8] million are legally required to adopt the structural regime ("*structuurregeling*"). Group finance companies and international holding companies are exempted by law, whereas Dutch

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multi-nationals can choose to apply the regime voluntarily in their Dutch subsidiaries. Firms operating under the structural regime have a two-tier control mechanism with a management board and a supervisory board. The latter reduces the power of shareholders significantly. The supervisory board appoints its own members by co-optation (*i.e.* existing members select the new members) provided that the general assembly of shareholders or the mandatory works council does not oppose a certain candidate. Given that the supervisory board comprises "professional managers", which can and often do include past members of management, there is plenty of scope for executives to influence the nomination of supervisory board members, and hence indirectly the nomination of its own management board members. In firms not operating under the structural regime, shareholders retain their power to hire and fire executives and to appoint at least ⅔ of all non-executive board members of the firm. By law, the statutes cannot require more than a ⅔ majority for such (and other) decisions.

Shareholders rights are also curtailed by a common practice of Dutch firms to set up a trust office ("A*dministratiekantoor*") that holds the original shares and issues share certificates instead. A certificate holder has dividend rights, can freely trade certificates and can attend and has the right to speak at the general meeting of shareholders, but the voting rights are held by the trust office, which is not accountable to certificate holders and is usually on friendly terms with the management board of the company. In most cases, certificates have limited exchangeability, implying that the trust office grants proxies to certificate holders to exercise their voting rights only up to a certain percentage of the company's share capital. Certification can be used as a powerful instrument for protection against hostile takeovers, effectively capping the voting power of larger shareholders. Other takeover defence instruments that allow a company to deviate from the "one share – one vote" principle include the issuing of priority shares (granting voting privileges to priority shareholders) and preference shares to a friendly trust office or outside investor during a hostile takeover threat. The latter only pays 25 per cent of the amount up front but enjoys all the voting rights. When used, the dilution of voting power for existing shareholders can be up to 50 per cent.

In some countries, the presence of controlling or block shareholders has acted as a mechanism to ensure that the behaviour of managers does not conflict with the interests of the shareholders. As in most continental European countries, there is a predominance of large voting blocks in the Netherlands, but a recent study commissioned by the Ministry of Finance has indicated that insurance companies, banks and other financial institutions hardly ever exercise their albeit limited rights at the general assembly of shareholders of firms in which they hold large stakes. In sum, as a result of weak shareholder rights and the associated passivity of large institutional investors, managers of Dutch companies are poorly monitored and hence are able to pursue their own interests. This might explain why managers were successful in negotiating excessive remuneration packages

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leading to a misalignment in incentives *vis-à-vis* shareholders. The insider oriented incentive structure has been in turn a major factor behind the drive to expand firm size and to inflate share prices through excessive merger and acquisition activity, which has left many Dutch firms over-leveraged.

Corporate governance reforms should restore investors' confidence

Prompted by several corporate scandals and by the clear failure of an earlier voluntary code[9] a commission was set up in March 2003 to produce a new code of conduct with respect to corporate governance. The Tabaksblat code was finalized in December 2003 and entered into effect on 1 January of 2004. It is aimed at listed companies which have their headquarters in the Netherlandsand. It includes many recommendations based on international best practices with respect to the managerial and supervisory board, the general assembly of shareholders and financial reporting.

- For members of the managerial board, the code proposes a ceiling on termination payments equal to one year's salary, and requires that the height and structure of each executive's remuneration package be made public and that the variable component be linked to predetermined and measurable performance criteria. The supervisory board is given the responsibility to determine the remuneration of the individual members of the managerial board, taking into account the proposal by the remuneration committee and the remuneration policy adopted by the general assembly of shareholders. At the same time, the tax treatment of stock options will be modified such that stock options will always be taxed at the moment when they are exercised. If these checks and balances prove inefficient in keeping management remuneration packages in line with international practices, the government intends to make additional use of legal and fiscal instruments.
- The most important recommendations affecting the supervisory board are related to its independence from the managerial board, requiring that all but one non-executive director be independent according to a tightened set of criteria for independence. In addition, the duration of an appointment is limited to three terms of four years and the number of positions that a non-executive director may hold at any time to five (the position of "chairman" counts as two). If the supervisory board consists of more than four members, it must appoint from among its members an audit committee, a remuneration committee and a selection and appointments committee.
- In order to strengthen shareholders' rights, the code recommends that the participation of shareholders in decision-making be enhanced in several ways. First of all, measures to facilitate proxy voting are

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desirable. Secondly, the code stipulates that voting rights be returned unconditionally to certificate receipt holders upon request. Finally, institutional investors are required to disclose on their website their policy on the exercise of their voting rights for shares.

– The code also includes some recommendations with respect to the internal and external auditing functions.

Learning from past mistakes, compliance with the Tabaksblat code will be anchored by a requirement to "apply or explain" in a proposed new company law which has still to be approved by the upper house of parliament. In a first step, firms are required to develop a corporate governance structure and strategy outlining which best practices the firm intends to adhere to and which it seeks to deviate from with a motivation of the reasons in the latter case. The general assembly of shareholders approves the corporate governance structure and strategy and authorises the non-application of certain code provisions. In other words, an explanation for a deviation accepted by shareholders implies compliance with the code. In addition, firms are expected to include a chapter in their annual reports that describes the extent to which the agreed upon corporate governance strategy have been put into practice. The annual report is also subject to discussion by and approval of the general assembly of shareholders, which is responsible for monitoring compliance with the corporate governance strategy and can raise corporate governance issues (structure, strategy, observance, non-application of certain code provisions) annually and subject them to a vote if deemed desirable.

As a result of the new corporate governance code, a wider debate on the desirability of the structural regime has arisen and a decision about its future will be part of the new law. In its current form, the regime is not fully consistent with the principles of the Tabaksblat code because it limits the power of shareholders and makes it difficult to hold supervisory board members accountable due to the co-optation system. The proposed law will give shareholders (as well as the works council) more influence with respect to the nomination of supervisory board members,[10] and will abolish the co-optation system. An apparent inconsistency between the proposed law and the recommendations made by the Tabaksblat commission arises from the fact that the new law still allows for the requirement that certificate receipt holders are given unlimited and unconditional proxy rights to be breached under certain specified conditions.[11] Such restrictions should be removed.

Recent scandals in the Netherlands have also cast doubt on the accuracy of financial information given to boards and released to the markets. There is also legislation under way to strengthen supervision on financial reporting of listed companies and on accountancy organizations, as well as to introduce auditors' standards (covering issues such as integrity, independence, reliability and

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professional ability of both the individual accountants and their organizations). Supervision will be assigned to the Authority of Financial Markets (AFM), which can also initiate a procedure to review the quality of financial reporting. As part of this new task, the AFM will also verify whether a chapter describing the corporate governance structure and stating compliance with the Tabaksblat code is included in the annual report, and whether this description is consistent with the explanation of the code.

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Notes

1. The share of companies with no R&D expenditure rose from 60 per cent in the period 1996-98 to 66 per cent in 1998-2000.

2. ICT, sustainable systems innovation, and health, food, genetic and biotechnological research are the fields with the highest numbers of projects submitted. In all submissions, requests are made for a subsidy covering between two-fifths and three-fifths of the expected total project costs. A recent socio-economic evaluation found that the more promising projects belong to the fields ICT, microsystems and nanotechnology, and health, food and biotechnological research (CPB, 2004).

3. Especially in pharmaceuticals cross-border net outflows of R&D capital reflect internationalisation rather than a loss in competitiveness (Cornet and Rensman 2001).

4. Kox (2003, p. 13) finds that the maximum turnover per employee from client-specific services is achieved in firms with 20-100 employees, whereas for standardised services the most productive firm scale is 100-200 employees.

5. The current system of aid to students consists of a basic grant, a means-tested supplementary grant and an option to take up an interest-bearing loan (repayable at an income-dependent rate within fifteen years after leaving university), providing for about € 200 each of them per month per student on average (MinOCW 2003, p. 94).

6. A side effect of growing competitive pressures would be that universities would have stronger incentives to make more commercial use of their inventions and the public facilities set up to advise them on this issue.

7. General grants are inefficient because they are also paid to those who do not need them and are generally financed from distortionary taxation. Moreover, they often imply income redistribution from the average taxpayer to students who either come from wealthy families or will themselves earn high incomes in the future.

8. The new law (still under discussion) proposes to increase the capital requirement to € 16 million.

9. In 1997, the Peters commission drafted a first code of conduct with respect to corporate governance, but the choice to comply with the code was entirely voluntarily. It was found that less than half of the listed companies complied with the Peters committee recommendation that they "comply or explain".

10. The supervisory board has the binding right to propose candidates for nomination on the board, although it can only freely choose ⅔ of the nominees, the other ⅓ being nominated by the work council. The general meeting of shareholders must approve all nominations individually with a simple majority of the votes cast representing at least

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⅓ of the issued share capital. If this proportion of the capital was not represented at the meeting, a new meeting could be convened at which the nomination of a candidate may be overruled regardless of the proportion of capital represented.

11. These specific conditions are defined as: *a*) the presence or threat of a hostile takeover bid; *b*) the presence of two or more colluding certificate receipt holders representing at least 25 per cent of equity capital; or *c*) a perception by the trust office that the granting of proxy rights to a specific certificate receipt holder requesting it would conflict with the interests of the company (Article BW 118).

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7. Sustainable development in the Netherlands

There is a growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental and social outcomes both domestically and on a global basis. This section looks at three specific issues of sustainable development that are of particular importance for the Netherlands: *climate change*, *water pollution*, and *sustainable use of natural resources*. In each case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies in that area. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (see Box 7.1).

Climate change

Main issues

Given its potential to raise the sea level, climate change is seen as a major risk for future wellbeing in the Netherlands. As part of the EU's burden-sharing agreement, related to the Kyoto Protocol, the Netherlands accepted a target of reducing greenhouse gas emissions of 6 per cent between 1990 and 2008-12. The slightly lower target than the 8 per cent for the EU as a whole reflects the substantial shift from more carbon-intensive fuels to natural gas in electricity generation and household consumption that had already occurred before the 1990 base year so raising the marginal abatement costs for achieving additional abatement domestically above those of other countries that have not yet made the shift to natural gas. In this light, the main issues for the authorities are designing policies that minimise the cost to the economy of meeting the abatement target.

Performance

During the 1990s and reflecting a rapid expansion of the economy, GHG emissions grew by 4 per cent over the decade, more rapidly than in the rest of the EU (Figure 7.1, Table 7.1). However, emissions of carbon dioxide rose by as much as 8 per cent, principally driven by rising demand for transport and strong use of electricity (Table 7.2). Partially offsetting this, emissions of methane (which

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Box 7.1. The integration of policies across sustainable development areas*

Since 1989, National Environmental Policy Plans have provided the backbone to environmental policy, and have more recently incorporated long-term goals for achieving sustainable development. A specific National Strategy for Sustainable Development was also formulated in the run up to the UN Summit on Sustainable Development in 2002. These policies have increasingly targeted core domestic issues for the Netherlands such as water and energy use, health, the impacts of agriculture and the preservation of biodiversity. Sectoral programmes as part of the national action plans are based on a "transition" to a new state when addressing these concerns. The "transition" recognises the importance of promoting innovations and the involvement of stakeholders in solving major environmental problems. On the international front, the Dutch authorities base policies on the overarching goals of sustainable production and consumption and poverty eradication.

Policy integration in the Netherlands is promoted through widespread consultation and willingness to compromise in policy formation, though there are no formal structures to ensure policy integration across the pillars of sustainable development. However, where tensions are potentially greatest, legislation dictates that different parties consult with one another formally. The implementation of centrally agreed policies has on occasion been complicated at the local level, though more recently new structures, such as the Commission for Integrated Water Management, facilitate better co-ordination across levels of government (Bouder and Fink, 2002). A particular aspect of environmental related policy has been the heavy reliance on voluntary agreements, which have been reached with sectors for a large number of different policy areas. As a result the authorities have developed expertise in designing these agreements, recognizing the need for a credible threat of sanction to ensure compliance.

When evaluating policy options, cost-benefit analysis is not used systematically, though environmental impact assessments are undertaken for all policies and projects. Only the Ministry of Transportation, Infrastructure, and Water Management undertakes cost-benefit analysis for all policies and projects. However, parliamentarians are increasingly demanding that such analysis be prepared for proposed policies.

* The sections in this report dealing with climate change, water pollution, and sustainable use of natural resources are inputs into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council decision in May 2001.

accounted for one-eighth of GHG emissions in 1990) fell by almost one-quarter, mainly as a result of the introduction of emission capture technology at landfill sites and to a lesser extent to cuts in livestock numbers as part of water pollution policy (see below).

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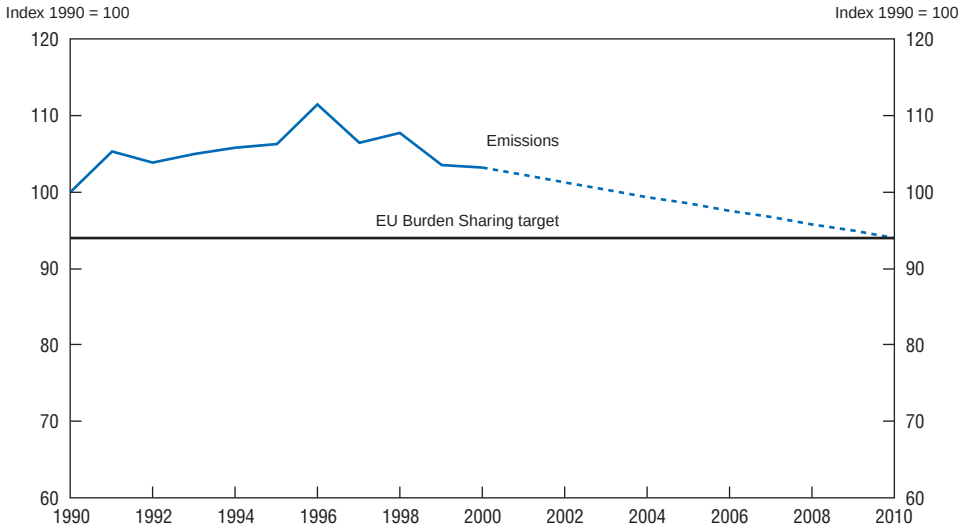
Figure 7.1. **Greenhouse gas emissions**



Source: UNFCCC.

Policies

In light of anticipated high costs of achieving abatement domestically, climate change policy adopted in 1999 explicitly set the target of achieving half of the necessary abatement with international measures, such as joint implementation and clean development mechanisms, and the purchase of emission permits.[1] The climate-change package also identified reserve polices, which will be introduced if the domestic abatement target is in danger of being missed. Periodic reviews of progress will allow some flexibility in implementation of the total package of measures.

An important policy included in the climate change package was a sharp increase in the tax on the use of electricity and natural gas (the regulatory energy tax), although additional revenues are recycled as a fixed annual rebate and support for energy efficiency policies, in addition to the reduction of direct taxes that accompanied the introduction of this tax in 1996. However, the tax rate on natural gas and electricity is inversely related to consumption with small consumers facing tax rates 130 and 19 times greater than large electricity and gas users, respectively (Table 7.3). Furthermore, the tax rates for smaller consumers are around 10 times greater than the likely price of an international emission permit. This was accepted as the revenues were to be recycled and in addition to

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Table 7.1. **Main indicators: climate change**

Indicators of greenhouse gas (GHG) emission intensity, grammes of CO_2 equivalent per $PPP of GDP, in 1995 prices

	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions	Total GHG emissions	CO_2 emissions, electricity	CO_2 emissions, transport	Other GHG emissions
	Level, 1999				Average annual percentage change 1990-99			
Australia	1 053	370	155	528	–2.07	–0.21	–1.93	–3.24
Austria	419	72	91	256	–1.87	–2.75	–0.52	–2.06
Belgium	617	97	101	419	–1.36	–2.12	0.16	–1.52
Canada	893	151	193	549	–0.98	–0.12	–0.36	–1.41
Czech Republic	1 058	457	88	513	–3.05	2.55	5.53	–6.93
Denmark	549	194	94	261	–1.64	–1.43	–1.49	–1.85
Finland	652	181	105	366	–1.88	–0.02	–1.29	–2.83
France	416	32	103	280	–1.69	–2.04	0.16	–2.26
Germany	536	169	96	271	–4.00	–3.86	–0.57	–5.05
Greece	813	275	130	408	–0.24	0.07	0.74	–0.73
Hungary	786	250	84	453	–2.33	1.44	0.38	–3.74
Iceland	395	4	88	303	–1.28	0.00	–2.31	0.81
Ireland	694	165	103	426	–4.27	–2.41	0.79	–5.75
Italy	439	105	92	242	–1.05	–0.82	0.37	–1.64
Japan	432	130	82	221	–0.30	–0.03	1.24	–0.99
Luxembourg	344	6	242	97	–11.46	–30.20	–0.45	–18.81
Netherlands	**573**	**138**	**82**	**352**	**–2.38**	**–1.03**	**–0.94**	**–3.15**
New Zealand	1 096	92	175	828	–2.28	4.58	0.65	–3.32
Norway	487	4	113	369	–2.54	1.31	–1.53	–2.87
Poland	1 195	481	90	624	–4.96	–6.63	0.50	–4.12
Portugal	540	149	106	285	0.41	2.58	3.37	–1.39
Slovakia	957	200	76	680	–4.47	–1.21	3.13	–5.78
Spain	537	127	130	280	0.41	1.12	1.28	–0.26
Sweden	358	41	112	204	–1.55	0.07	–0.65	–2.30
Switzerland	276	3	79	195	–0.62	–1.96	–0.28	–0.73
United Kingdom	526	132	108	287	–3.66	–5.30	–1.38	–3.61
United States	792	278	196	318	–1.89	–0.60	–1.18	–3.28
OECD total	649	196	140	312	–1.80	–0.98	–0.38	–2.83
EU	506	120	103	283	–2.36	–2.60	–0.16	–2.95

Source: Greenhouse gas emissions: national submissions to the UNFCCC and national publications. Carbon dioxide emissions for electricity and transport: IEA (2001). GDP: OECD, SNA database.

addressing climate change the tax is used to raise general revenues and promote energy saving. However, the deliberate introduction of different tax rates for different energy users means that the policy is inefficient in attaining its climate change objectives. From the beginning of 2003 the regulatory energy tax has also been applied (at a lower rate) to renewable energy sources, which had previously been exempt from this tax.

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Table 7.2. **GHG emissions and sector indicators**

	Total GHG emissions		CO_2 emissions per kWh electricity	Manufacturing CO_2 emissions per unit of output	Residential CO_2 emissions per unit of private consumption	Road transport CO_2 emissions per vehicle-kilometre	Electricity use per unit of GDP	Industrial output per unit of GDP
	Level million tonnes CO_2 equivalent 2000	Annual average percentage change						
		1990-2000	1990-2000	1990-2000[1]	1990-2000[2]	1990-99	1990-2000	1990-2000[1]
Australia	502	1.7	0.1	–0.7	–1.1	–0.8	–0.6	–1.6
Austria	80	0.3	–2.2	–1.7	–2.4	–1.8	0.1	0.3
Belgium	152	0.7	–1.6	–0.2	–1.1	–0.8	–0.2	–0.5
Canada	726	1.8	1.0	–2.6	–2.0	–0.4	–0.5	1.0
Czech Republic	147	–2.7	–0.2	–7.7	–15.4	4.1	2.8	0.2
Denmark	69	–0.1	–3.4	–1.7	–4.0	–0.2	0.7	–0.6
Finland	74	–0.4	–0.9	–7.2	–7.0	–1.4	0.7	3.2
France	550	–0.2	–2.8	–2.4	–1.7	–0.5	0.6	0.3
Germany	991	–2.1	–1.3	–1.1	–4.0	0.7	–2.1	–1.8
Greece	130	2.2	–2.0	–0.4	2.6	–4.4	2.1	–1.7
Hungary	84	–1.8	–1.3	–12.7	5.4	–3.3	0.2	5.3
Iceland	3	0.7	4.9	n.a.	–7.7	0.3	2.6	n.a.
Ireland	67	2.2	–1.3	n.a.	–6.2	4.8	–1.8	n.a.
Italy	547	0.5	–1.2	–2.0	–1.4	–1.1	0.8	–0.2
Japan	1 386	1.1	–0.5	–0.8	0.1	0.3	1.0	–0.2
Luxembourg	6	–7.8	–24.5	–13.0	–2.3	3.8	–3.4	–2.2
Netherlands	**218**	**0.4**	**–2.8**	**–1.8**	**–2.9**	**–0.1**	**1.8**	**–0.6**
New Zealand	77	0.5	3.7	n.a.	–3.2	–2.1	–0.8	n.a.
Norway	55	0.6	0.5	0.8	–8.8	0.5	–1.9	–2.7
Poland	386	–3.7	0.5	–5.8	–6.7	–6.4	6.9	4.4
Portugal	85	2.7	–0.9	0.7	–1.0	–0.4	1.8	–0.6
Slovakia	49	–4.0	1.0	–4.7	–7.4	2.3	0.2	2.4
Spain	386	3.0	–0.2	–0.6	0.4	–0.3	1.2	–0.1
Sweden	69	–0.2	–2.3	–4.7	–4.1	–1.4	–0.5	2.5
Switzerland	53	–0.1	1.8	n.a.	–1.7	–0.3	1.1	n.a.
United Kingdom	649	–1.3	–4.1	–2.1	–1.9	–0.5	–0.7	–1.6
United States	7 001	1.3	–0.4	–4.0	–1.9	–0.2	–0.4	0.4

Table 7.2. **GHG emissions and sector indicators** (*cont.*)

	Total GHG emissions		CO_2 emissions per kWh electricity	Manufacturing CO_2 emissions per unit of output	Residential CO_2 emissions per unit of private consumption	Road transport CO_2 emissions per vehicle-kilometre	Electricity use per unit of GDP	Industrial output per unit of GDP
	Level million tonnes CO_2 equivalent 2000	Annual average percentage change						
		1990-2000	1990-2000	1990-2000[1]	1990-2000[2]	1990-99	1990-2000	1990-2000[1]
Total of above OECD countries	14 543	0.6	–0.7	–2.2	–1.8	–0.3	–0.2	0.0
OECD excluding US	7 542	–0.1	–1.0	–1.5	–1.7	–0.4	0.0	–0.3
EU countries	4 073	–0.3	–2.1	–1.8	–2.4	–0.4	–0.2	–0.7
	Total CO_2 emissions							
Other OECD countries								
Korea	444	6.5	–0.7	–2.3	–7.9	–4.3	5.2	2.0
Mexico	369	2.2	1.0	–6.9	–2.7	1.5	1.7	0.9
Turkey	206	4.1	0.2	1.2	–2.6	–5.0	4.3	0.4

1. 1991-2000 for Germany and Hungary; 1992-2000 for Poland; 1993-2000 for Slovakia; 1990-99 for Portugal; no data for Iceland, Ireland, New Zealand and Switzerland.
2. 1991-98 for Czech Republic; 1993-2000 for Slovakia.

Source: GHG National submissions to UNFCCC, national sources and UNFCCC; carbon dioxide data, IEA; industrial production, private consumption, OECD.

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Table 7.3. **The regulatory energy tax**

Electricity				Natural gas		
kWh	Tax rate			Cubic metres	Tax rate	
	€ per kWh	€ per tonne of CO_2			€ per cubic metre	€ per tonne of CO_2
		Gas displacement	Coal displacement			
< 10 000	0.0654	176	89	< 5 000	0.1429	74
10 000-50 000	0.0212	57	29	5 000-170 000	0.0727	37
50 000-10 mn	0.0065	18	9	170 000-1 mn	0.0113	6
> 10 mn	0.0005	1	1	> 1 mn	0.0075	4

Source: Ministry of Finance (2003).

Part of the authorities' rationale to avoid taxing large energy users at the same rate was the authorities' concern about the international competitiveness of energy intensive enterprises and because they were already subject to separate voluntary agreements on energy efficiency.[2] Nearly all energy-intensive enterprises in the Netherlands have renewed such agreements with the national authorities (so-called "benchmark covenants" and "long term agreements"). In return for signing these covenants and agreements, enterprises would not face additional, specific, national measures with respect to climate change. The target stipulated in the covenants, which was proposed by the energy intensive sectors themselves, is that energy efficiency should be within 10 per cent of companies identified as international best practice. Most enterprises are already meeting this standard, suggesting that the agreements require limited additional abatement effort from a business as usual perspective. However, some estimates suggest that the benchmark covenants in comparison with requirements set in previous energy efficiency voluntary agreements could contribute to additional abatement of between 5 to 15 per cent (Phylipsen *et al.*, 2002). In an attempt to address the shortcoming of using a downstream energy tax to promote abatement efforts in the power generation sector, the government has also agreed a voluntary agreement with coal-fired power stations. The coal input for power plants co-firing coal and biomass gains a tax exemption, while under certain conditions biomass is taxed as a renewable.

The authorities considered introducing a domestic emission trading system (CO_2 Trading Commission, 2002), but suspended this out of concern that the costs to the economy would be excessive if implemented unilaterally (Mulder, 2002). However, the introduction of an EU-wide emission trading scheme will mandate that many firms covered by the benchmark covenants will be included in the EU scheme. The authorities have chosen to make the allocation of allowances

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in line with the current benchmark covenants and long term agreements (SER, 2003). Such an allocation is based on the actual energy efficiency performance of the individual enterprise. However, this will grant these enterprises a windfall scarcity rent. This arises as the proposed EU emission trading scheme grandfathers at least 90 per cent of allowances, whereas estimates suggest that only a small share of allowances need to be grandfathered to compensate for loss of shareholder value.[3]

Like many other European countries and also in response to indicative targets at the EU level, the government has set separate targets for renewable energy. These are to increase the supply of renewable energy to 9 per cent in 2010 and 17 per cent of domestic electricity consumption by 2020. This target was met in 2002, when 13 per cent of electricity consumption came from renewable sources with 75 per cent being imported (van Damme, 2003). Such a policy success was bought at the cost of tax breaks initially, which were later combined with production subsidies.[4]

In mid 2003, the renewable energy market support system was reformed, shifting support towards domestic renewable energy production. Renewable energy producers now receive three potential revenue streams. Their output is sold to the wholesale market at normal prices, domestic producers receive a feed-in tariff that varies according to the type of technology used to produce the electricity and finally they receive a tradable green production certificate. Once the marketing company[5] has purchased sufficient production certificates to match the outstanding amount of its green supply contracts sold to retail consumers, it can reclaim approximately half of the specific tax on electricity. This effectively caps the price of a certificate at the amount of the tax refund. The sum of the feed-in tariff and electricity tax exemption for the domestic producers reaches a maximum of € 0.078 per kWh for expensive renewable energies (wind and photovoltaic). For these sources, the reform means that there was only a slight decrease from the pre-reform level of support, with the current level equivalent to € 195 per avoided tonne of carbon dioxide emissions when the renewable energy is produced by wind and biomass, falling to € 188 per tonne once allowance is made for reduced nitrogen dioxide emissions.[6]

The overall subsidy for renewable energy is set on the basis of the extra costs of different types of renewable energy sources and is thus inefficient as marginal abatement costs vary according to the energy source. The official calculations estimate the overall premium (the sum of the feed-in tariff and the electricity tax rebate) necessary for producers to obtain a market rate of return on their investment (van Sambeek, 2003). In practice, there is evidence that the price has appeared to be sufficient in the past to generate above-normal returns and that competition has ensured that the subsidy has been split between producers, distributors, consumers and the transmission companies. Three factors point to

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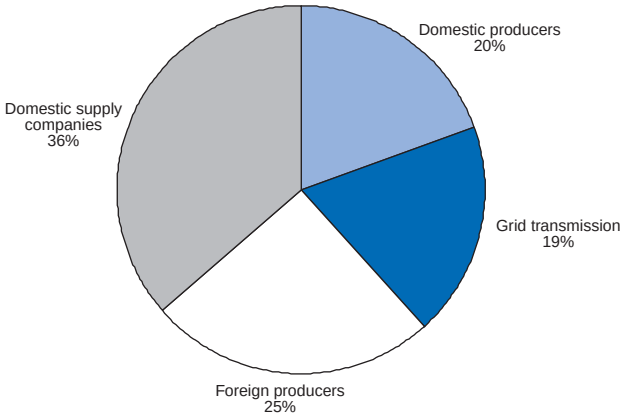
this conclusion, first, the price for imported renewable energy certificates was only € 0.01 per kWh in 2002 (van Damme, 2003) – one-eigth the total subsidy then available for green energy[7] – and well below the estimate of 6 cents per kWh used in studies to evaluate possible support systems (Junginger, 2004). Secondly, the demand for imported green electricity has outstripped the capacity on the interconnection links between the Netherlands and Germany, from where most renewable power was imported. This has driven up the price of capacity on this link by € 0.015 per kWh, allowing the grid company to capture a significant part of the renewable energy subsidy (Market Surveillance Committee, 2002). However, under the new regime imported electricity is treated less generously and will not be eligible for support from the beginning of 2005. Thirdly, the open nature of the electricity market in the Netherlands has meant that consumers have been free to switch to companies that market green energy and which have been offering to pass-on part of the reduction in the electricity tax to the consumers. This factor has only come into play in 2003 as supply companies tried to expand the number of green contracts. Overall in 2002, more than half of the subsidy was paid to supply and transmission companies rather than to producers (Figure 7.2). Moreover, in the future when the European Trading System for Greenhouse gases is functioning, renewable producers will receive part of the marginal value of the allowances to the extent that the wholesale price of electricity is pushed up.

While the reforms introduced in 2003 reduced the level of support available for some sources of renewable energy and cut the tax rebate available for imported renewable electricity, the present levels of support for renewable

Figure 7.2. **The estimated recipients of Dutch renewable energy subsidies in 2002**

Percentage of total



Source: Van Damme (2003).

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energy remain substantial and lead to high costs both for the government (through the tax rebate) and the consumer of ordinary electricity (through the cross-subsidy paid to finance the feed-in premium). Indeed, there is evidence from renewable markets that a lower premium is sufficient to create significant supply. For example, the price of renewable certificates in Sweden (the source of a significant amount of renewable energy imported to the Netherlands) has been only 2 cents per kilowatt hour since the inception of the market in May 2003. Restricting the premium in the Netherlands to this level would reduce costs by almost 0.2 per cent of GDP. However, other factors such as reducing air pollution and ensuring security of supply should be considered in determining the level of support. In this context, it would likely be cheaper to reduce carbon emissions by purchasing permits on the European greenhouse gas market than by purchasing renewable energy, even after allowing for the ancillary benefits such as reduced air pollution.[8]

Targets for the expansion of combined heat and power (CHP) have been particularly ambitious, setting as an objective a quadrupling of capacity between 1990 and 2010, which would provide 30 per cent of power generation (IEA, 2000). Support for this sector was offered through a refund on the energy tax (Newbery *et al.*, 2002). Following the liberalisation of the electricity market coupled with rising natural gas prices, CHP installations became less competitive with conventional sources of electricity providing base load power (such as coal). As a result, CHP only operates at full capacity during peak hours, when the marginal fuel being replaced is likely to be natural gas, and thus the marginal abatement costs will be significantly higher than if the marginal fuel being replaced were brown coal.

While the agriculture sector represents an important source on GHG emissions, direct policy initiatives in this area have been limited but substantial falls in emissions have occurred as the result of other policies in this sector. In particular polices to restrict fertiliser input have led to a lower nitrogen content of the soil and hence lower nitrous oxide emissions. At the same time, the need for a manure disposal policy has led to fall in the attractiveness of livestock farming. The number of animals has fallen and with the extent of methane emissions.[9] Some scope exists for changing the feedstock mix for cattle and breeding more efficient cattle but the possible benefits from such policies to farmers are already substantial (Bates, 2001) and so specific polices have not been introduced. Voluntary covenants have been signed with the horticulture sector in order to reduce energy in that sector.

As in all other OECD countries, abating greenhouse gas emissions in the transport sector has proven to be difficult. Agreements forged at the European level will result the gradual renewal of vehicles with more fuel efficient models, leading to some decoupling of emissions from transport use, but at a high resource cost for the economy. At the national level some opportunities are

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available to address the externalities associated with vehicle transportation, although the implicit tax on carbon levied by the vehicle fuel tax and excise duties is already substantial.[10] The government is studying converting all fixed automobile taxes into distance-based taxes. Such a measure would increase the marginal cost of transport and might reduce vehicle emissions by between 0.7 and 1.5 million tonnes (equivalent to between 0.3 and 0.7 per cent of total GHG emissions). Such a tax will have the advantage of targeting transport-related externalities more efficiently than a flat-rate tax (VROM, 2002).

Conclusions

Climate change policy has imposed very high costs on smaller energy consumers, principally due to the high rates of the energy tax, though the revenues of this tax are recycled. At the same time, large energy intensive users have been shielded from the full impact of greenhouse gas mitigation measures, though they have participated in voluntary agreements. This arose because the government felt that it was unable to offset the impact of the energy tax by a reduction of other taxes for energy intensive companies. The introduction of the EU emission trading scheme will require the participation of these large energy users. This could serve as an opportunity to rebalance climate change policies to equalise abatement costs across sectors. Such rebalancing is particularly important for small energy consumers who have borne a high cost due to the elevated regulatory energy tax that they face. This tax should be replaced by an explicit carbon tax for the sectors not covered by the EU emission trading scheme, with the tax rate consistent with the expected price of an emission permit. This would reduce revenues from carbon taxes and may require offsetting increases in revenue from other sources. In any case, the support for renewable energy should be further reduced and the method of support reformed. A system based just on renewable certificates, which lets the market determine the appropriate premium for producers, would likely reduce expenditure by three-quarters given the price of such certificates on major exporting markets and would increase efficiency by unifying marginal abatement costs across renewable energy sources. The authorities should also investigate the feasibility of widening the proposal for distance-based charging for the transportation sector to cover all infrastructure costs and externalities not directly linked to fuel consumption, and consider the implications of such a policy for the harmonisation of the vehicle fuel and excise tax with the proposed carbon tax. Climate change policy has pragmatically included international measures as means to attain a large share of the necessary abatement needed to meet the EU Burden-Sharing target. As more evidence emerges from the planned reviews of the actual costs of climate change mitigation measures, the authorities should reallocate resources to the most cost-effective measures, irrespective of whether this approach breaches the ambition to achieve half of the necessary abatement domestically.

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Water pollution

Main issues

The pollution of water resources is one of the major environmental challenges in the Netherlands. While policy has developed with respect to point source polluters since the 1980s, it is only more recently that the authorities have started to address the intense pressure from agriculture, which is the largest domestic source of nutrient pollution. The main issue in this area is ensuring that water quality is improved in a cost efficient way.

Performance

Nutrient pollution presents an important environmental challenge to the Dutch authorities. A large share (70 per cent) of the nutrients in Dutch waters arrives in the major watercourses that traverse the country. Of domestic sources agriculture is the predominant source (15 per cent of total nutrients), though pollution from households and industry are also significant (10 and 5 per cent, respectively). Due to the very intensive livestock farming practised in the Netherlands, the production of manure is a particular problem contributing to very high concentrations of nitrates in water bodies. In the second half of the 1990s, the production of manure fell by one-sixth and manure was spread over a larger area of farmland than before. However, emissions from agricultural land to surface water are estimated to have remained largely unchanged at under 86 million kilograms of nitrogen a year, though accumulation in groundwater and soil have fallen. The nitrate content of water arriving in the Netherlands fell sharply, contributing to a significant reduction in nitrate concentrations in some rivers (Table 7.4). Despite this improvement, such concentrations in rivers and particularly groundwater in sandy areas remain above those seen in most EU member states. Indeed, most excess nitrogen appears to be transmitted to aquifers. In consequence of this pressure, the Netherlands applied its nitrates Action Programme to its whole territory. In the latter part of the 1990s, the annual external costs of eutrophication associated with nitrate emissions was estimated to be slightly over € 600 million (0.2 per cent of GDP) (Howarth, Pearce *et al.*, 2001).

Policies

Due to the intensive nature of agriculture, the reductions of nitrate and phosphorus pollution are key challenges for the Dutch authorities. The EC Nitrate Directive is particularly demanding for the Netherlands, requiring marked reductions in nitrate concentrations in groundwater, while the implementation of the Framework Water Directive and meeting the internationally-agreed targets of reducing pollution loads in the North Sea will also require additional effort to reduce pollution to surface water (Table 7.5).[11] Water quality management

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Table 7.4. **Performance indicators: water pollution**
Selected rivers

	Biochemical oxygen demand		Nitrates		Total phosphorus	
	mg O_2/litre		mg N/litre		mg P/litre	
	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years
Australia	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Austria	2.2	2.2	1.2	1.3	0.2	0.1
Belgium	6.6	n.a.	n.a.	n.a.	0.7	n.a.
Canada	n.a.	n.a.	0.1	0.1	0.1	0.0
Czech Republic	8.5	4.6	5.1	3.5	n.a.	0.3
Denmark	3.9	2.0	3.5	2.6	0.3	0.1
Finland	n.a.	n.a.	0.2	0.3	0.0	0.0
France	5.2	3.2	2.8	3.1	0.4	0.4
Germany	3.6	2.4	3.5	3.3	0.5	0.2
Greece	n.a.	n.a.	1.3	1.5	0.4	0.4
Hungary	4.4	2.9	2.2	1.6	0.4	0.3
Iceland	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Ireland	1.7	1.8	2.2	2.9	0.1	0.1
Italy	n.a.	n.a.	2.0	2.1	0.3	0.2
Japan	2.3	1.4	n.a.	n.a.	n.a.	n.a.
Korea	n.a.	2.8	n.a.	2.6	n.a.	0.1
Luxembourg	3.8	2.7	4.1	4.1	0.6	0.4
Mexico	3.7	15.9	1.9	0.6	n.a.	0.1
Netherlands	**2.8**	**3.1**	**4.4**	**3.3**	**0.5**	**0.2**
New Zealand	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Norway	n.a.	n.a.	0.3	0.3	0.0	0.0
Poland	5.0	4.3	1.8	1.9	0.3	0.3
Portugal	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Slovak Republic	5.1	3.3	2.0	2.2	0.1	0.2
Spain	4.7	3.6	1.7	3.1	0.6	0.2
Sweden	n.a.	n.a.	0.4	0.5	0.0	0.1
Switzerland	n.a.	n.a.	1.5	1.5	0.1	0.1
Turkey	2.1	2.7	1.3	1.0	0.3	0.2
United Kingdom	3.4	2.9	4.6	5.2	0.8	0.9
United States	1.8	1.6	1.1	n.a.	0.2	0.1

Source: OECD.

involves a mixture of government bodies and the very long-standing Water Boards.[12] The national government determines standards that must be adhered to by provincial authorities and the Water Boards who are responsible for groundwater and surface water, respectively.[13] Municipal authorities are generally responsible for sewerage services.

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Table 7.5. **Nitrogen and phosphorus pollution and targets**

	Actual	National target		International target
	2000	2006	Long-term	
Groundwater concentrations, mg per litre				
Nitrogen	130[1]			50[2]
Surface water concentrations, mg per litre				
Nitrogen	3.1-3.7	2.2	1	
Phosphorus	0.11-0.22	0.15	0.05	
Pollution load into the North Sea, thousand tonnes				
Nitrogen	114 414			82 900[3]
Phosphorus	8 875			15 308[3]

1. The nitrogen concentration is for groundwater under agricultural holdings in sandy areas. Concentrations in other soil types generally meet the EU target.
2. Nitrates Directive.
3. OSPAR agreement, by 2005.

Source: OECD (2003), RIVM (2002).

The system for controlling point source water pollution involves both licensing and a pollution discharge tax. Point source polluters require a permit to operate. While licences generally stipulate best available technology, the financial consequences are considered if the pollutant involved is not regarded as a serious problem. A discharge fee is applied for substances that consume oxygen and for heavy metals. Rates vary across the country, also depending on whether the water body is under the authority of a Water Board or the state. In the case of Water Boards the pollution levy is set close to the rate that covers the operating costs of meeting water quality targets, principally incurred by wastewater treatment plants (Dutch Association of Water Boards, 2003). The pollution charge on industry has contributed to a two-thirds reduction in pollution loads from industry since its introduction in the 1970s. Notwithstanding the impact of the pollution levy, the implementation of the EC directive on wastewater treatment will require additional investment in treatment plants in order to conform to nitrate removal standards. It would, however, be much cheaper to implement localised preventative measures in the Dutch agricultural sector where groundwater is used for the public supply as these are estimated to cost only 10 per cent of equivalent drinking water denitrification measures (European Commission, 2002).

In addressing the problems created by agriculture, principally manure, the authorities have introduced various policy measures (Ministry of Agriculture, 2001). First, the government has set a number of standards for the application of nutrients in manure to agricultural land. Second, measures have been used to reduce the number of animals. The Pig Restriction Act of 1997, for example, mandated a 25 per cent reduction of pig numbers (later reduced to 10 per cent) following over a decade of negotiation by the government trying to reach

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voluntary agreements with farmers (Bouder and Fink, 2002). The authorities also contribute to the reduction in the number of pigs by buying up production rights and "creaming off" a proportion of these rights when they are traded among farmers. Third, various administrative restrictions have been imposed on farmers, including storage requirements and limiting application of manure to times of the year when the risk of run-off is lower.

A major policy innovation in 1998 was the establishment of a mineral accounting system (MINAS) and a manure transfer system.[14] MINAS requires farmers to account for the nitrogen and phosphorus content of inputs and outputs, so creating nitrogen and phosphorus balances for their farm. Before the introduction of this system, the nutrient surpluses in the Netherlands were extremely high by international standards reflecting both cheap animal feed imports and high prices in the dairy sector making intensive dairy operations profitable (Table 7.6). The authorities set permitted nutrient surpluses, which are being gradually reduced over time with the aim of complying with the water quality standards contained in the Nitrates Directive by 2008. Farmers are required to ensure that nitrogen balances are adhered to, inducing farmers to implement measures on the farm to reduce the nitrogen and phosphorus surpluses and if that is insufficient to contract with farmers with lower nutrient surpluses (typically arable farmers) to dispose of excess manure. In the latter case, farmers must have proof of a contract with a registered manure transport company. Surpluses are also differentiated by soil type, such that permitted surpluses are lower on sandy soils where water pollution is of greater concern (Verschuur *et al.*, 2003). If the farmer's actual surplus is above this standard, a tax is levied on the excess. While the tax has been prohibitive for many farmers, simulations suggest that the tax rate on excess nitrogen surpluses has been too low in relation to expected manure disposal costs for some of the most intensive farms (Berentsen and Tiessink, 2003), though preliminary evaluation of an increase in the tax rate in 2002 suggests that the tax is now sufficient to alter practice on these farms.

Table 7.6. **Dairy farm nutrient balances**
1994-97

	New Zealand	France	**The Netherlands**
Cows per hectare	2.6	1.4	1.64
Kg of nitrogen per hectare			
Nitrogen input	167	267	486
of which: Fertilisers	50	200	291
Nitrogen output	61	61	78
Nitrogen balance	106	206	407
Nitrate leaching	30	60	124

Source: Legard *et al.*, 2000.

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The costs of implementing MINAS, the manure trading system, and also providing subsidies for reductions in livestock numbers are substantial. Administration costs for officials implementing the scheme are estimated to amount to over € 24 million, subsidies amounted to € 56 million in 2001, the administrative burden to farmers in complying with MINAS was over € 160 million, while the costs to farmers for transporting excess manure exceeded € 200 million. These costs are likely to be significantly higher than the benefits resulting from the resulting emission reductions from the agricultural sector. While this scheme is expected to reduce pollution loads in the most severely affected regions, where the most intensive livestock farming is practiced, the application of manure in other areas has increased. Due to the soil characteristics in these areas being more susceptible to run-off, there is a danger that pollution loads in surface water in these areas may rise.

Conclusions

The mineral balancing system has introduced economic incentives for farmers to reduce the application of excess nitrogen and phosphorus to the soil. The operation of the system is expensive, and the authorities should consider means to simplify and minimise administrative burdens. While soil characteristics are taken into consideration in setting the targeted surpluses, the water boards or provincial authorities may require stricter balances in some areas in order to meet the water quality standards they are responsible for. The authorities should examine as intended the implementation of MINAS for reforms due in 2006 to see whether there is a case for modifying the tax rates on nutrient surpluses to those that are consistent with meeting water quality objectives.

Sustainable use of natural resources

Main issues

Due to the low lying nature of much of the country, saltwater intrusion, which can irreversibly damage land, is a serious threat. Sustainable management of fresh water resources is important in mitigating this threat. Apart from this critical challenge, the Netherlands is endowed with substantial hydrocarbon reserves, which require a balanced policy approach to ensure that current exploitation is not to the detriment of future generations' welfare.

Performance

The intensity of water use in the Netherlands is well below the OECD average (Table 7.7), though this does not prevent over-exploitation of groundwater. Part of this over-exploitation arises from agricultural irrigation, which is among the most extensive in the world, covering almost 60 per cent of agricultural land. As a result of past uncontrolled water abstraction from aquifers, groundwater depletion and saltwater intrusion present significant threats to a large part of the country that

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Table 7.7. **Main indicators: water withdrawal and land degradation**

	Water withdrawals		Irrigation coverage, per cent of farmland	Land degradation	
	Per cent of available resources	Relative to GDP (1980 = 100)		Per cent of total area	Per cent due to agriculture
Australia	4.3	n.a.	5	3	59
Austria	2.7	72	0	38	83
Belgium	42.5	n.a.	5	65	37
Canada	1.7	80	3	2	100
Czech Republic	15.6	57	2	100	93
Denmark	15.7	45	17	0	0
Finland	2.2	41	2	11	100
France	23.9	n.a.	10	9	0
Germany	24.4	69	3	31	67
Greece	12.1	n.a.	36	48	5
Hungary	5.0	103	4	65	45
Iceland	0.1	93	n.a.	52	0
Ireland	2.6	n.a.	n.a.	0	0
Italy	32.2	n.a.	24	28	12
Japan	20.8	61	53	0	0
Korea	35.6	41	58	42	41
Luxembourg	3.4	n.a.	n.a.	51	0
Mexico	17.4	n.a.	24	44	51
Netherlands	**4.9**	**n.a.**	**56**	**5**	**96**
New Zealand	0.6	n.a.	6	7	30
Norway	0.7	n.a.	n.a.	3	0
Poland	18.7	61	1	83	18
Portugal	15.0	n.a.	24	21	100
Slovakia	1.4	44	6		
Spain	36.8	65	19	38	16
Sweden	1.5	48	n.a.	25	29
Switzerland	4.9	78	5	0	0
Turkey	15.2	100	14	99	4
United Kingdom	14.6		2	19	100
United States	19.9	55	12	30	75

Source: OECD, FAO, International Commission on Irrigation and Drainage.

lies below sea level. Groundwater depletion affects around 15 per cent of the land (OECD, 2003a) and perhaps up to 5 per cent of the surface area has been irreversibly damaged by saltwater intrusion (Oude Essink, 2001). A possible rise in the sea level as a consequence of climate change could lead to further saltwater intrusion over the coming century (Oude Essink and Schaars, 2002).

Natural gas reserves in Dutch fields are among the most important in the EU, accounting for around 20 per cent of supply in the EU natural gas market. Production has declined to three-quarters of its peak in the mid-1970s and expected remaining reserves have also contracted over the same period by

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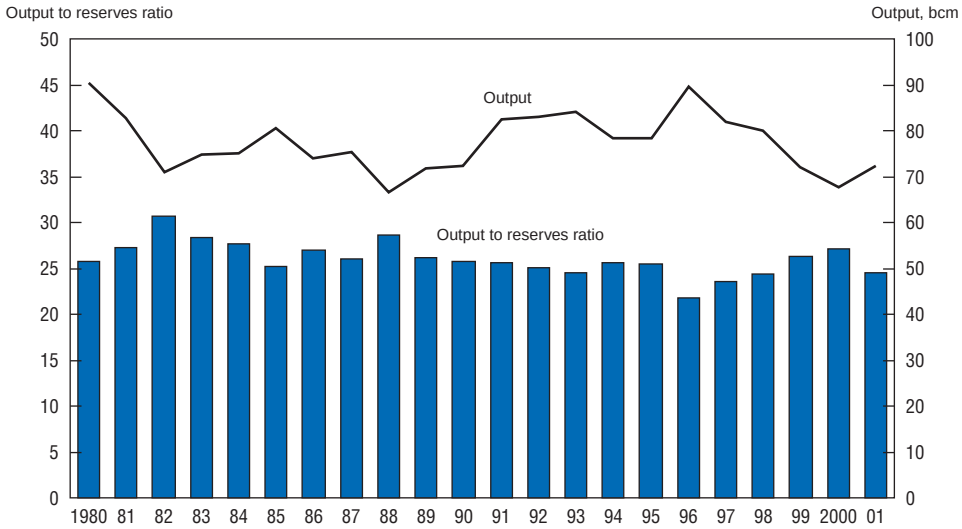
Figure 7.3. **Natural gas output and output to reserves ratio**



Source: Ministry of Economic Affairs (2002).

similar magnitude. As a result, the reserves-to-output ratio has remained relatively stable over this period (Figure 7.3). The value of the depletion is small (0.3 per cent of net national product on average in the 1990-99 period) compared with net domestic saving (17.4 per cent).

Policies

Given the considerable importance of preventing water depletion and saltwater intrusion, government policy in this area has evolved over the past decade to meet these challenges. In 1995, the Commission for Integrated Water Management was created, bringing together the various bodies involved, to ensure that policy is co-ordinated in this field. Using a mixture of regulations and subsidies, policy has paid increasing attention to "more space for water" in an attempt to prevent the water table from falling further. Since 2001 projects are subject to a "water test" that evaluates whether the adverse consequences of a project can be avoided or compensated (Netherlands Commission for Integrated Water Management, 2002). In this light, policy has attempted to reverse some of the egregious practices in the past, such as tightening land use regulation to prevent land being drained for agricultural use. A central goal has been to recover one-quarter of the land that had been subject to groundwater depletion through

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support for projects that recreate wetlands and ensure that management of drainage ditches reduce pressure on the water table. However, progress has been slow in redressing groundwater depletion, with only a small fraction of land so far recovered (OECD, 2003a).

As groundwater depletion and saltwater intrusion presents an extremely serious threat to land adjacent to coastal waters, the national and regional authorities have introduced groundwater abstraction taxes in an attempt to prevent over-abstraction from aquifers. The rate for the national groundwater abstraction tax is € 0.17 per cubic metre. The rate of provincial levy on groundwater abstraction varies across the country, but is generally much lower, yielding one-eighth of the revenue of the national tax. In part, increasing the relative price of groundwater to surface water encourages household water suppliers and industrial users to make greater use of the latter source. The incentives for farmers to change their water sources are much weaker. Only around 2 per cent of farmers are liable to the national tax and many farmers also avoid paying the lower provincial tax (Hellegers *et al.*, 2001).[15] As a result of continuing pressures, major capital investment has been made to replenish groundwater in areas where pressure is greatest. These can include pumping treated surface water to replenish the depleted aquifer and relocating the source of water abstraction to an area not suffering from groundwater depletion. However, responding to these pressures requires consideration of the likely high costs of transporting water over long distances and the administrative burden of monitoring many small extraction sources.

The government influences natural gas extraction through taxation and regulation. The revenues from natural gas amounted to between € 5 to € 10 billion in the late 1990s and early 2000s (around 1 to 2 per cent of GDP). These revenues are composed of royalty payments and shares of profits as well as corporate income taxes, and have ensured a comparatively high take of the resource rent by the government (IEA, 2000). The authorities set production ceilings, which are targeted at the optimal rate of extraction. The authorities also give priority to the production of gas from small fields, through obliging Gasunie to guarantee the purchase of gas for a market price if the small fields producer wants to sell. In this context, the main Groningen field is used as a "balancing field" with its output adjusted to allow for fluctuations in gas demand and for production from small fields.

The "small fields policy" including the balancing role of Groningen has encouraged exploration activity and has brought more reserves on stream. An important consideration for the authorities has been that the marginal cost of gas from the small fields would be higher if they were only exploited in the future as it would be necessary to re-build pipelines.[16] In addition, small field reserves add to security of supply and a shift of production from the Netherlands may incur environmental externalities at the global level, notably from pipeline losses and the energy used in transporting gas over long distances. This has meant that high

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cost reserves (small fields) are being exploited first, leaving low-cost reserves (Groningen) in the ground.[17] Normally, such an approach would reduce the net present value of resource rents. With the ongoing liberalisation and development of the European gas market various market participants could perform the balancing role hitherto played by the Groningen field as part of the small-fields policy. Rising interconnection capacity is reducing barriers to competition and initiatives facilitating the emerging European gas market are developing (for example, markets developing at the natural gas hubs at Zebrugge and Groningen).[18] As a result, the government should evaluate the net present value of gas resource rents taking these factors into account so as to be able to decide whether or not it is worth maintaining the small-fields policy.

Unlike Norway, the Netherlands does not have an explicit fund to ensure that future generations also benefit from current exploitation of hydrocarbon reserves. Instead, a substantial proportion of the revenues from the commercial exploitation of gas reserves (41.5 per cent) is hypothecated to finance major infrastructure projects and education (F*onds* E*conomische* S*tructuur*). Whether this policy ensures that future generations in fact benefit from the gas revenues depends on how the remainder of government spending reacts to the use of the hypothetical revenue. More generally, a sustainable fixed policy – meaning that government programmes can be financed going forward at constant tax rates and with stable public debt to GDP in the long run – ensures that natural resource rents are shared with future generations. In this regard, putting public finances on a sustainable path is a major policy goal of the government. A series of consolidation packages has been announced that is aimed at increasing the structural budget balance to 0.5 per cent by 2007, which the authorities judge to be sufficient to put public finances on a sustainable path (see Chapter 3).

Conclusions

Given the threat of saltwater intrusion any overexploitation of aquifers can result in very serious consequences for the land. In this context, the authorities should ensure that all abstractions of groundwater incur the groundwater abstraction charges levied by the central and provincial governments, which should take into account the full cost of the associated externalities, unless the administration costs outweigh the benefits. In the area of resource depletion, the government has consistently favoured the production of high-cost reserves before low-cost reserves as it considers that the factors outlined above support such a policy. The authorities should evaluate the net present value of different resource management approaches to assess whether or not the small-fields policy should be maintained in the future. After a long period when gas revenues were used to finance immediate consumption, efforts to place fiscal policy on a more sustainable basis are welcomed.

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Notes

1. Early results show that abatement costs associated with international measures are low compared with domestic ones. For example, abatement costs reported from the World Bank Carbon Fund are estimated at between $11 and $18 per tonne of carbon (VROM, 2002).
2. The government did not apply the same tax rates for energy intensive users because it felt that the impact of the energy tax could not be offset for these companies by a reduction of other taxes or through rebates owing to EU-wide state aid rules.
3. For example, CBO (2001) estimate that fewer than 20 per cent of allowances are required to be grandfathered to protect shareholder value in estimates for the United States. Quirion (2003) estimates that in the EU the iron and steel sector would only require 10 per cent of allowances grandfathered to offset the impact on profits.
4. A production subsidy of € 0.02 per kWh and tax break of € 0.06 per kWh were paid, regardless of where the production occurred, though foreign small hydro electricity was not eligible for the tax break. In the second half of 2003, part of the tax breaks were ended and replaced by a subsidy scheme, which are positively related to the costs of the production of the type of renewable energy.
5. The electricity market is open to competition from third party suppliers.
6. This estimate is made on the basis that renewable energy displaces a gas-fired power station with characteristics similar to those in the ExternE reports for Netherlands and United Kingdom.
7. The exporter also received the production subsidy in 2002.
8. The Swedish premium is equivalent to a carbon abatement cost of € 43 per tonne of carbon dioxide, if the electricity is produced from gas.
9. Agricultural emissions dropped by 8 per cent between 1990 and 2000.
10. The vehicle fuel tax is around € 19 per tonne of carbon for both diesel and gasoline. The implicit tax on carbon inherent in other excise duties raises the tax rate per tonne of carbon to over € 400 for diesel and € 900 for gasoline.
11. The OSPAR convention requires a halving of pollution loads flowing into the North Sea that contribute to eutrophication. Agriculture has been responsible for limited progress in reducing nitrate loads.
12. The Water Boards are locally elected bodies responsible for surface water quality and management and flood protection. These bodies have been in existence since the 13th century.
13. The central government is responsible for water quality in the so called “state waters”: major water bodies and rivers flowing through the Netherlands.

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14. Prior to this, a tax on phosphorus was levied. In order to stimulate lower nutrients excretions, farmers could lower the tax rate they faced by proving the nitrogen and phosphate content of their feed was lower than average (OECD, 2003b).
15. Exemption from the provincial tax is available if the pumping facilities do not exceed 10 cubic metres per second and total annual abstractions are less than one million cubic metres.
16. The effects of gas market liberalisation and lower natural gas prices in the late 1990s renewed fears that with depressed exploration activity not all the reserves in smaller fields would be exploited. The government responded *inter alia* by exempting small fields from the royalties' regime, though not from profit sharing agreements, in order to encourage the exploitation of these fields further (IEA, 2000).
17. In 1998, marginal extraction costs were 0.45 eurocents per cubic metre in the Groningen field against 1.8 and 3.2 eurocents for onshore and offshore small fields, respectively (Correlje, 2000).
18. A spot market has already developed at Zebrugge and is linked to forwards markets in the United Kingdom, offering a liquid market for small producers to sell output and hedge risks. Excluding periods of outage of the interconnector linking Belgium to the United Kingdom, the International Petroleum Exchange natural gas futures contracts (the United Kingdom's National Balancing Point, NBP) can be used as an effective hedge for Zeebrugge price exposure. The correlation between the NBP and Zeebrugge price movements is 0.9.

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Glossary of acronyms

ABP	Public Sector Pension Fund
ACTAL	Advisory Board on the Screening of Administrative Costs
ALM	Asset liability management
AOW	Public Pension Scheme Act (Algemene Ouderdomswet)
APB	Average production worker
AWBZ	Long-term Health Care Insurance Act (Algemene Wet Bijzondere Ziektekosten)
CBS	National statistical office (Centraal Bureau voor de Statistiek)
CHP	Combined Heat and Power
CPB	Netherlands Bureau of Economic Policy Analysis (Centraal Planbureau)
CWI	Centres for Work and Income (Centrum voor Werk en Inkomen)
DB	Disability benefit
DNB	Dutch National Bank (De Nederlandsche Bank)
DTC	Diagnosis Treatment Combinations system
EGB	Separate Work Injury Insurance (Extra Garantieregeling Beroepsrisico's)
EPL	Employment projection legislation
EPO	European Patent Office
EU	European Union
FDI	Foreign direct investment
GDP	Gross domestic product
GHG	Greenhouse gas
GP	General Practitioner
HBO	Higher vocational education (*Hoger beroepsonderwijs*)
IAS	International Accounting Standards
ICT	Information and communication technology
IEA	International Energy Agency
ILO	International Labour Organisation
LF	Labour force
LTI	Leading Technological Institute
MBO	Upper secondary vocational education (*Middelbaar beroepsonderwijs*)
MEV	Macroeconomic Projections (*Macro Economische Verkenning*)
MFP	Multi-factor productivity
MINAS	Minimum Accounting System
MinOCW	Ministry of Education, Culture and Science (*Ministerie van Onderweis, Cultuur en Wetenschap*)
MinEZ	Ministry of Economic Affairs (Ministerie van Economische Zaken)
MinSZW	Ministry of Social Affairs and Employment (*Ministerie voor Sociale Zaken en Werkgelegenheid*)
NAIRU	Non-accelerating inflation rate of unemployment

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NMa	Dutch Competition Agency (Nederlandse Mededingingsagentschap)
OAP	Old-age pension
OD	Occupational disease
PAYG	Pay-as-you-go
PBB	Performance-based budgeting
PGGM	Health Sector Pension Fund
PIA	Public Purchasing and Public Procurement Agency
PSW	Pension and Savings Fund Act (*Pensioen- en Spaarfondsenwet*)
PVK	Pensions and Insurance Supervisory Authority (*Pensioen- en Verzekeringskamer*)
SB	Sickness benefit
SER	Social and Economic Council (*Sociaal-Economische Raad*)
SGP	Stability and Growth Pact
SME	Small-and-medium-sized enterprise
UB	Unemployment benefit
UMTS	Universal Mobile Telecommunications Systems
UWV	UB and DB agency (Uitvoeringsinstituut Werknemersverzekeringen)
VMBO	Pre-vocational secondary education (*Voorbereidend middelbaar beroepsonderwijs*)
VROM	Netherlands Ministry of Spatial Planning, Housing and the Environment (*Ministerie van Volkshuisvesting, Ruimtelijke Ordening en Milieubeheer*)
VUT	Voluntary Early Retirement Scheme (*Vervroegde Uittreding*)
WAO	Disability Insurance Act (Wet op de Arbeidsongeschiktheidsverzekering)
WAZ	Disability Insurance Act for Self-employed (Wet Arbeidsongeschiktheidsvoorziening zelfstandigen)
WGA	Law on labour participation of the partly disabled (*Werkhervatting Gedeeltelijk Arbeidsgeschikten*)
WI	Work injury
ZBO	Semi-independent public entity (*Zelfstandig bestuursorgan*)

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Annex A

Follow-up on OECD recommendations for structural reform since 2002

A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
The labour market		
1. *Reforming the disability scheme*		
– Implement the new system of screening disability claims and start reintegration already during the first year of sick leave. – Introduce a re-examination process for all benefit recipients based on more stringent requirements.	– The Gatekeeper Act came into force, stipulating stronger mutual reintegration obligations during the sickness benefit (SB) period. UWV may prolong the SB period if efforts have been insufficient. – The SB period has been extended to two years and stricter criteria for still feasible jobs/functions have taken effect. The stock of beneficiaries is being re-examined.	– The inflow into the new WAO should be limited to fully and permanently disabled persons, as planned. – Implement the reform of the partial disability scheme (WGA, planned for 2006), with stricter eligibility criteria, a lower second-stage DB for non-working persons and freedom of choice between private and public insurance for employers. – The UWV should further enhance its efficiency as a gatekeeper. Its performance targets should be enforced through a system of benefits and sanctions.
2. *Promoting work incentives and opportunities for unemployed persons*		
– To reduce unemployment traps, replace income-related special subsidies with higher tax credits and enforce reintegration requirements strictly.	– The general earned-income tax credit and the working-spouse credit are being raised and all supplemental subsidy schemes put under central regulation.	– Reduce the replacement rate for social benefit recipients, especially those without children, and further increase earned-income tax credits.
– Increase the role of private placement agencies in the activation process.	– Municipalities have got full responsibility over their social benefit budget. They have to contract out 70 per cent of their reintegration and budget pay back unused amounts.	– To stimulate competition among providers, UWV and municipalities should contract out more homogenous groups and intensify *ex post* evaluation. Payment of unemployment benefits (UB) should also be contracted out to boost providers' reintegration incentives.
– New proposal.		– Shorten the duration of UB and base it on the actual employment record. The unemployment assistance should be abolished, as announced. – Increase the minimum work record required for access to UB to 39 out of the past 52 weeks and abolish the short-term benefit.

A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
3. *Closing routes to early retirement*		
– Speed up the process of phasing out fiscal incentives for early retirement.	– The agreement to phase out existing early-retirement PAYG schemes (VUT) by 2022 still exists but legislation has been delayed.	– The recommendation is maintained. End the VUT scheme by 2006, as announced, to raise the employment rate of older workers more significantly. In this regard it needs to be ensured that the "lifecycle" scheme currently under discussion is not used as a new route to early retirement.
– Reinstate normal job search requirements for unemployed 57½ and older.	– Job search requirements for older unemployed have been reinstated.	– The job-search requirements should be enforced. To limit the number of persons falling on welfare benefits, contributions could be lowered for firms keeping older employees.
4. *Stimulating more labour inputs by women*		
– Create adequate childcare facilities.	– The number of childcare facilities has doubled from 1998 to 2003. As prices have also increased, demand bottlenecks have almost disappeared.	– Implement the unified subsidy to working parents paying for childcare and replace municipal zoning rules with national regulation as planned. The latter should allow for a diversified supply of childcare facilities, making childcare more affordable.
5. *Employment protection legislation*		
– New proposal.		Limit redundancy pay to 12 months of salary as intended. Reduce the weight of third parties in dismissals *via* the Labour Office.
Education and innovation policies		
– Make the educational and vocational system more demand-oriented.	– Job qualification profiles have become more demand-oriented.	– No further action needed.
	– In tertiary education recognition of partial diplomas and competencies obtained outside school/university has been facilitated and a bachelor-master structure introduced. Results of performance assessments are published.	– Give universities more freedom in raising fees to foster competition. Extend the system of student loans repayable at income-dependent rates. Public funding should become still more dependent on performance.
– New proposal.	– The government has stepped up R&D tax-expenditure schemes for science-industry or private-public co-operation to speed up the knowledge transfer.	– Closely monitor the effects of this expenditure. – Introduce TechnoPartner, the generic tax facility for innovative start-ups co-operating with universities.

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A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
	– Direct support to specific research projects involving private-public co-operation has been increased.	– Evaluate the comparative performance of generic versus specific R&D support schemes. – Target public support to start-up finance to innovative firms and reduce the administrative burden for start-ups.
Product markets		
1. *Competition law and related regulation*		
– Make the competition authority (NMa) fully independent.	– A proposal is before Parliament to make the NMa legally independent.	– The proposal should be passed into law soon, increasing the NMa's credibility. – Give the NMa greater investigative powers and sanctioning powers, notably for non-co-operation, as planned and the power to impose structural remedies in appropriate cases.
– Finalise "high-profile" Competition, Deregulation, Legislative Quality (MDW) projects.	– The MDW project was terminated in 2002.	– Examples of results achieved by the MDW project include: opening up markets containing exclusive rights; allowing new forms of competition; and further harmonisation of laws and regulations and updating outdated rules. – The authorities should re-examine laws and regulations governing the liberal professions with a view to eliminating unwarranted anti-competitive practices, as planned. – Ease planning restrictions so that large-format retail outlets, which have higher productivity, can enter the market or expand. Ensure that local governments are not unduly responsive to incumbent interests in deciding on the location of large retail outlets.
2. *Network industries*		
– Give small and medium-sized electricity users the choice of supplier by 2004 and eliminate infrastructure bottlenecks.	– Small and medium-sized electricity users will have choice of supplier from 1 July 2004. The authorities plan to increase interconnection capacity subject to satisfactory co-operation with networks in neighbouring countries.	– Local government should privatise their retail activities after electricity and gas distribution networks have been split off and their networks after network quality has been safeguarded as these holdings may deter entry and the development of competition in these markets.

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A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
– New proposal.		– In the roll out of broadband internet infrastructure, the government should leave to private parties the responsibility for development and investment and making the associated technological and architectural choices.
– New proposal.		– The government should end the control it maintains over KPN and the postal operator (TPG) through its golden shares.
– Inject more market forces in public transport.	– Ten regional railways services have been decentralised and another 16 soon will be. Provinces have to tender these services. So far, six services have been tendered. Legislation covering railways, which enters into force in 2004, requires that all concessions must be tendered except that for the national railway network. Many bus services have also been tendered.	– Ensure that the ticketing system, which is owned by the Dutch railways (NS) does not act as a barrier to the development of competition on regional services.
3. *Consumer protection*		
– New proposal.		– Sectoral Disputes Committees should exist in more sectors and should settle disputes more quickly. The authorities should also consider creating a small claims tribunal.
4. *Housing market*		
– New proposal		– Ease tough zoning restrictions on residential building sites and simplify building regulations to increase the price elasticity of housing supply. – Phase out tax subsidies for owner-occupied housing by shifting incremental housing purchases, to the third box in the personal income tax system; in this way, tax subsidies on existing mortgage loans would be granted further.

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A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
The health care system		
– Pursue the plan for a full overhaul of the healthcare system		– The authorities aim to increase efficiency and responsiveness to patient demands by moving to a managed competition system by 2006. The government should develop quality indicators to help consumers to put pressure on insurers to deliver better services. The costs to consumers of switching insurers should be as low as possible.
– Define standard procedures and treatment in health care and rely on a DRG-based (or therapy-based) payment system for in-patients.	– The Diagnosis Treatment Combinations system (DTC, which is broader than DRG because it includes related specialist outpatient services) has recently been introduced.	– Assignment of DTCs to patients by specialists should be closely monitored to reduce the risk of creep (the deliberate and systematic shift in the reported case mix in order to improve reimbursement).
Financial markets		
1. *Corporate governance*		
– Increase transparency in corporate governance and strengthen shareholder rights.	– The Tabaksblat code on corporate governance entered into effect.	– Implement proposed legislation which will anchor the Tabaksblat code by a requirement to comply or explain.
– Implement proposed legislation to remove all obstacles to proxy voting.	– Legislation is still pending.	– Implement legislation which grants certificate holders unlimited and unconditional proxy rights. – Take further action if shareholders prove ineffective in moderating remuneration packages of board members. – Implement proposed legislation to extend the tasks of the AFM to include supervision of accountancy agencies and financial statements issued by listed companies.
2. *Pension funds*		
– New proposal.	– Parliament has approved principles of a new regulatory framework for pension fund supervision that will be incorporated in the Pension Act planned for 2006.	– Pension Act should include guidelines on reporting standards and transparency in communication and establish some supervision on reporting.
Public sector		
– New proposal.		– Stick to the target to reduce the administrative burden on firms by one quarter by the end of 2007 and implement all the initiatives as planned.

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A. Previous recommendations	B. Action taken	C. Recommendations for further structural reform
– New proposal.	– Performance based budgeting was introduced in 2002 and performance contracts have been concluded with agencies and ZBOs.	– Invest in the generation of more and better information on performance. – Make more use of benchmarking (*e.g.* police forces). – Reduce the cost of making civil servants and managers redundant. – Increase the accountability of managers by relaxing restrictions on their dismissal.
Sustainable development		
– New proposal.		– Climate change policy should be rebalanced to equalise marginal abatement costs. Specifically, the regulatory energy tax should be changed to an explicit carbon tax for energy users not included in the EU's trading scheme with the rate set at the level of the expected price of an emission permit. The authorities should also ensure that support offered to renewable energy and combined heat and power does not exceed the carbon tax rate.
– New proposal.		– The authorities should reduce the significant costs of the MINAS scheme aimed at lowering nitrogen and phosphorous pollution from agriculture. Allowable nitrogen and phosphorous balances should be tightened for soil types that are more susceptible to run-off.
– New proposal.		– The authorities should ensure that all abstractions of groundwater, including for agriculture, incur the groundwater abstraction charges levied by the central and provincial governments and reflect the full cost of the associated externalities, unless the administrative costs outweigh the benefits.
– New proposal.		– In order to maximise the present value of the surplus extracted from the Netherlands' gas fields, the government should assess the net present value of different resource management approaches to assess whether or not the "small fields" policy should be maintained in the future.

Source: OECD.

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